



Queensland University of Technology
Brisbane Australia

INVERSE KINEMATICS & ROBOT MOTION

Inverse kinematics

- Set of joint angles given the robot tip pose

$$\xi_N = \mathcal{K}(\mathbf{q})$$

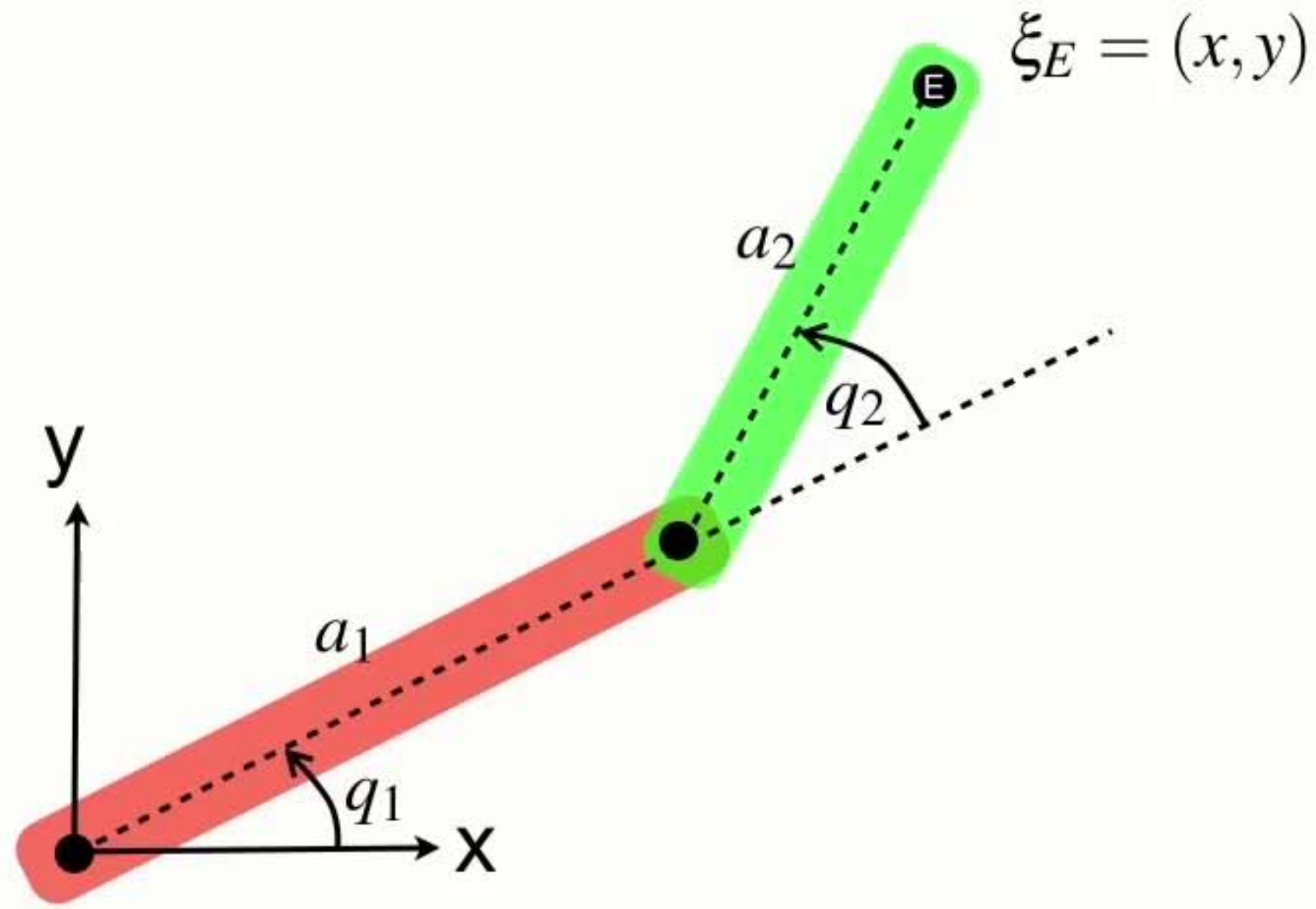
$$\mathbf{q} = \{q_j, j \in [1 \cdots N]\}$$

$$\mathbf{q} = \mathcal{K}^{-1}(\xi_N)$$

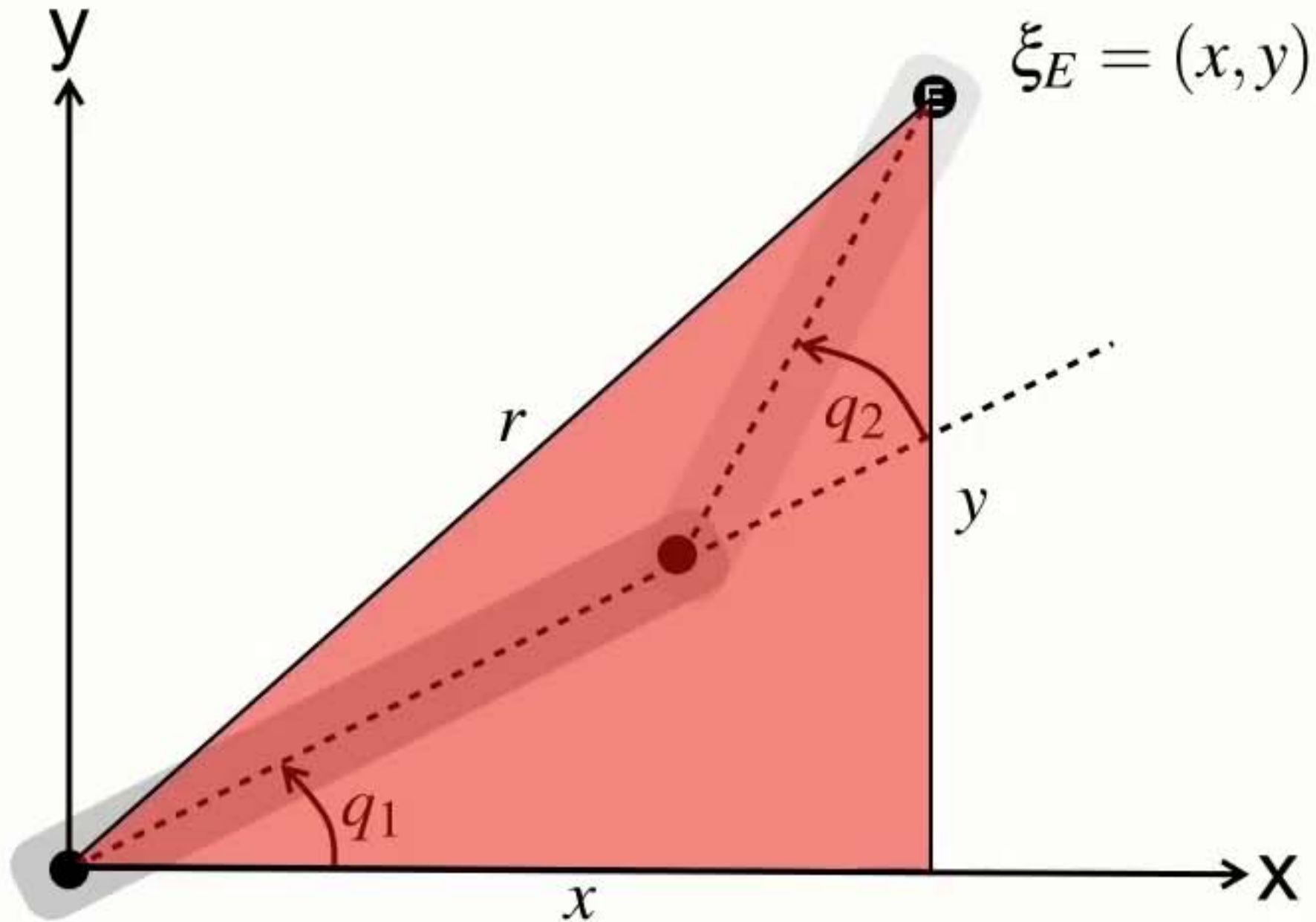
Section 1

Geometric solution for a 2-link planar robot

Tool tip pose



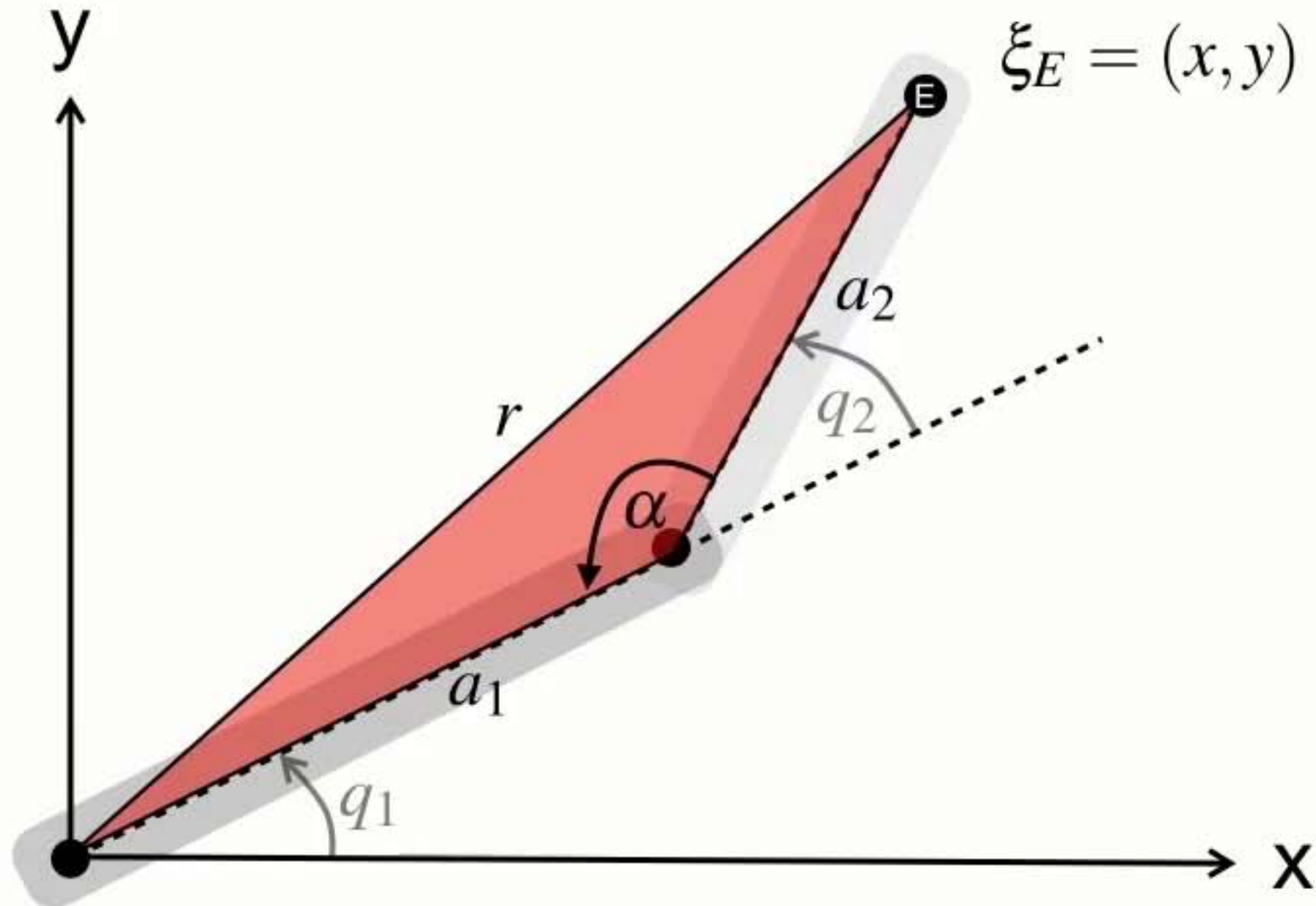
Finding the **joint angles**



$$r^2 = x^2 + y^2$$

■ Apply Pythagoras

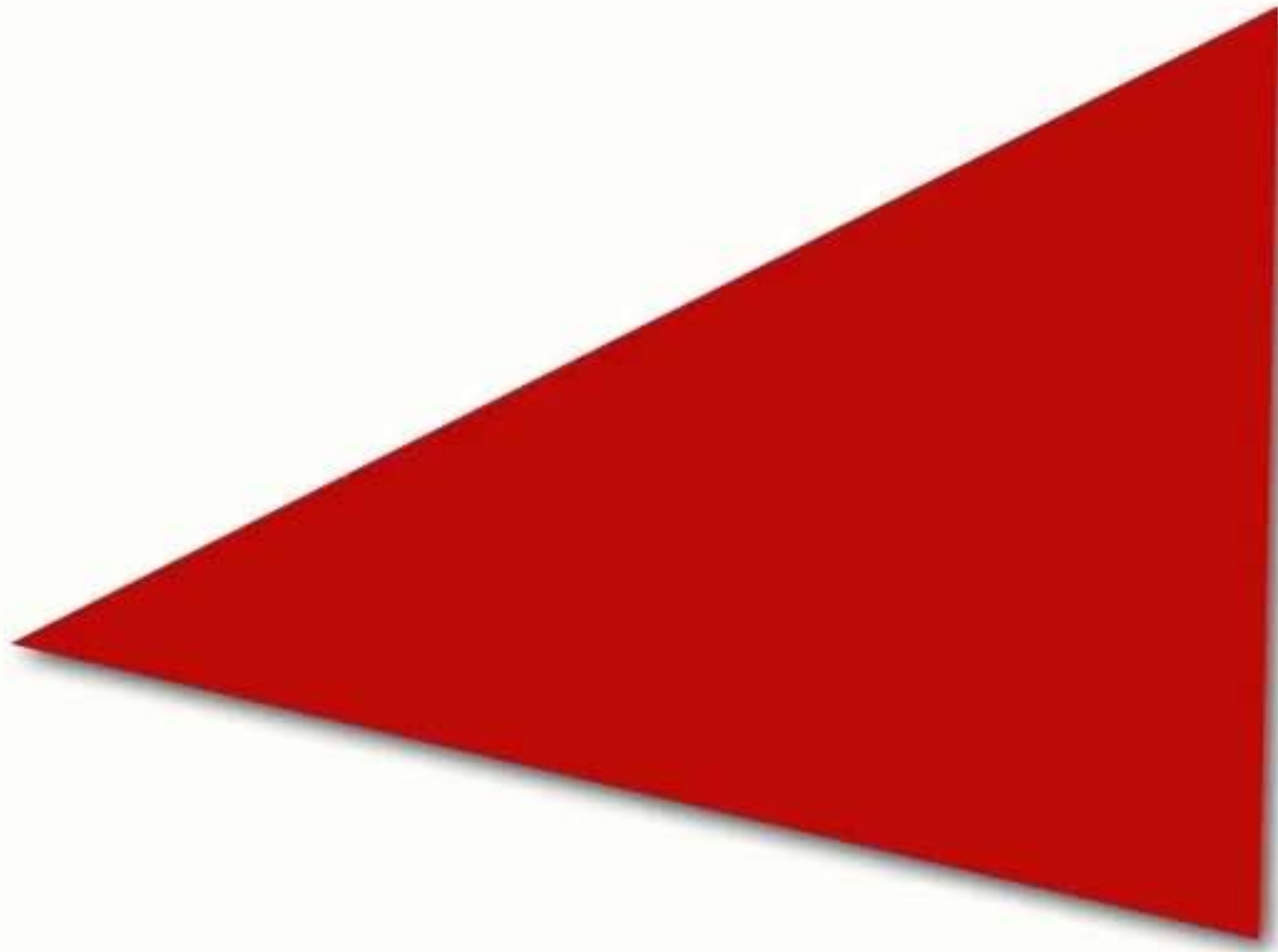
Finding the **joint angles**



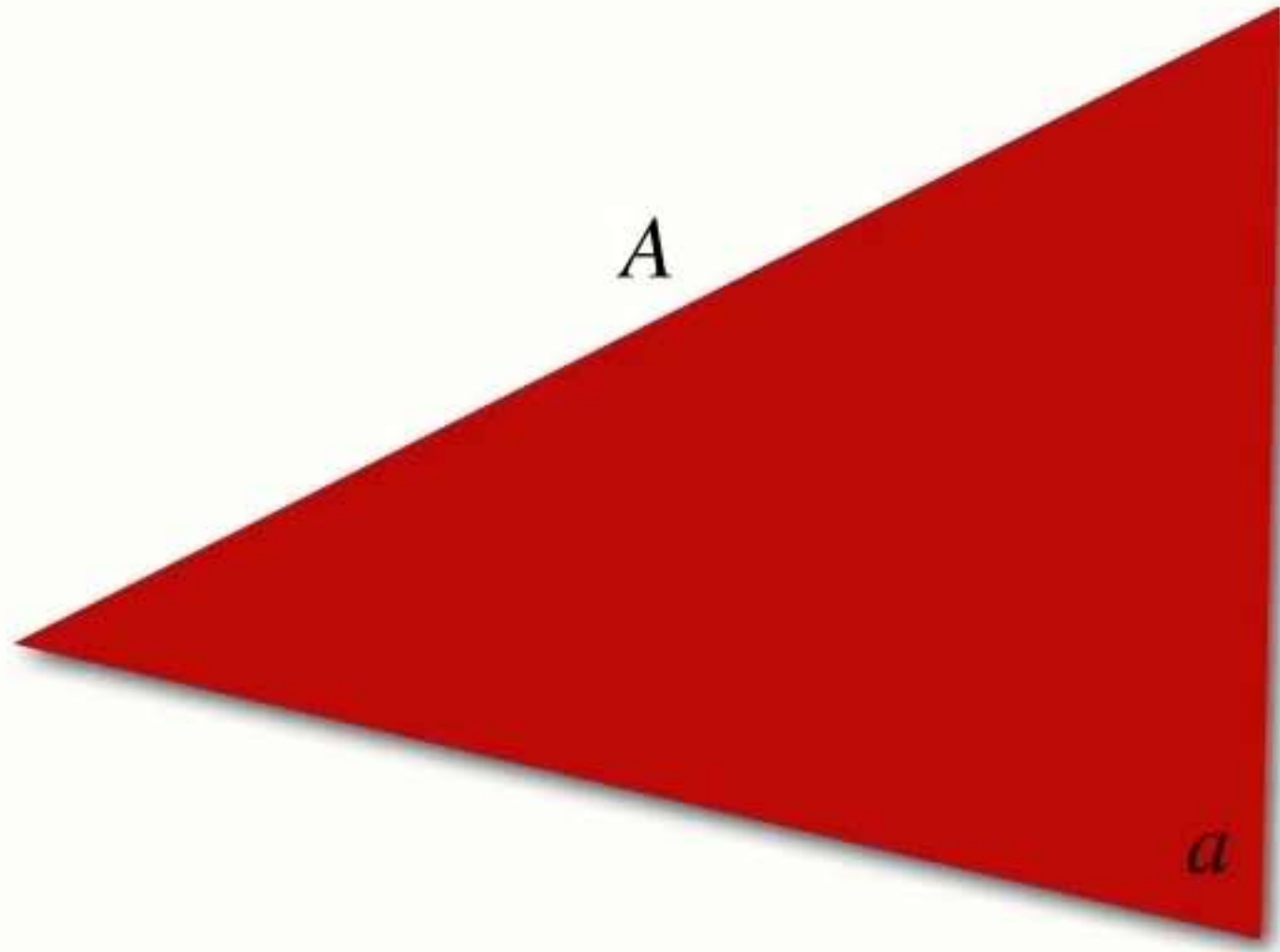
$$r^2 = x^2 + y^2$$

- Apply the cosine rule

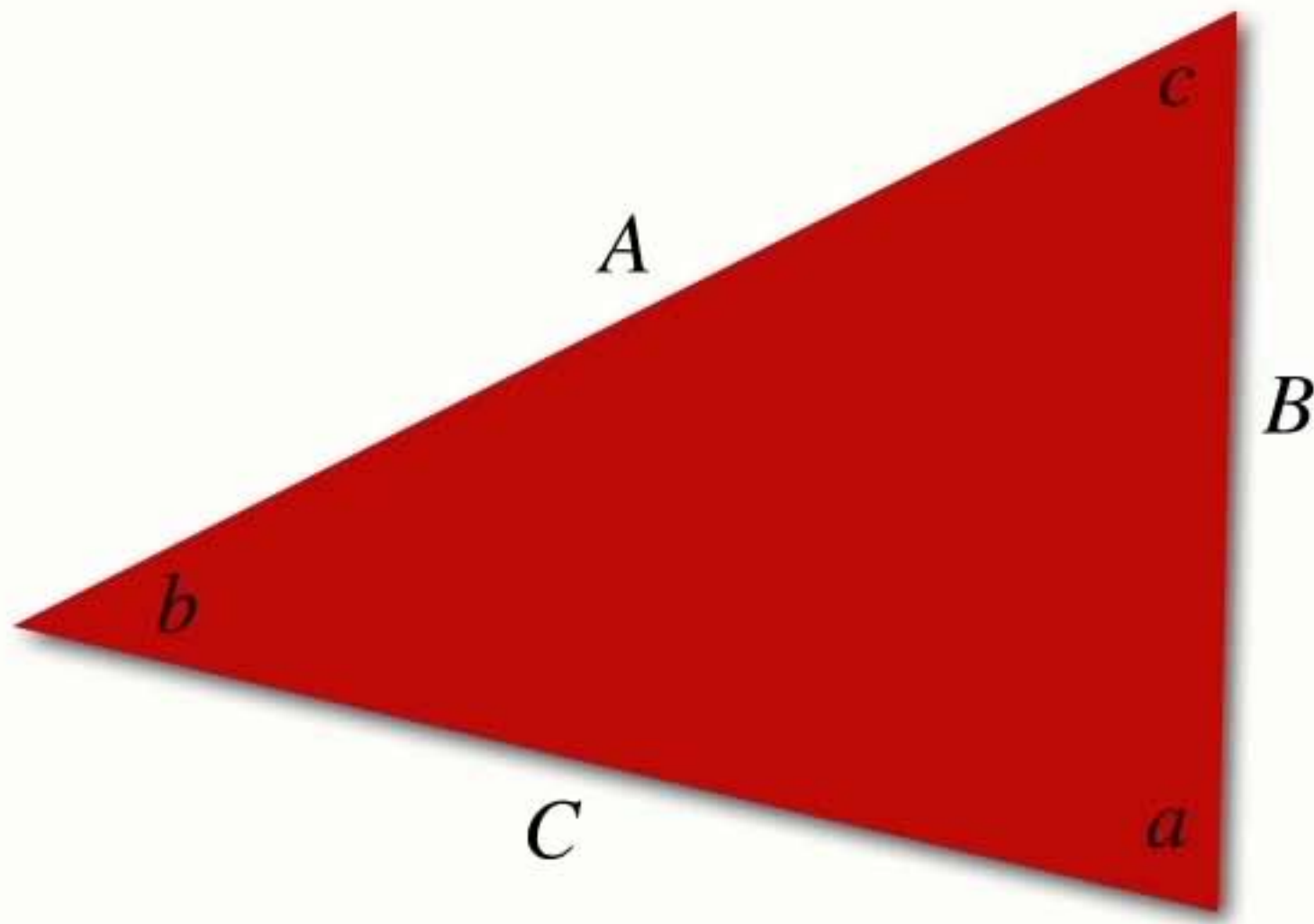
Recap: **the cosine rule**



Recap: **the cosine rule**

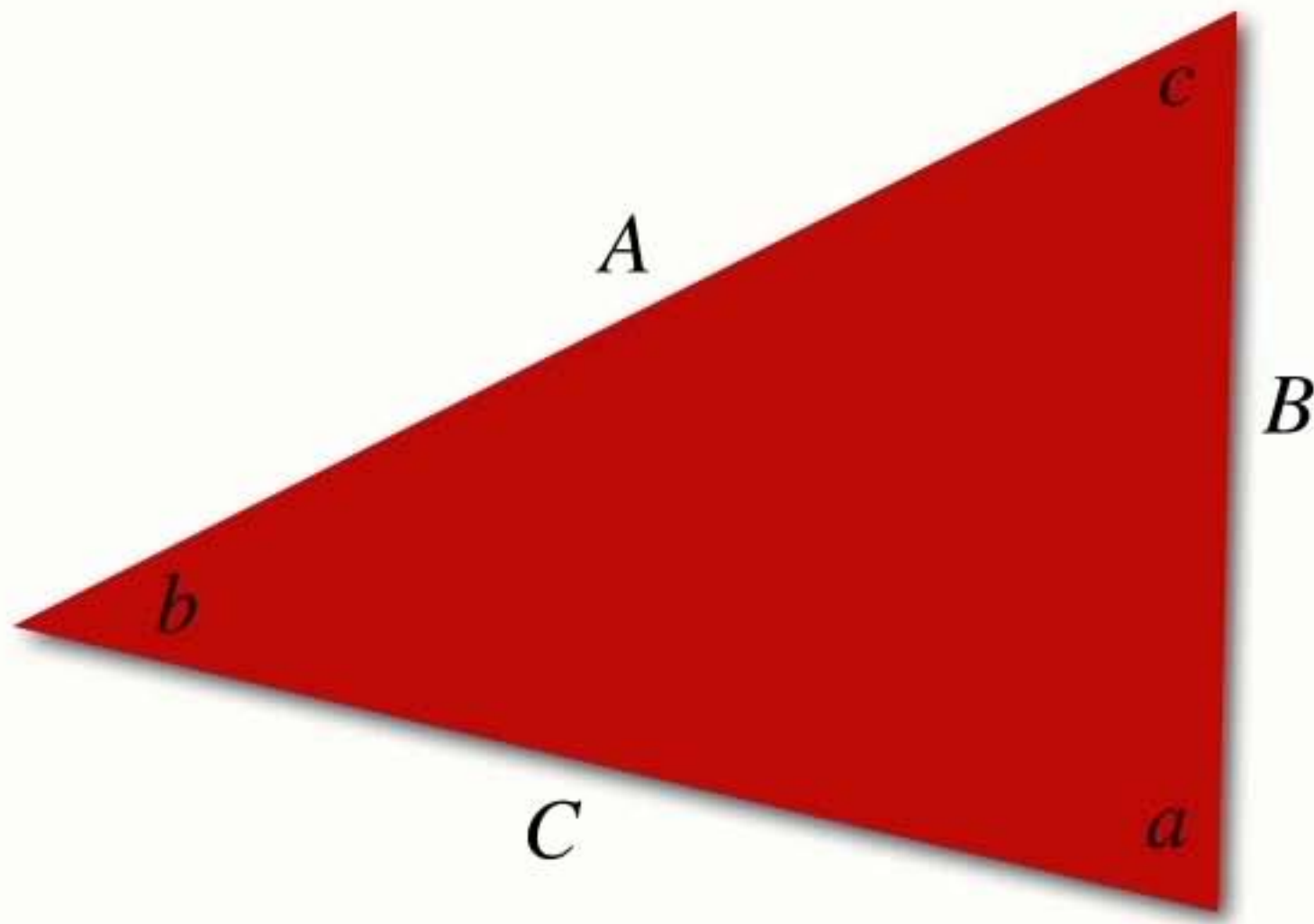


Recap: **the cosine rule**



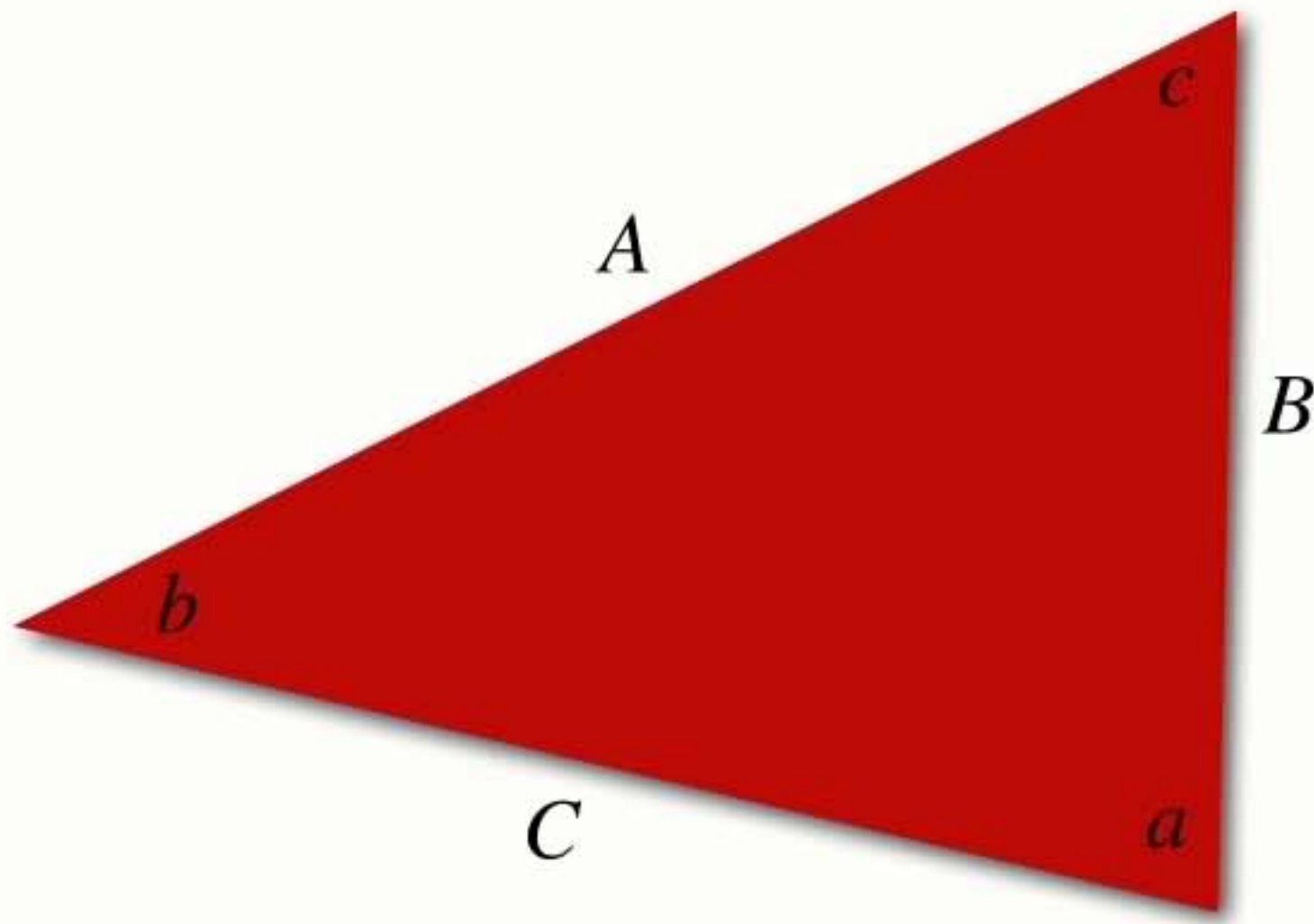
Sides: A, B, C
Angles: a, b, c

Recap: **the cosine rule**



Sides: A, B, C
Angles: a, b, c

Recap: **the cosine rule**

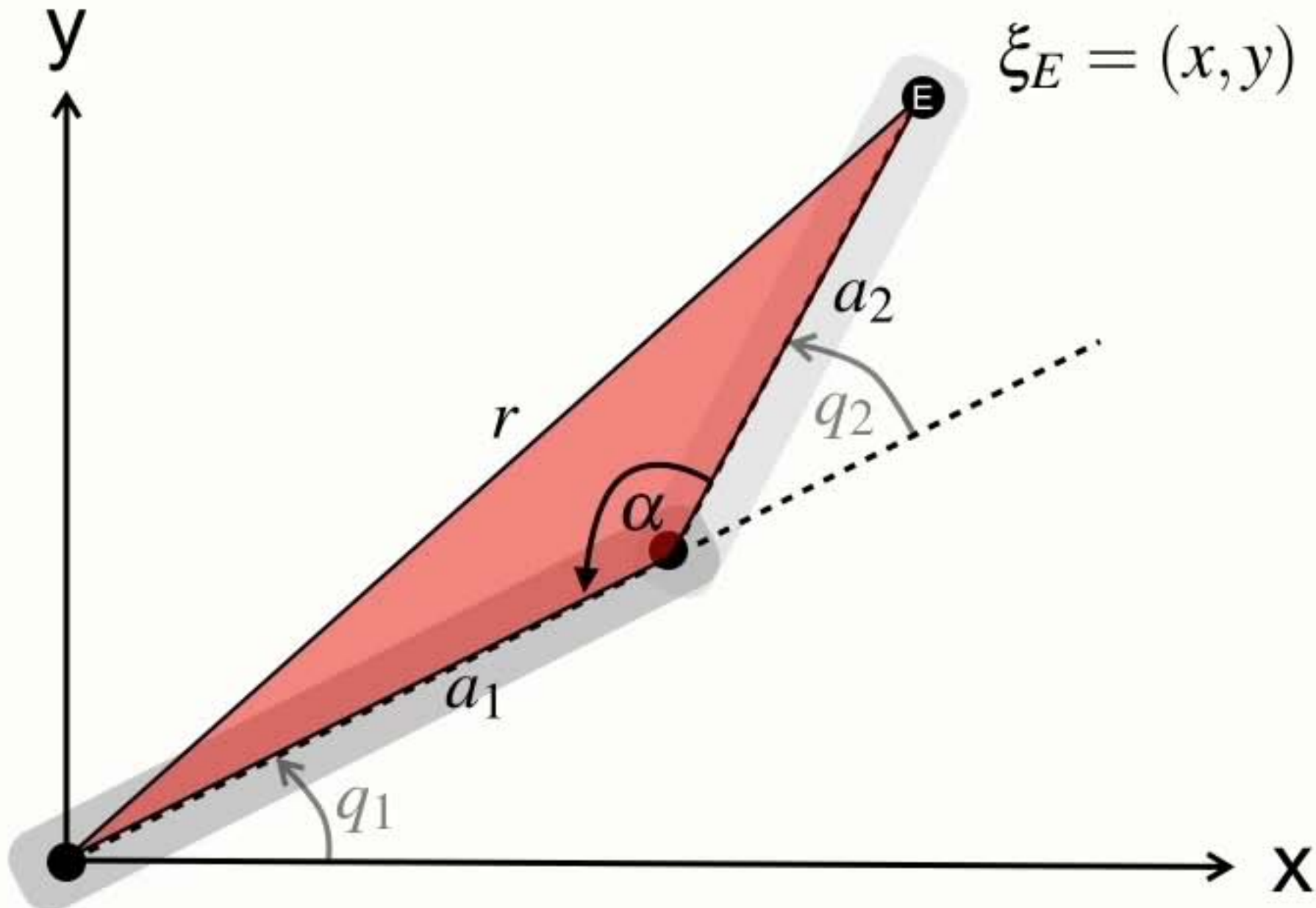


Sides: A, B, C
Angles: a, b, c

$$A^2 = B^2 + C^2 - 2BC \cos a$$

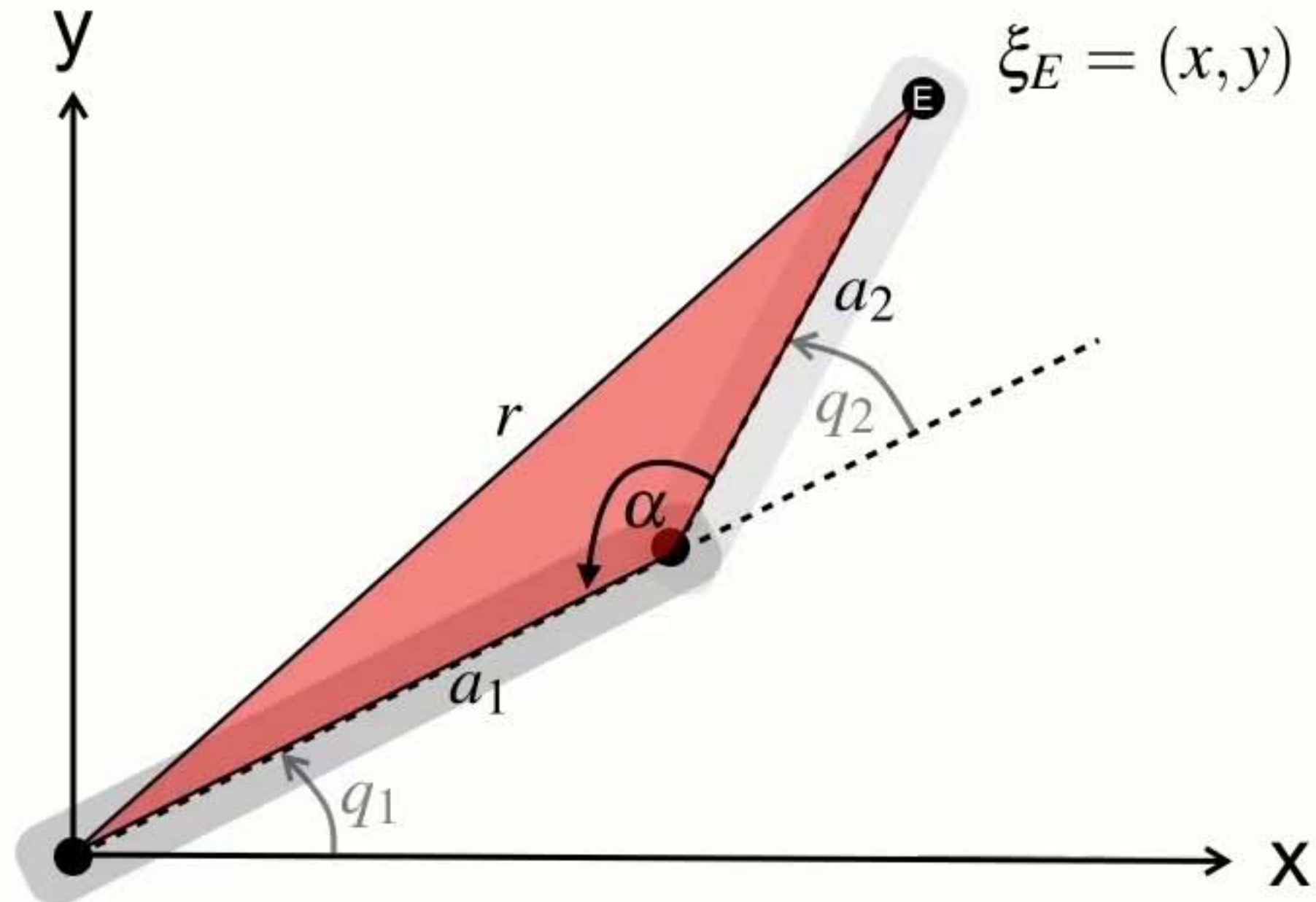
Finding the **joint angles**

$$r^2 = x^2 + y^2$$



■ Apply the cosine rule

Finding the **joint angles**

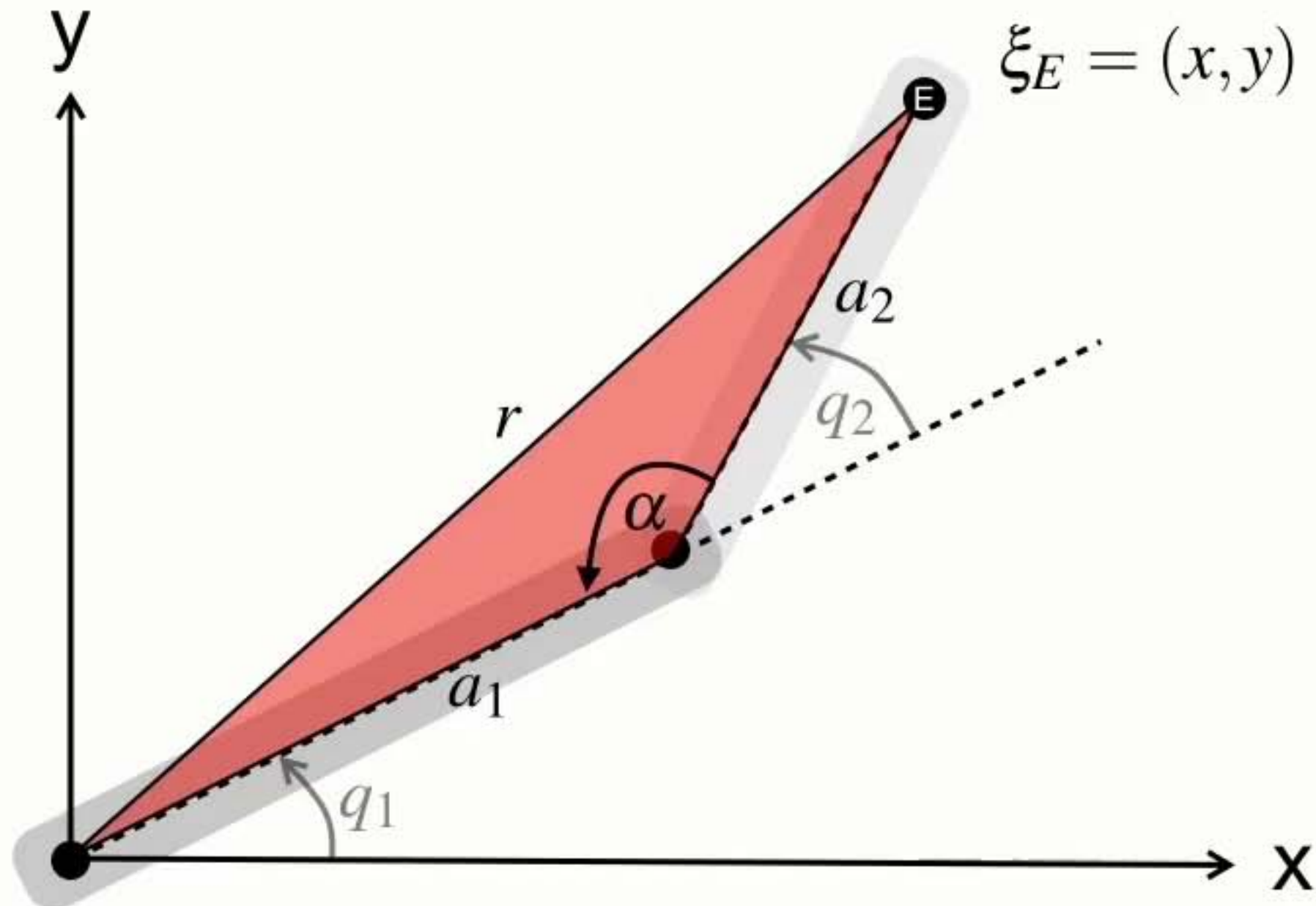


$$r^2 = x^2 + y^2$$

$$r^2 = a_1^2 + a_2^2 - 2a_1a_2 \cos \alpha$$

■ Apply the cosine rule

Finding the **joint angles**



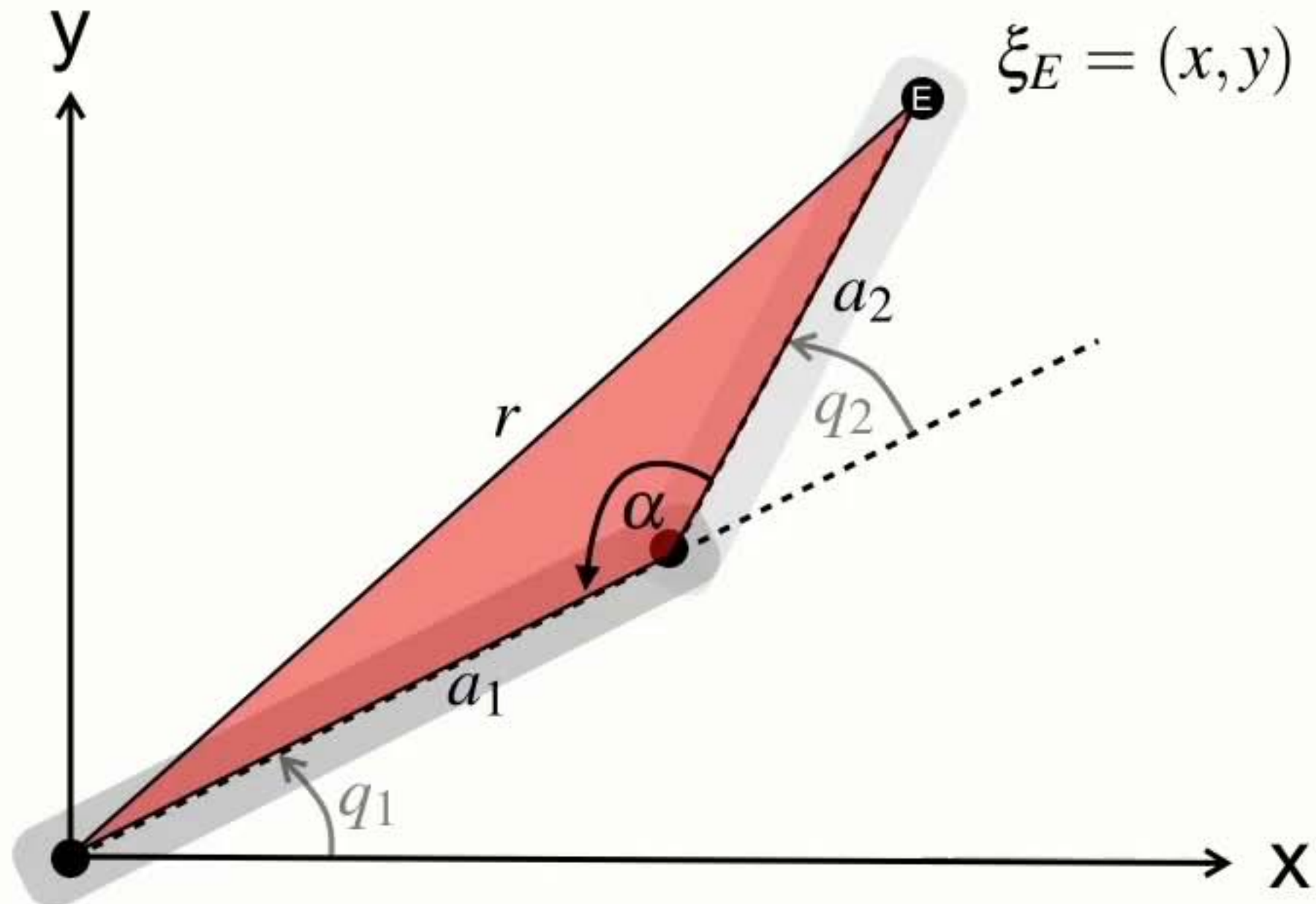
■ Apply the cosine rule

$$r^2 = x^2 + y^2$$

$$r^2 = a_1^2 + a_2^2 - 2a_1a_2 \cos \alpha$$

$$\begin{aligned} \cos \alpha &= \frac{a_1^2 + a_2^2 - r^2}{2a_1a_2} \\ &= \frac{a_1^2 + a_2^2 - x^2 - y^2}{2a_1a_2} \end{aligned}$$

Finding the **joint angles**



■ Apply the cosine rule

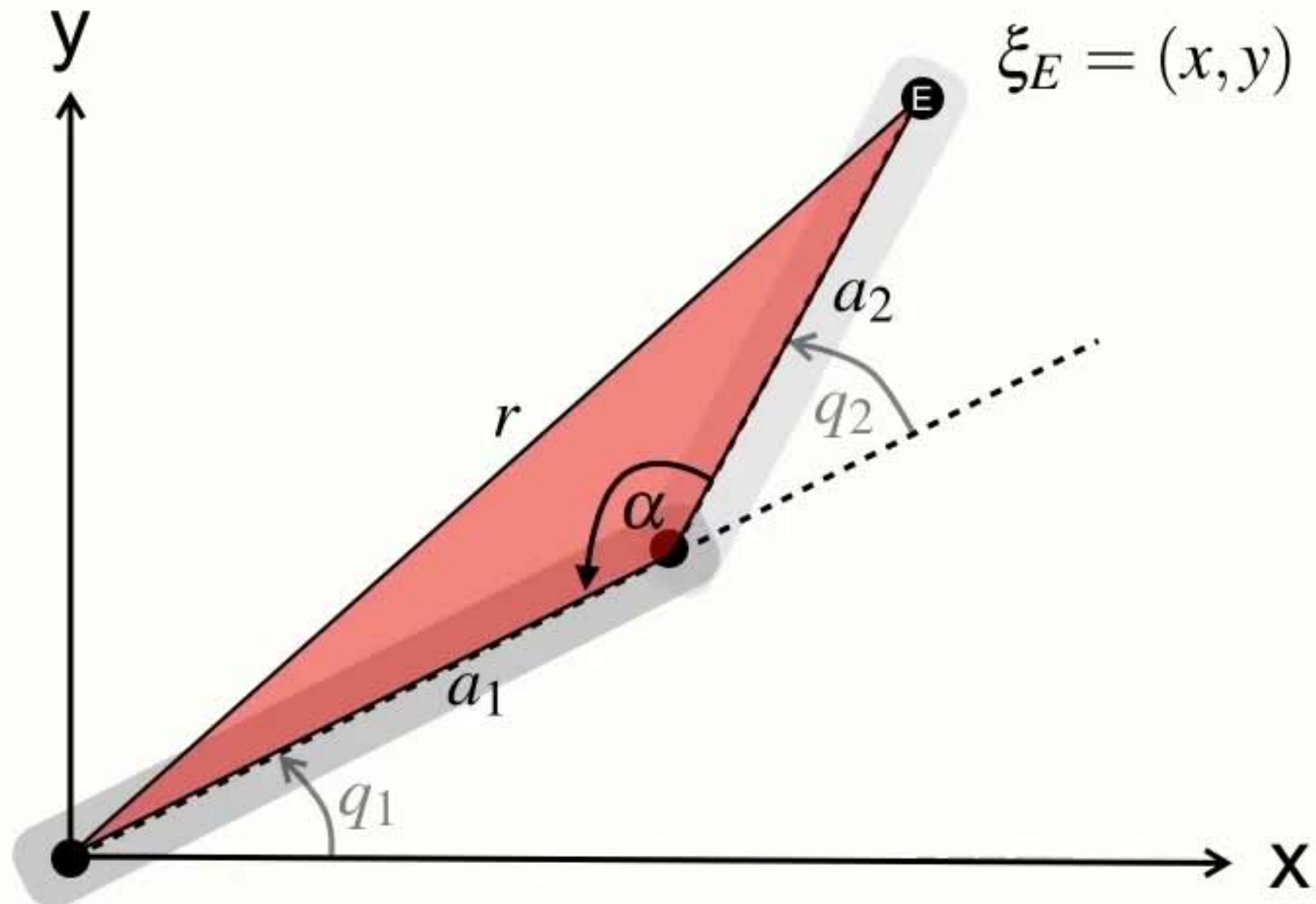
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$$q_2 = \pi - \alpha$$

Finding the **joint angles**



■ Apply the cosine rule

$$r^2 = x^2 + y^2$$

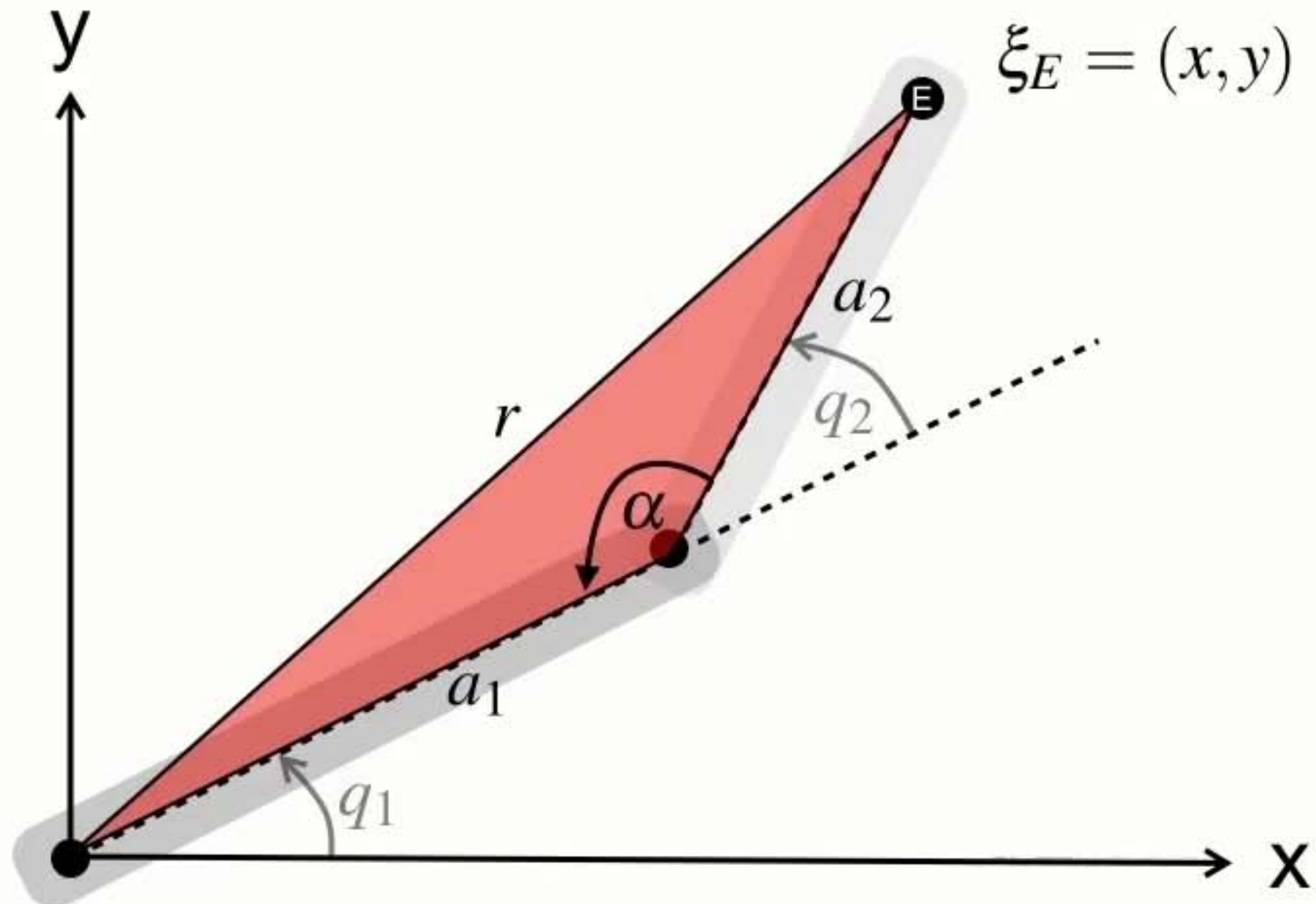
$$r^2 = a_1^2 + a_2^2 - 2a_1a_2 \cos \alpha$$

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$$q_2 = \pi - \alpha$$

$$\cos q_2 = -\cos \alpha$$

Finding the **joint angles**



■ Apply the cosine rule

$$r^2 = x^2 + y^2$$

$$r^2 = a_1^2 + a_2^2 - 2a_1a_2 \cos \alpha$$

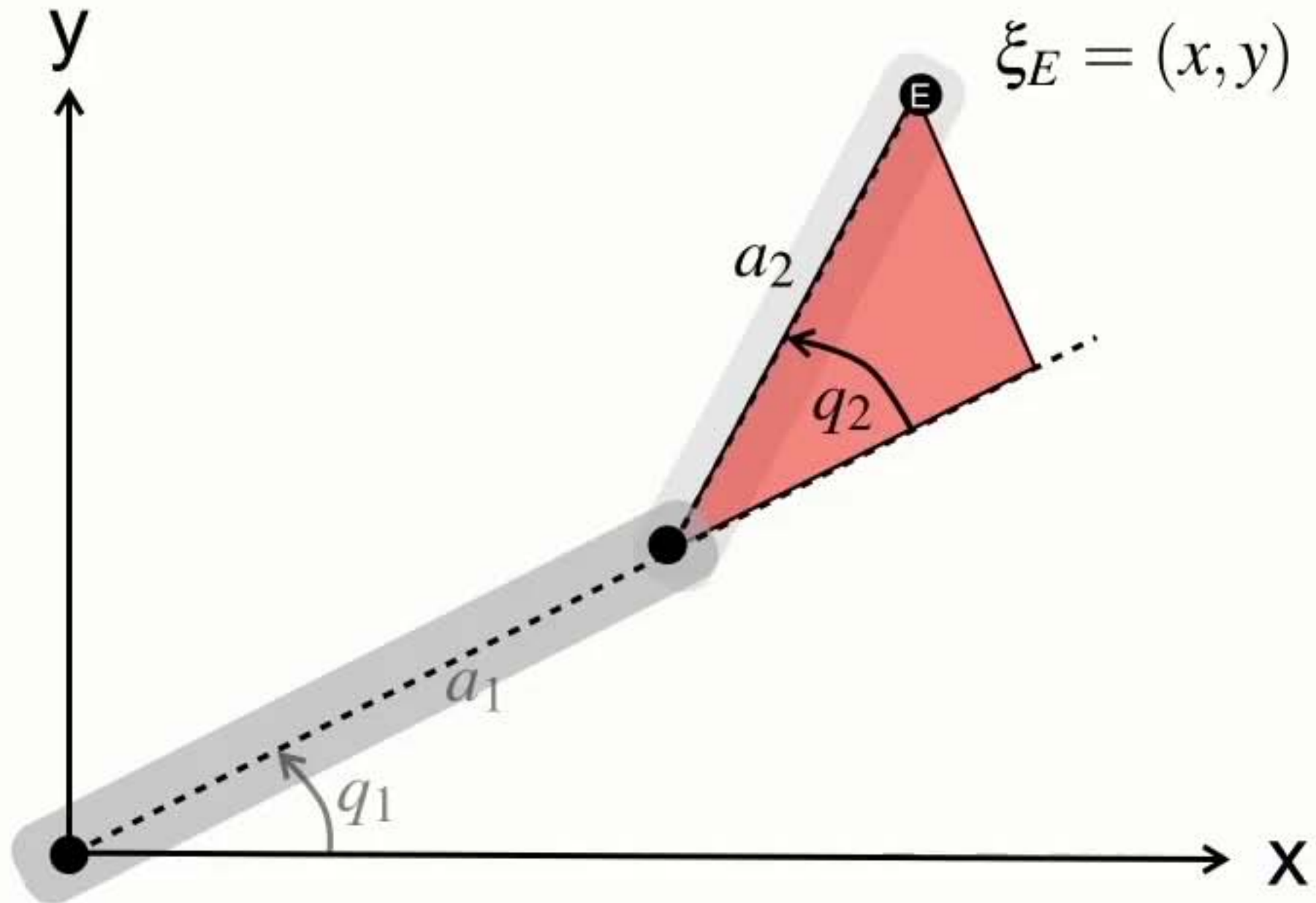
$$\begin{aligned} \cos \alpha &= \frac{a_1^2 + a_2^2 - r^2}{2a_1a_2} \\ &= \frac{a_1^2 + a_2^2 - x^2 - y^2}{2a_1a_2} \end{aligned}$$

$$q_2 = \pi - \alpha$$

$$\cos q_2 = -\cos \alpha$$

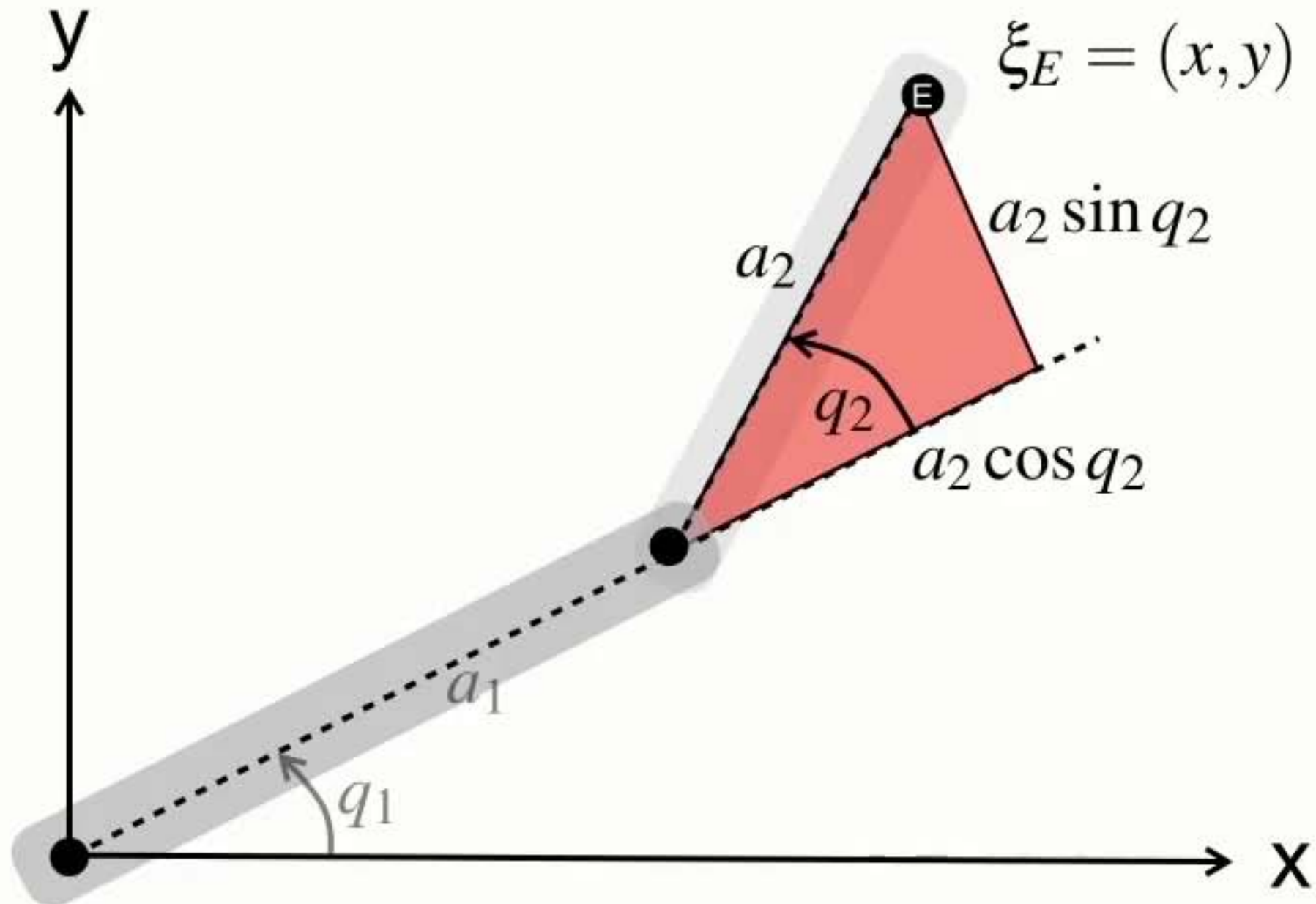
$$\cos q_2 = \frac{x^2 + y^2 - a_1^2 - a_2^2}{2a_1a_2}$$

Finding the **joint angles**



- Apply simple trigonometry

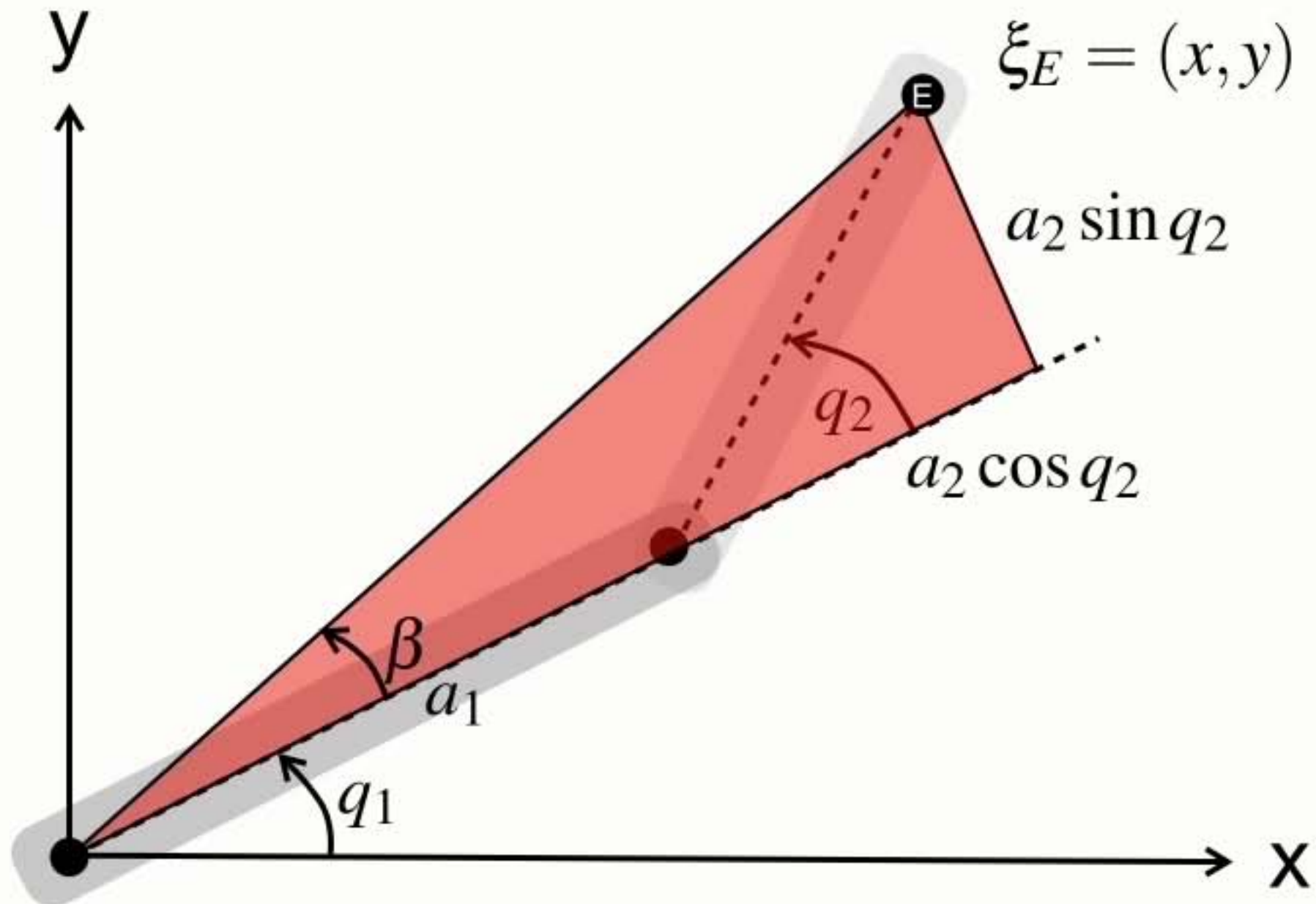
Finding the **joint angles**



$$\sin q_2 = \sqrt{1 - \cos^2 q_2}$$

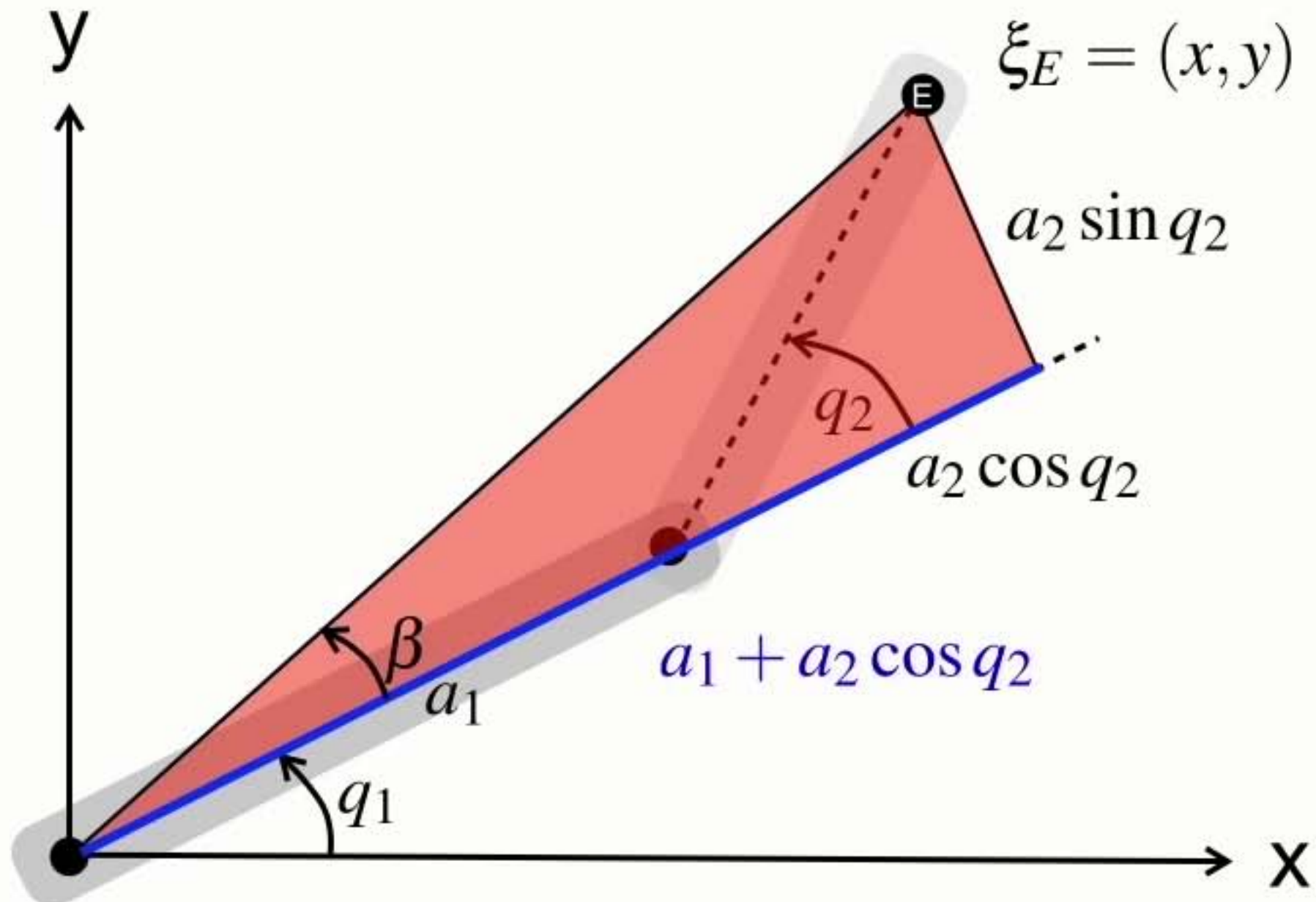
- Apply simple trigonometry

Finding the **joint angles**



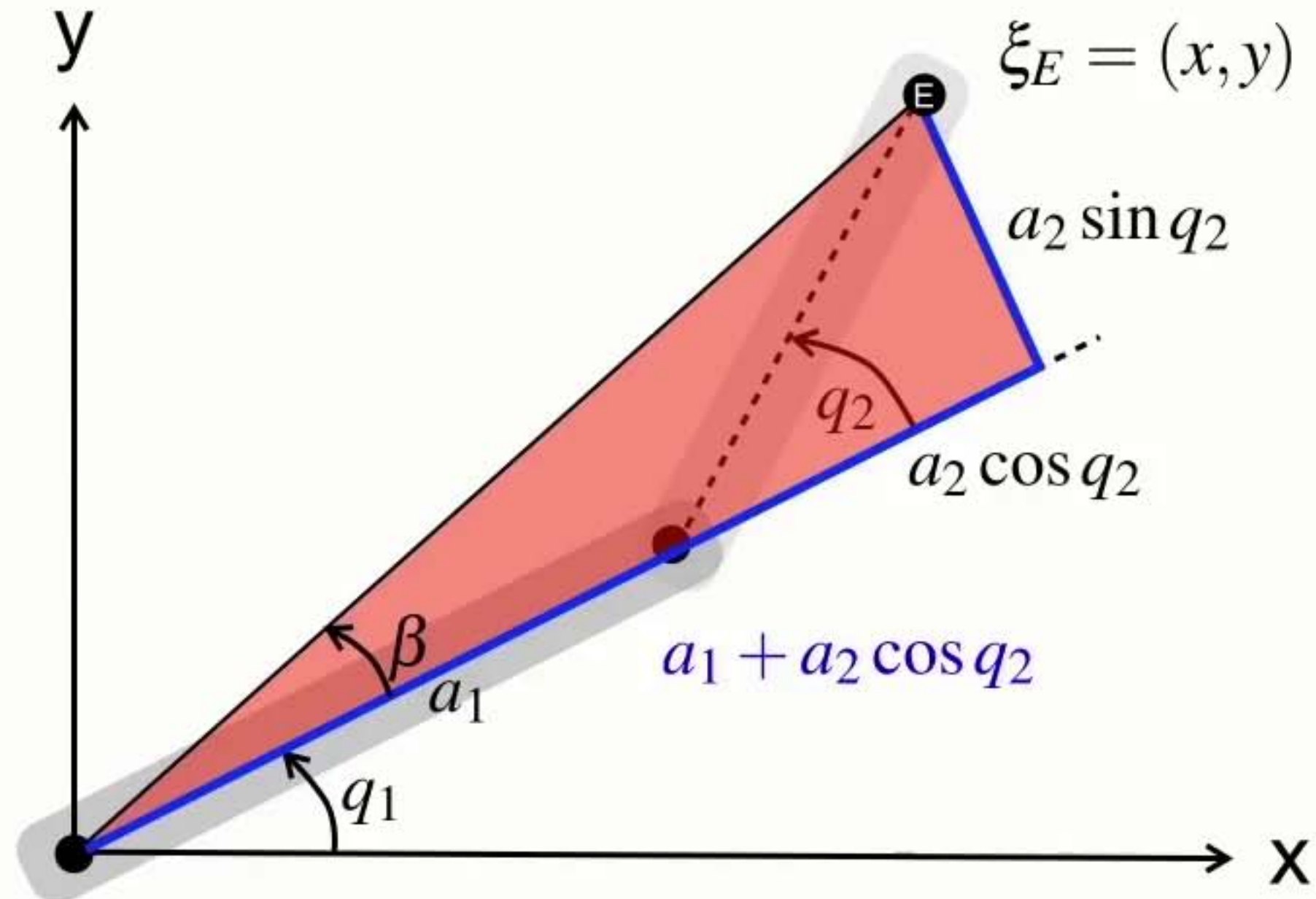
- Apply simple trigonometry

Finding the **joint angles**



- Apply simple trigonometry

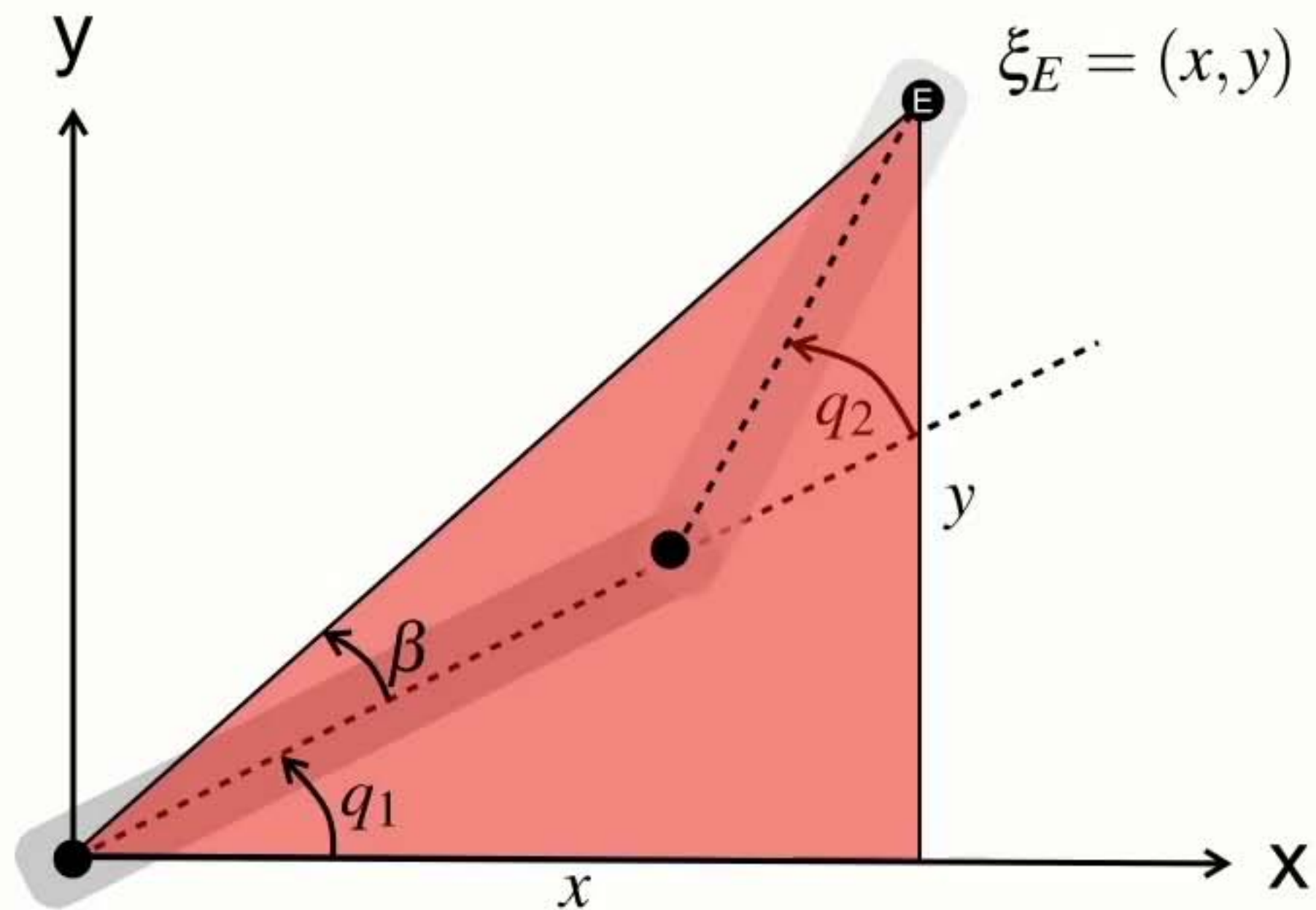
Finding the **joint angles**



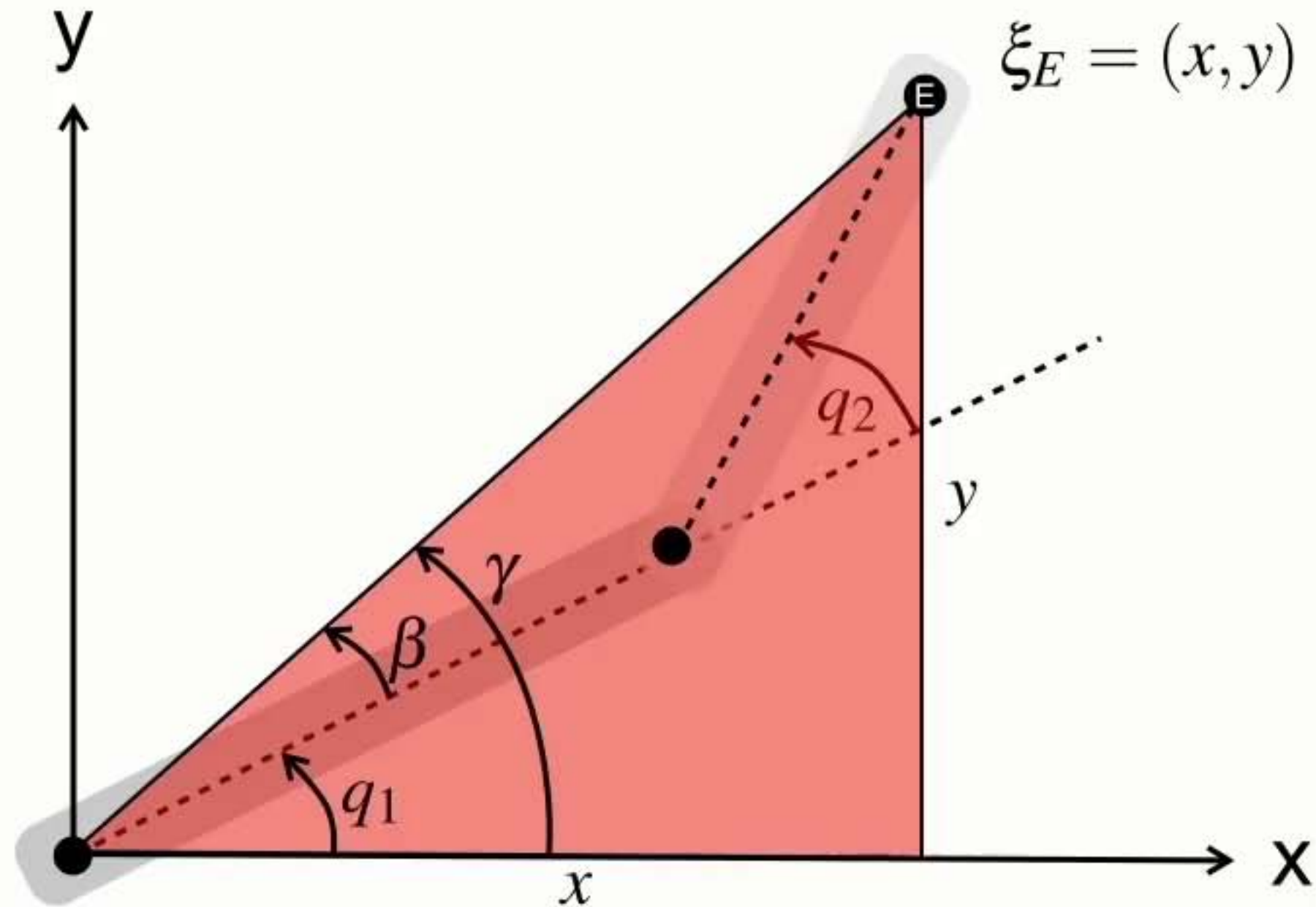
■ Apply simple trigonometry

$$\beta = \tan^{-1} \frac{a_2 \sin q_2}{a_1 + a_2 \cos q_2}$$

Finding the **joint angles**

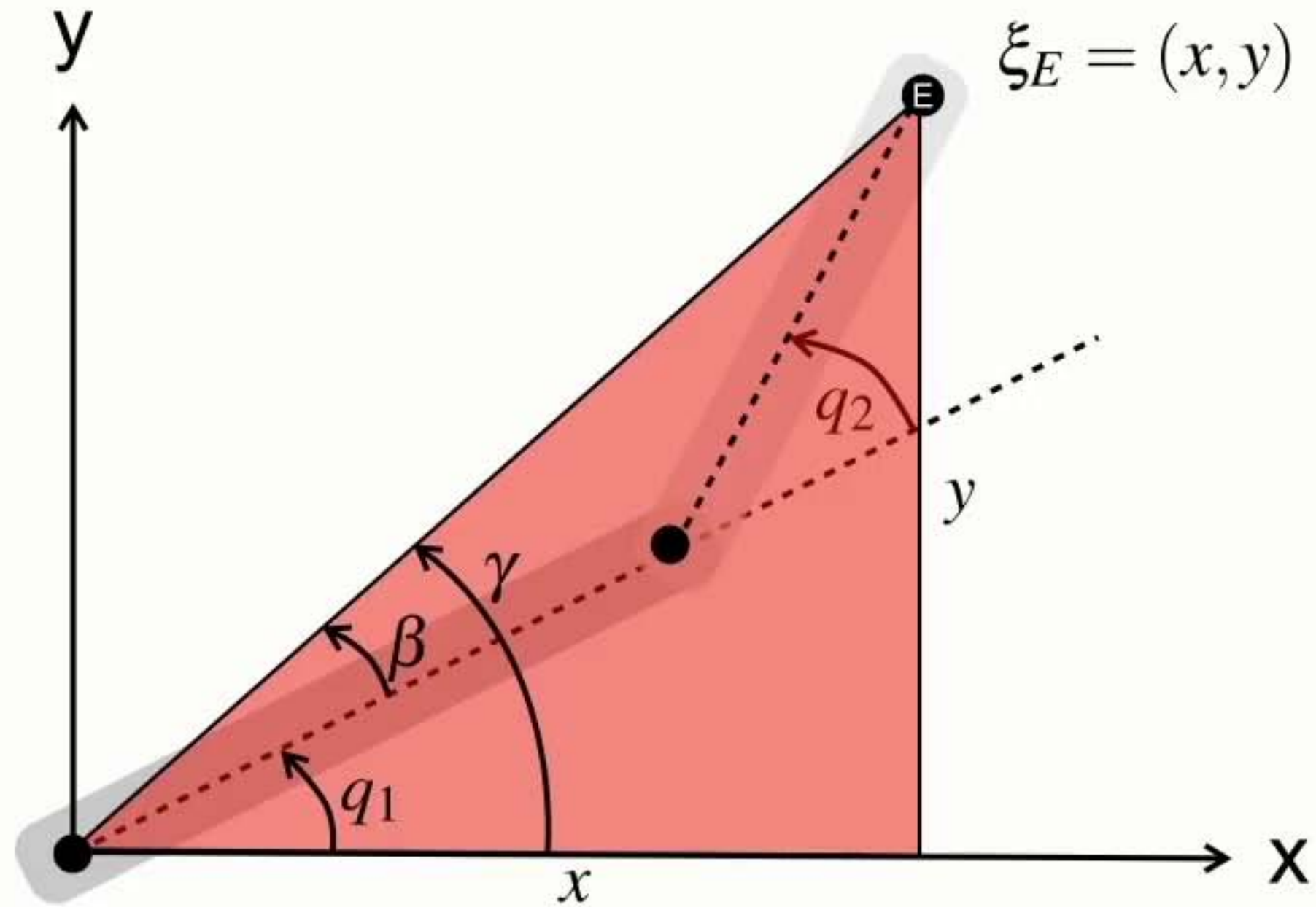


Finding the **joint angles**



$$\gamma = \tan^{-1} \frac{y}{x}$$

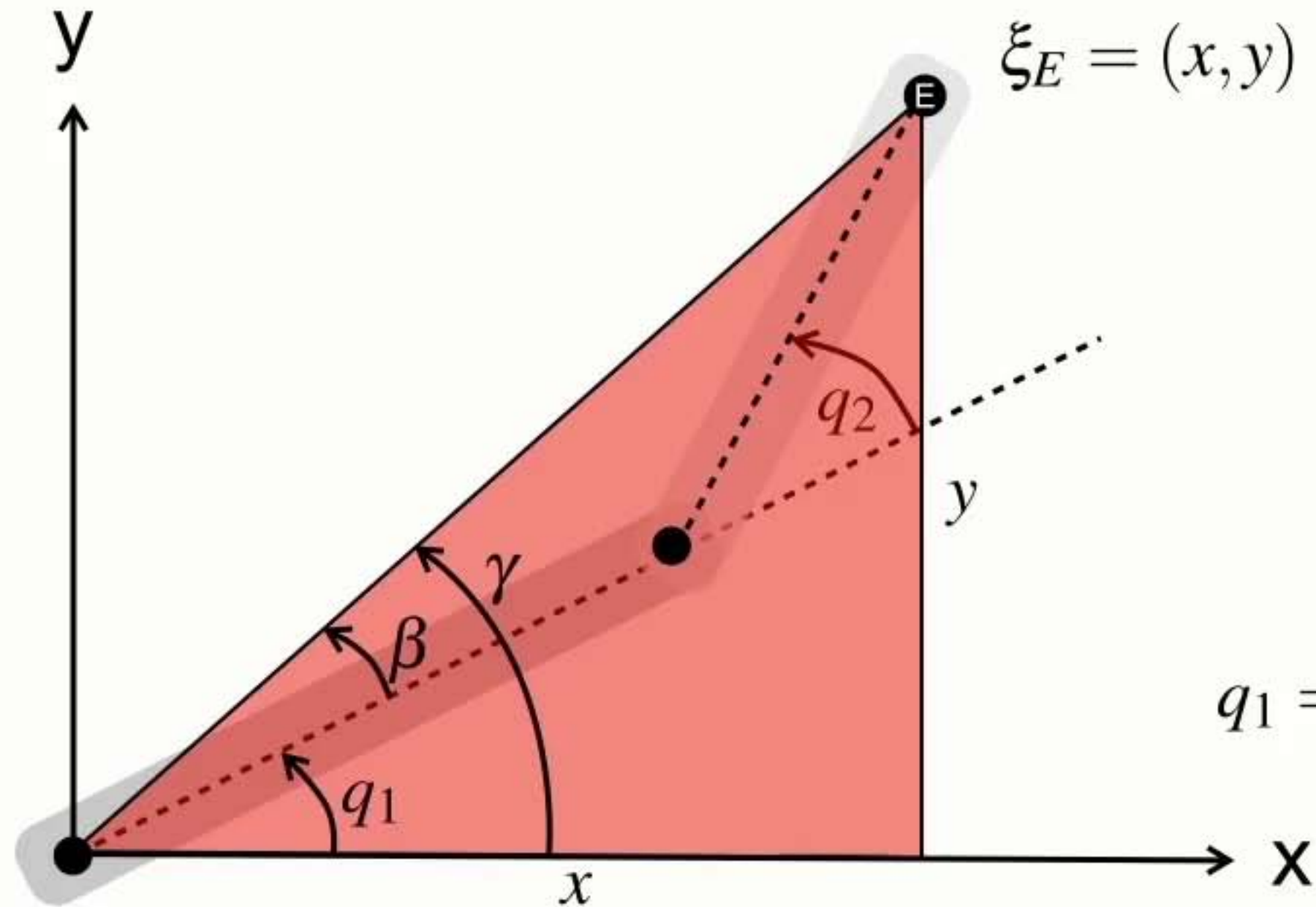
Finding the **joint angles**



$$\gamma = \tan^{-1} \frac{y}{x}$$

$$q_1 = \gamma - \beta$$

Finding the **joint angles**

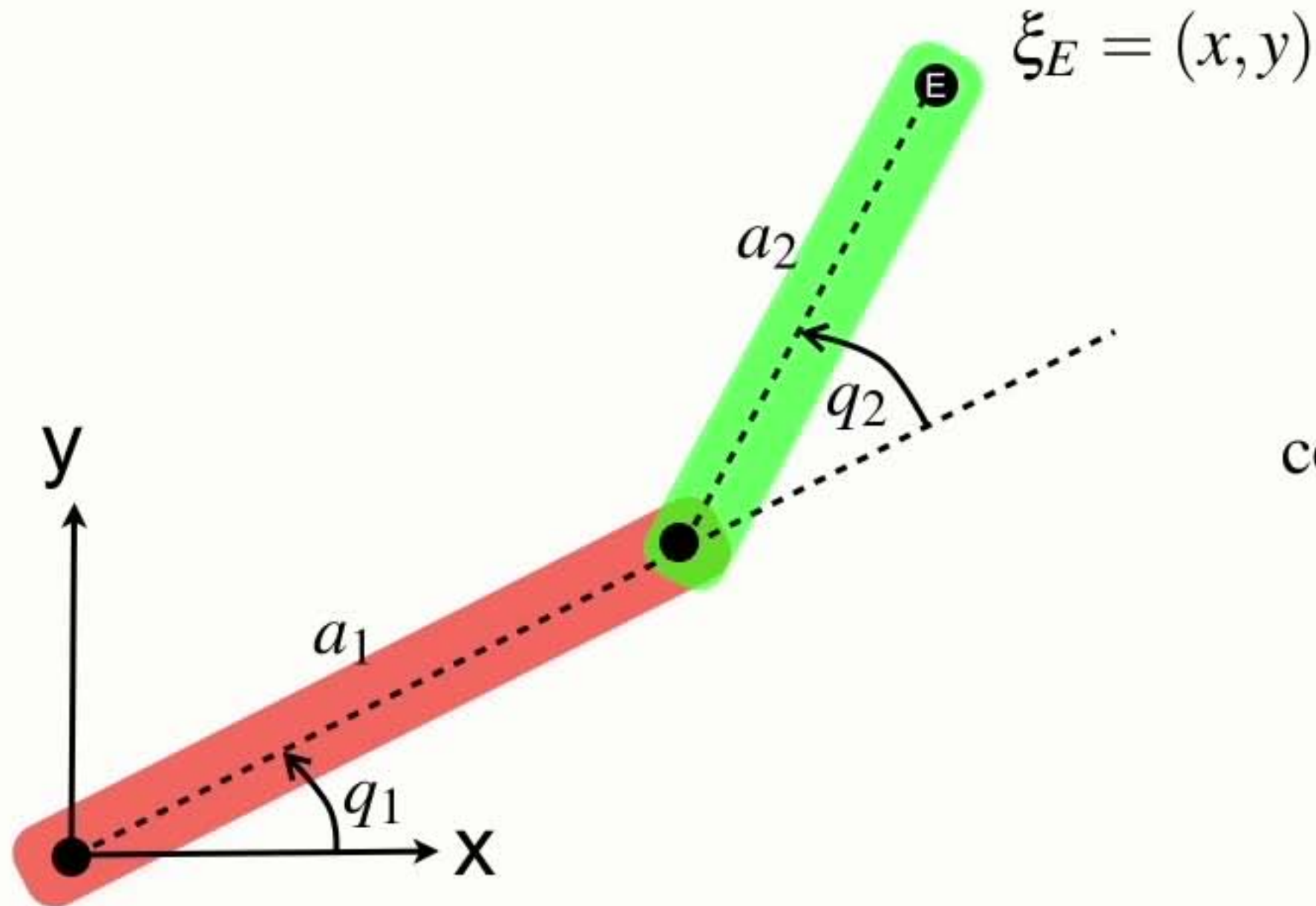


$$\gamma = \tan^{-1} \frac{y}{x}$$

$$q_1 = \gamma - \beta$$

$$q_1 = \tan^{-1} \frac{y}{x} - \tan^{-1} \frac{a_2 \sin q_2}{a_1 + a_2 \cos q_2}$$

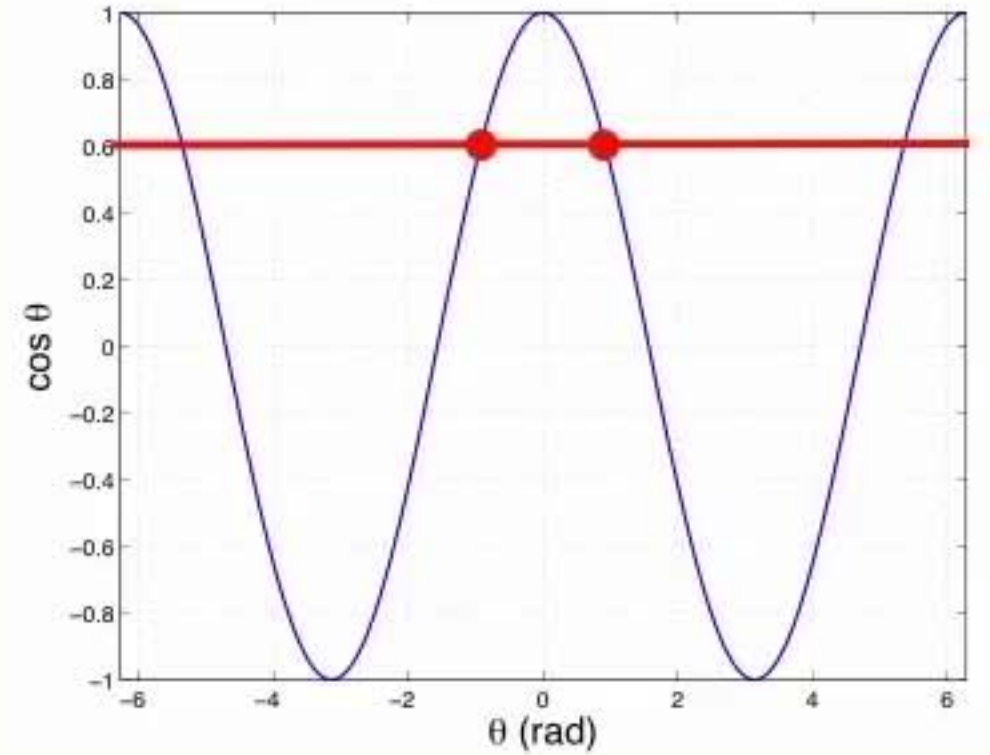
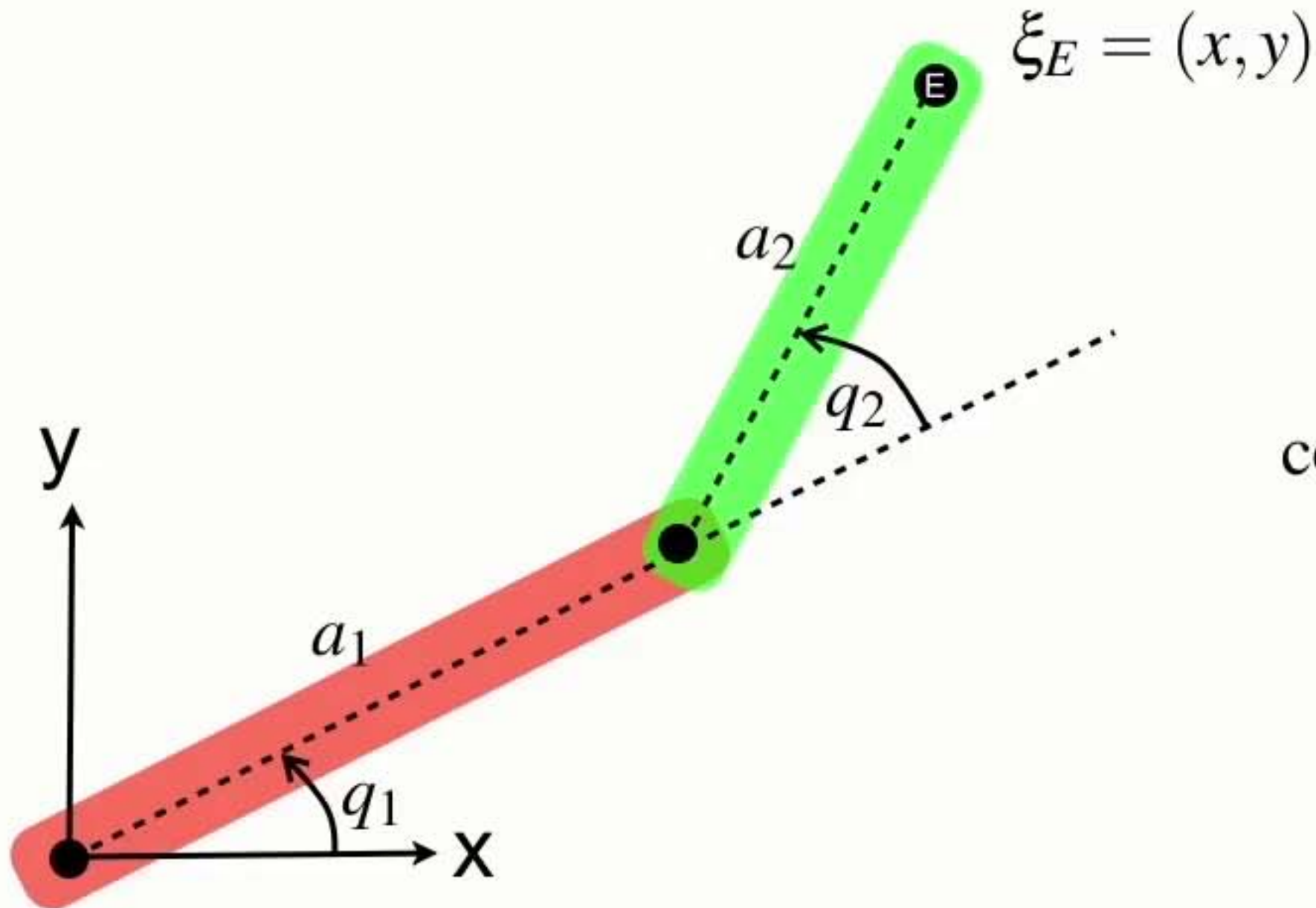
Tool tip pose



$$\cos q_2 = \frac{x^2 + y^2 - a_1^2 - a_2^2}{2a_1a_2}$$

$$q_1 = \tan^{-1} \frac{y}{x} - \tan^{-1} \frac{a_2 \sin q_2}{a_1 + a_2 \cos q_2}$$

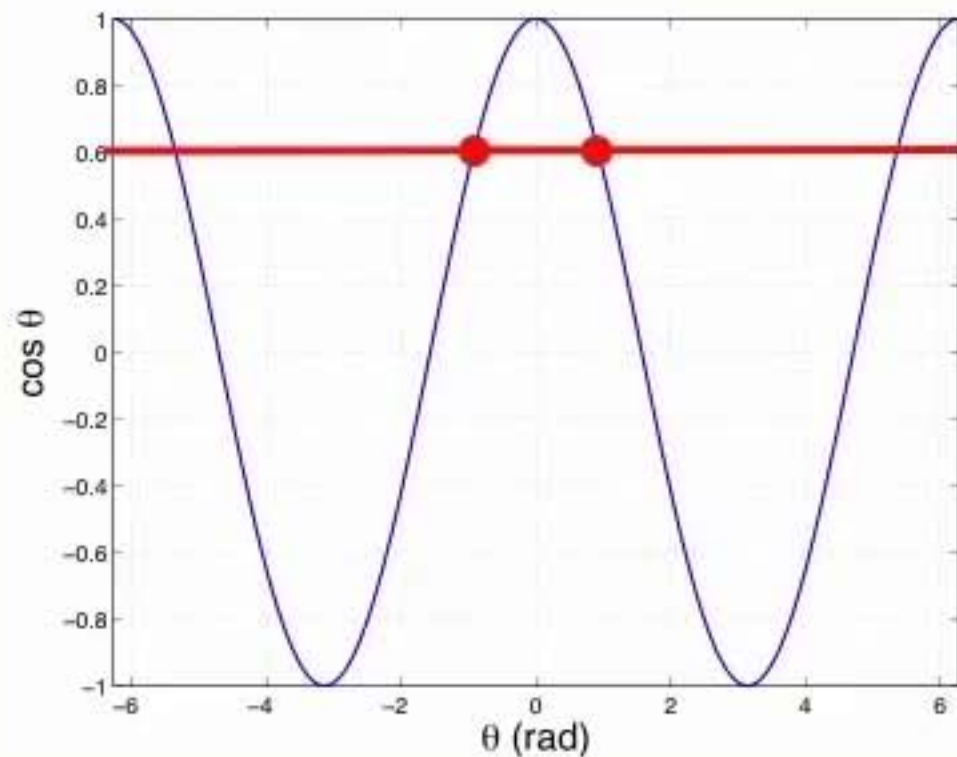
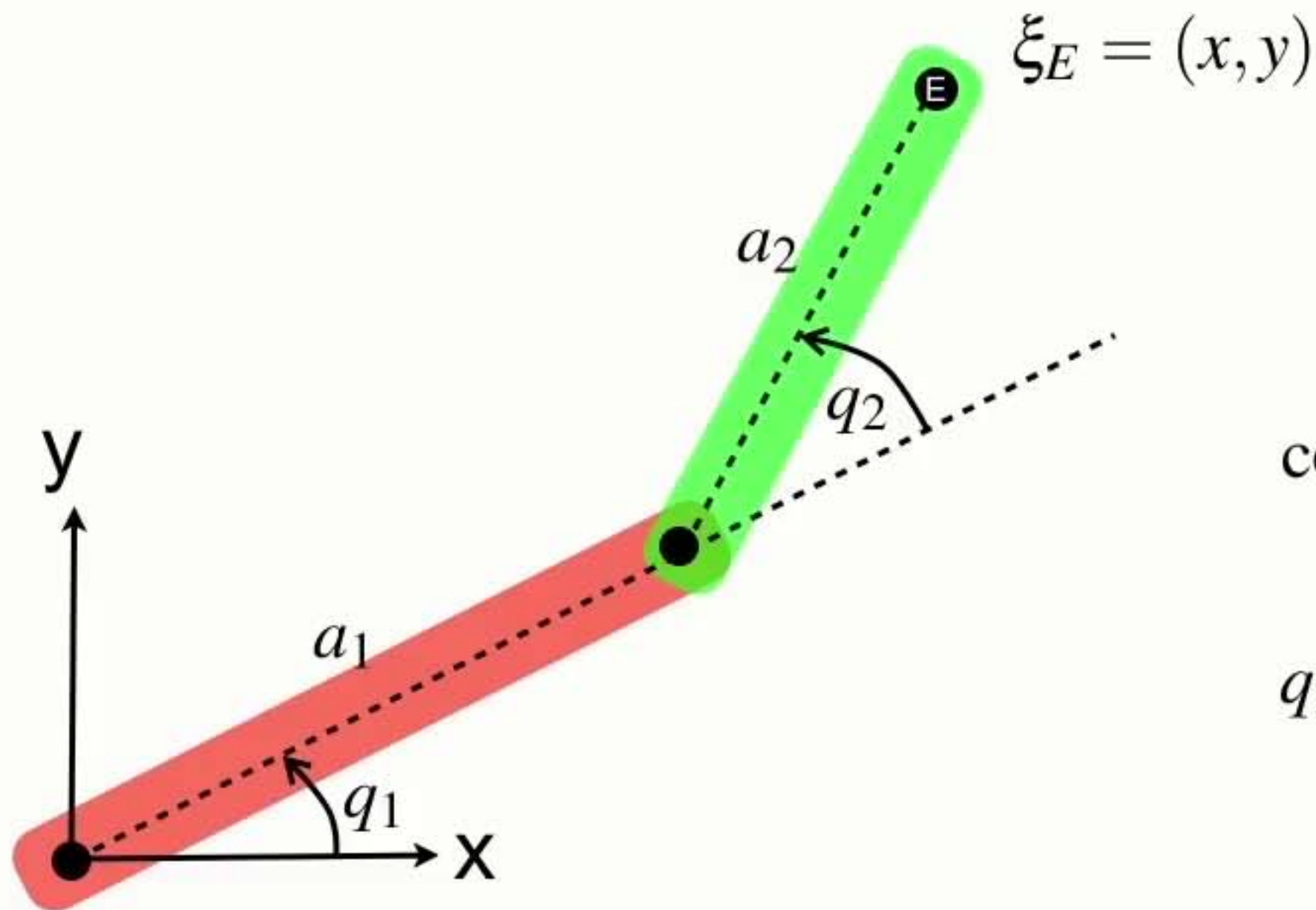
Tool tip pose



$$\cos q_2 = \frac{x^2 + y^2 - a_1^2 - a_2^2}{2a_1a_2}$$

$$q_1 = \tan^{-1} \frac{y}{x} - \tan^{-1} \frac{a_2 \sin q_2}{a_1 + a_2 \cos q_2}$$

Tool tip pose

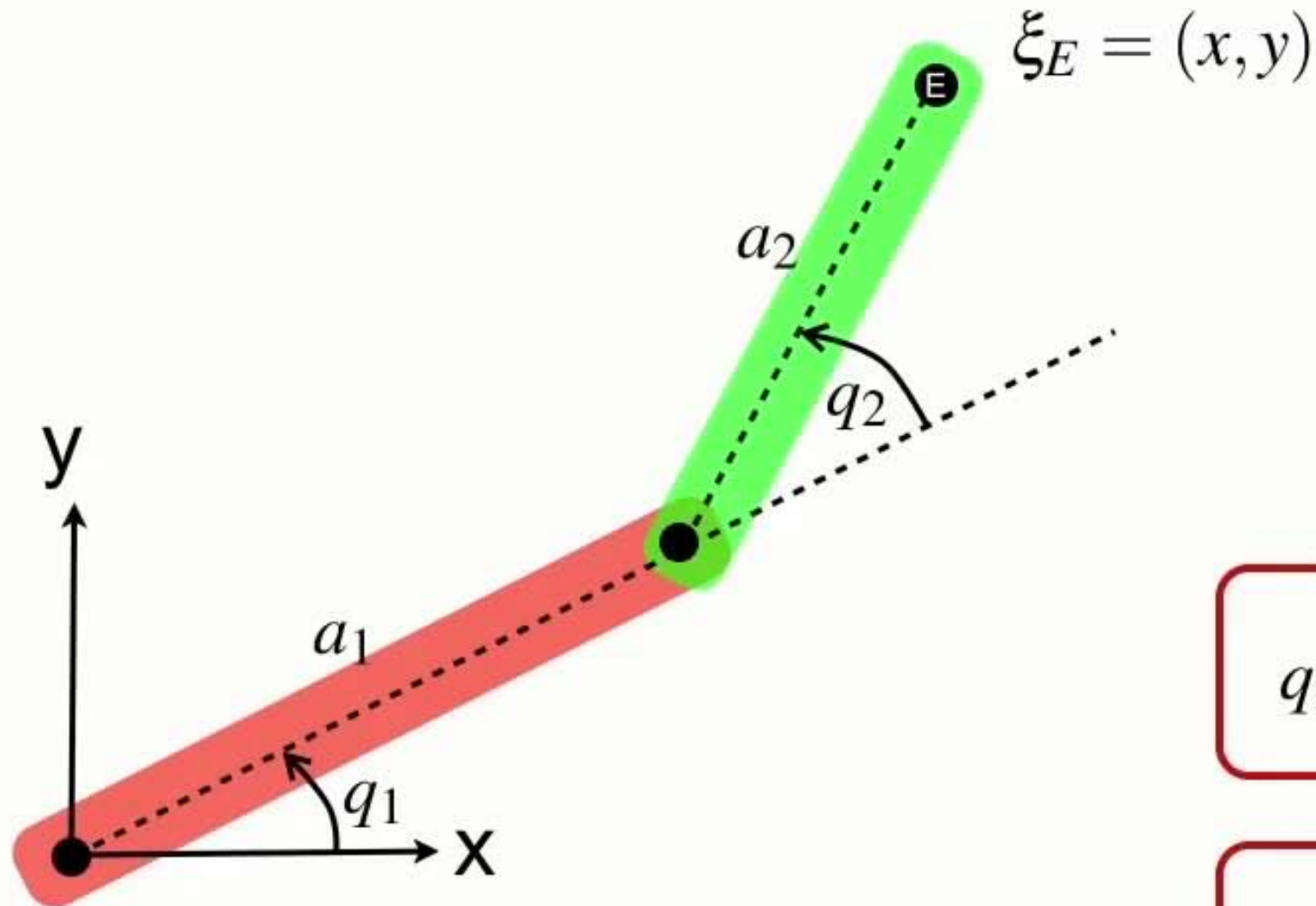


$$\cos q_2 = \frac{x^2 + y^2 - a_1^2 - a_2^2}{2a_1a_2}$$

$$q_2 = \cos^{-1} \frac{x^2 + y^2 - a_1^2 - a_2^2}{2a_1a_2}$$

$$q_1 = \tan^{-1} \frac{y}{x} - \tan^{-1} \frac{a_2 \sin q_2}{a_1 + a_2 \cos q_2}$$

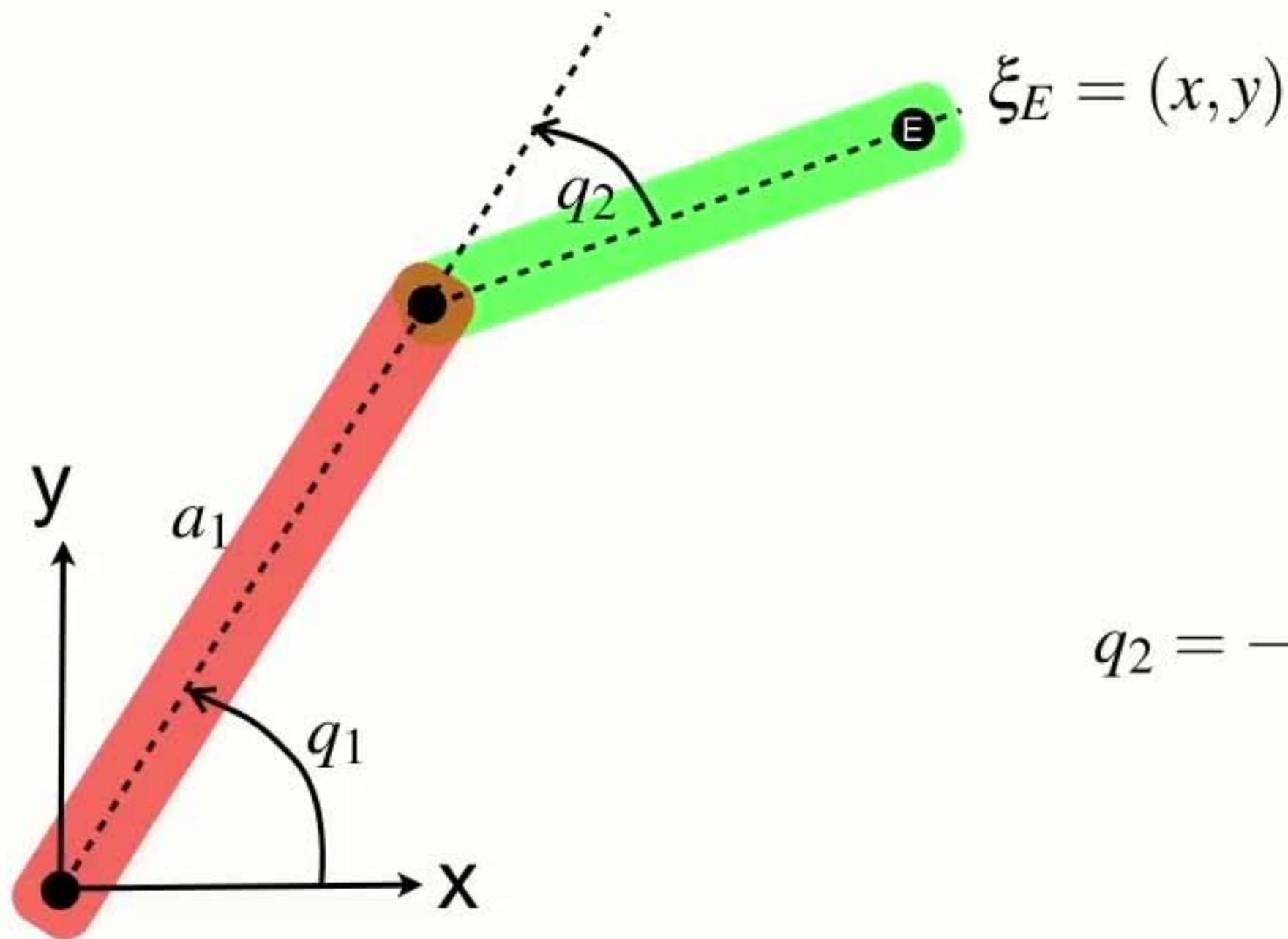
Tool tip pose



$$q_2 = \cos^{-1} \frac{x^2 + y^2 - a_1^2 - a_2^2}{2a_1a_2}$$

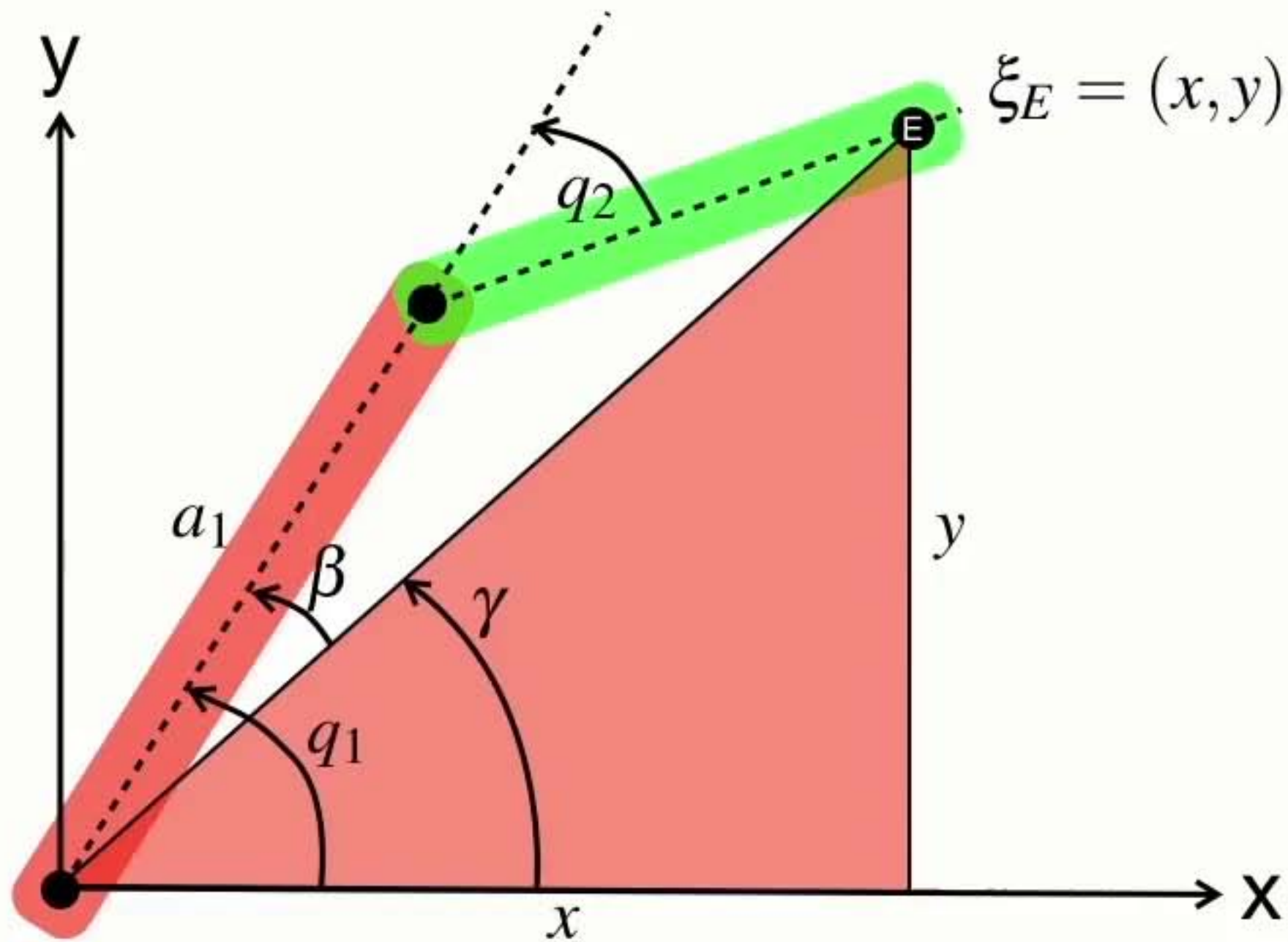
$$q_1 = \tan^{-1} \frac{y}{x} - \tan^{-1} \frac{a_2 \sin q_2}{a_1 + a_2 \cos q_2}$$

Tool tip pose



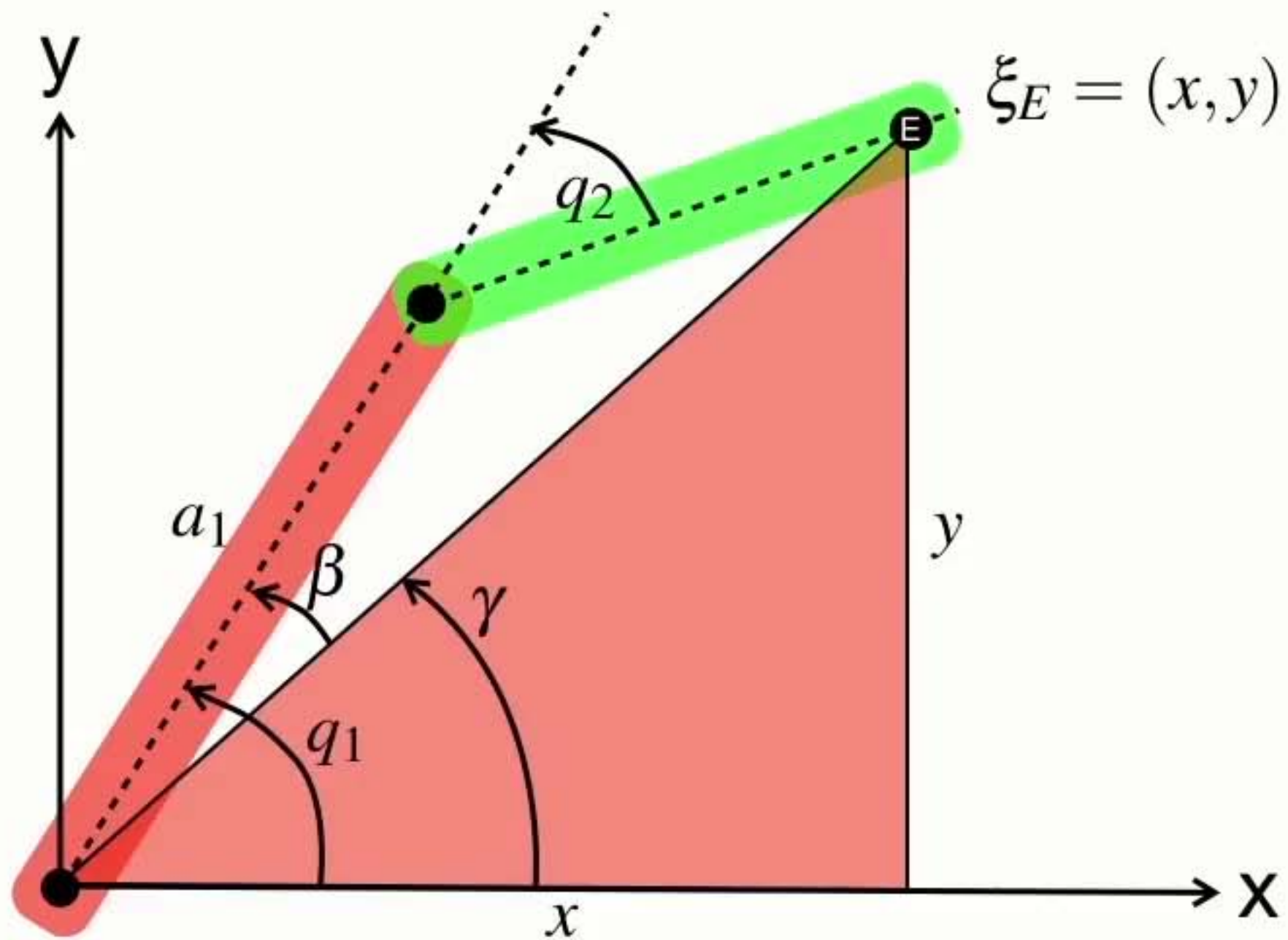
$$q_2 = -\cos^{-1} \frac{x^2 + y^2 - a_1^2 - a_2^2}{2a_1a_2}$$

Tool tip pose



$$\gamma = \tan^{-1} \frac{y}{x}$$

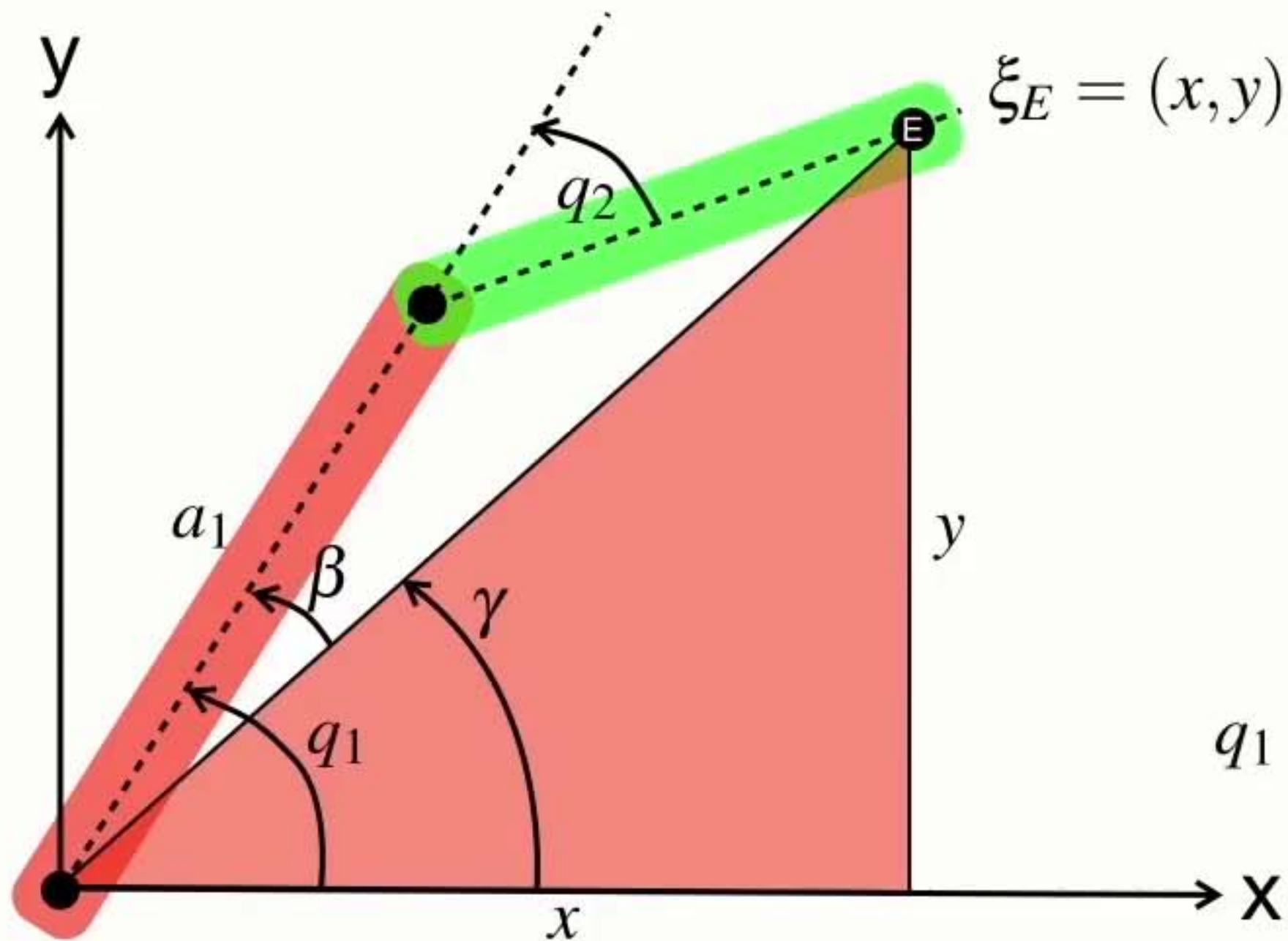
Tool tip pose



$$\gamma = \tan^{-1} \frac{y}{x}$$

$$q_1 = \gamma + \beta$$

Tool tip pose

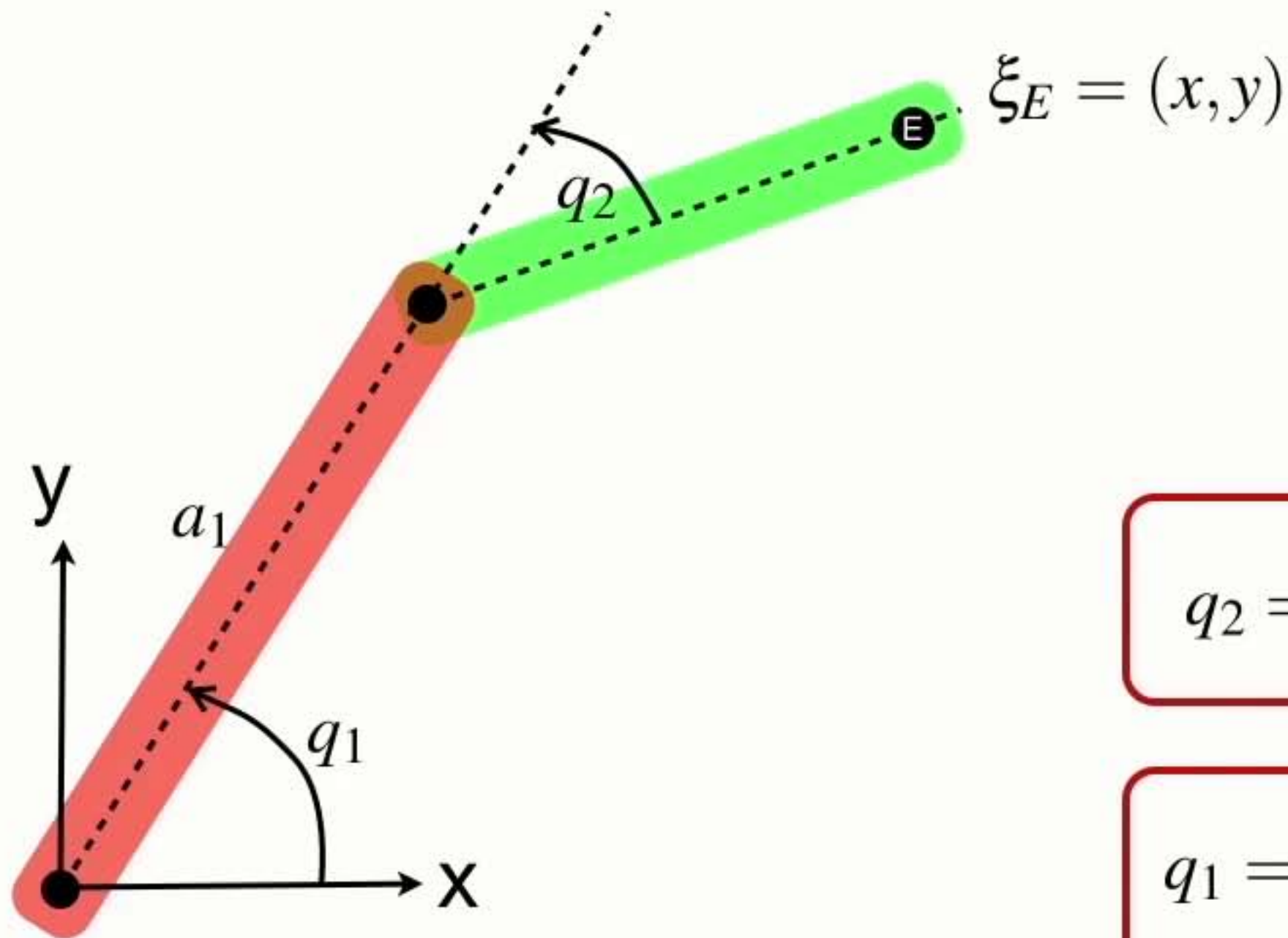


$$\gamma = \tan^{-1} \frac{y}{x}$$

$$q_1 = \gamma + \beta$$

$$q_1 = \tan^{-1} \frac{y}{x} + \tan^{-1} \frac{a_2 \sin q_2}{a_1 + a_2 \cos q_2}$$

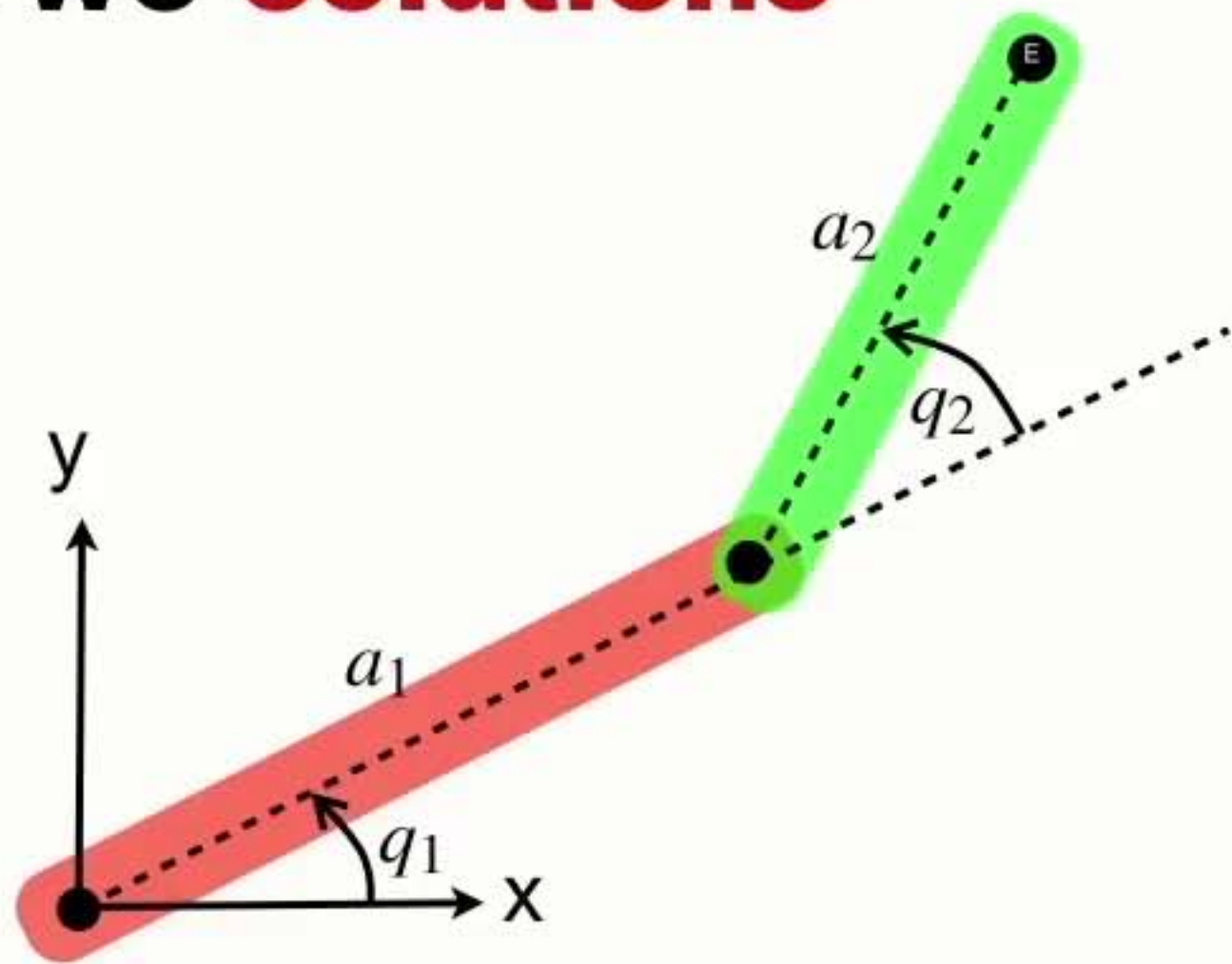
Tool tip pose



$$q_2 = -\cos^{-1} \frac{x^2 + y^2 - a_1^2 - a_2^2}{2a_1a_2}$$

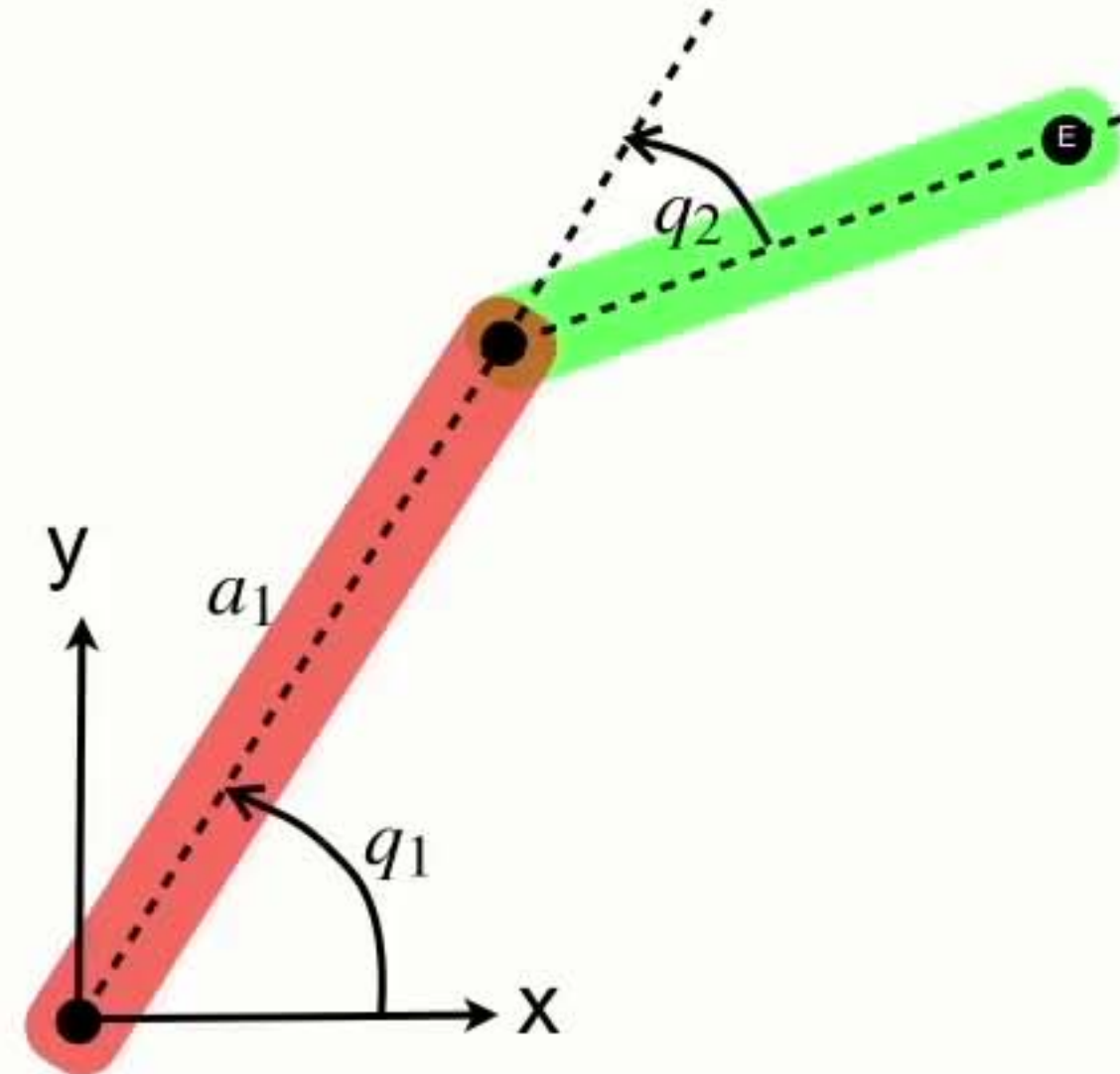
$$q_1 = \tan^{-1} \frac{y}{x} + \tan^{-1} \frac{a_2 \sin q_2}{a_1 + a_2 \cos q_2}$$

Two solutions



$$q_2 = \cos^{-1} \frac{x^2 + y^2 - a_1^2 - a_2^2}{2a_1a_2}$$

$$q_1 = \tan^{-1} \frac{y}{x} - \tan^{-1} \frac{a_2 \sin q_2}{a_1 + a_2 \cos q_2}$$



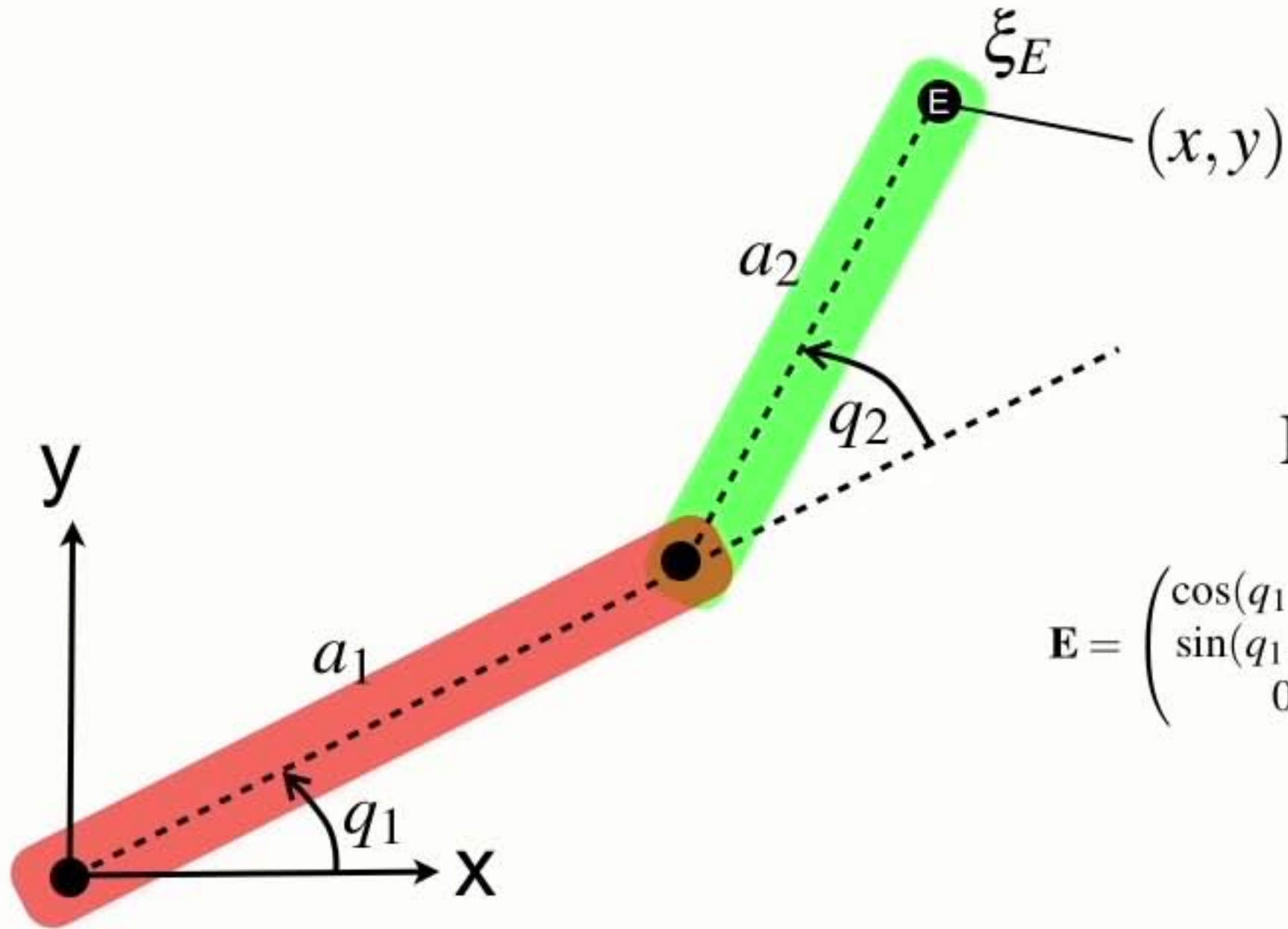
$$q_2 = -\cos^{-1} \frac{x^2 + y^2 - a_1^2 - a_2^2}{2a_1a_2}$$

$$q_1 = \tan^{-1} \frac{y}{x} + \tan^{-1} \frac{a_2 \sin q_2}{a_1 + a_2 \cos q_2}$$

Section 2

Analytic solution for a 2-link planar robot

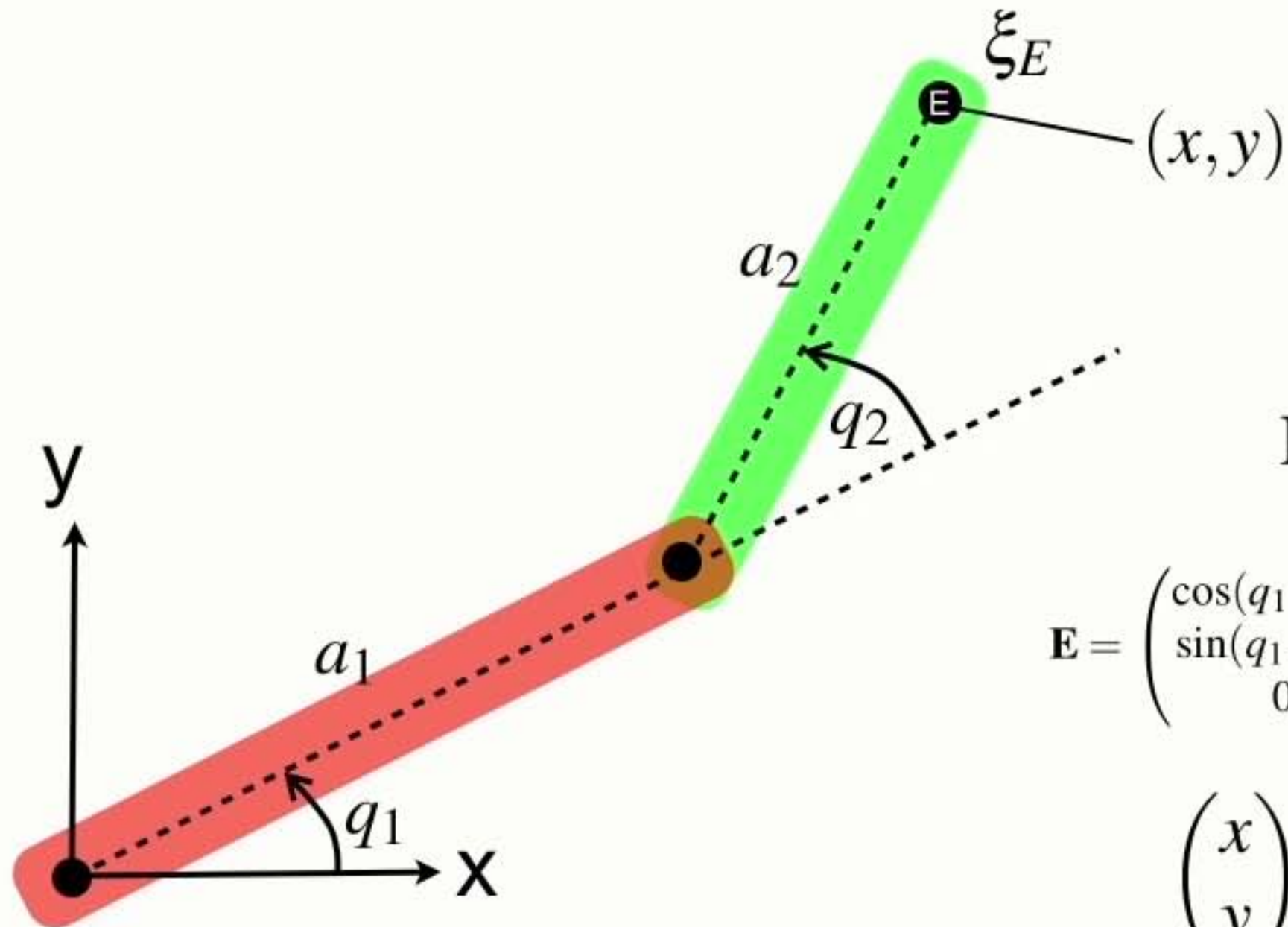
Tool tip pose



$$\mathbf{E} = \mathbf{R}(q_1) \mathbf{T}_x(a_1) \mathbf{R}(q_2) \mathbf{T}_x(a_2)$$

$$\mathbf{E} = \begin{pmatrix} \cos(q_1 + q_2) & -\sin(q_1 + q_2) & a_2 \cos(q_1 + q_2) + a_1 \cos q_1 \\ \sin(q_1 + q_2) & \cos(q_1 + q_2) & a_2 \sin(q_1 + q_2) + a_1 \sin q_1 \\ 0 & 0 & 1 \end{pmatrix}$$

Tool tip pose

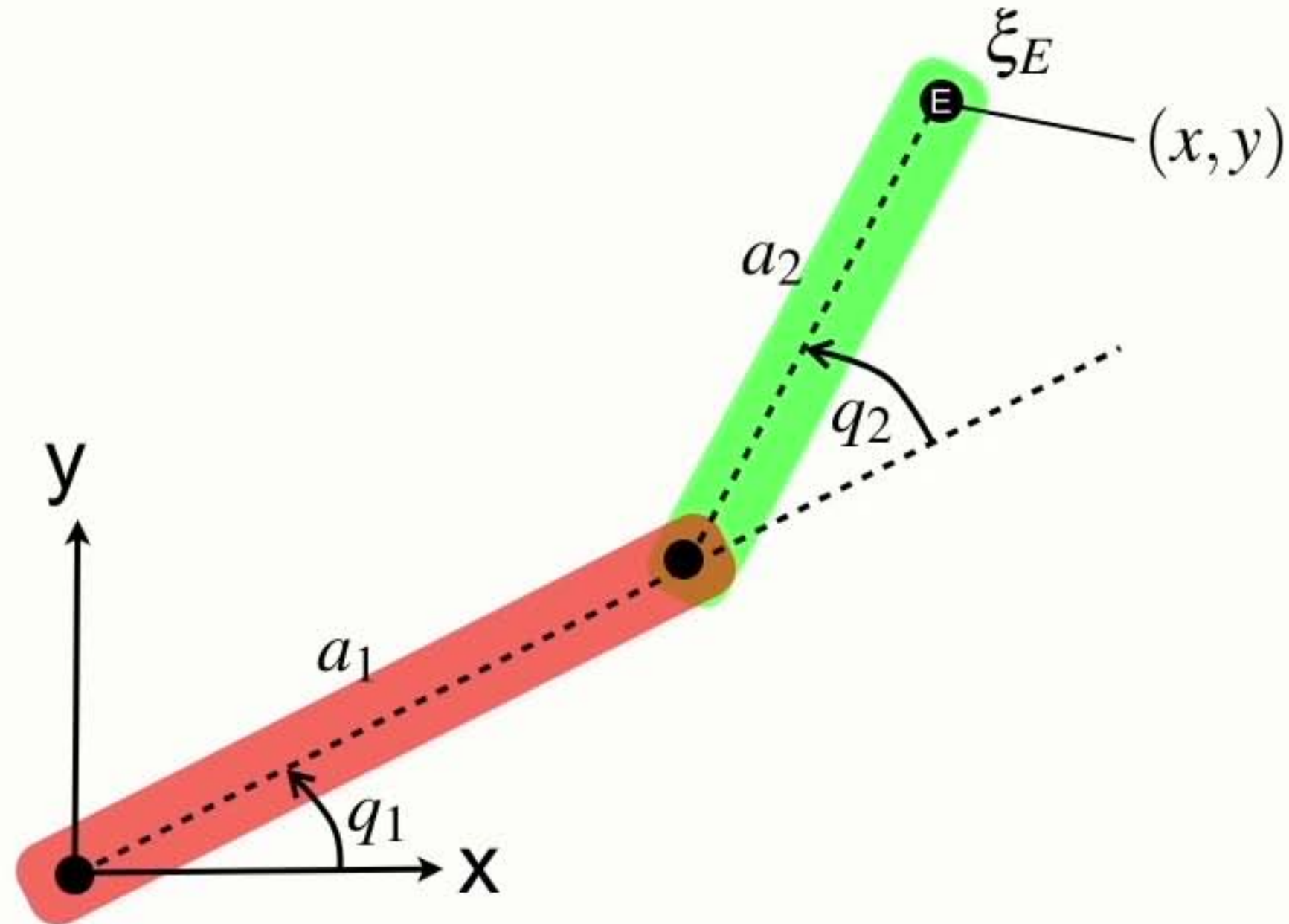


$$\mathbf{E} = \mathbf{R}(q_1) \mathbf{T}_x(a_1) \mathbf{R}(q_2) \mathbf{T}_x(a_2)$$

$$\mathbf{E} = \begin{pmatrix} \cos(q_1 + q_2) & -\sin(q_1 + q_2) & a_2 \cos(q_1 + q_2) + a_1 \cos q_1 \\ \sin(q_1 + q_2) & \cos(q_1 + q_2) & a_2 \sin(q_1 + q_2) + a_1 \sin q_1 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} a_2 \cos(q_1 + q_2) + a_1 \cos q_1 \\ a_2 \sin(q_1 + q_2) + a_1 \sin q_1 \end{pmatrix}$$

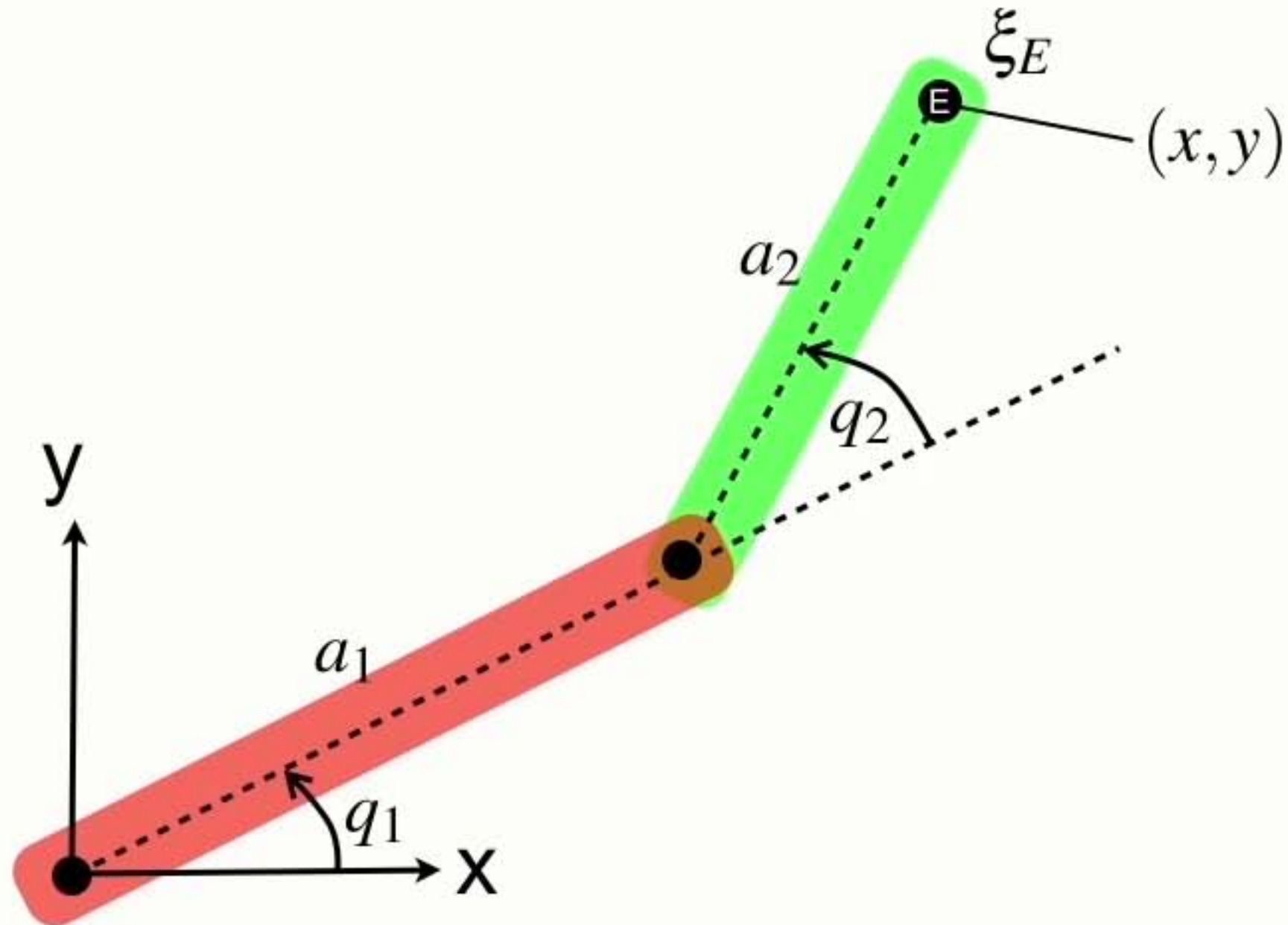
Finding the **joint angles**



$$x = a_2 \cos(q_1 + q_2) + a_1 \cos q_1$$
$$y = a_2 \sin(q_1 + q_2) + a_1 \sin q_1$$

■ Square and add

Finding the **joint angles**



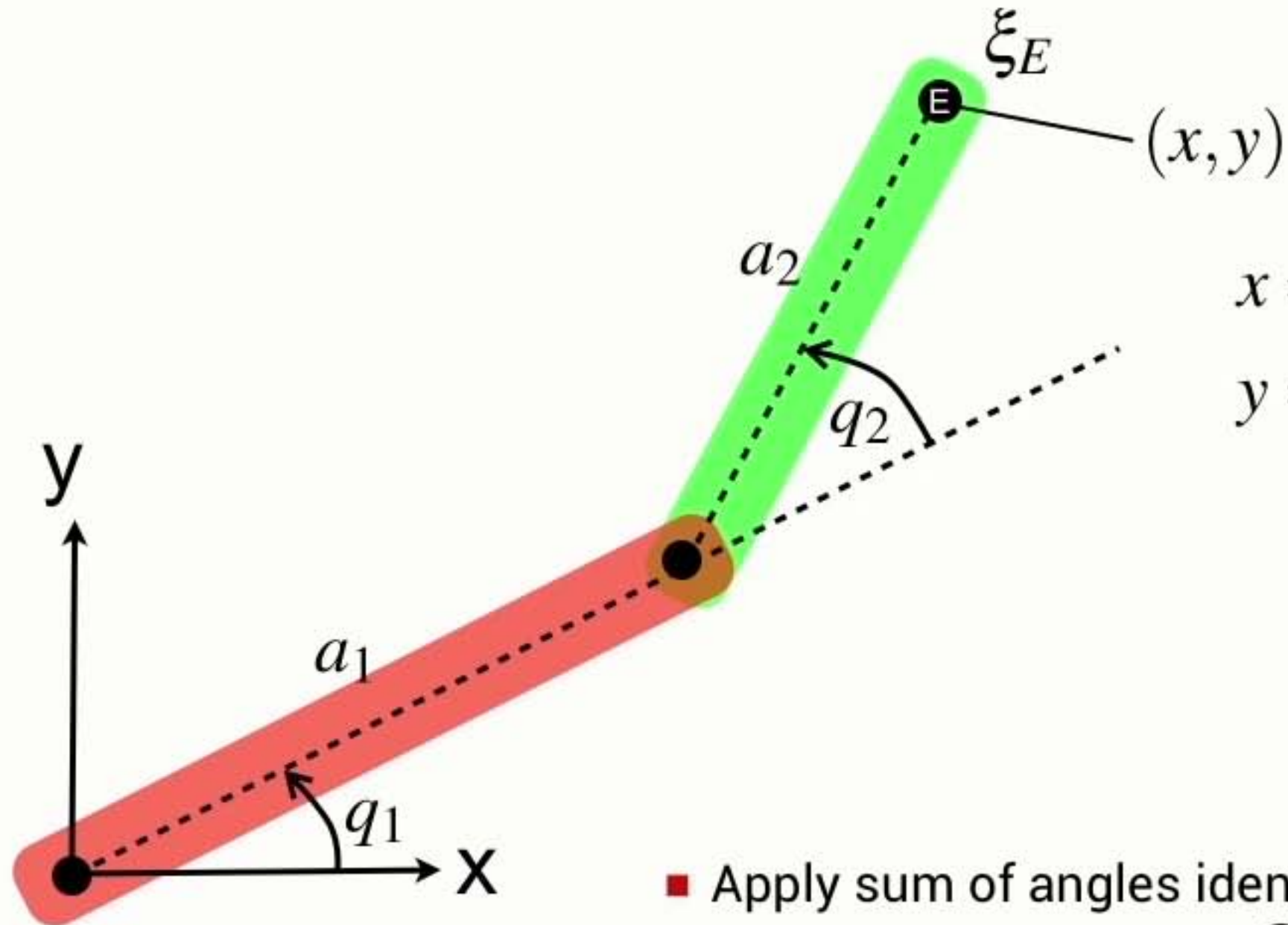
■ Square and add

$$\begin{aligned}x &= a_2 \cos(q_1 + q_2) + a_1 \cos q_1 \\y &= a_2 \sin(q_1 + q_2) + a_1 \sin q_1\end{aligned}$$

$$x^2 + y^2 = a_1^2 + a_2^2 + 2a_1a_2 \cos q_2$$

$$\cos q_2 = \frac{x^2 + y^2 - a_1^2 - a_2^2}{2a_1a_2}$$

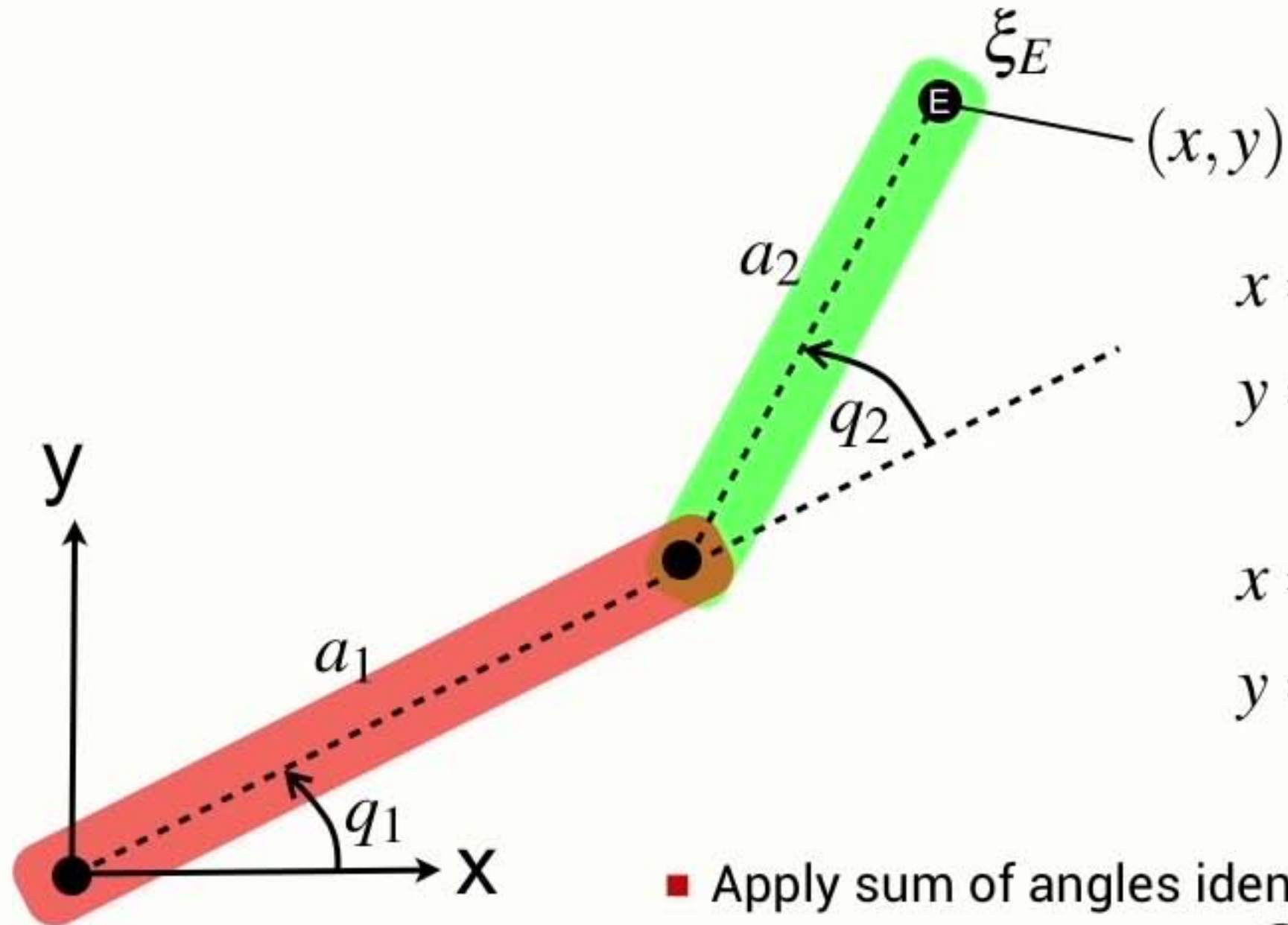
Finding the **joint angles**



$$x = a_2 \cos(q_1 + q_2) + a_1 \cos q_1$$
$$y = a_2 \sin(q_1 + q_2) + a_1 \sin q_1$$

- Apply sum of angles identities
- Substitute $\cos q_2 \rightarrow C_2, \sin q_2 \rightarrow S_2$

Finding the **joint angles**



$$x = a_2 \cos(q_1 + q_2) + a_1 \cos q_1$$

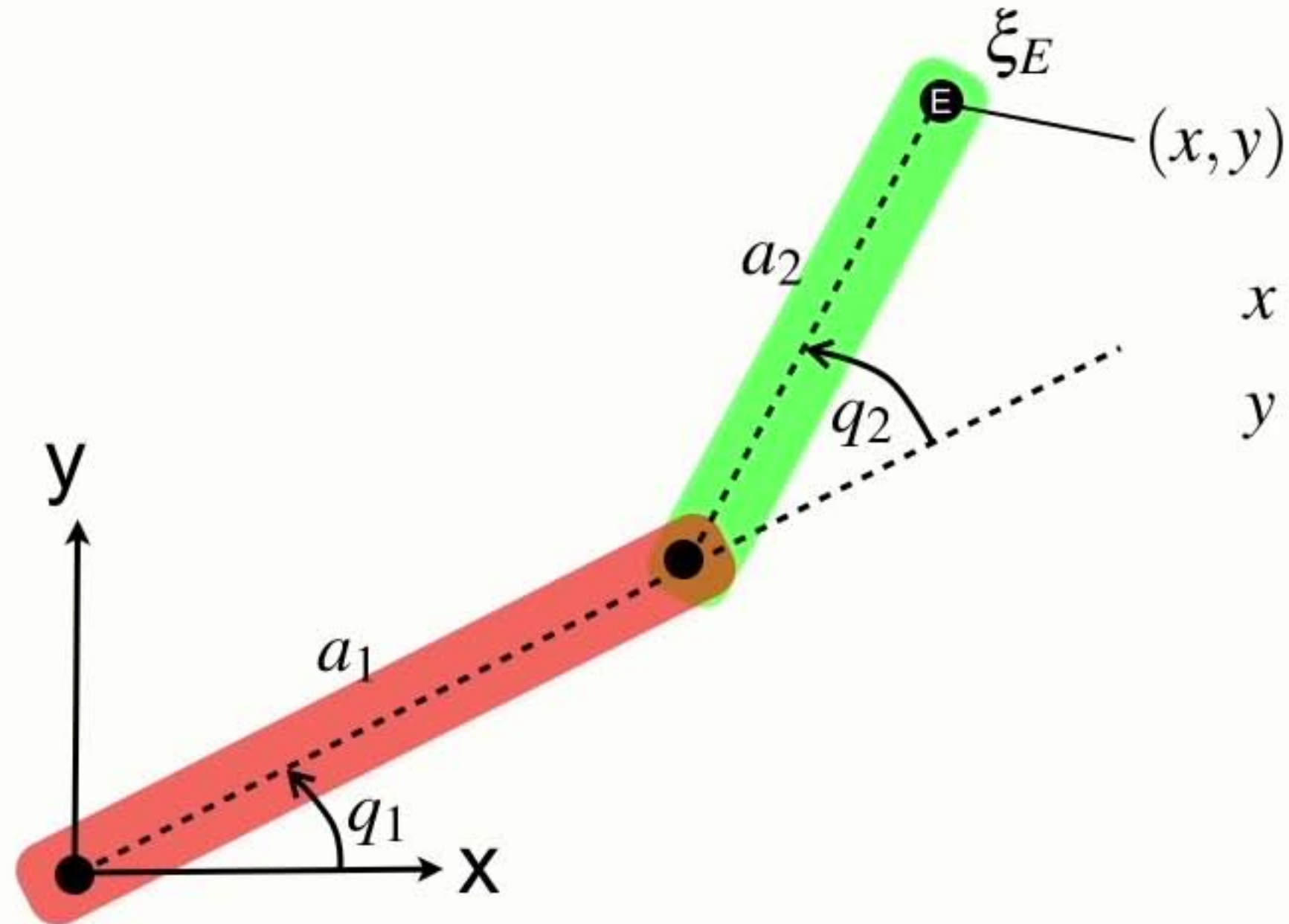
$$y = a_2 \sin(q_1 + q_2) + a_1 \sin q_1$$

$$x = (a_1 + a_2 C_2) \cos q_1 - a_2 S_2 \sin q_1$$

$$y = (a_1 + a_2 C_2) \sin q_1 + a_2 S_2 \cos q_1$$

- Apply sum of angles identities
- Substitute $\cos q_2 \rightarrow C_2, \sin q_2 \rightarrow S_2$

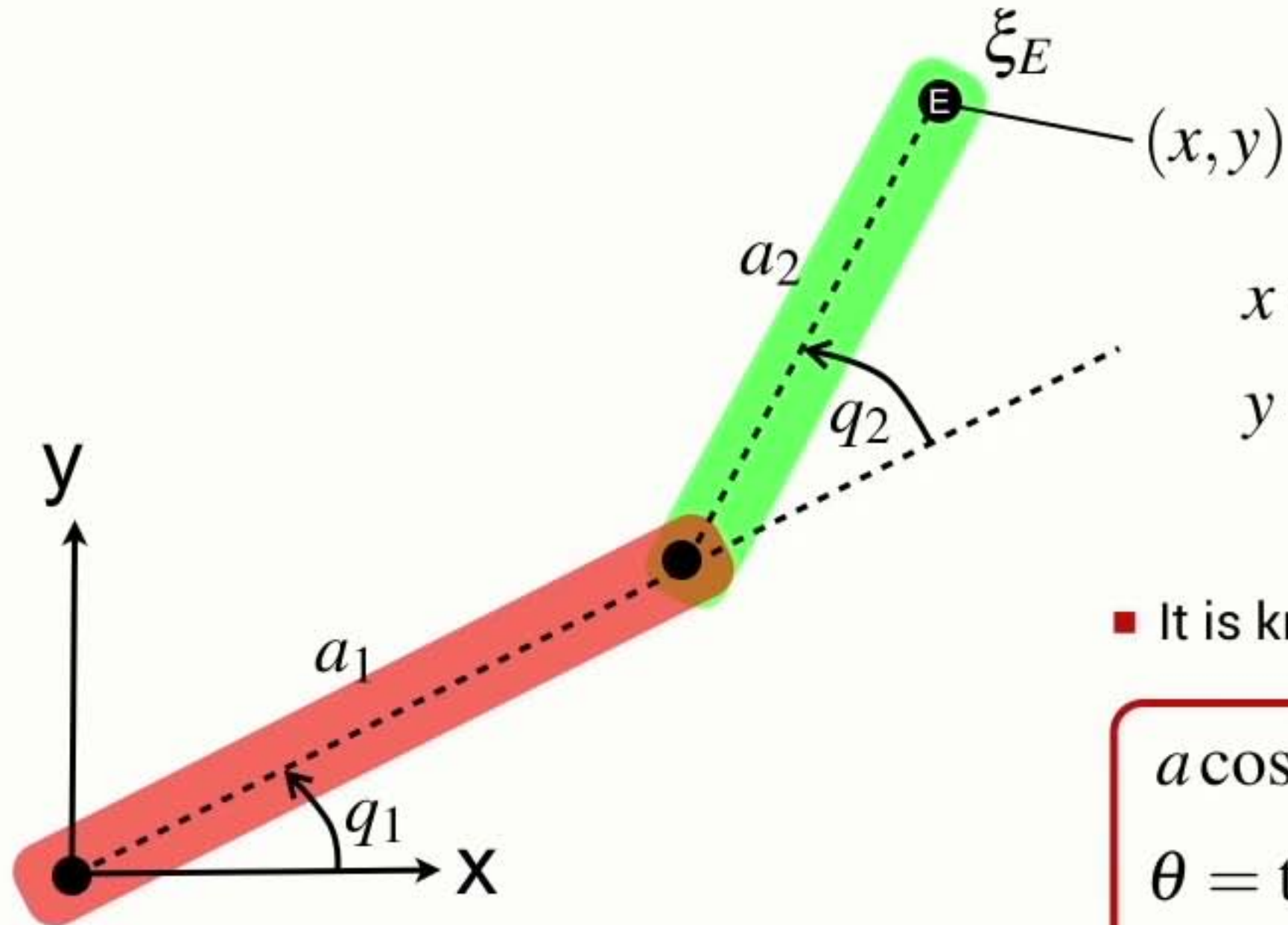
Finding the **joint angles**



$$x = (a_1 + a_2 C_2) \cos q_1 - a_2 S_2 \sin q_1$$

$$y = (a_1 + a_2 C_2) \sin q_1 + a_2 S_2 \cos q_1$$

Finding the **joint angles**



$$x = (a_1 + a_2 C_2) \cos q_1 - a_2 S_2 \sin q_1$$
$$y = (a_1 + a_2 C_2) \sin q_1 + a_2 S_2 \cos q_1$$

■ It is known that

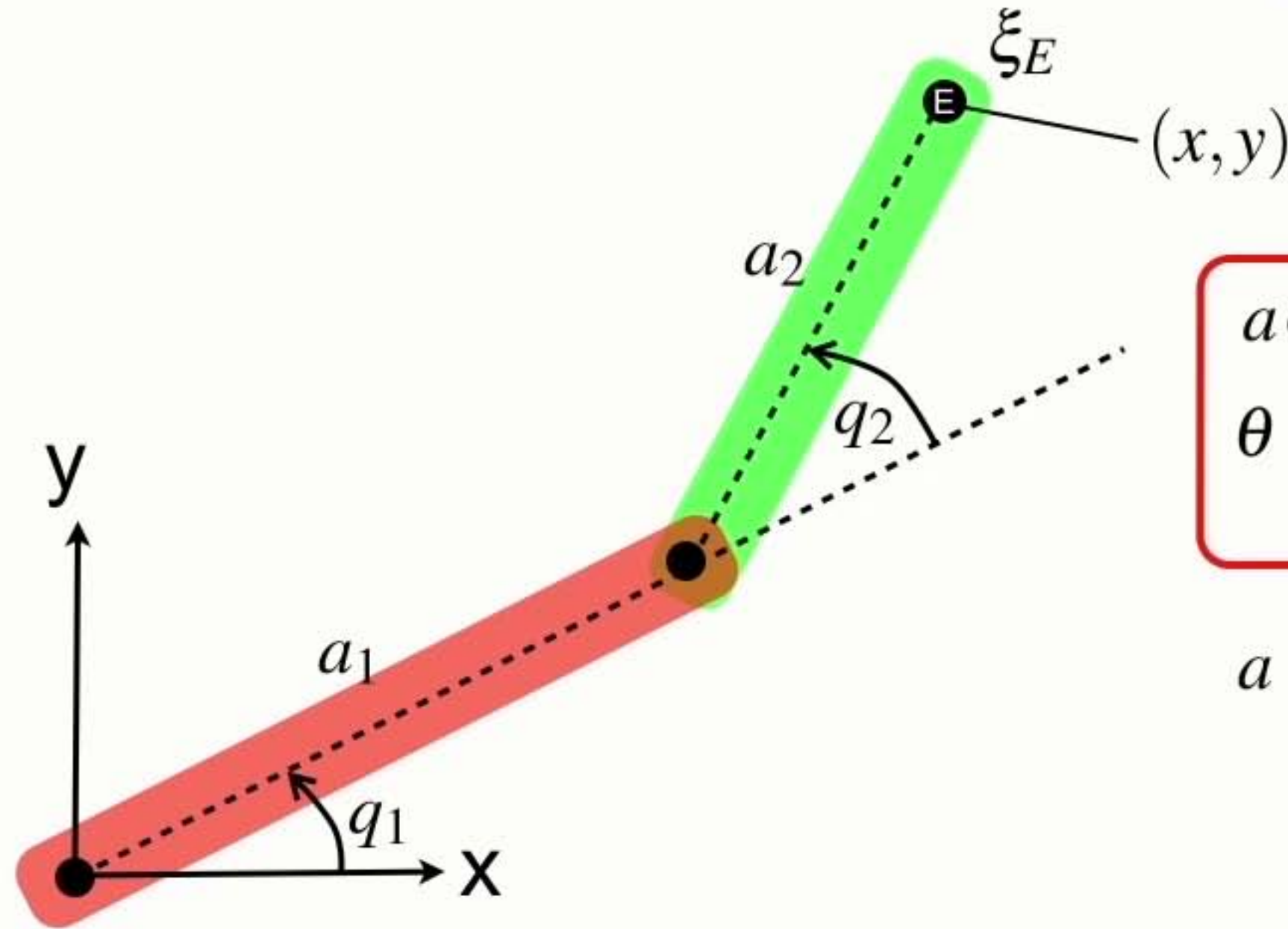
$$a \cos \theta + b \sin \theta = c$$

$$\theta = \tan^{-1} \frac{c}{\pm \sqrt{a^2 + b^2 - c^2}} - \tan^{-1} \frac{a}{b}$$

Finding the **joint angles**

$$x = (a_1 + a_2 C_2) \cos q_1 - a_2 S_2 \sin q_1$$

$$y = (a_1 + a_2 C_2) \sin q_1 + a_2 S_2 \cos q_1$$



$$a \cos \theta + b \sin \theta = c$$

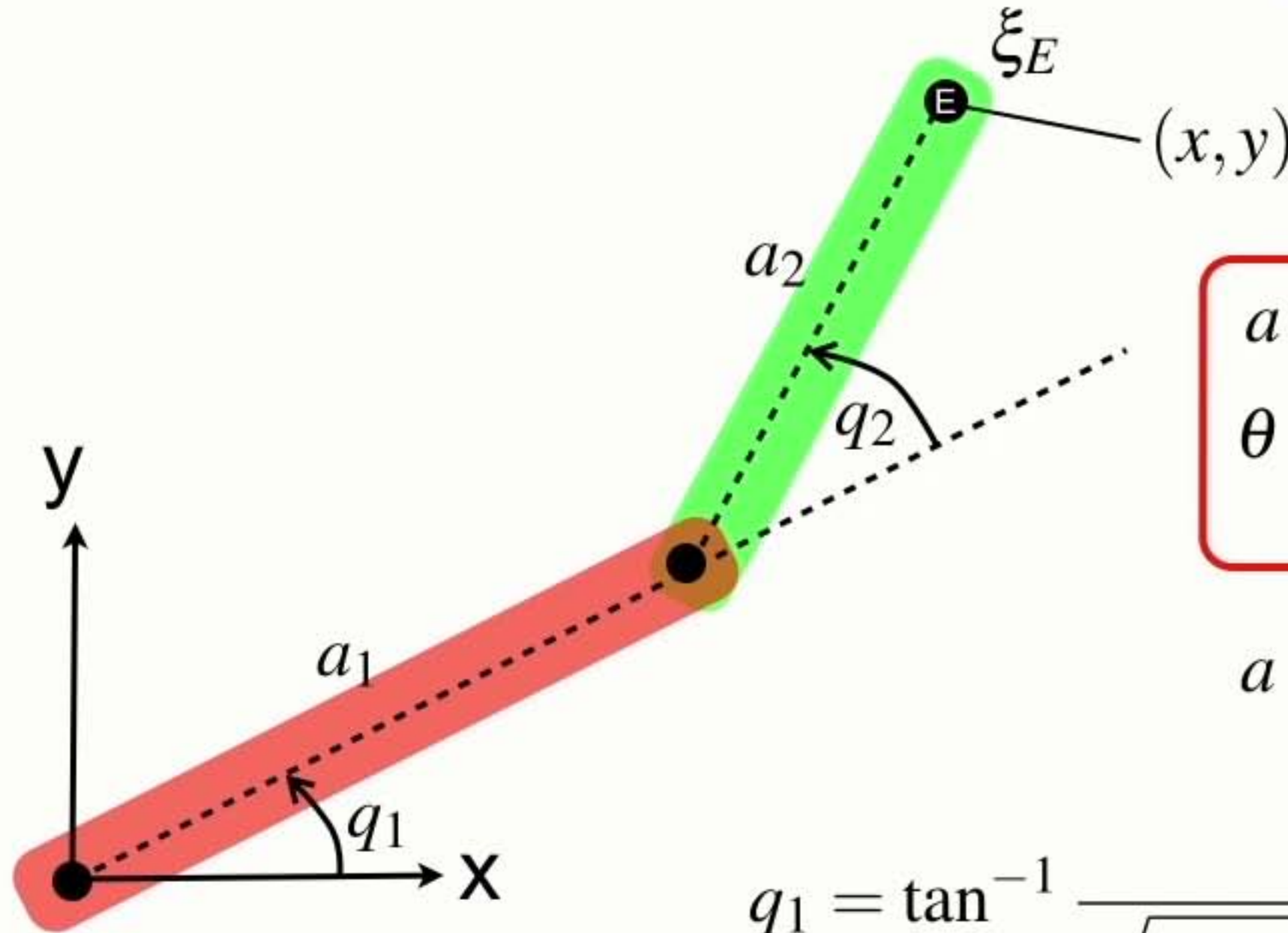
$$\theta = \tan^{-1} \frac{c}{\pm \sqrt{a^2 + b^2 - c^2}} - \tan^{-1} \frac{a}{b}$$

$$a = a_2 S_2, b = a_1 + a_2 C_2, c = y$$

Finding the **joint angles**

$$x = (a_1 + a_2 C_2) \cos q_1 - a_2 S_2 \sin q_1$$

$$y = (a_1 + a_2 C_2) \sin q_1 + a_2 S_2 \cos q_1$$



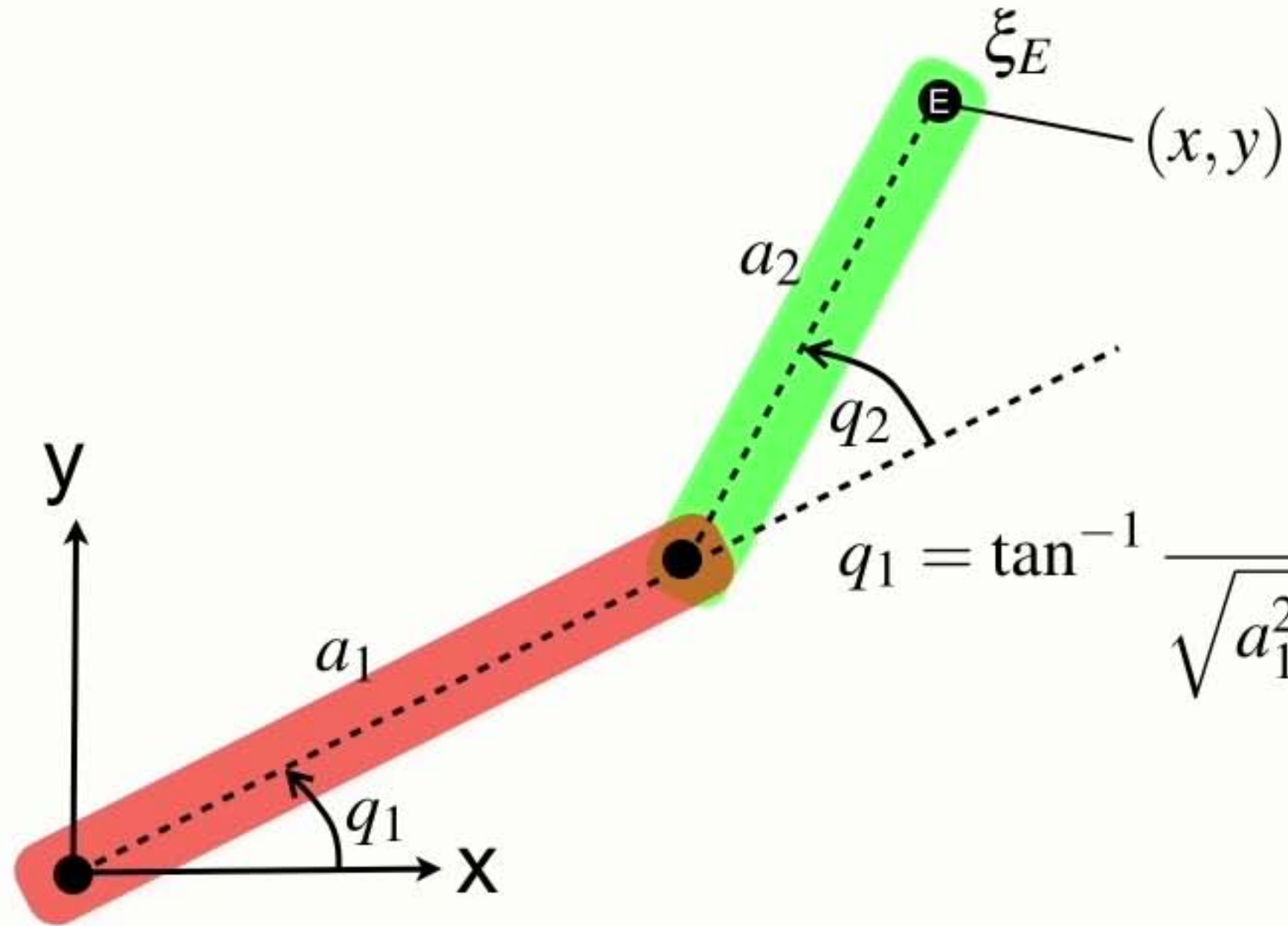
$$a \cos \theta + b \sin \theta = c$$

$$\theta = \tan^{-1} \frac{c}{\pm \sqrt{a^2 + b^2 - c^2}} - \tan^{-1} \frac{a}{b}$$

$$a = a_2 S_2, b = a_1 + a_2 C_2, c = y$$

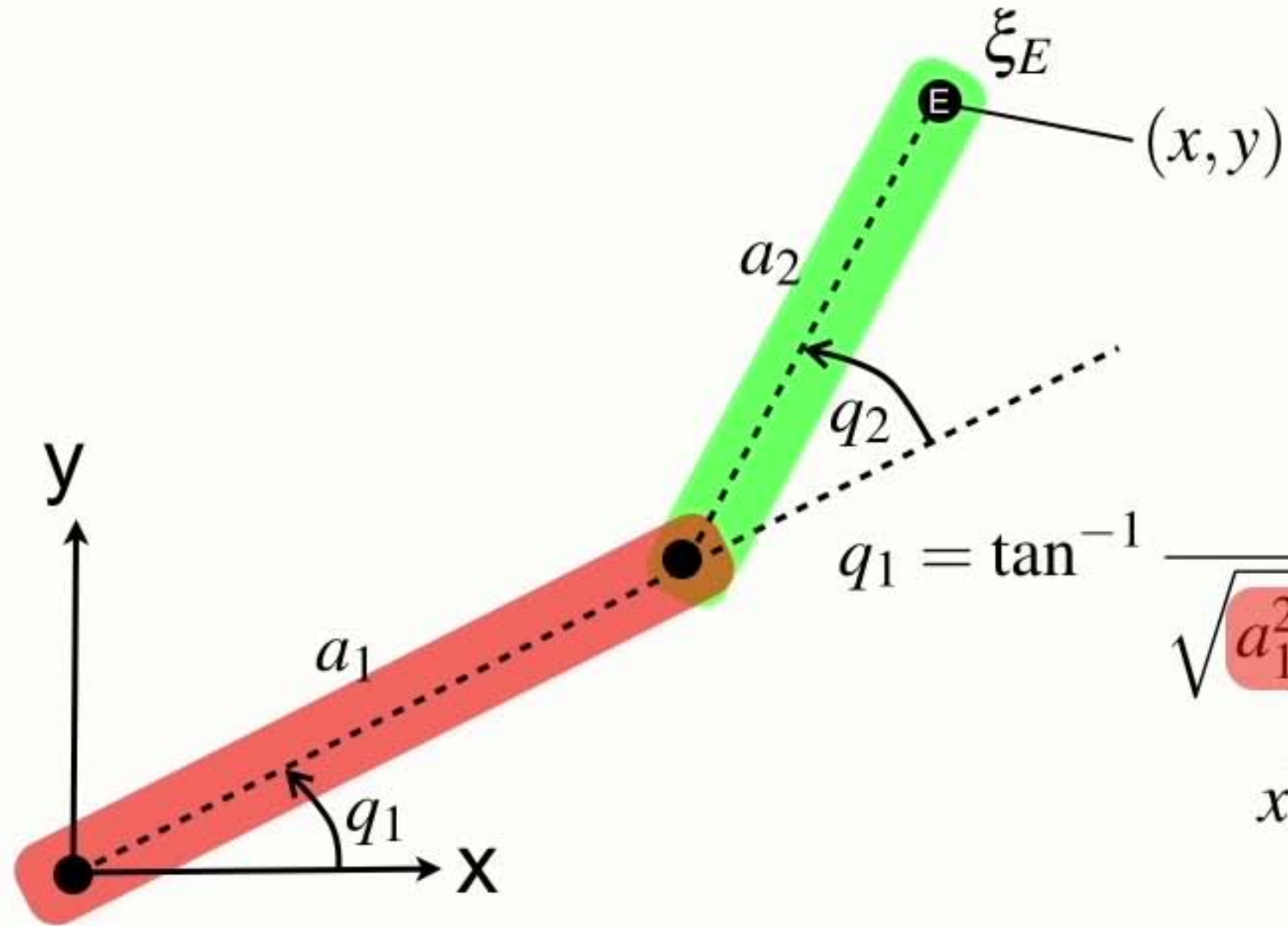
$$q_1 = \tan^{-1} \frac{y}{\sqrt{a_1^2 + a_2^2 + 2a_1 a_2 C_2 - y^2}} - \tan^{-1} \frac{a_2 S_2}{1 + a_2 C_2}$$

Finding the **joint angles**



$$q_1 = \tan^{-1} \frac{y}{\sqrt{a_1^2 + a_2^2 + 2a_1a_2C_2 - y^2}} - \tan^{-1} \frac{a_2S_2}{1 + a_2C_2}$$

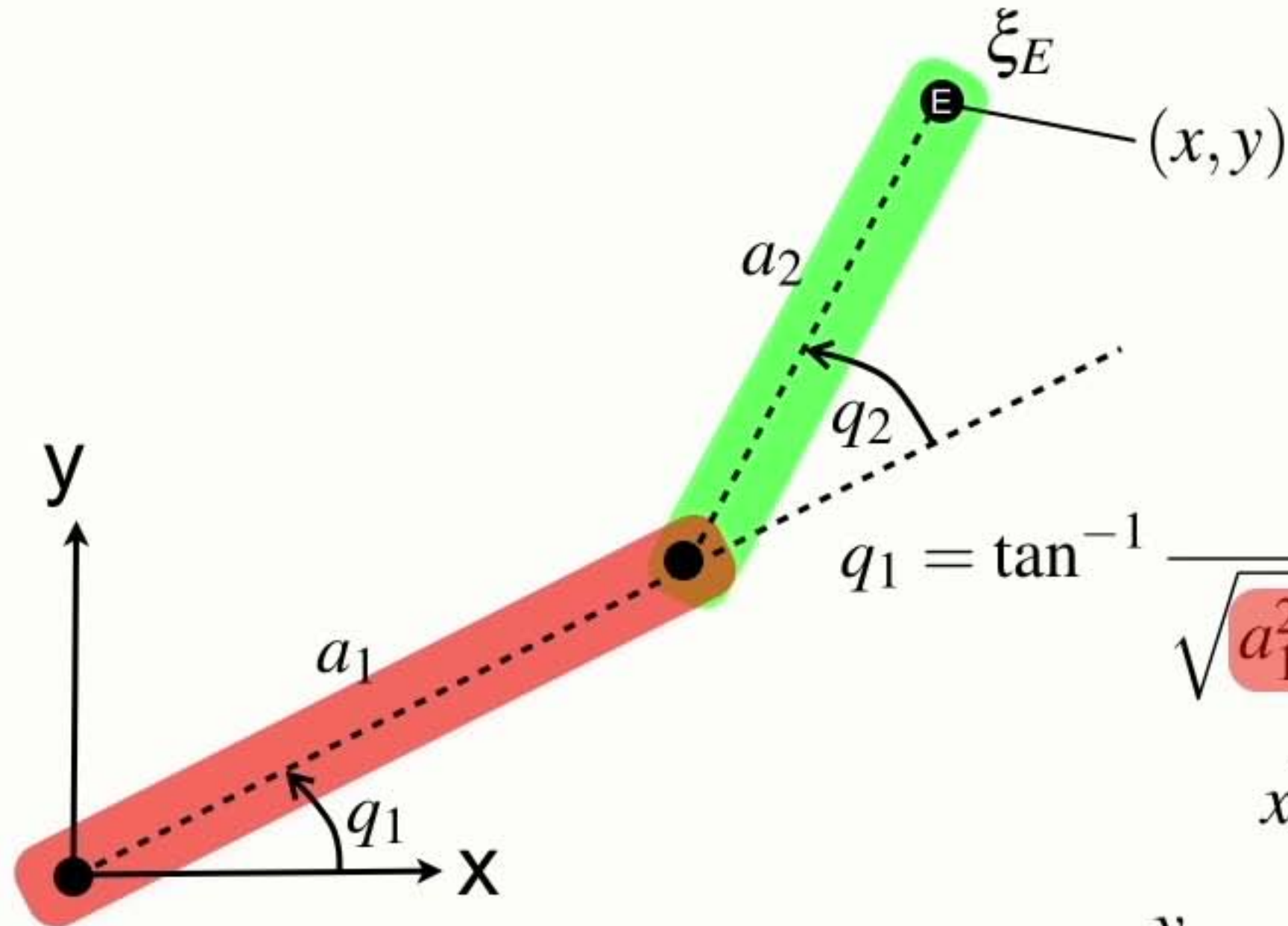
Finding the **joint angles**



$$q_1 = \tan^{-1} \frac{y}{\sqrt{a_1^2 + a_2^2 + 2a_1a_2C_2} - y^2} - \tan^{-1} \frac{a_2S_2}{1 + a_2C_2}$$

$$x^2 + y^2 = a_1^2 + a_2^2 + 2a_1a_2 \cos q_2$$

Finding the **joint angles**



$$q_1 = \tan^{-1} \frac{y}{\sqrt{a_1^2 + a_2^2 + 2a_1a_2C_2} - y^2} - \tan^{-1} \frac{a_2S_2}{1 + a_2C_2}$$

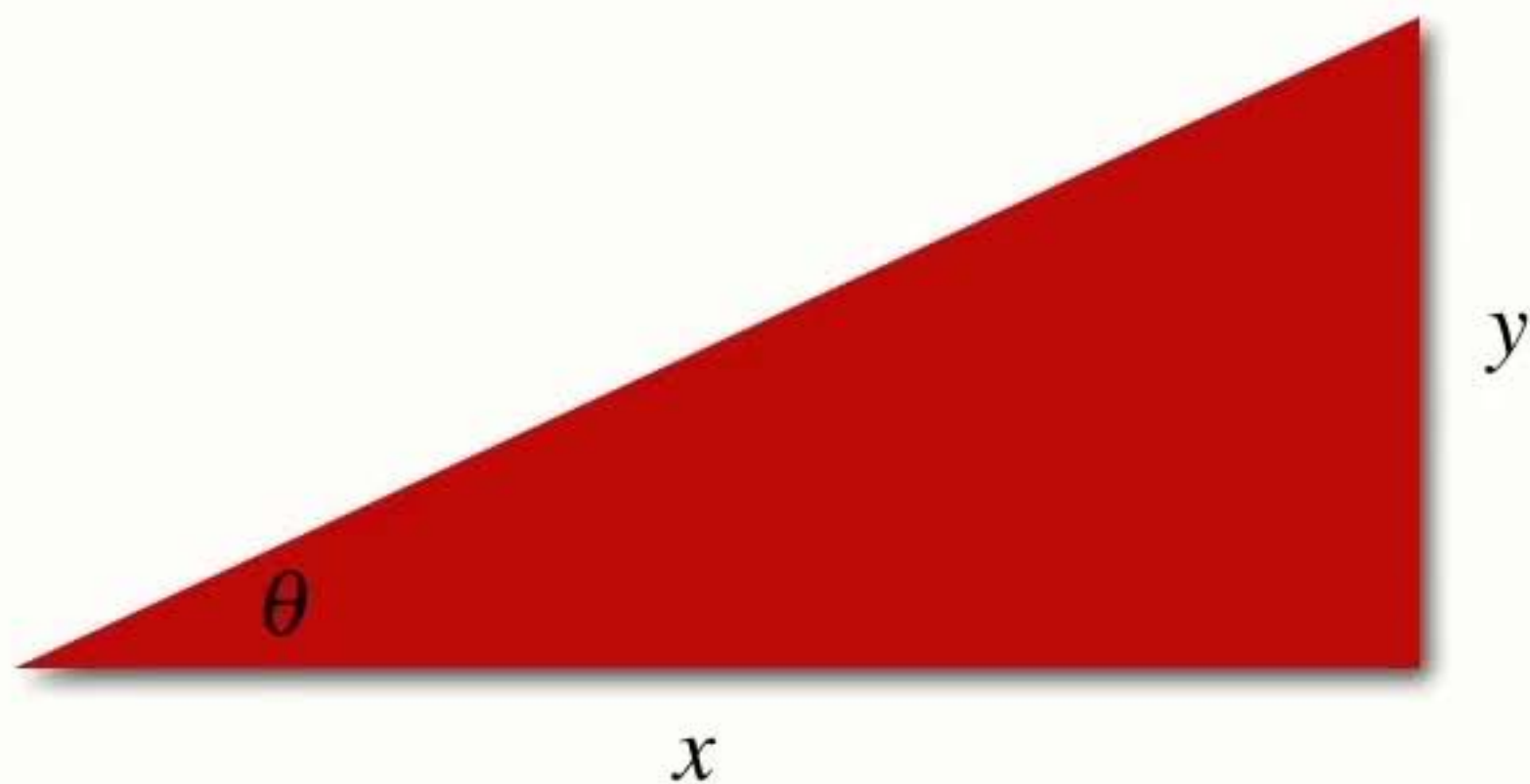
$$x^2 + y^2 = a_1^2 + a_2^2 + 2a_1a_2 \cos q_2$$

$$q_1 = \tan^{-1} \frac{y}{x} - \tan^{-1} \frac{a_2S_2}{1 + a_2C_2}$$

Section 3

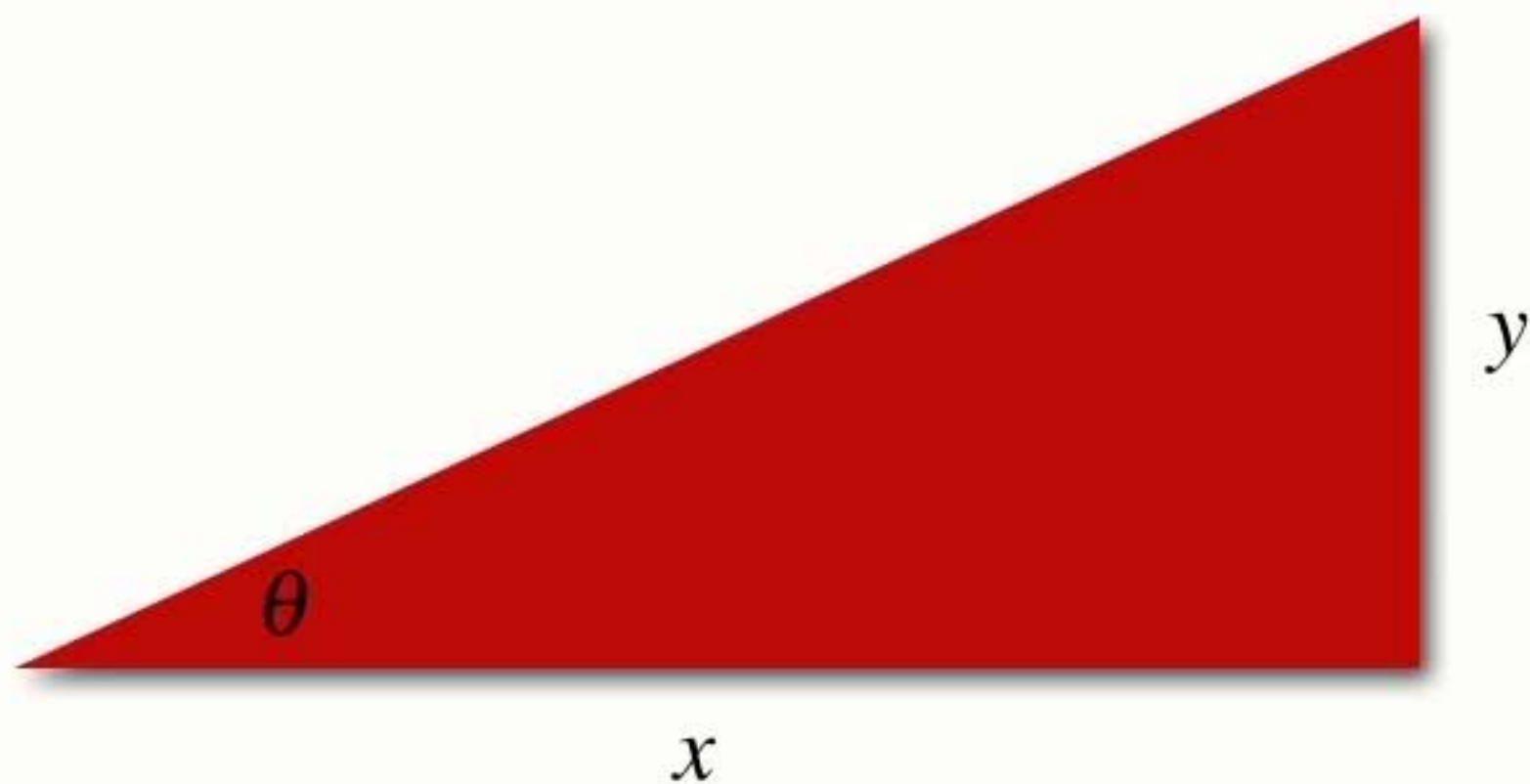
Recap: Inverse tangent function

The **tangent** function



$$y = x \tan \theta$$

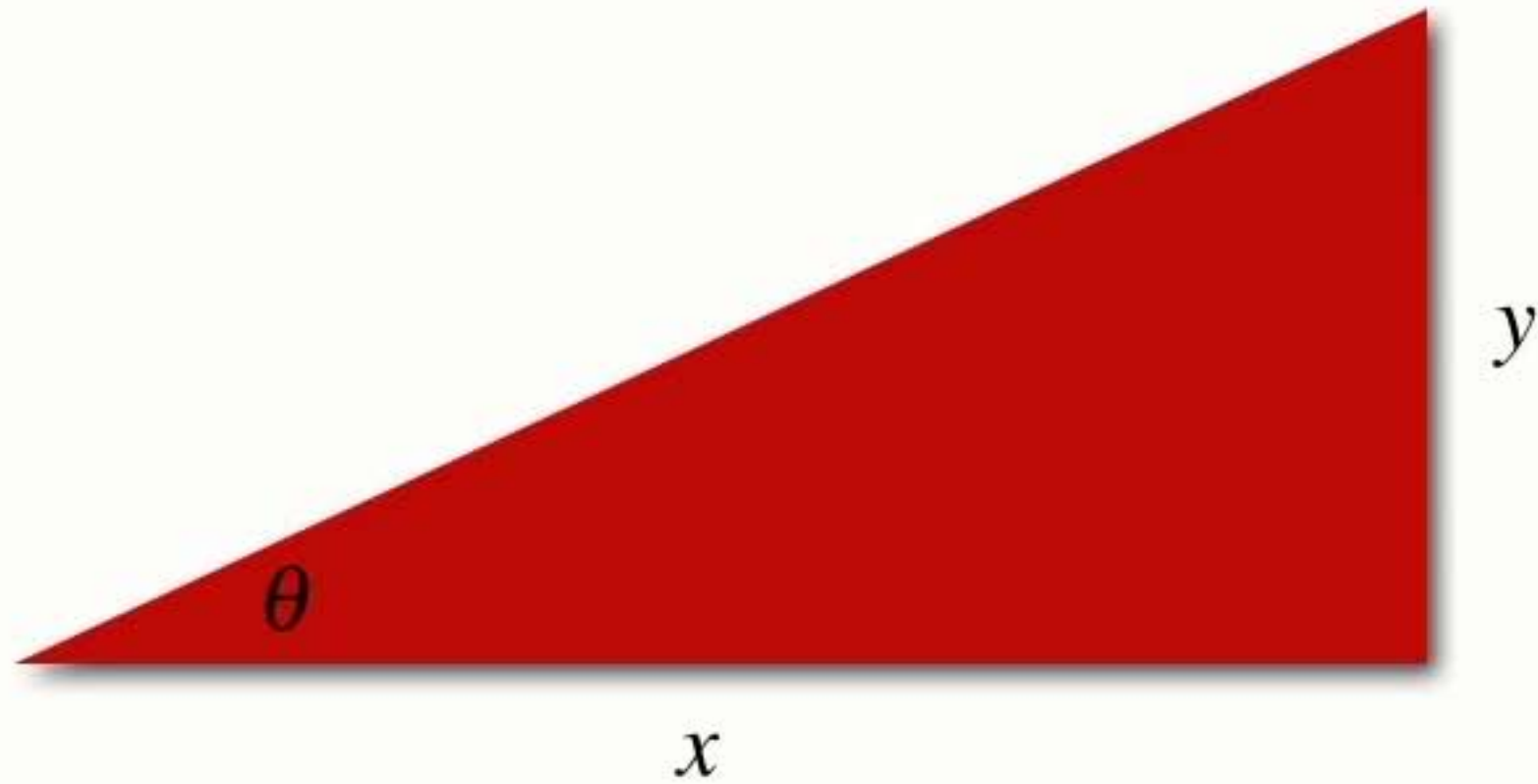
The **tangent** function



$$y = x \tan \theta$$

$$\tan \theta = \frac{y}{x}$$

The **tangent** function

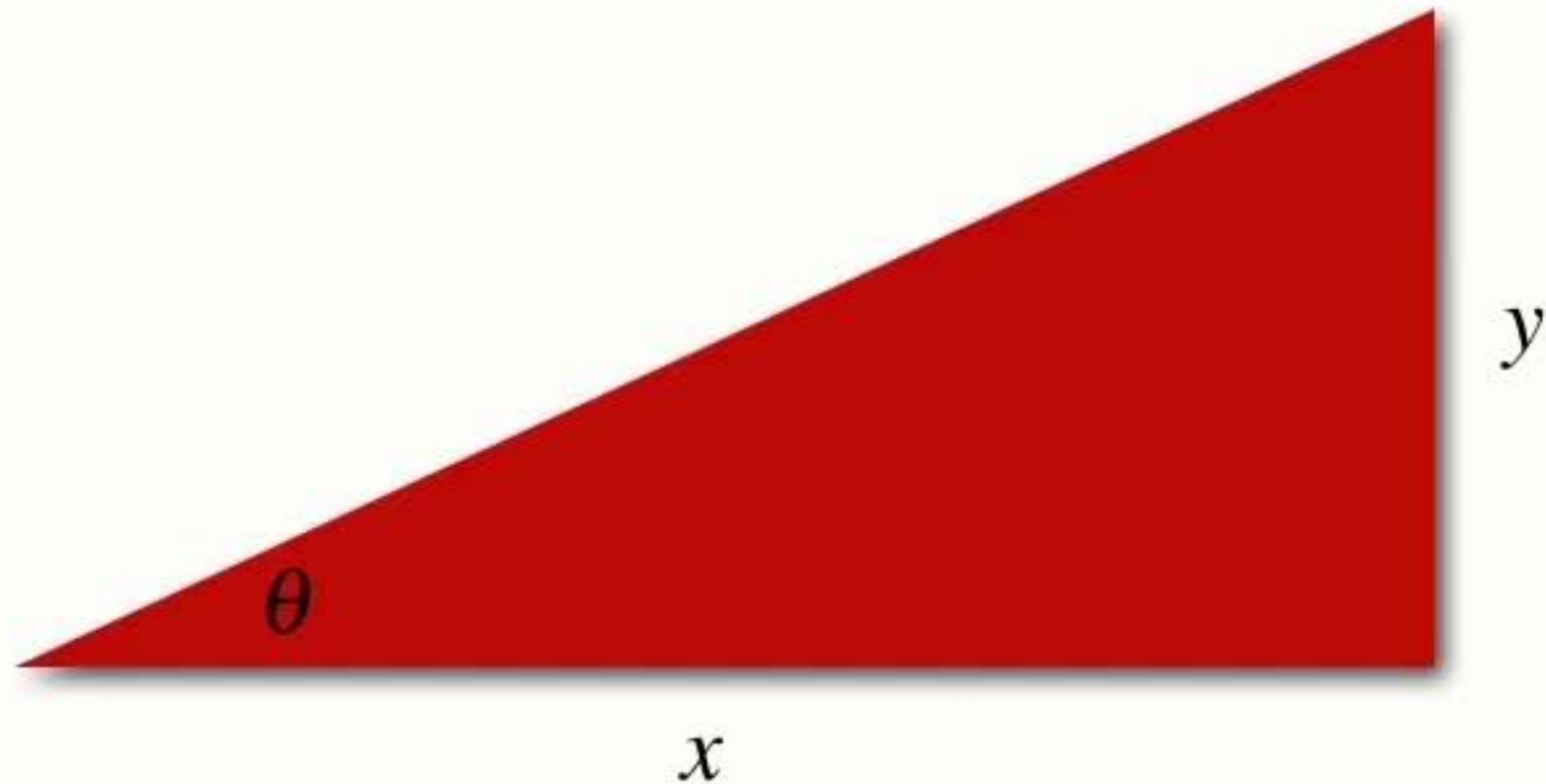


$$y = x \tan \theta$$

$$\tan \theta = \frac{y}{x}$$

$$\theta = \tan^{-1} \frac{y}{x}$$

The **tangent** function



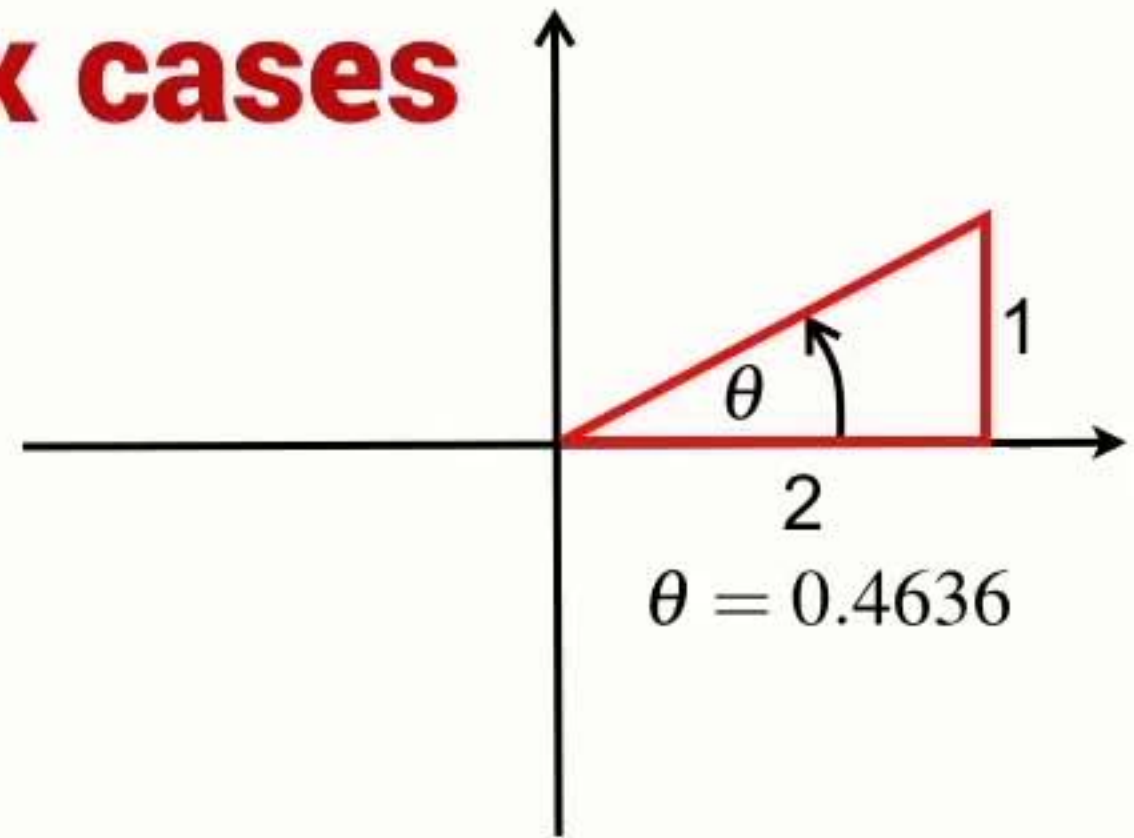
$$y = x \tan \theta$$

$$\tan \theta = \frac{y}{x}$$

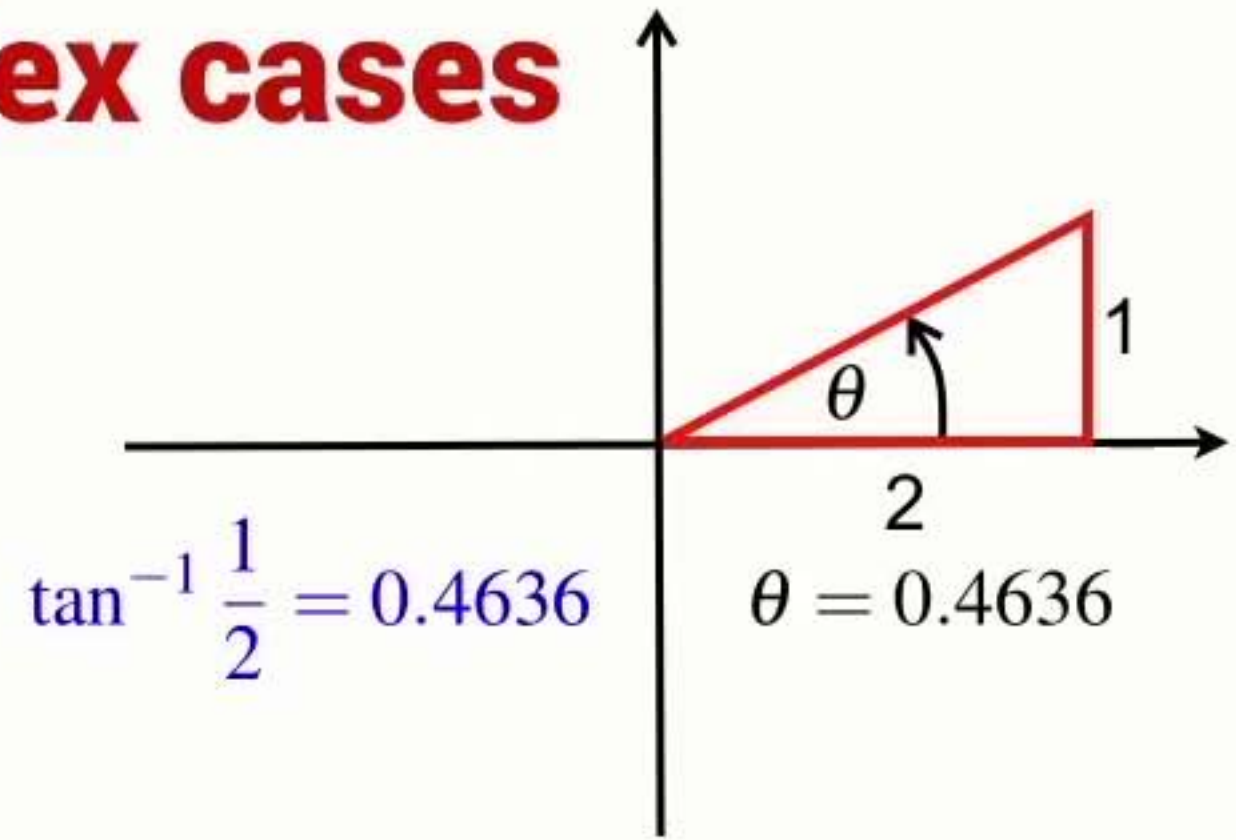
$$\theta = \tan^{-1} \frac{y}{x}$$

- Various names $\tan^{-1} \sim \text{atan} \sim \text{arctan}$

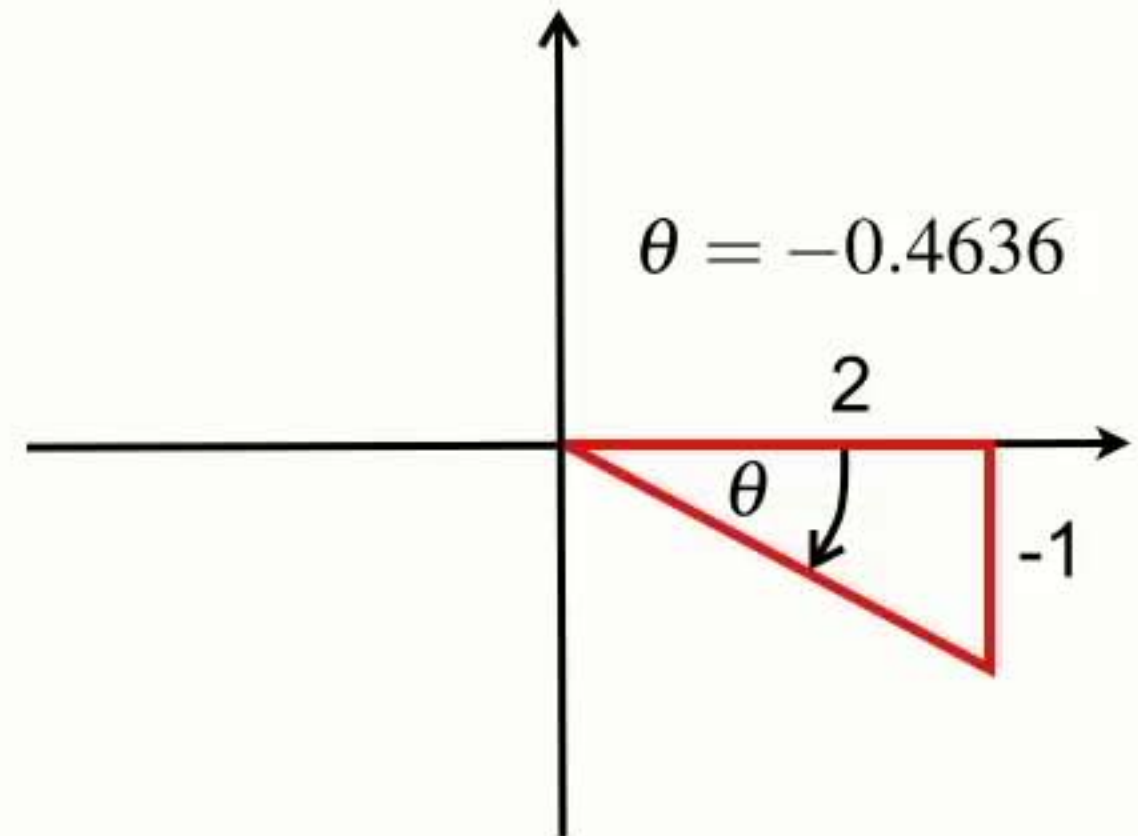
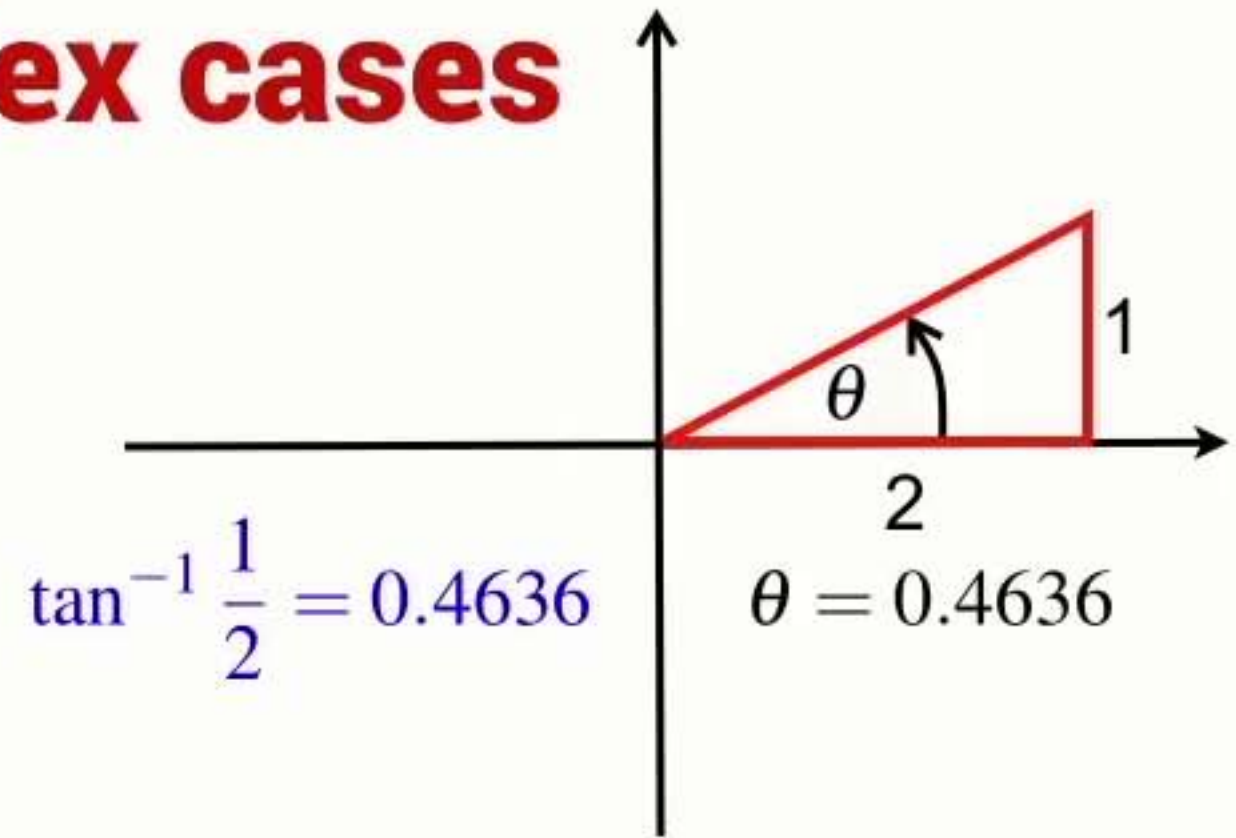
More complex cases



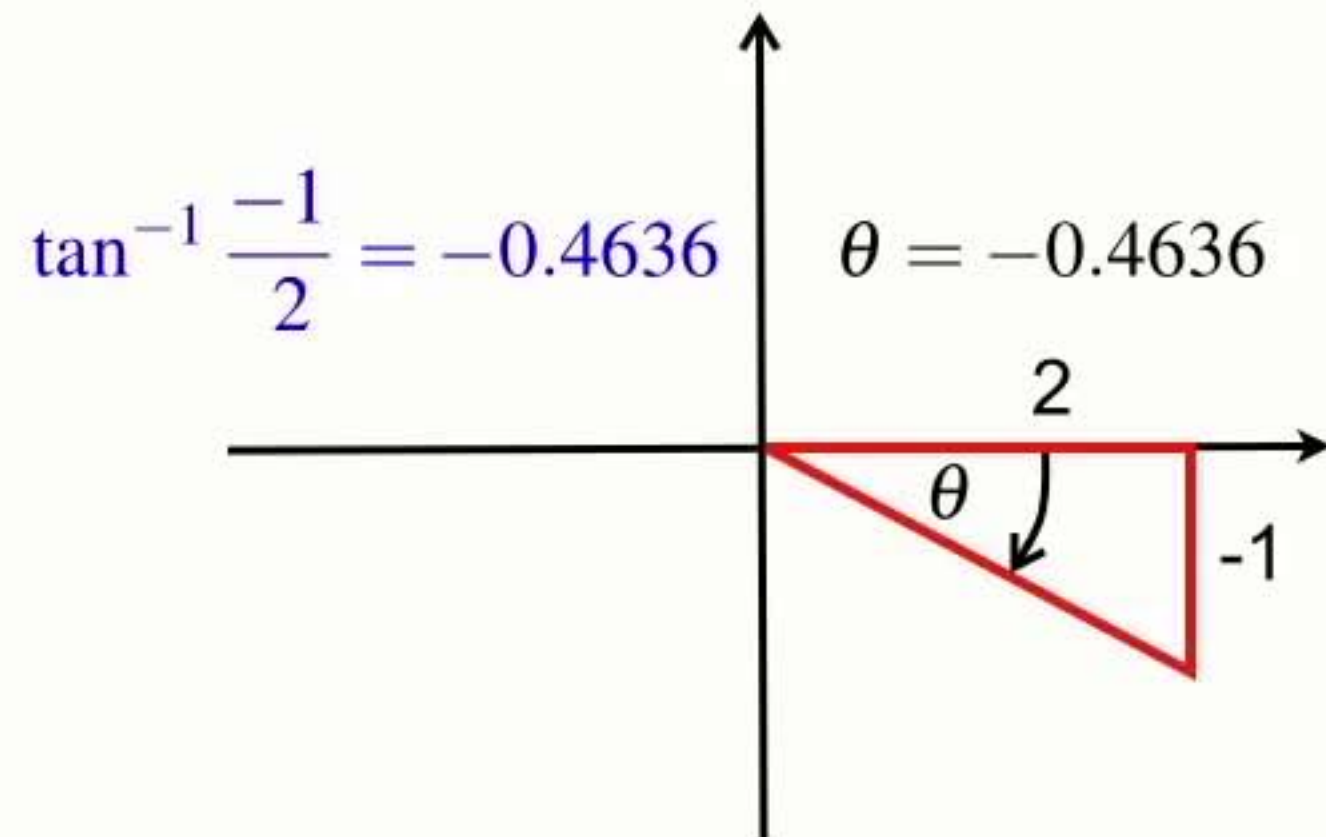
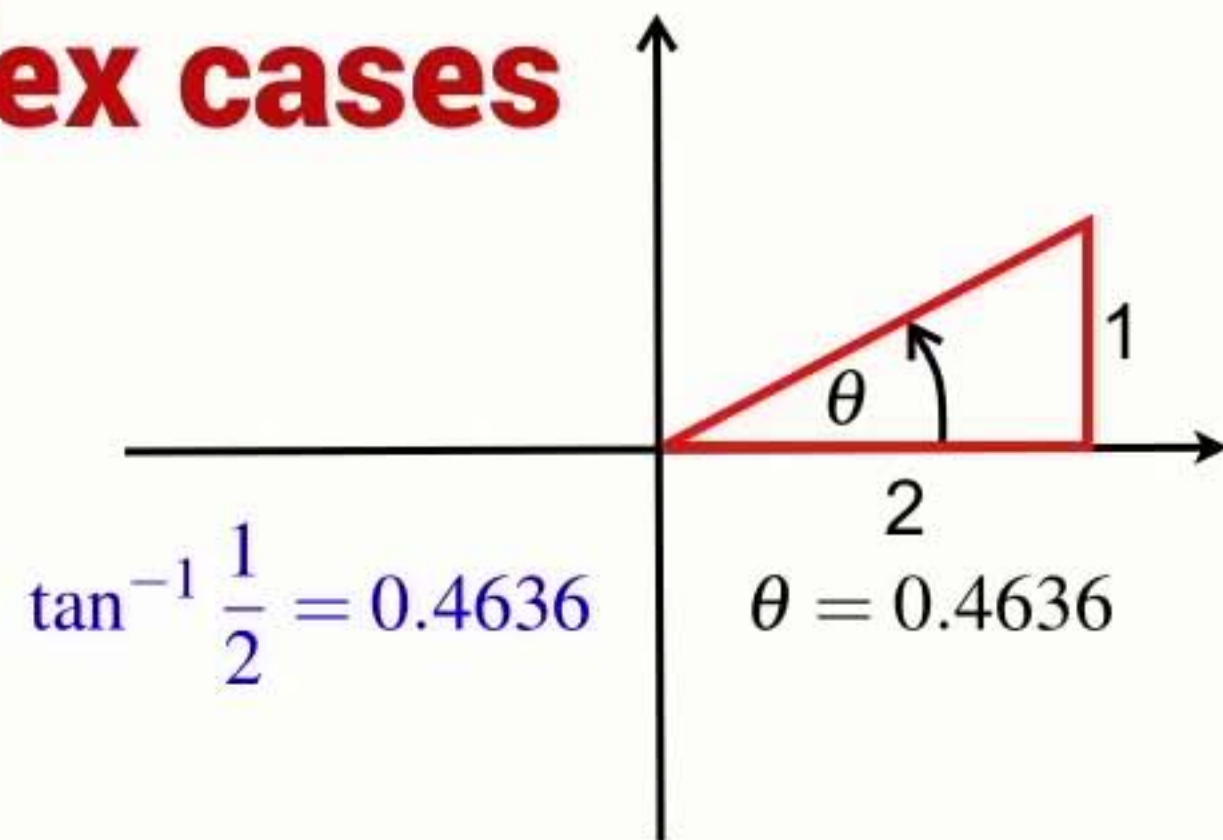
More complex cases



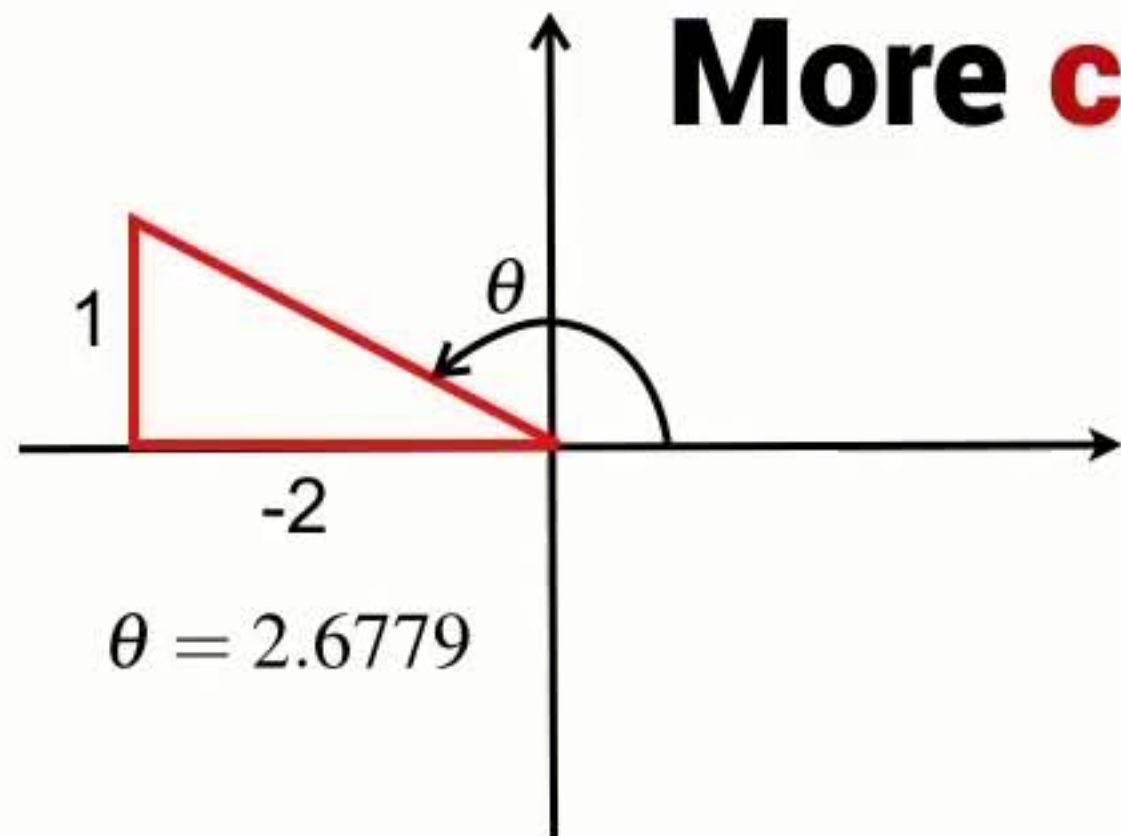
More complex cases



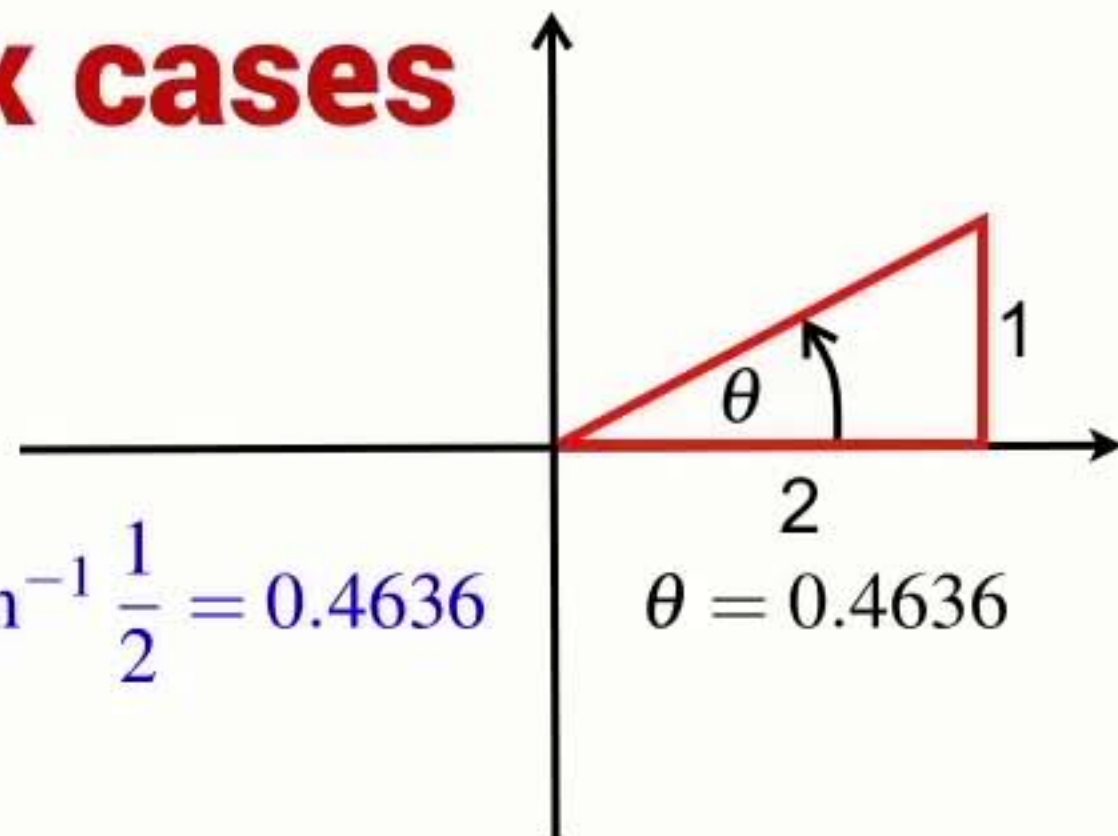
More complex cases



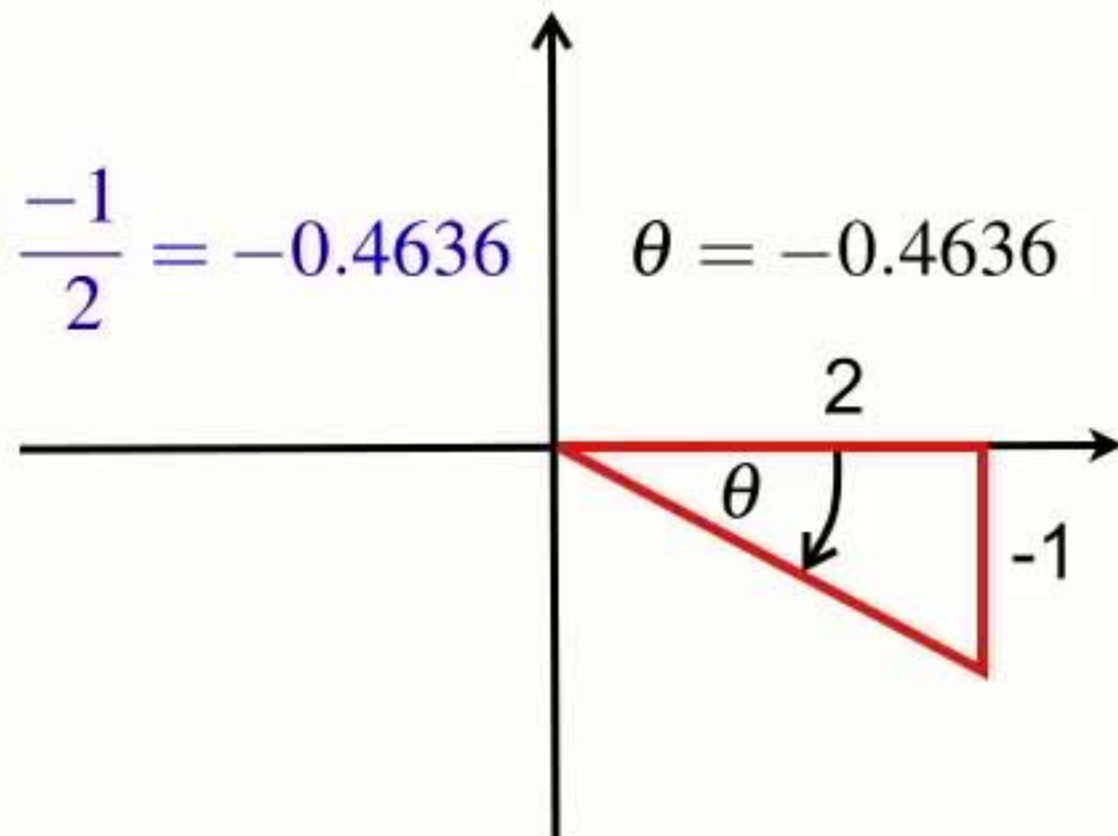
More complex cases



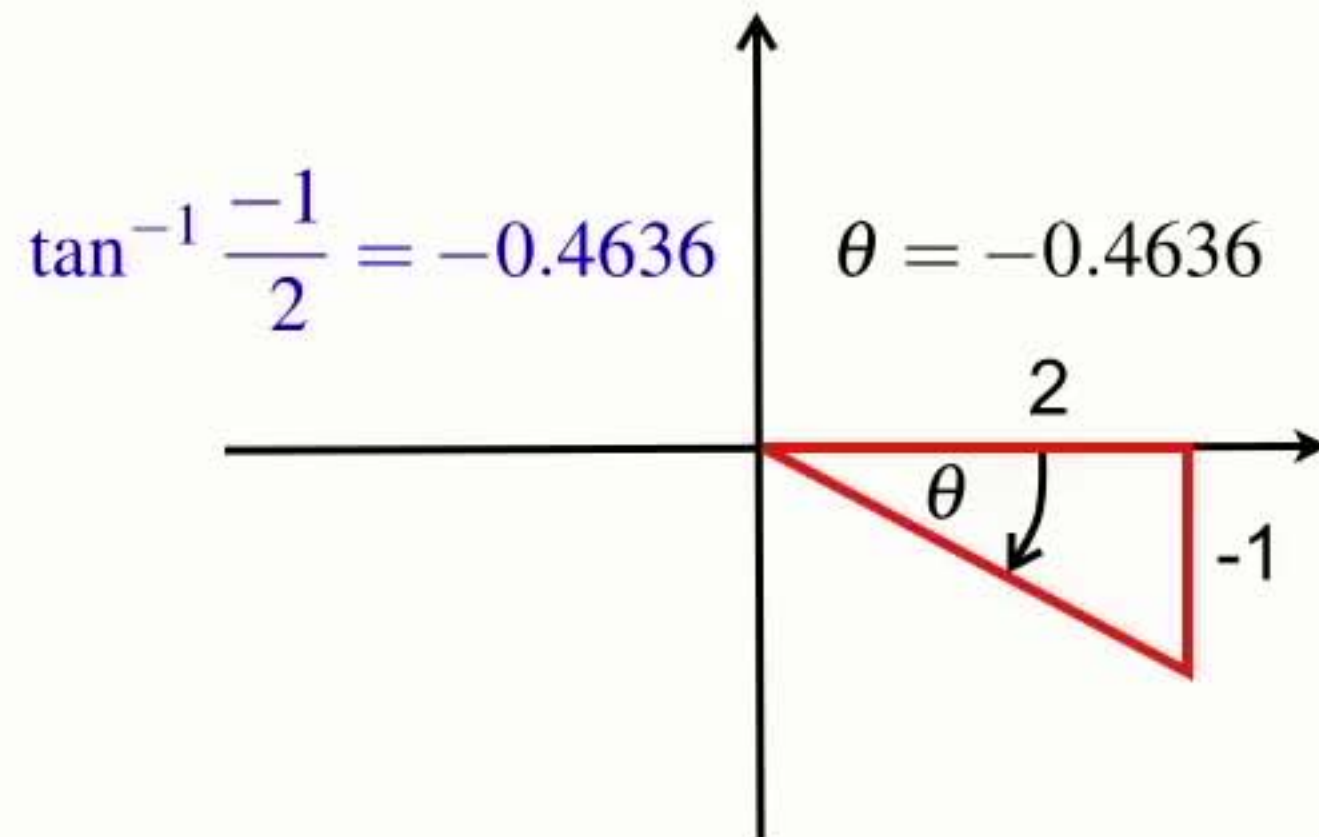
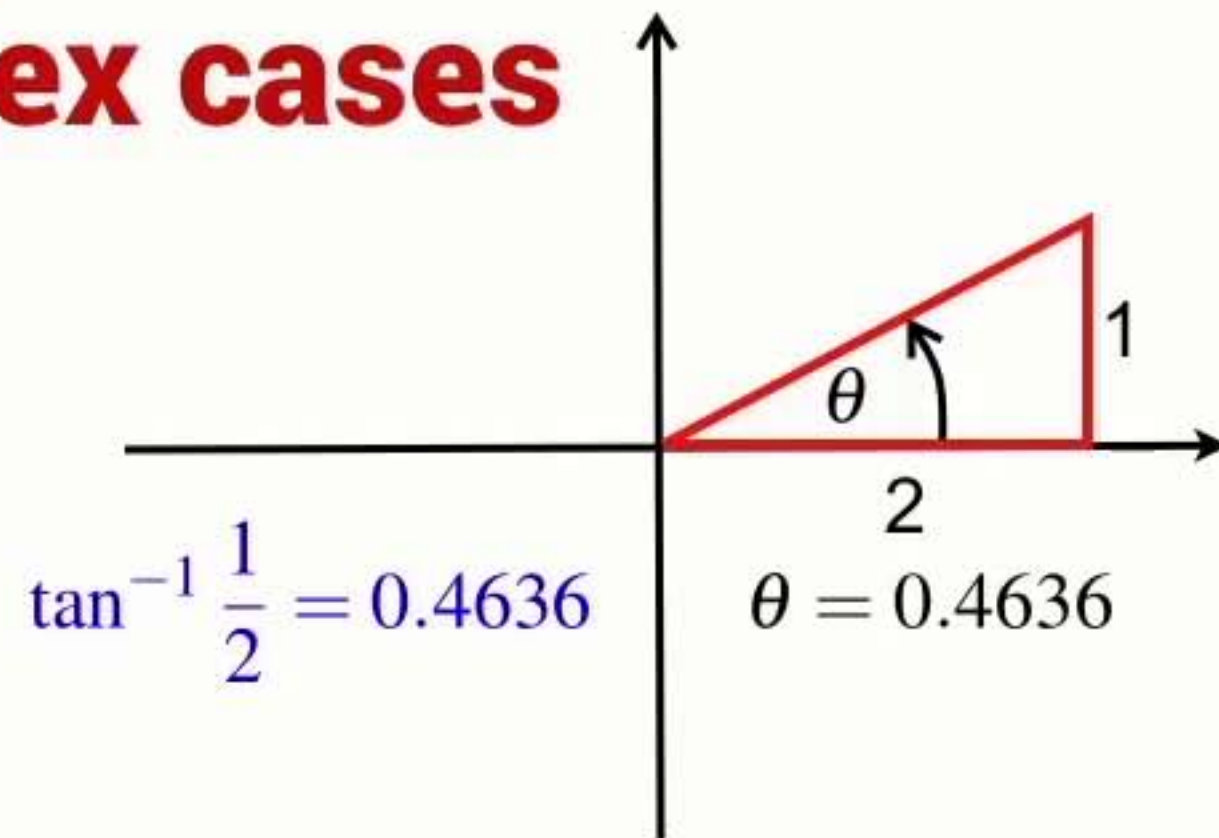
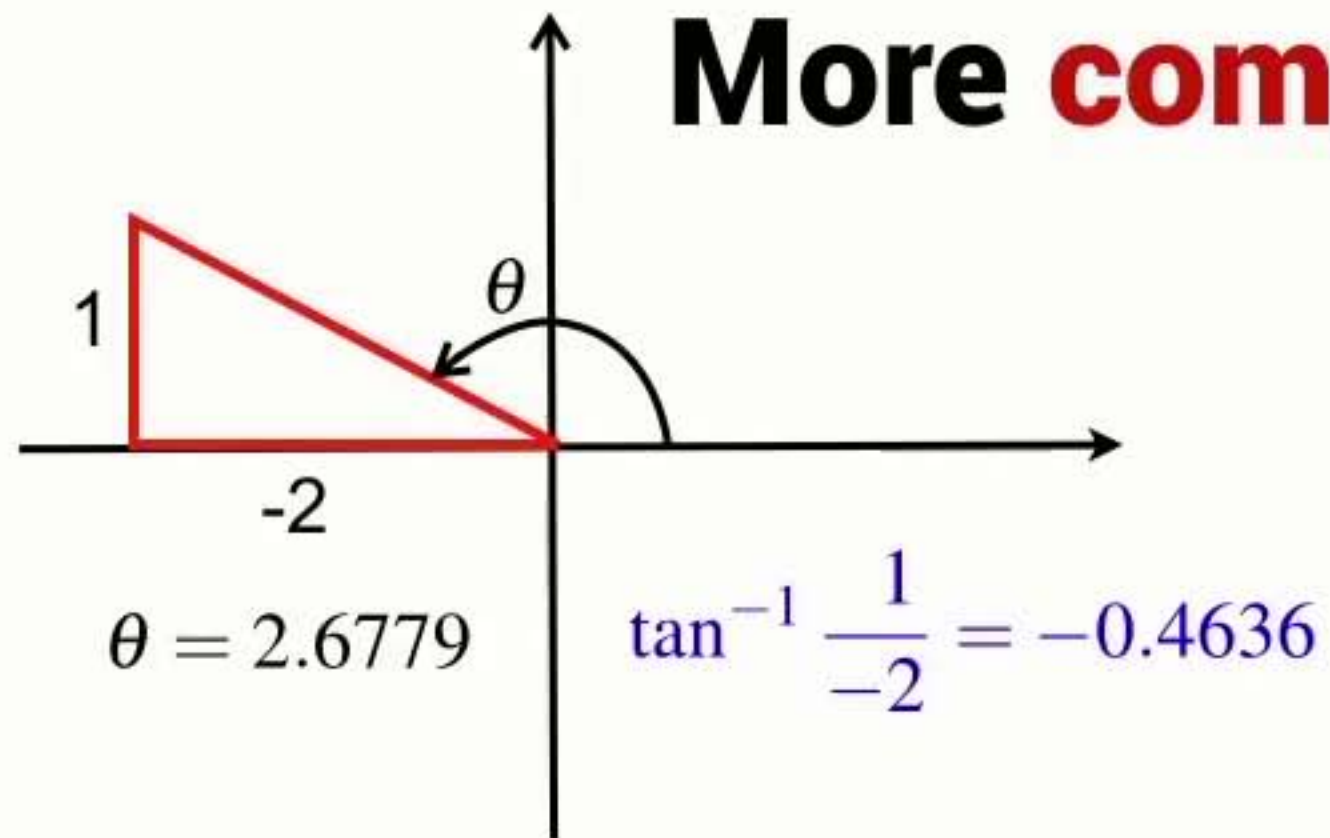
$$\tan^{-1} \frac{1}{2} = 0.4636$$



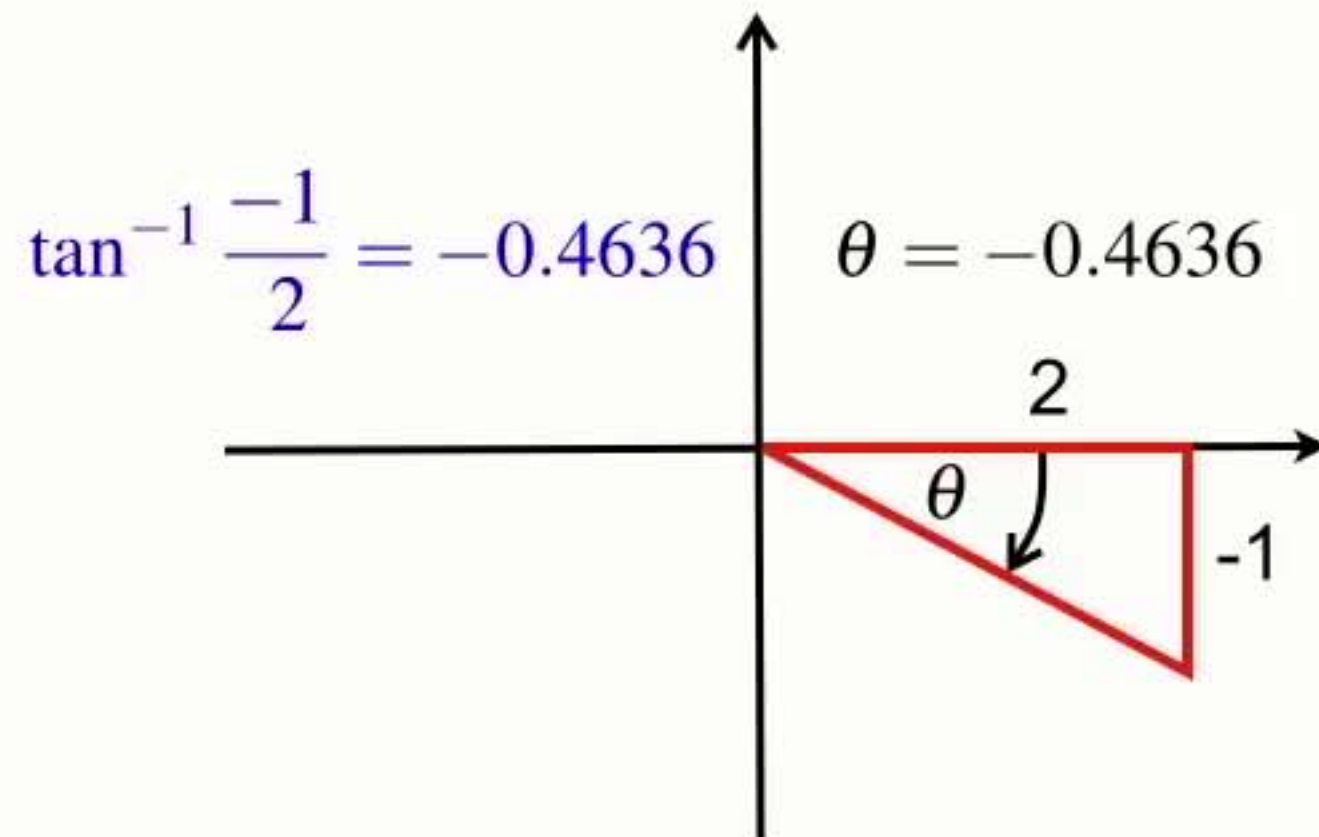
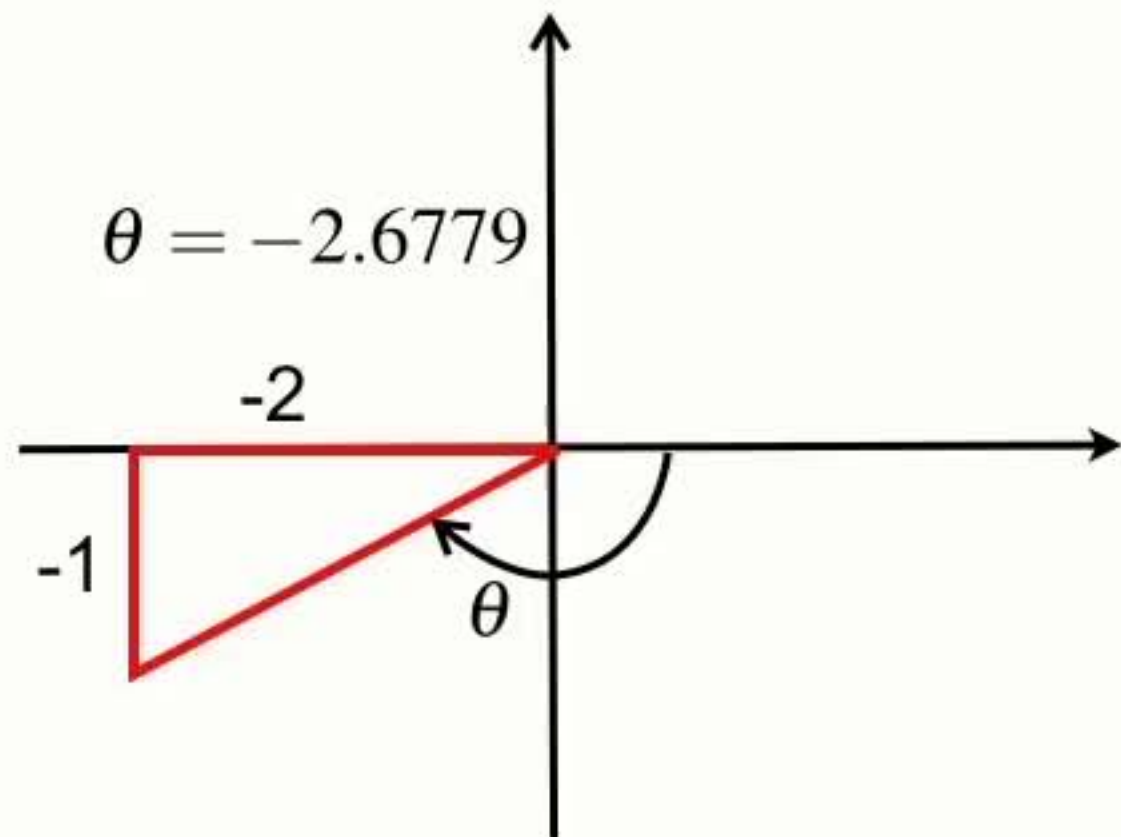
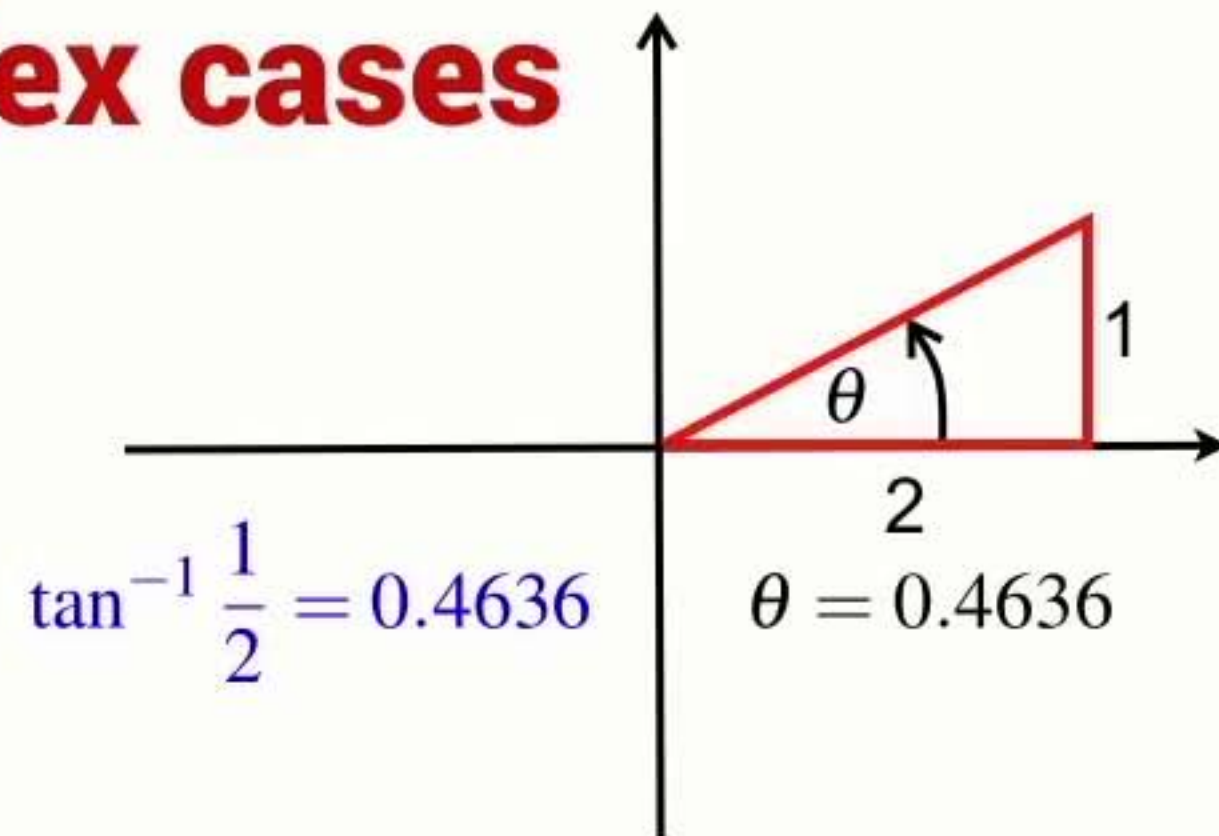
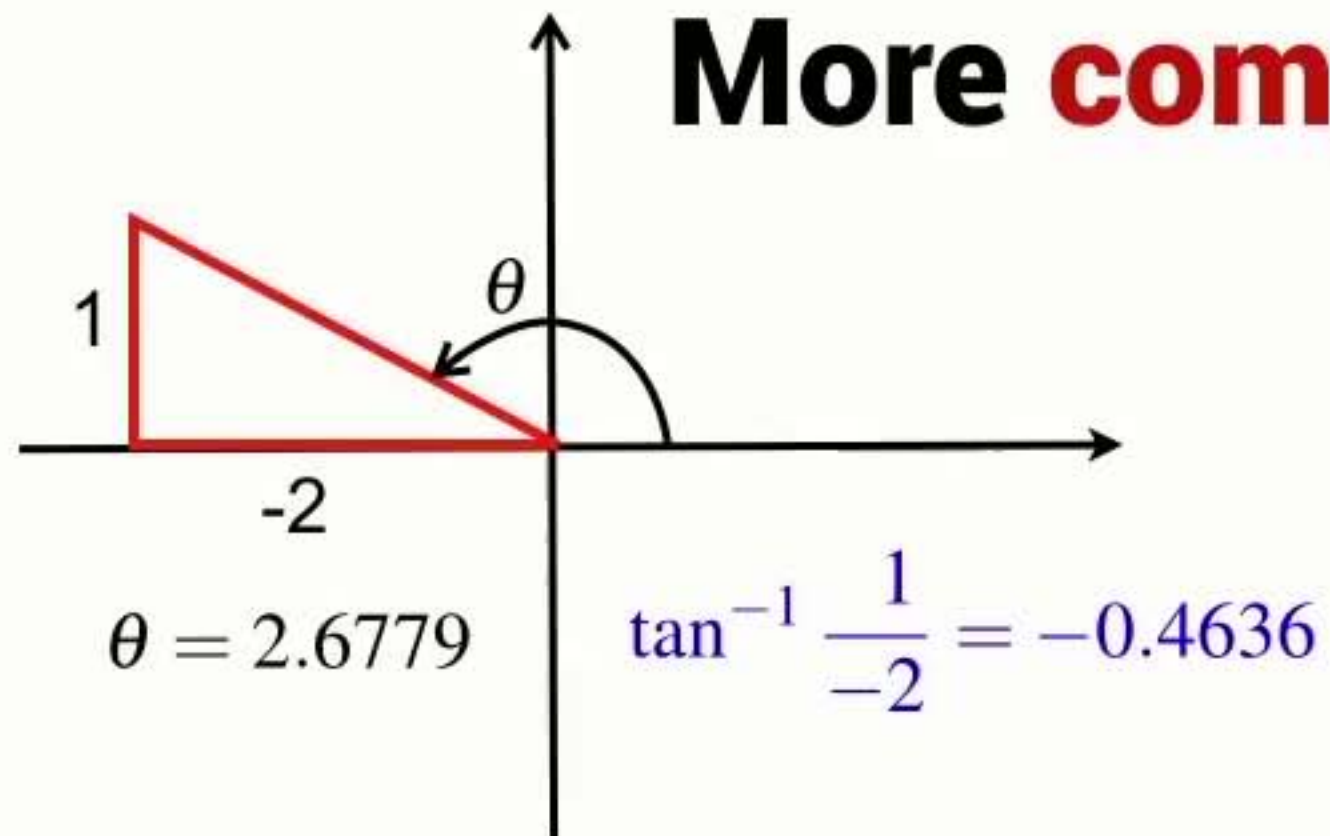
$$\tan^{-1} \frac{-1}{2} = -0.4636$$



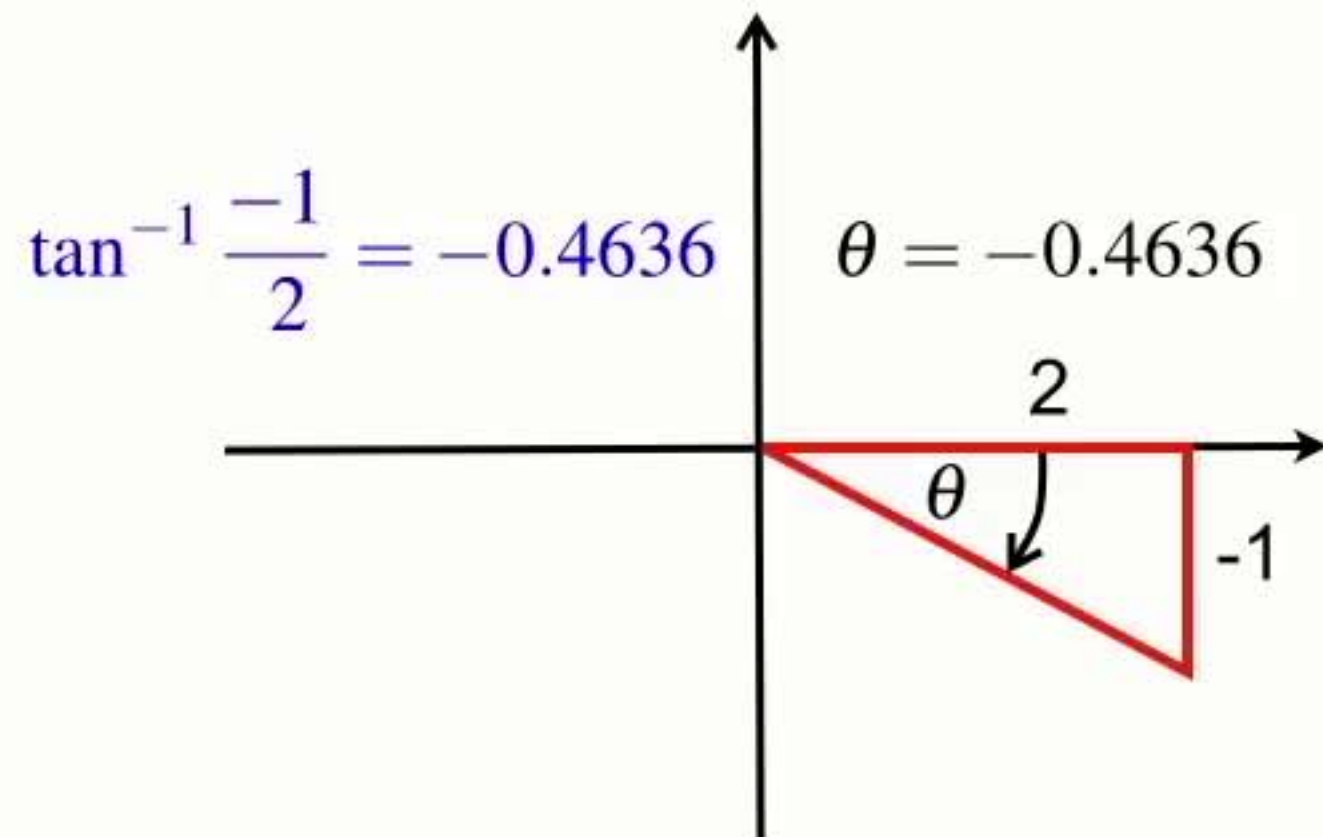
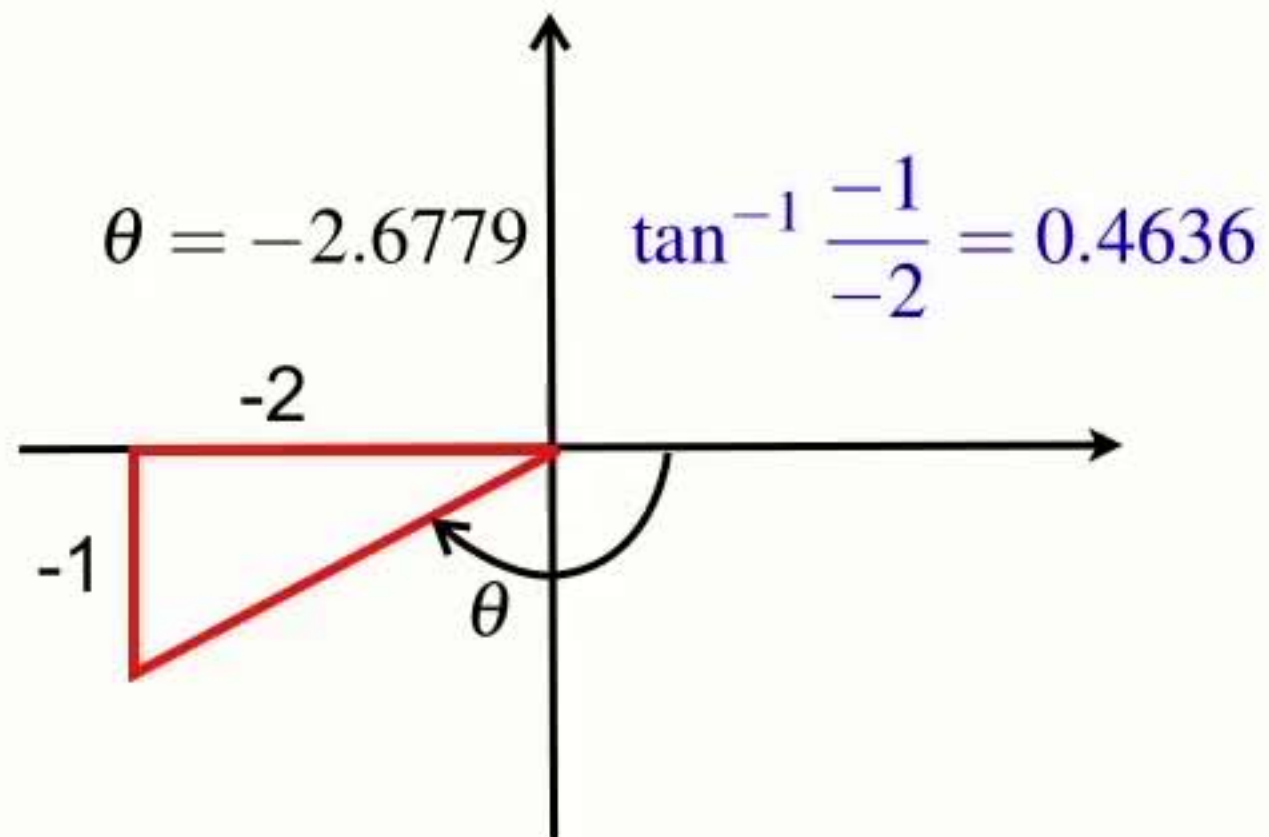
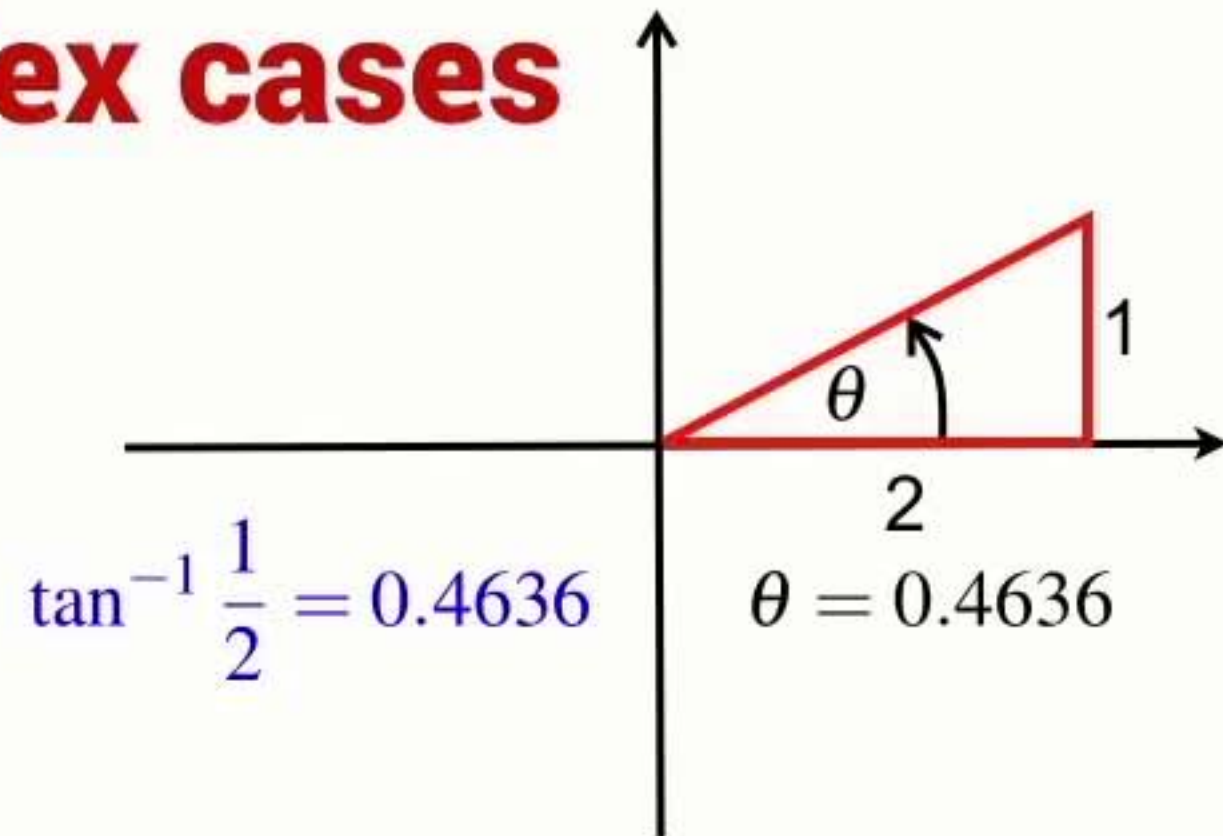
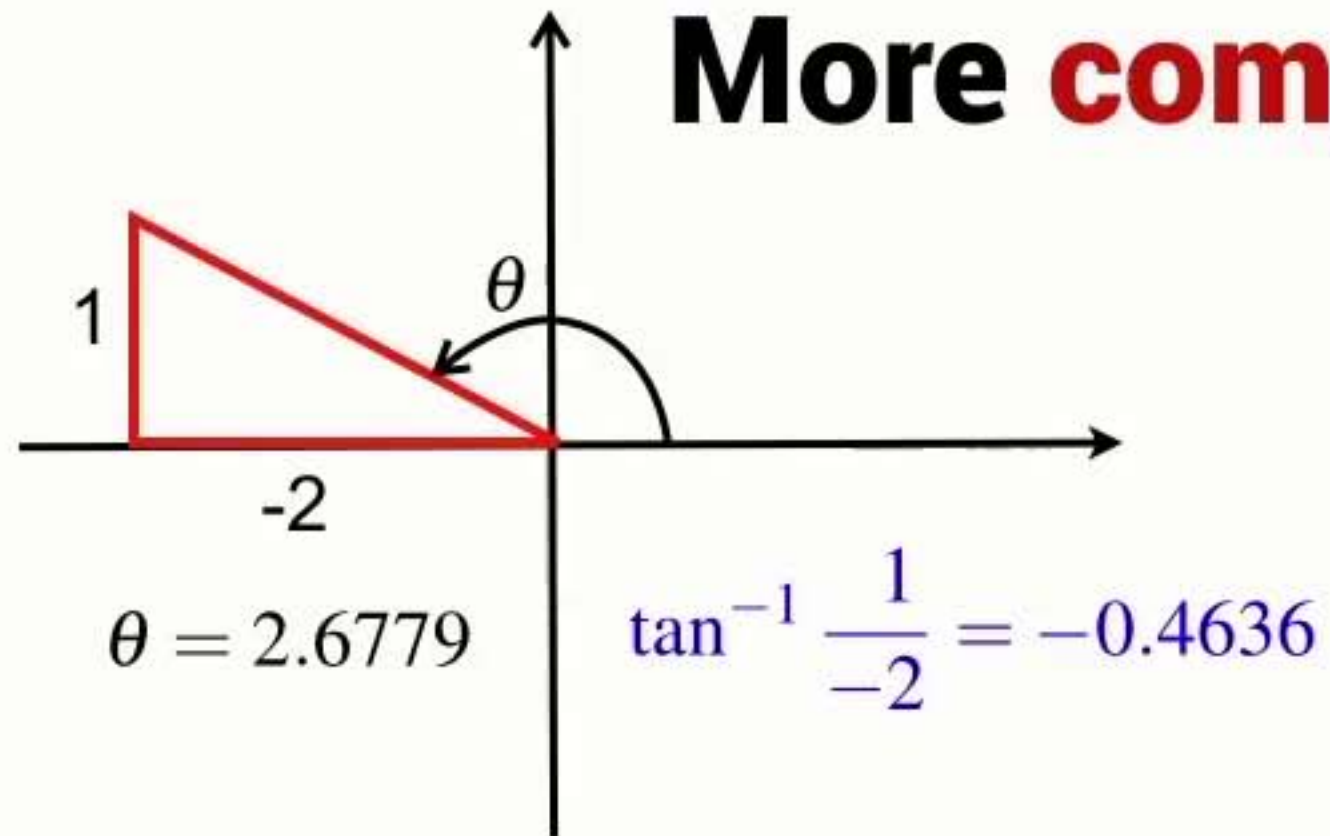
More complex cases



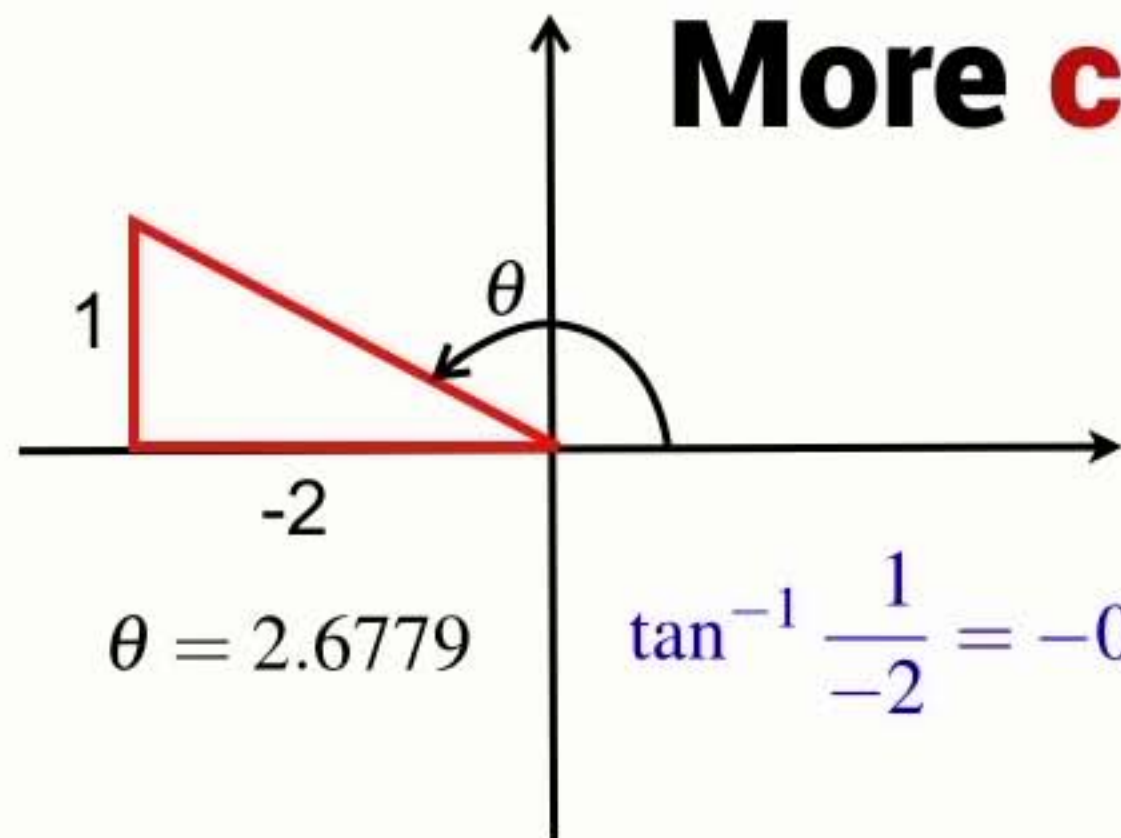
More complex cases



More complex cases

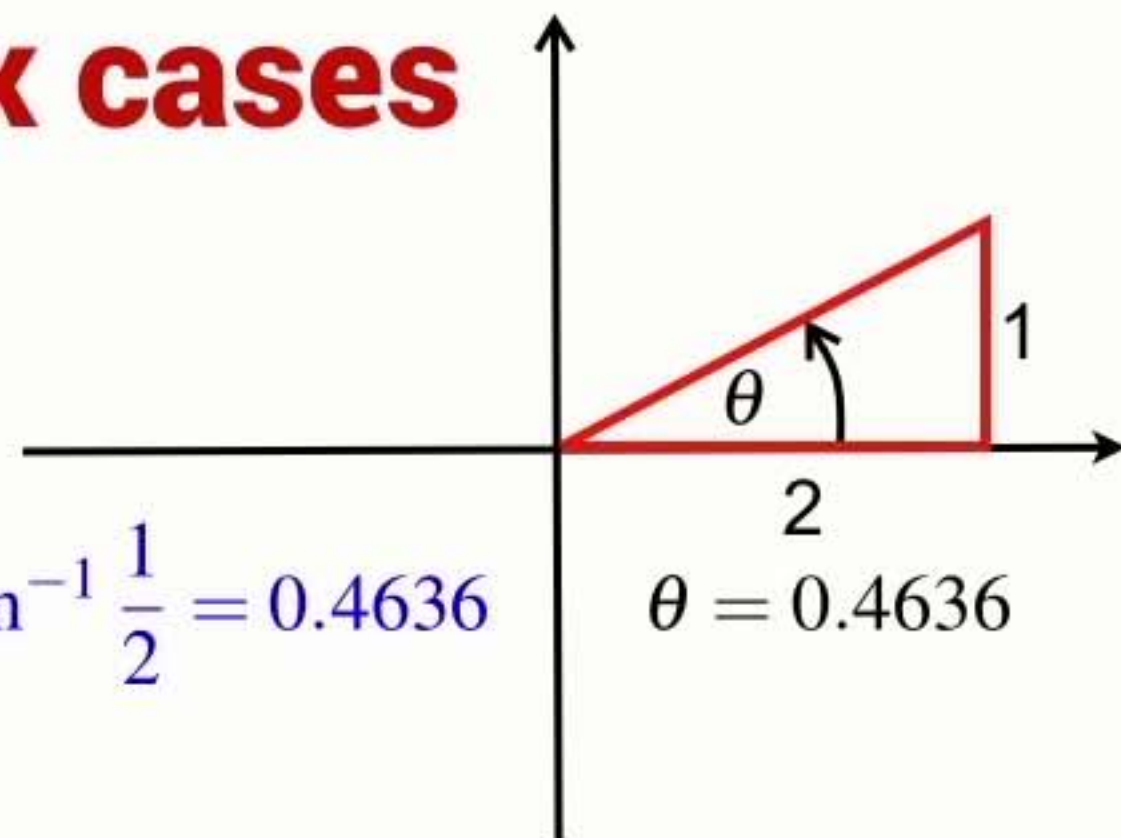


More complex cases



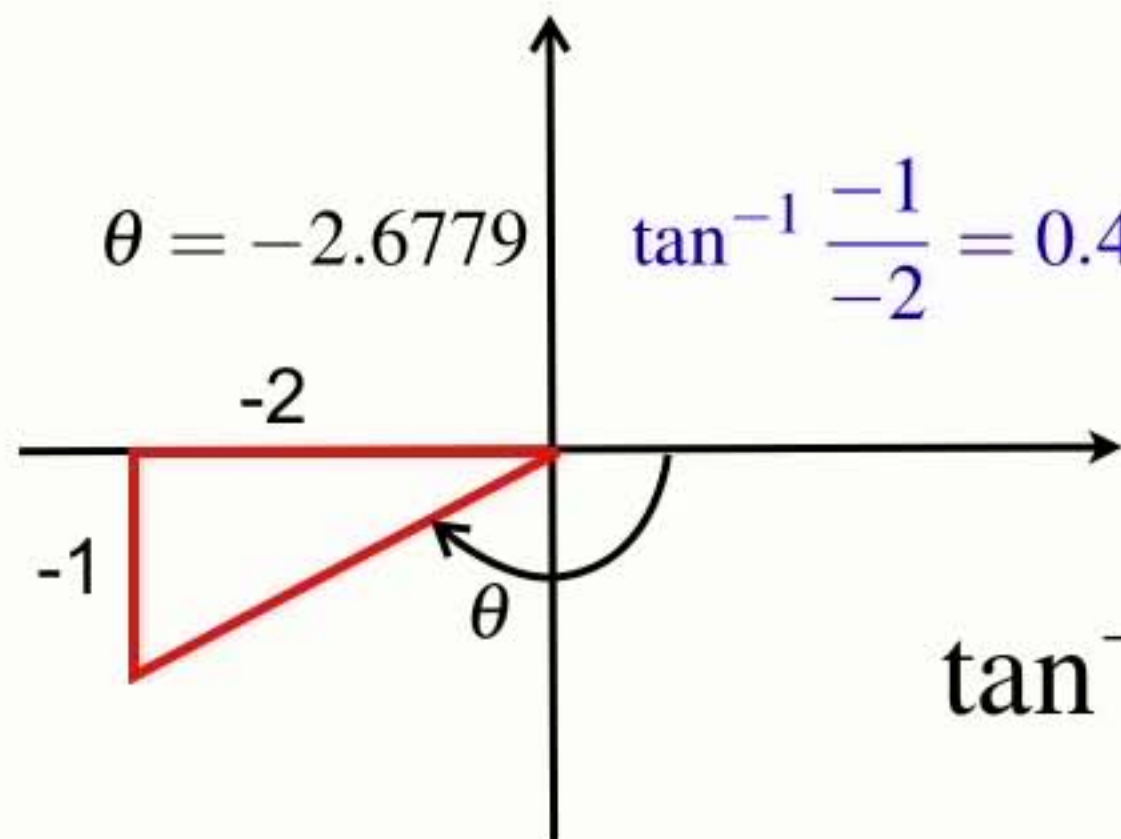
$$\theta = 2.6779$$

$$\tan^{-1} \frac{1}{-2} = -0.4636$$



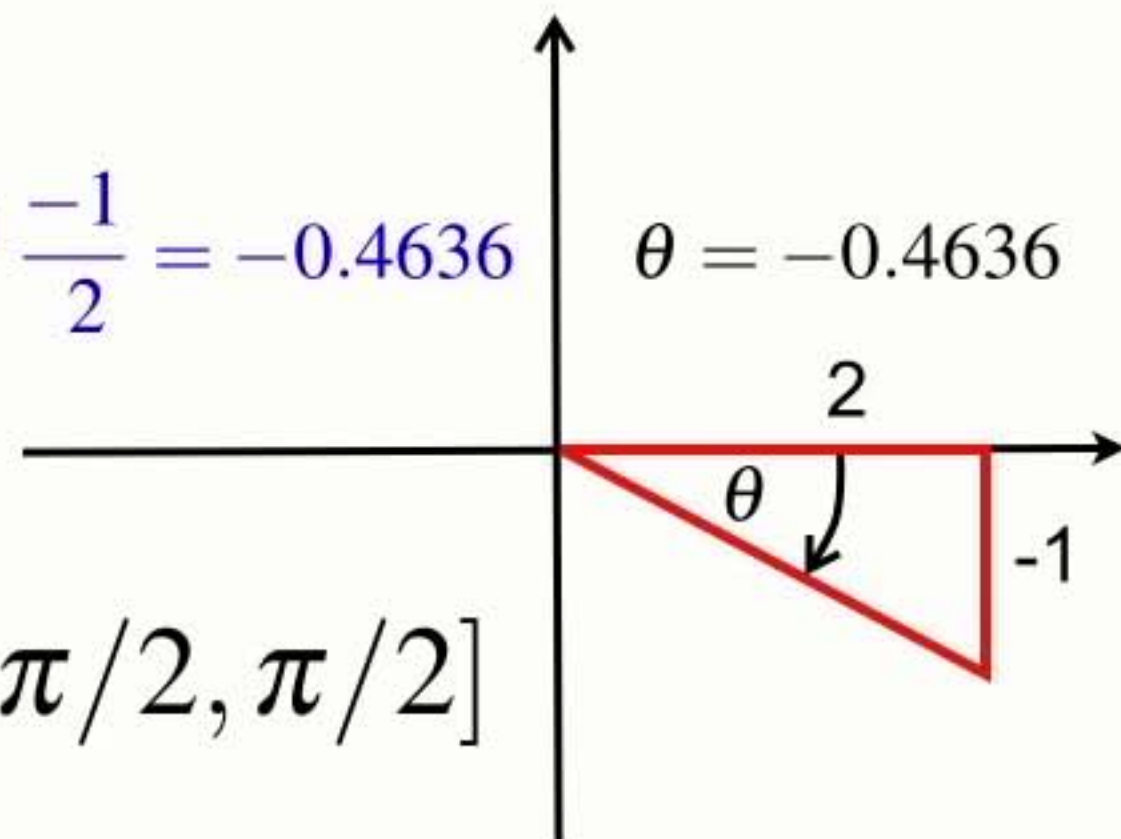
$$\tan^{-1} \frac{1}{2} = 0.4636$$

$$\theta = 0.4636$$



$$\theta = -2.6779$$

$$\tan^{-1} \frac{-1}{-2} = 0.4636$$



$$\tan^{-1} \frac{-1}{2} = -0.4636$$

$$\theta = -0.4636$$

$$\tan^{-1} x \in [-\pi/2, \pi/2]$$

The **atan2** function

- To get around the sign ambiguity problem we introduce a new function

$$\text{atan2}(a, b) = \tan^{-1} \frac{a}{b}$$

The **atan2** function

- To get around the sign ambiguity problem we introduce a new function

$$\text{atan2}(a, b) = \tan^{-1} \frac{a}{b} \in [-\pi, \pi]$$

The **atan2** function

- To get around the sign ambiguity problem we introduce a new function

$$\text{atan2}(a, b) = \tan^{-1} \frac{a}{b} \in [-\pi, \pi]$$

```
>> atan2(1, -2)  
ans =  
      2.6779
```

Section 4

6-link 3D robot

Inverse kinematics **for a 6-DOF robot**



Unimate Puma 500
Oussama Khatib | Used with permission

Command Window

```

>> mdl_puma560
fx >>
  
```

Workspace

Name ▲	Value	Class
deg	0.0175	double
p560	<1x1 SerialLink>	SerialLink
qn	[0,0.7854,3.1416,0,0.78...	double
qr	[0,1.5708,-1.5708,0,0,0]	double
qs	[0,0,-1.5708,0,0,0]	double
qz	[0,0,0,0,0,0]	double

Command Window

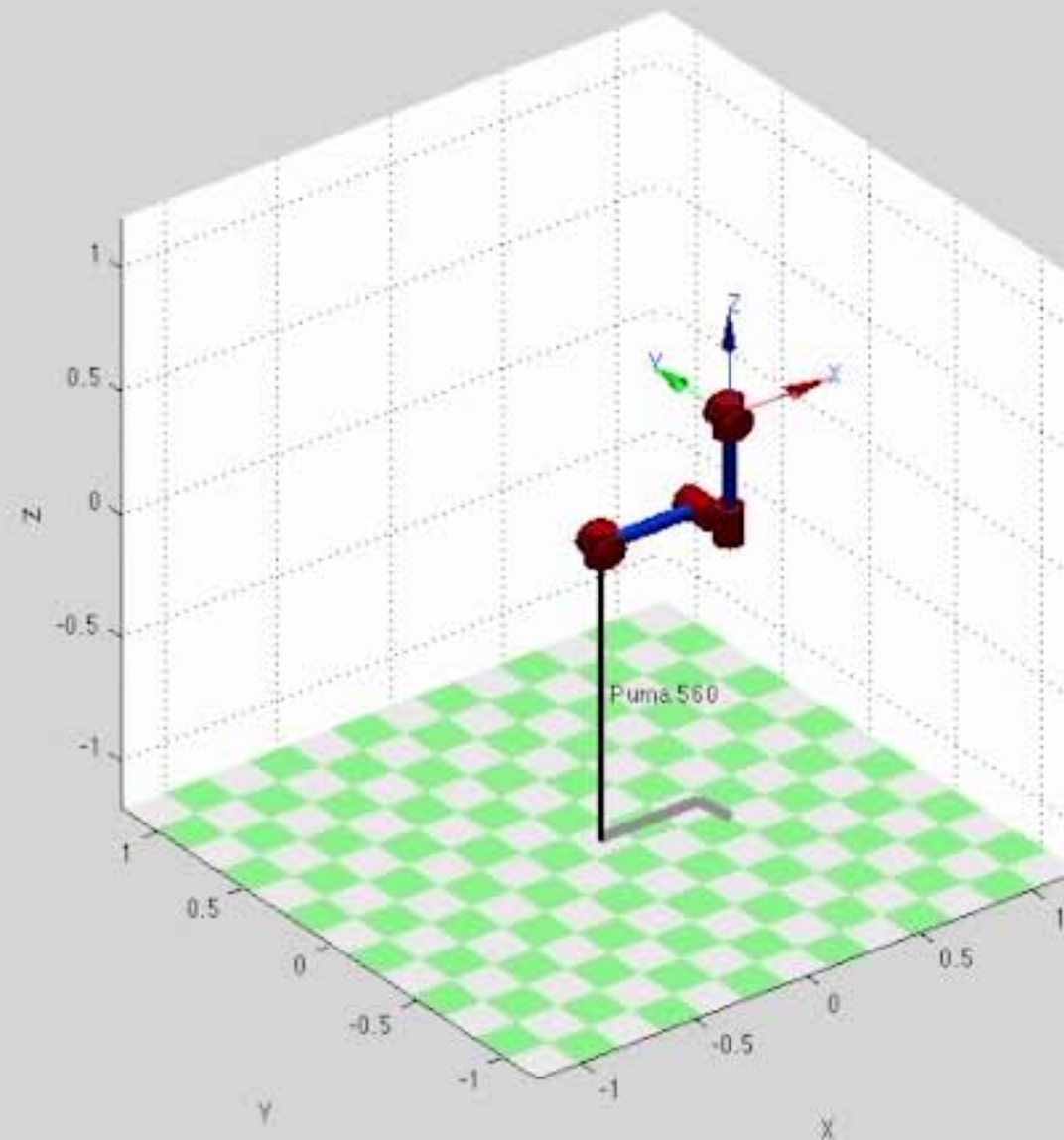
```
>> mdl_puma560
>> p560.plot(qz)
fx >>
```

Workspace

Name	Value	Class
deg	0.0175	double
p560	<1x1 SerialLink>	SerialLink
qn	[0,0.7854,3.1416,0,0.78...	double
qr	[0,1.5708,-1.5708,0,0,0]	double
qs	[0,0,-1.5708,0,0,0]	double
qz	[0,0,0,0,0,0]	double

Figure 1

Current Fol



Command Window

```
>> mdl_puma560
>> p560.plot(qz)
>> T = transl(0.6, 0.1, 0) * rpy2tr(0, 180, 0, 'deg')
```

T =

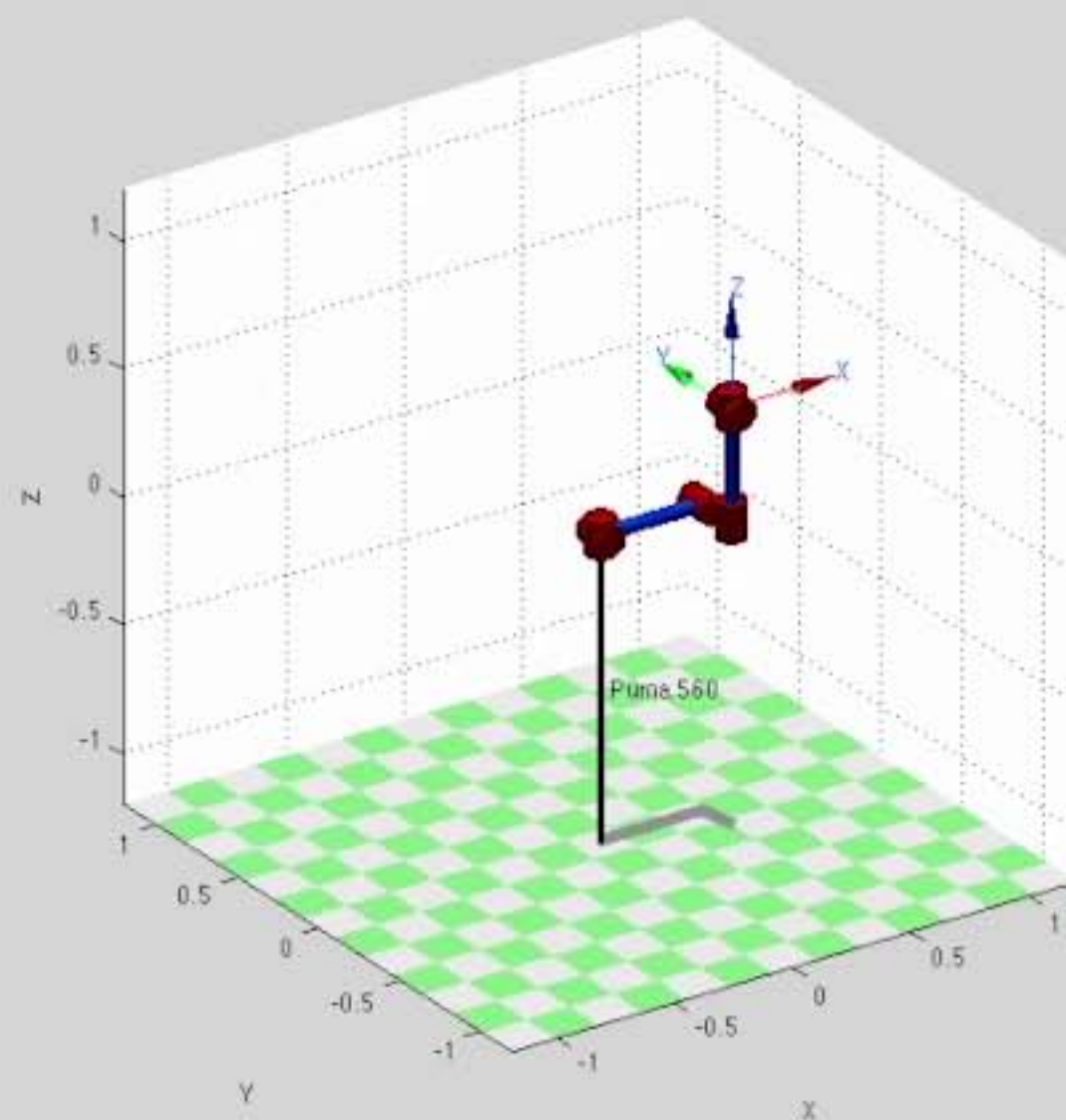
```
-1.0000    0    0.0000    0.6000
     0    1.0000     0    0.1000
-0.0000     0   -1.0000     0
     0     0     0    1.0000
```

fx >> |

Workspace

Name	Value	Class
T	<4x4 double>	double
deg	0.0175	double
p560	<1x1 SerialLink>	SerialLink
qn	[0,0.7854,3.1416,0,0.78...	double
qr	[0,1.5708,-1.5708,0,0,0]	double
qs	[0,0,-1.5708,0,0,0]	double
qz	[0,0,0,0,0,0]	double

Figure 1



Command Window

```
>> mdl_puma560
>> p560.plot(qz)
>> T = transl(0.6, 0.1, 0) * rpy2tr(0, 180, 0, 'deg')
```

T =

```
-1.0000    0    0.0000    0.6000
     0    1.0000     0    0.1000
-0.0000     0   -1.0000     0
     0     0     0    1.0000
```

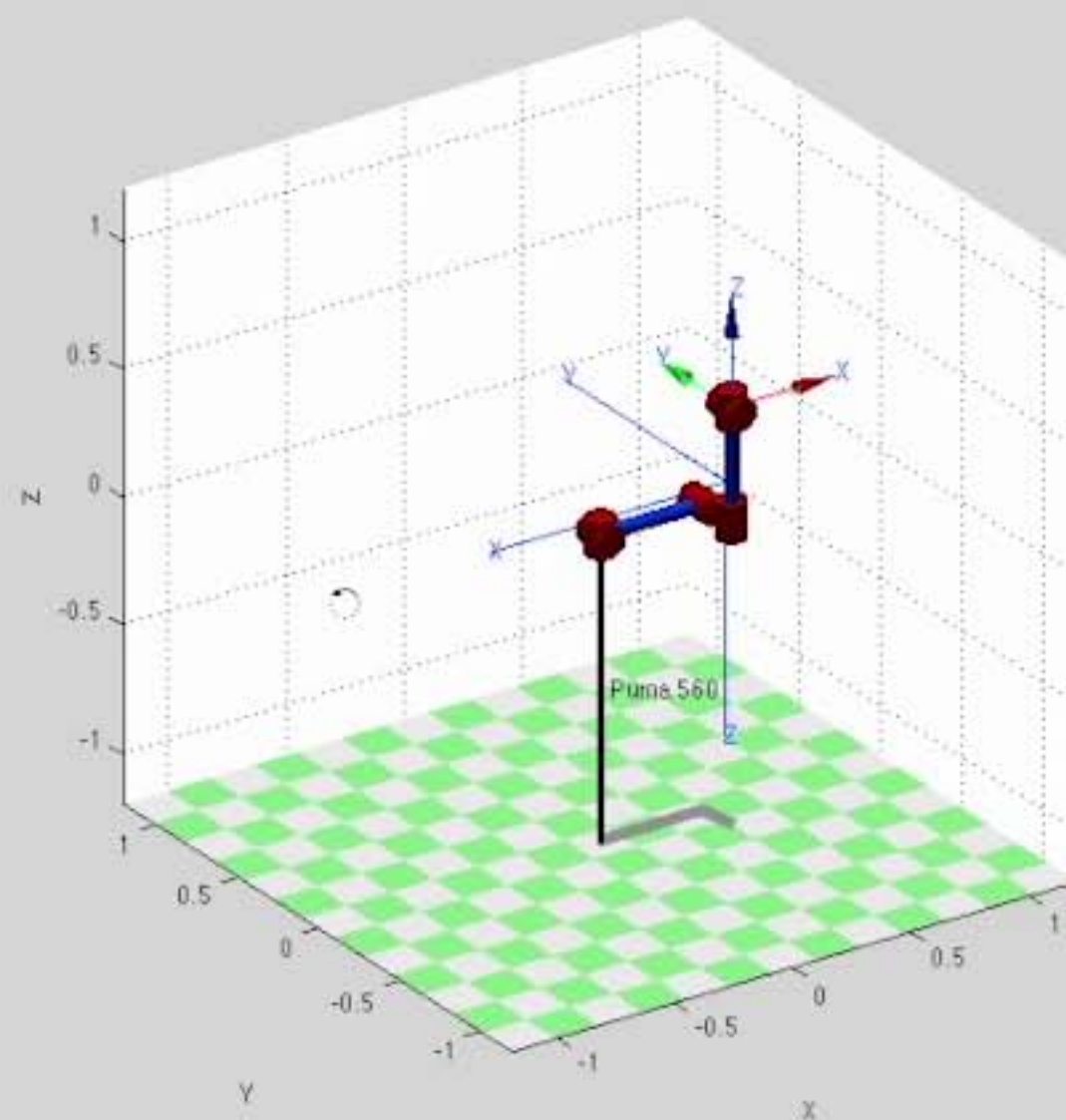
```
>> hold on
>> trplot(T)
```

fx >>

Workspace

Name	Value	Class
T	<4x4 double>	double
deg	0.0175	double
p560	<1x1 SerialLink>	SerialLink
qn	[0,0.7854,3.1416,0,0.78...	double
qr	[0,1.5708,-1.5708,0,0,0]	double
qs	[0,0,-1.5708,0,0,0]	double
qz	[0,0,0,0,0,0]	double

Figure 1



Command Window

```
>> mdl_puma560
>> p560.plot(qz)
>> T = transl(0.6, 0.1, 0) * rpy2tr(0, 180, 0, 'deg')
```

T =

```
-1.0000    0    0.0000    0.6000
     0    1.0000     0    0.1000
-0.0000     0   -1.0000     0
     0     0     0    1.0000
```

```
>> hold on
```

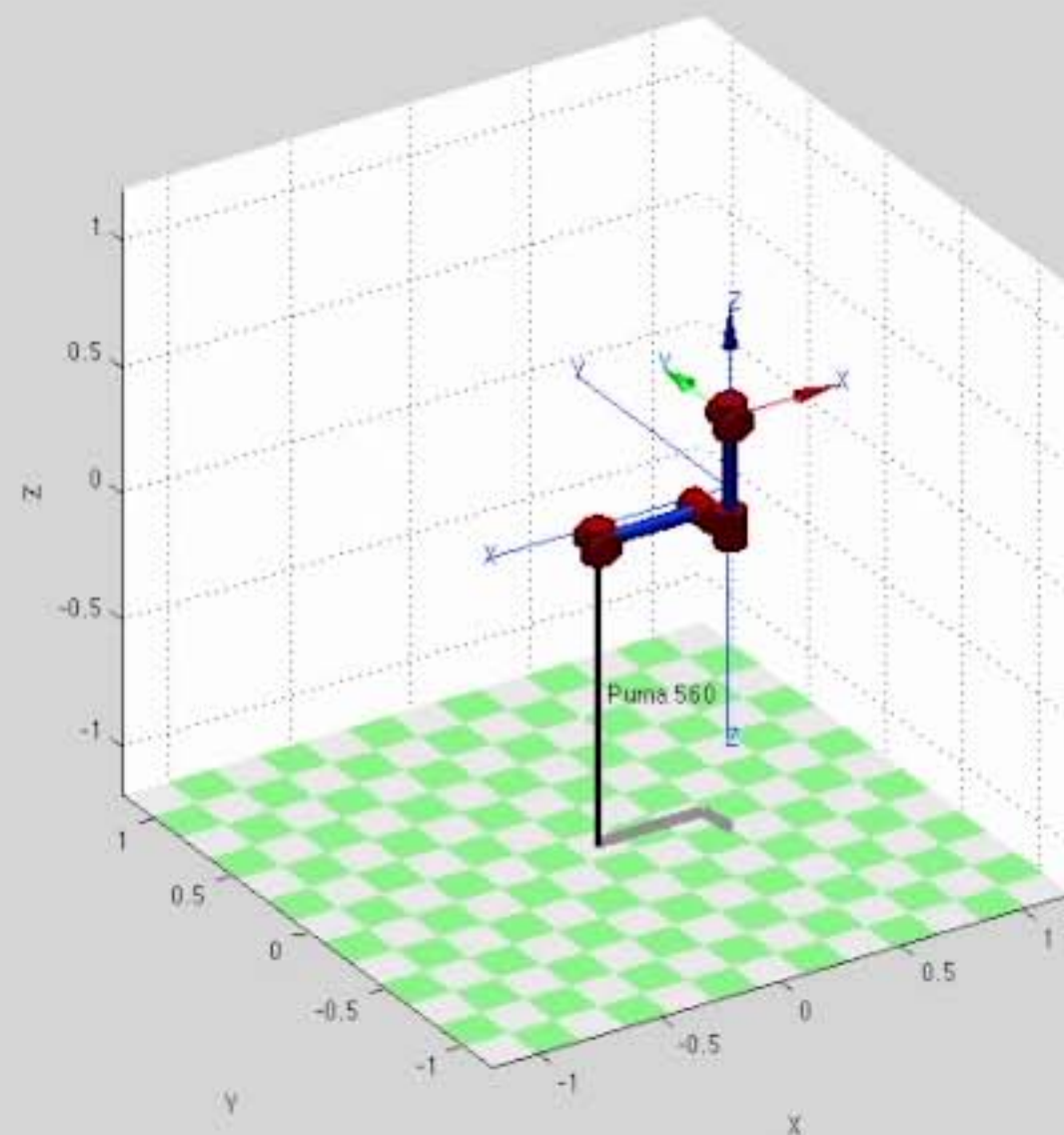
```
>> trplot(T)
```

```
>> q = p560.ikine6s(T)
```

Workspace

Name	Value	Class
T	<4x4 double>	double
deg	0.0175	double
p560	<1x1 SerialLink>	SerialLink
qn	[0,0.7854,3.1416,0,0.78...	double
qr	[0,1.5708,-1.5708,0,0,0]	double
qs	[0,0,-1.5708,0,0,0]	double
qz	[0,0,0,0,0,0]	double

Figure 1



Command Window

T =

```
-1.0000    0    0.0000    0.6000
         0    1.0000    0    0.1000
-0.0000    0   -1.0000    0
         0    0    0    1.0000
```

```
>> hold on
>> trplot(T)
>> q = p560.ikine6s(T)
```

q =

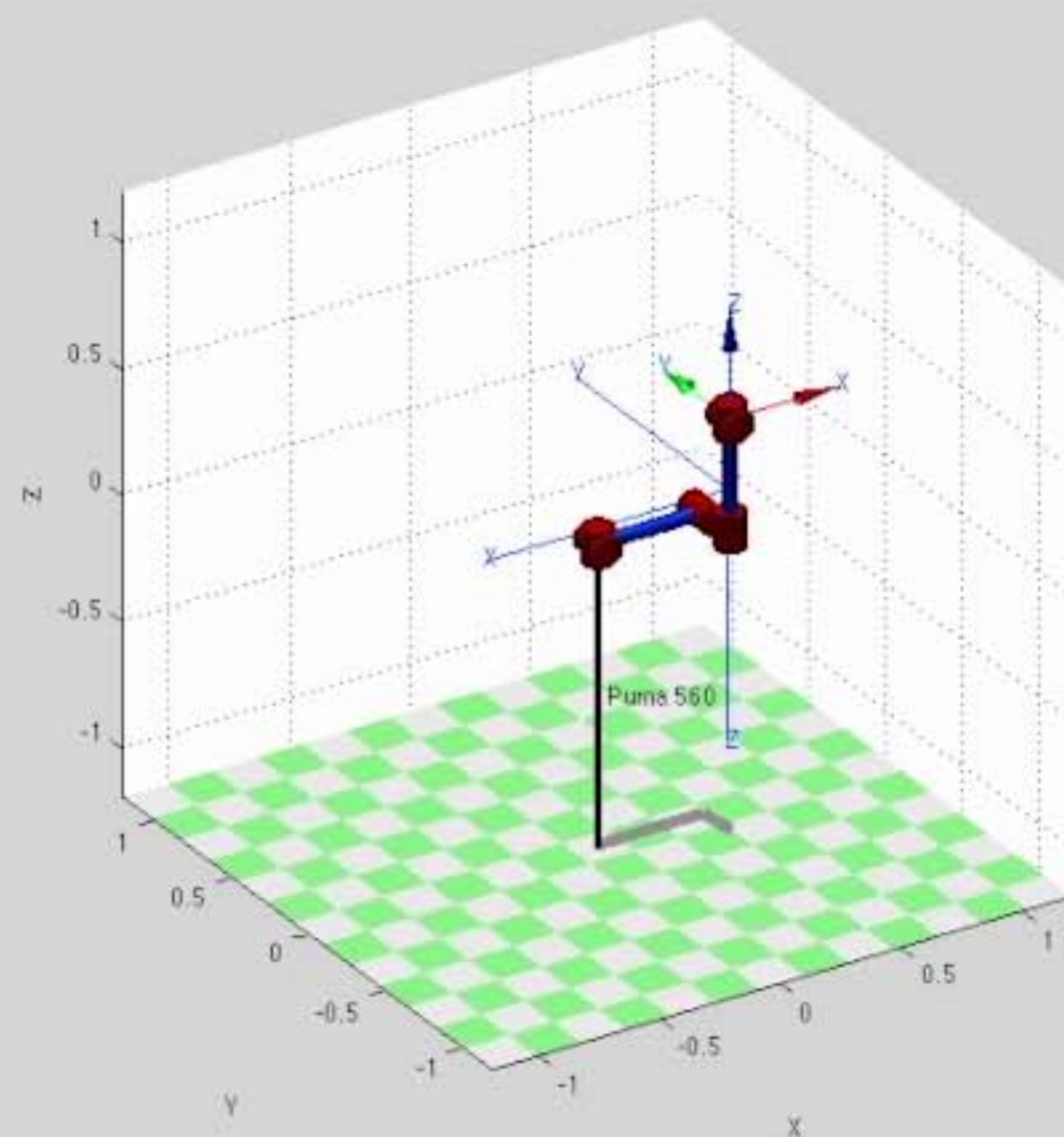
```
3.0575    2.3210    0.1162   -0.0000    0.7044    3.0575
```

fx >> |

Workspace

Name	Value	Class
T	<4x4 double>	double
deg	0.0175	double
p560	<1x1 SerialLink>	SerialLink
q	[3.0575,2.3210,0.1162,-...	double
qn	[0,0.7854,3.1416,0,0.78...	double
qr	[0,1.5708,-1.5708,0,0,0]	double
qs	[0,0,-1.5708,0,0,0]	double
qz	[0,0,0,0,0,0]	double

Figure 1



Command Window

```
-1.0000    0    0.0000    0.6000
    0    1.0000    0    0.1000
-0.0000    0   -1.0000    0
    0    0    0    1.0000
```

```
>> hold on
>> trplot(T)
>> q = p560.ikine6s(T)
```

q =

```
3.0575    2.3210    0.1162   -0.0000    0.7044    3.0575
```

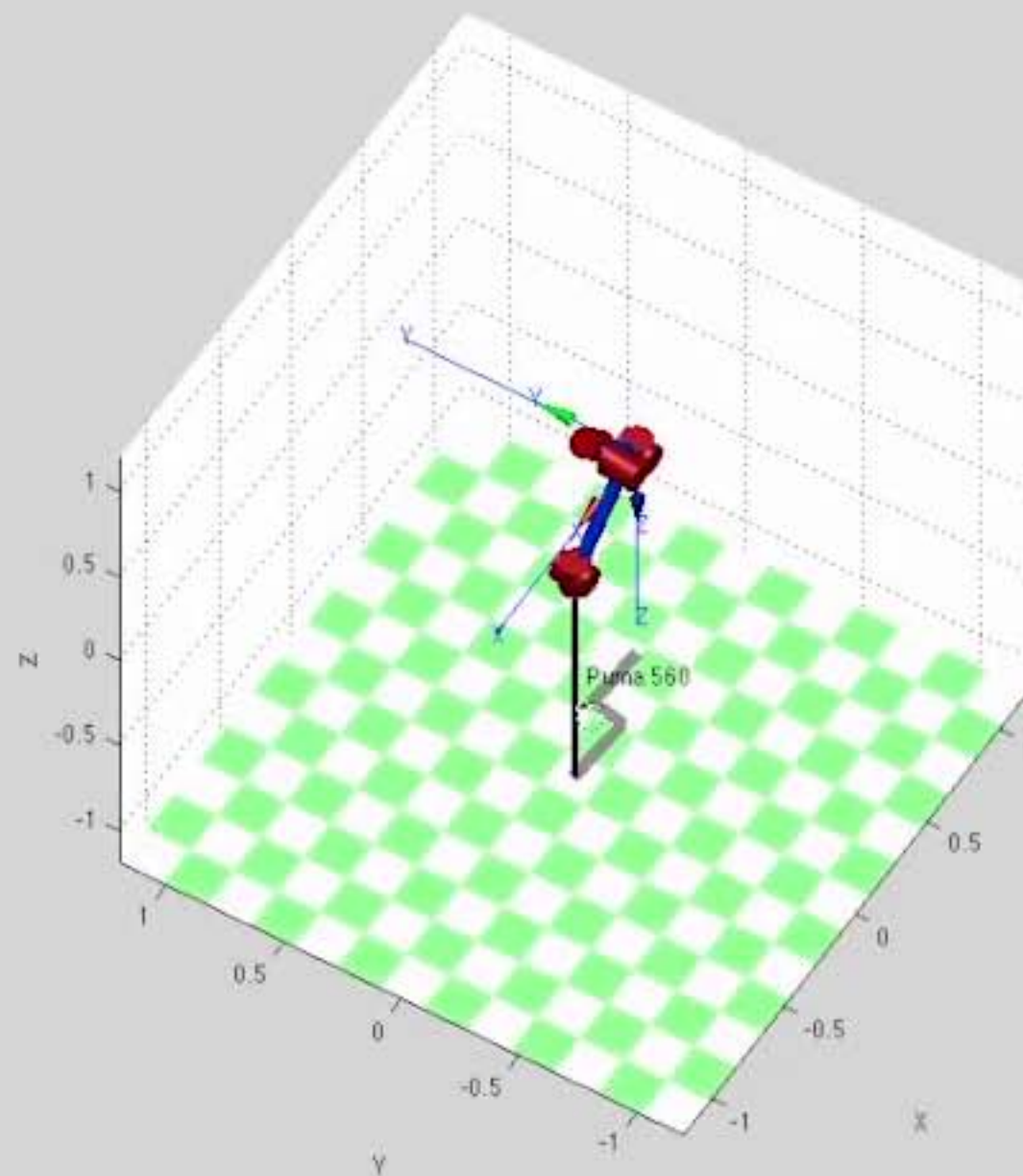
```
>> p560.plot(q)
```

fx >>

Workspace

Name	Value	Class
T	<4x4 double>	double
deg	0.0175	double
p560	<1x1 SerialLink>	SerialLink
q	[3.0575,2.3210,0.1162,-...	double
qn	[0,0.7854,3.1416,0,0.78...	double
qr	[0,1.5708,-1.5708,0,0,0]	double
qs	[0,0,-1.5708,0,0,0]	double
qz	[0,0,0,0,0,0]	double

Figure 1



Command Window

```

-1.0000    0    0.0000    0.6000
    0    1.0000    0    0.1000
-0.0000    0   -1.0000    0
    0    0    0    1.0000

>> hold on
>> trplot(T)
>> q = p560.ikine6s(T)

q =

    3.0575    2.3210    0.1162   -0.0000    0.7044    3.0575

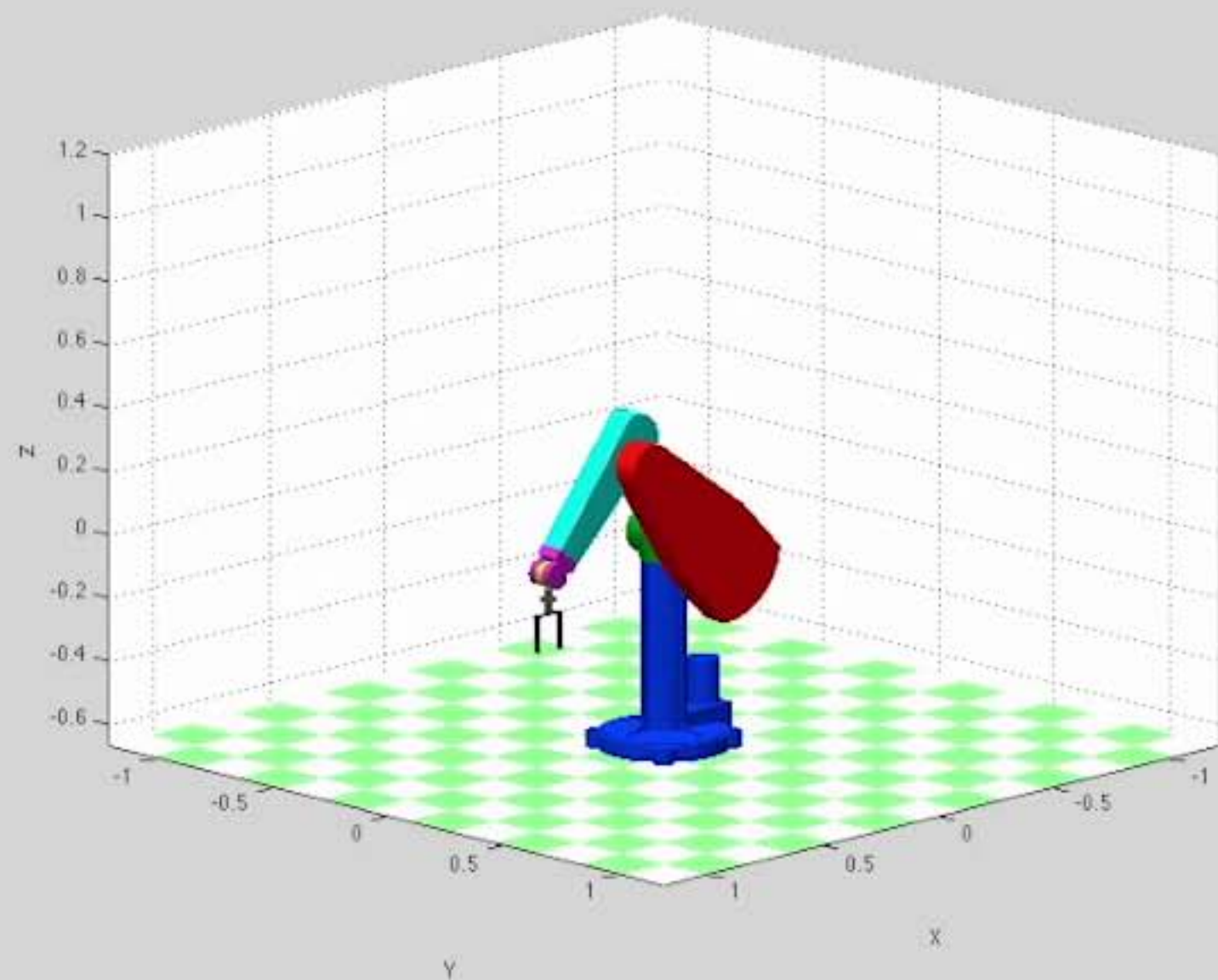
>> p560.plot(q)
>> p560.plot3d(q)
fx >>

```

Workspace

Name	Value	Class
T	<4x4 double>	double
deg	0.0175	double
p560	<1x1 SerialLink>	SerialLink
q	[3.0575,2.3210,0.1162,-...	double
qn	[0,0.7854,3.1416,0,0.78...	double
qr	[0,1.5708,-1.5708,0,0,0]	double
qs	[0,0,-1.5708,0,0,0]	double
qz	[0,0,0,0,0,0]	double

Figure 1



Command Window

```
>> trplot(T)
>> q = p560.ikine6s(T)

q =

    3.0575    2.3210    0.1162   -0.0000    0.7044    3.0575

>> p560.plot(q)
>> p560.plot3d(q)
>> q = p560.ikine6s(T, 'r')

q =

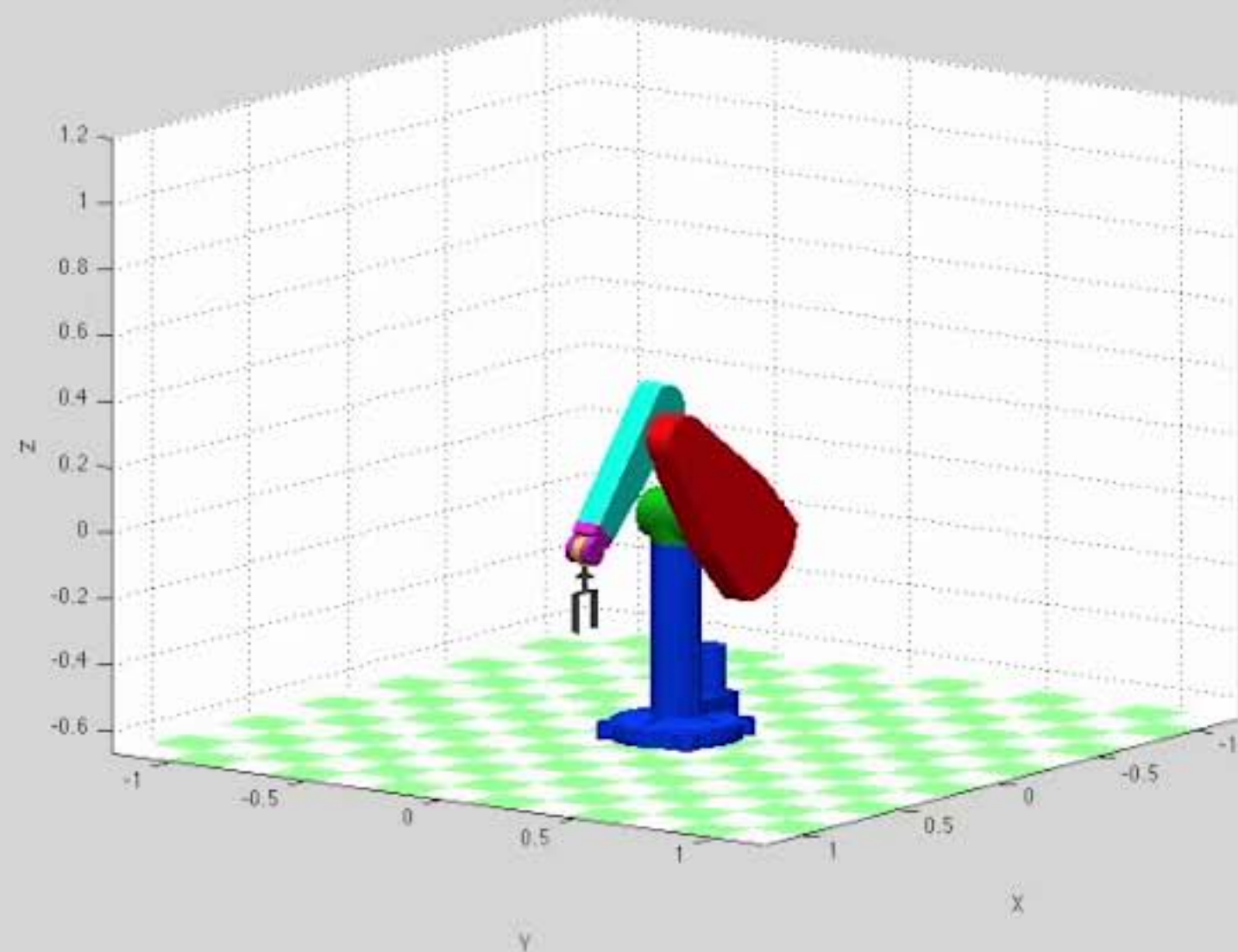
    0.4144    0.8206    3.1193   -3.1416    0.7983   -2.7272

fx >> |
```

Workspace

Name	Value	Class
T	<4x4 double>	double
deg	0.0175	double
p560	<1x1 SerialLink>	SerialLink
q	[0.4144,0.8206,3.1193,-...	double
qn	[0,0.7854,3.1416,0,0.78...	double
qr	[0,1.5708,-1.5708,0,0,0]	double
qs	[0,0,-1.5708,0,0,0]	double
qz	[0,0,0,0,0,0]	double

Figure 1



Command Window

```
>> q = p560.ikine6s(T)

q =

    3.0575    2.3210    0.1162   -0.0000    0.7044    3.0575

>> p560.plot(q)
>> p560.plot3d(q)
>> q = p560.ikine6s(T, 'r')

q =

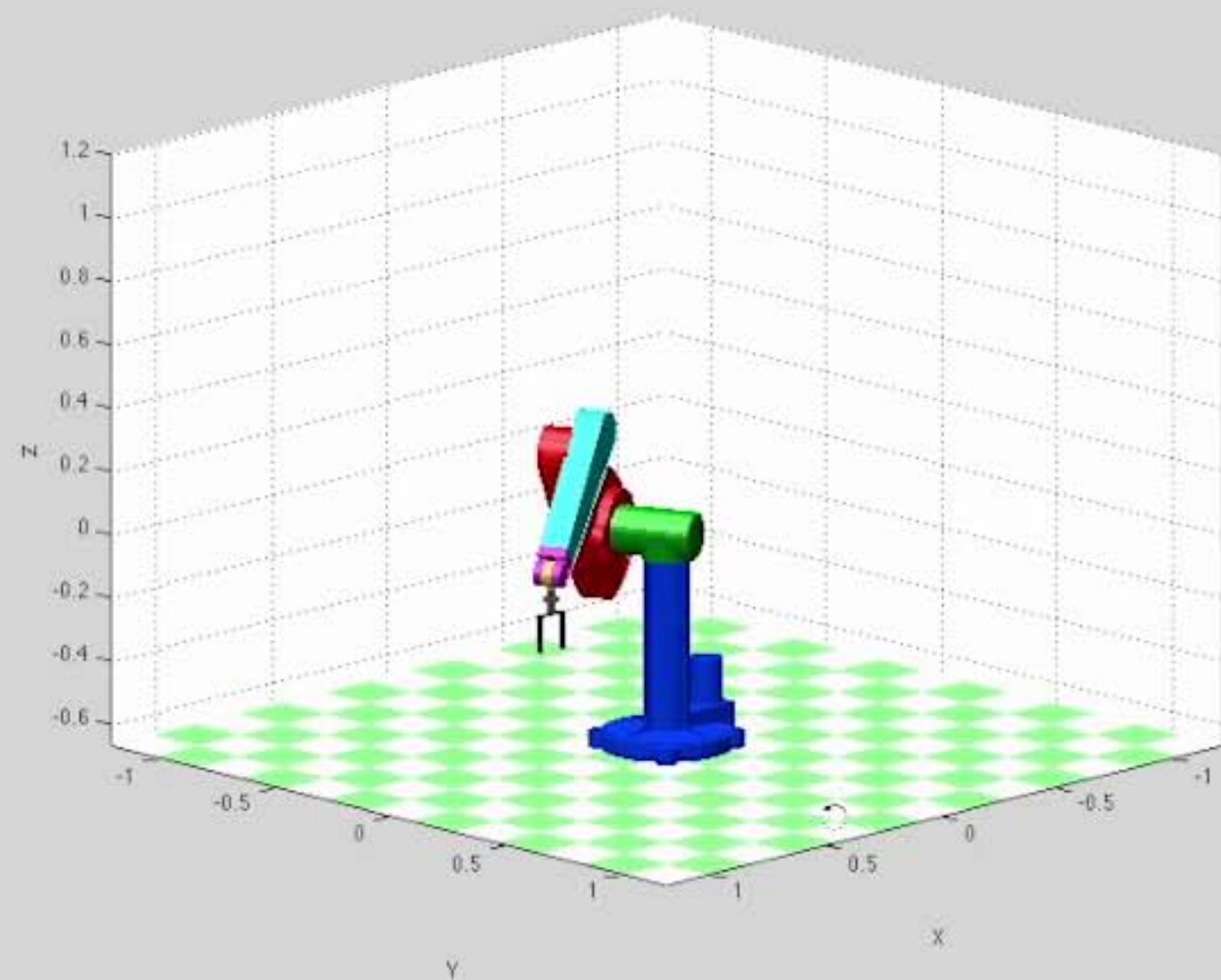
    0.4144    0.8206    3.1193   -3.1416    0.7983   -2.7272

>> p560.plot3d(q)
fx >>
```

Workspace

Name	Value	Class
T	<4x4 double>	double
deg	0.0175	double
p560	<1x1 SerialLink>	SerialLink
q	[0.4144,0.8206,3.1193,-...	double
qn	[0,0.7854,3.1416,0,0.78...	double
qr	[0,1.5708,-1.5708,0,0,0]	double
qs	[0,0,-1.5708,0,0,0]	double
qz	[0,0,0,0,0,0]	double

Figure 1



Command Window

```
>> p560.plot(q)
>> p560.plot3d(q)
>> q = p560.ikine6s(T, 'r')

q =

    0.4144    0.8206    3.1193   -3.1416    0.7983   -2.7272

>> p560.plot3d(q)
>> q = p560.ikine6s(T, 'rd')

q =

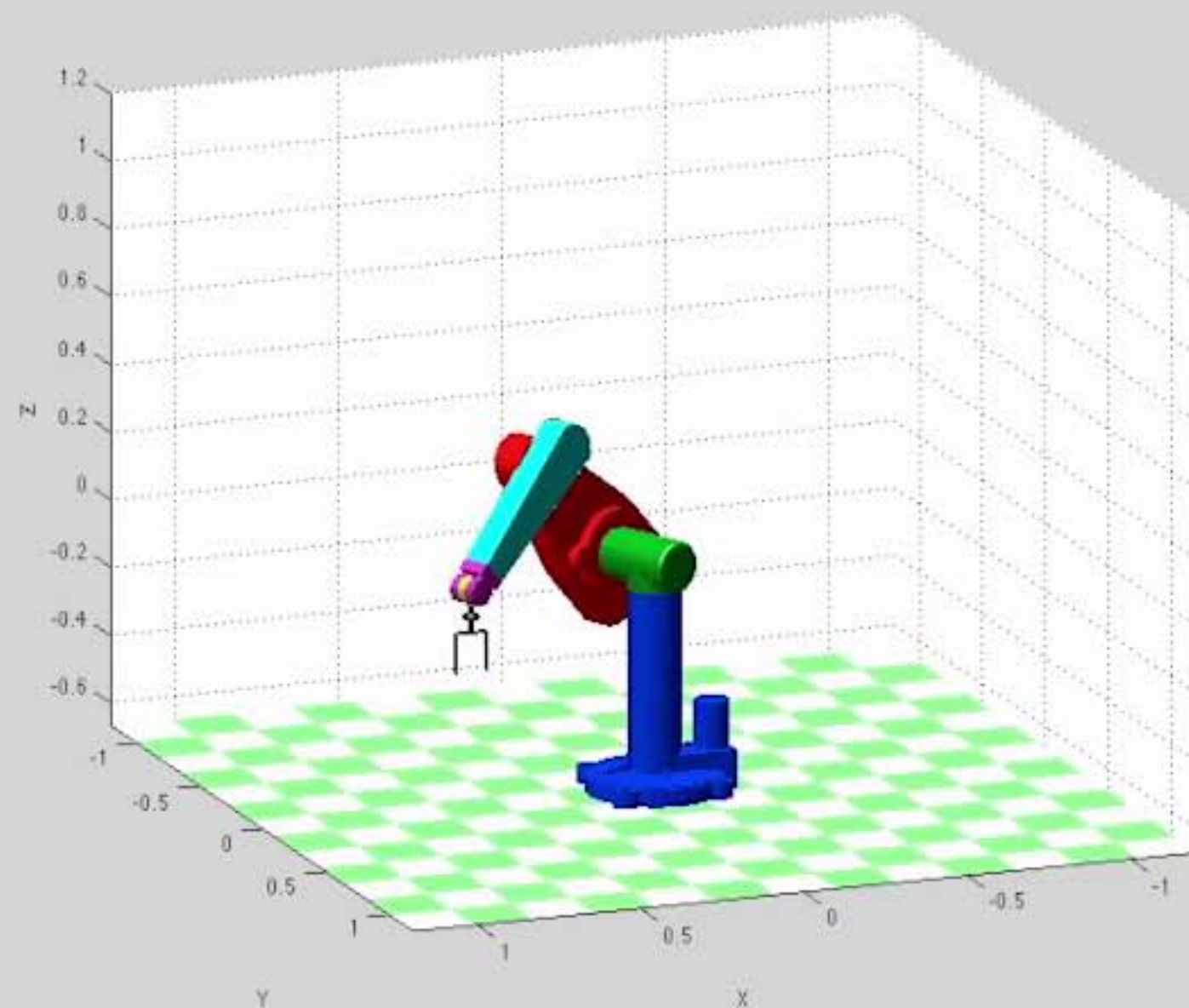
    0.4144   -0.8206    0.1162    3.1416    2.4372   -2.7272
```

fx >>

Workspace

Name	Value	Class
T	<4x4 double>	double
deg	0.0175	double
p560	<1x1 SerialLink>	SerialLink
q	[0.4144,-0.8206,0.1162,...	double
qn	[0,0.7854,3.1416,0,0.78...	double
qr	[0,1.5708,-1.5708,0,0,0]	double
qs	[0,0,-1.5708,0,0,0]	double
qz	[0,0,0,0,0,0]	double

Figure 1



Command Window

```
>> p560.plot3d(q)
>> q = p560.ikine6s(T, 'r')

q =

    0.4144    0.8206    3.1193   -3.1416    0.7983   -2.7272

>> p560.plot3d(q)
>> q = p560.ikine6s(T, 'rd')

q =

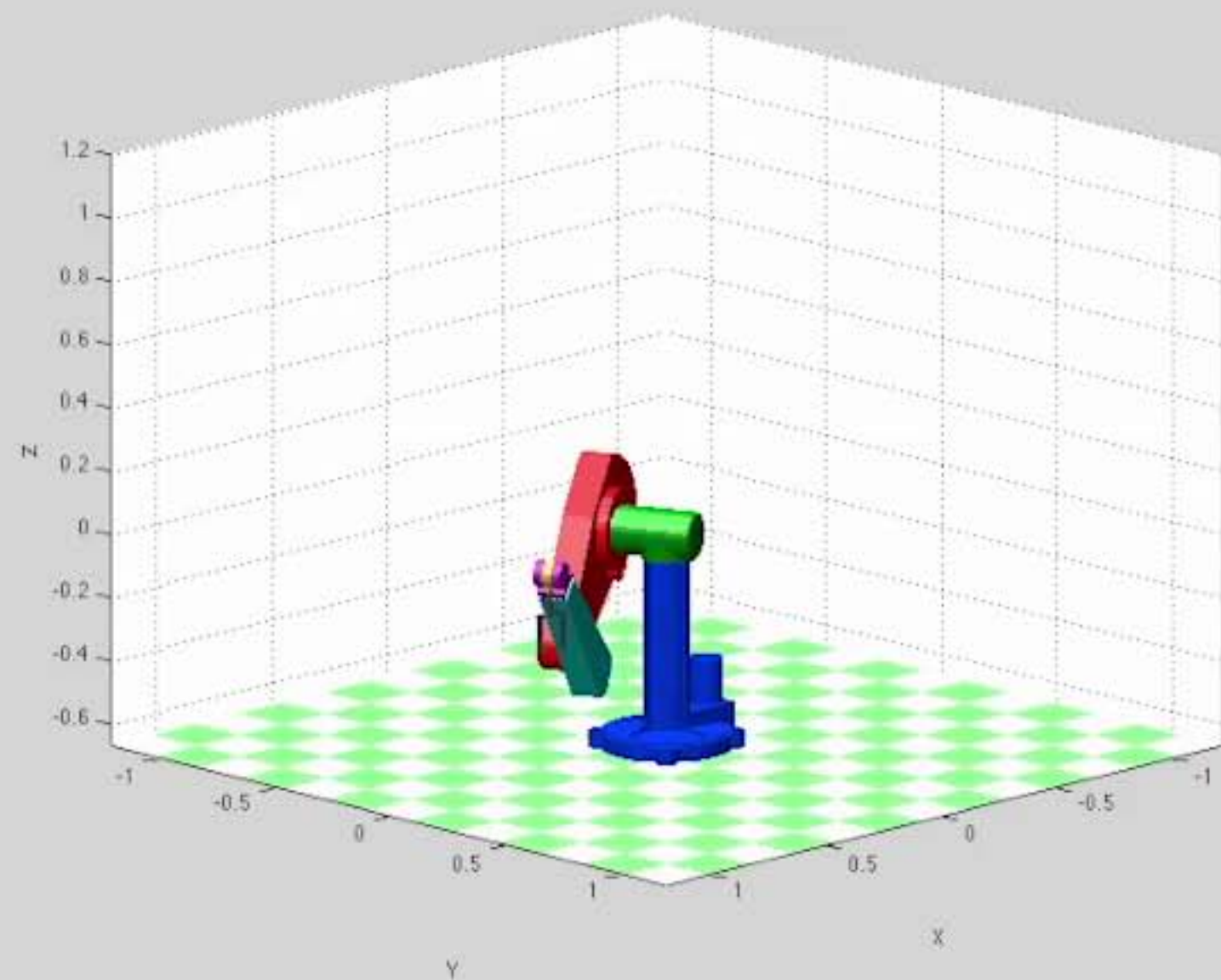
    0.4144   -0.8206    0.1162    3.1416    2.4372   -2.7272

>> p560.plot3d(q)
fx >> |
```

Workspace

Name	Value	Class
T	<4x4 double>	double
deg	0.0175	double
p560	<1x1 SerialLink>	SerialLink
q	[0.4144,-0.8206,0.1162,...	double
qn	[0,0.7854,3.1416,0,0.78...	double
qr	[0,1.5708,-1.5708,0,0,0]	double
qs	[0,0,-1.5708,0,0,0]	double
qz	[0,0,0,0,0,0]	double

Figure 1



Command Window

```
>> p560.plot3d(q)
>> q = p560.ikine6s(T, 'rd')

q =

    0.4144   -0.8206    0.1162    3.1416    2.4372   -2.7272

>> p560.plot3d(q)
>> q = p560.ikine6s(T, 'ld')

q =

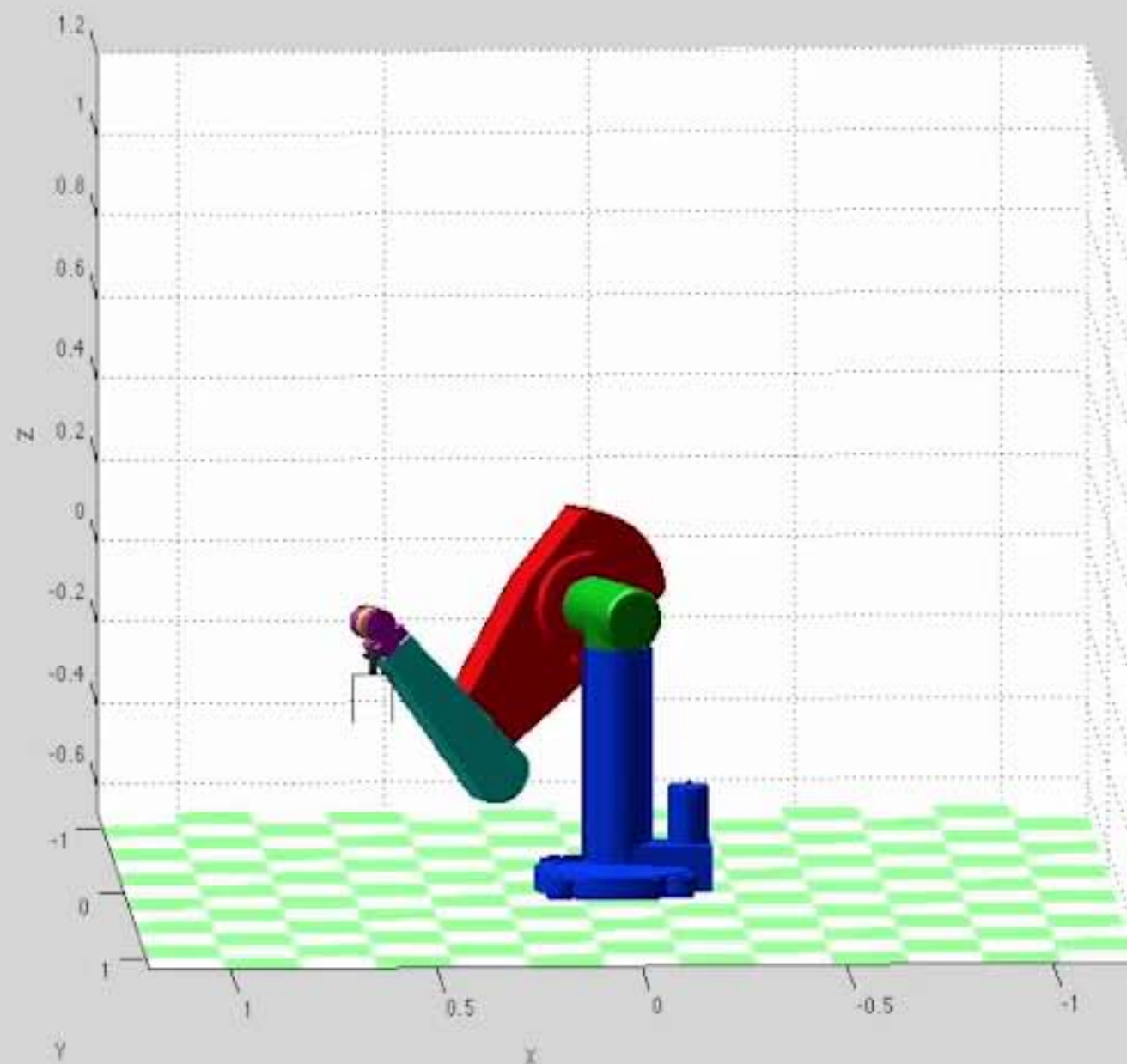
    3.0575    3.9622    3.1193   -0.0000    2.3432    3.0575
```

fx >>

Workspace

Name	Value	Class
T	<4x4 double>	double
deg	0.0175	double
p560	<1x1 SerialLink>	SerialLink
q	[3.0575,3.9622,3.1193,-...	double
qn	[0,0.7854,3.1416,0,0.78...	double
qr	[0,1.5708,-1.5708,0,0,0]	double
qs	[0,0,-1.5708,0,0,0]	double
qz	[0,0,0,0,0,0]	double

Figure 1



Command Window

```
>> p560.plot3d(q)
>> q = p560.ikine6s(T, 'rd')

q =

    0.4144   -0.8206    0.1162    3.1416    2.4372   -2.7272

>> p560.plot3d(q)
>> q = p560.ikine6s(T, 'ld')

q =

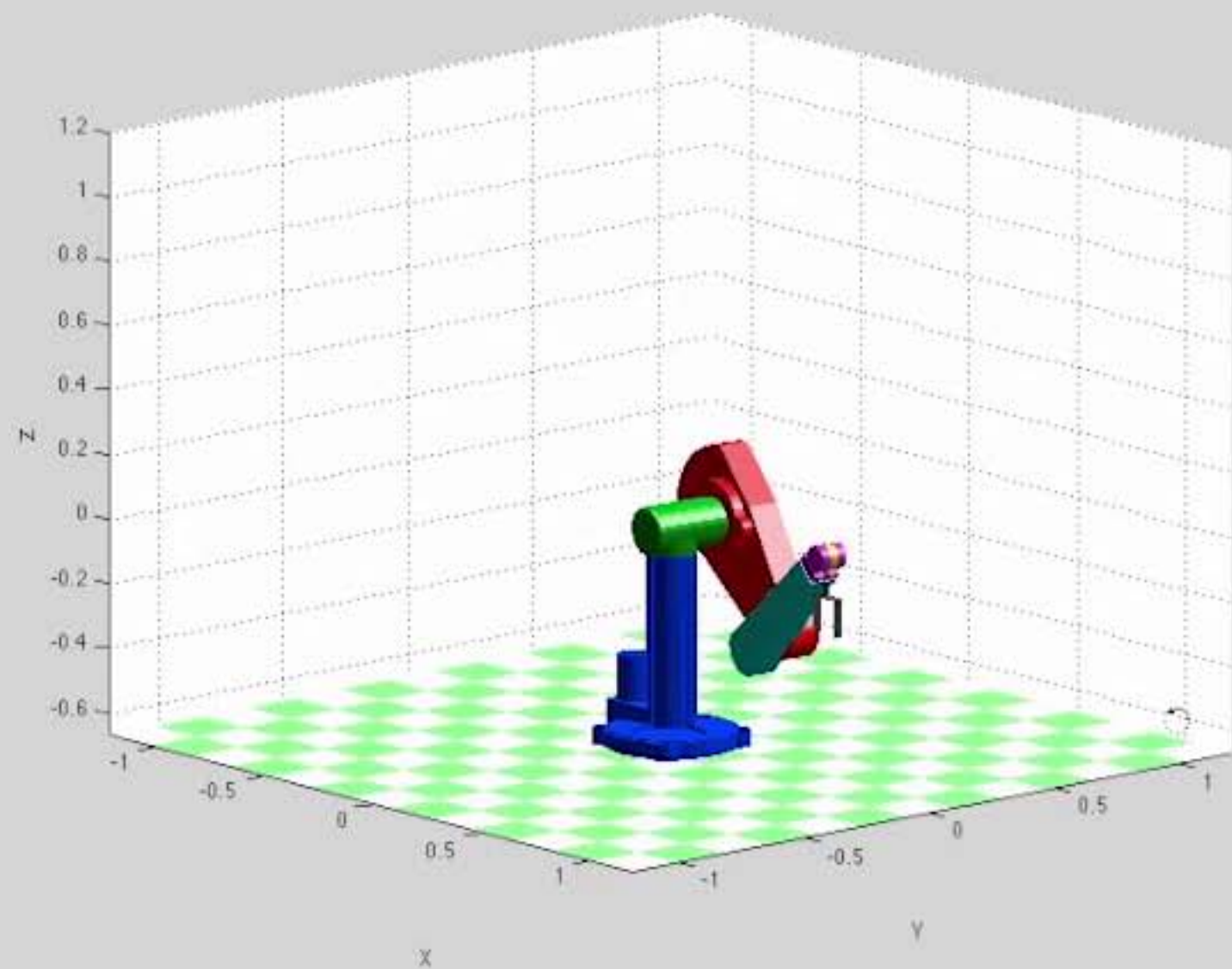
    3.0575    3.9622    3.1193   -0.0000    2.3432    3.0575

>> p560.plot3d(q)
fx >>
```

Workspace

Name	Value	Class
T	<4x4 double>	double
deg	0.0175	double
p560	<1x1 SerialLink>	SerialLink
q	[3.0575,3.9622,3.1193,-...	double
qn	[0,0.7854,3.1416,0,0.78...	double
qr	[0,1.5708,-1.5708,0,0,0]	double
qs	[0,0,-1.5708,0,0,0]	double
qz	[0,0,0,0,0,0]	double

Figure 1



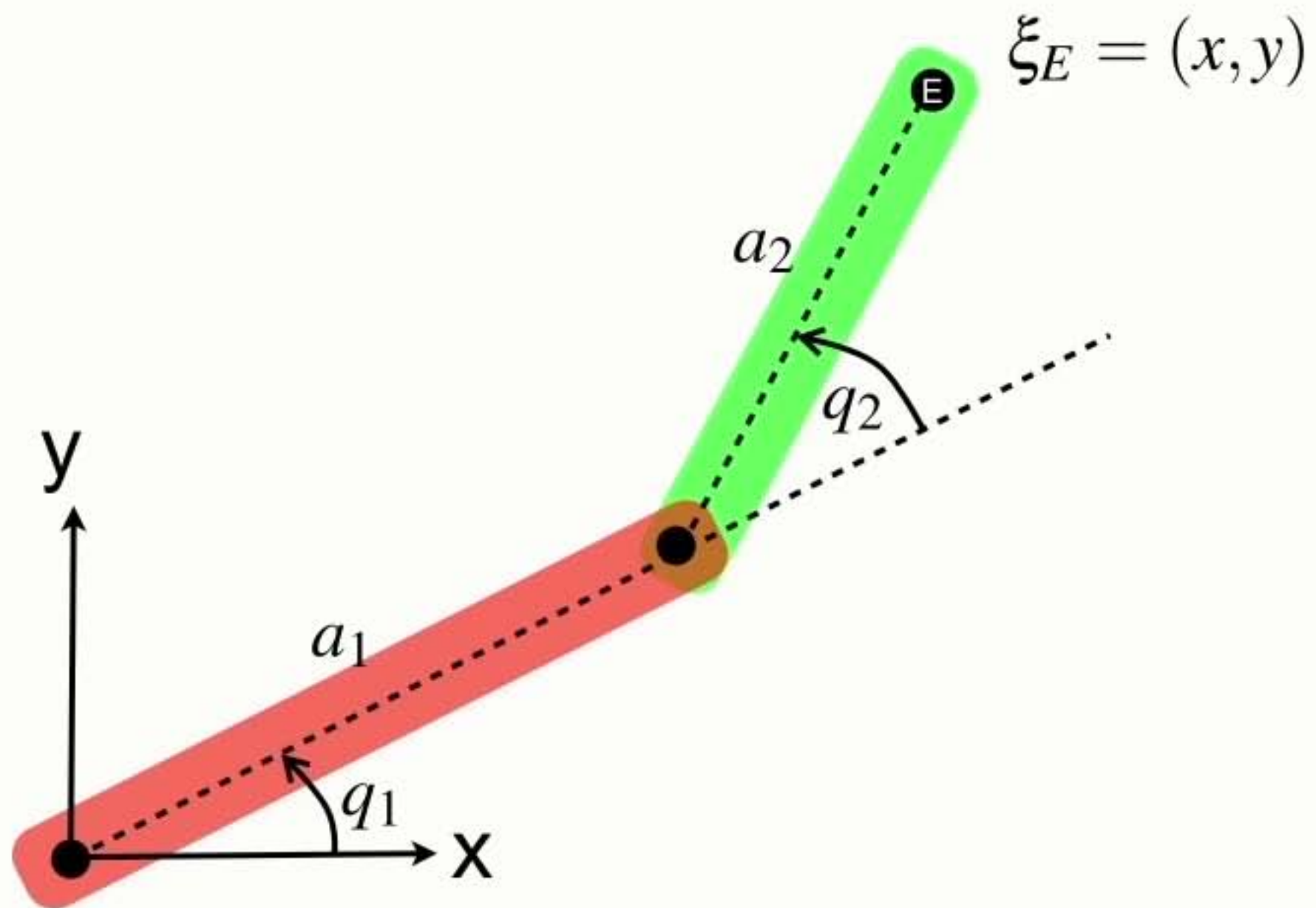
Inverse kinematics **for a 6-DOF robot**



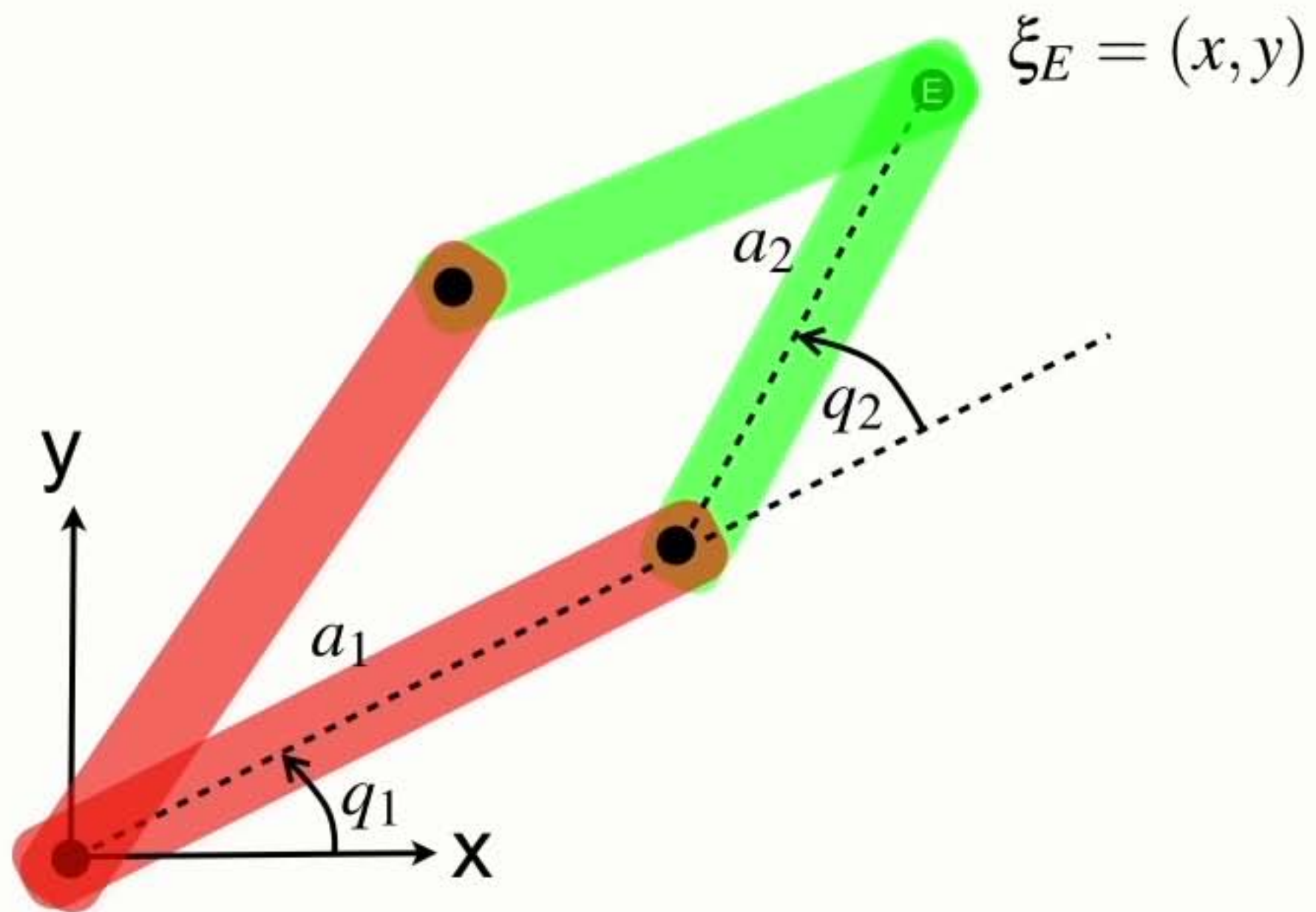
Unimate Puma 500
Oussama Khatib | Used with permission

- There are eight solutions:
 - ➔ left and right handed
 - ➔ elbow up (above the shoulder) and elbow down (below the shoulder)
 - ➔ wrist flipped and not flipped

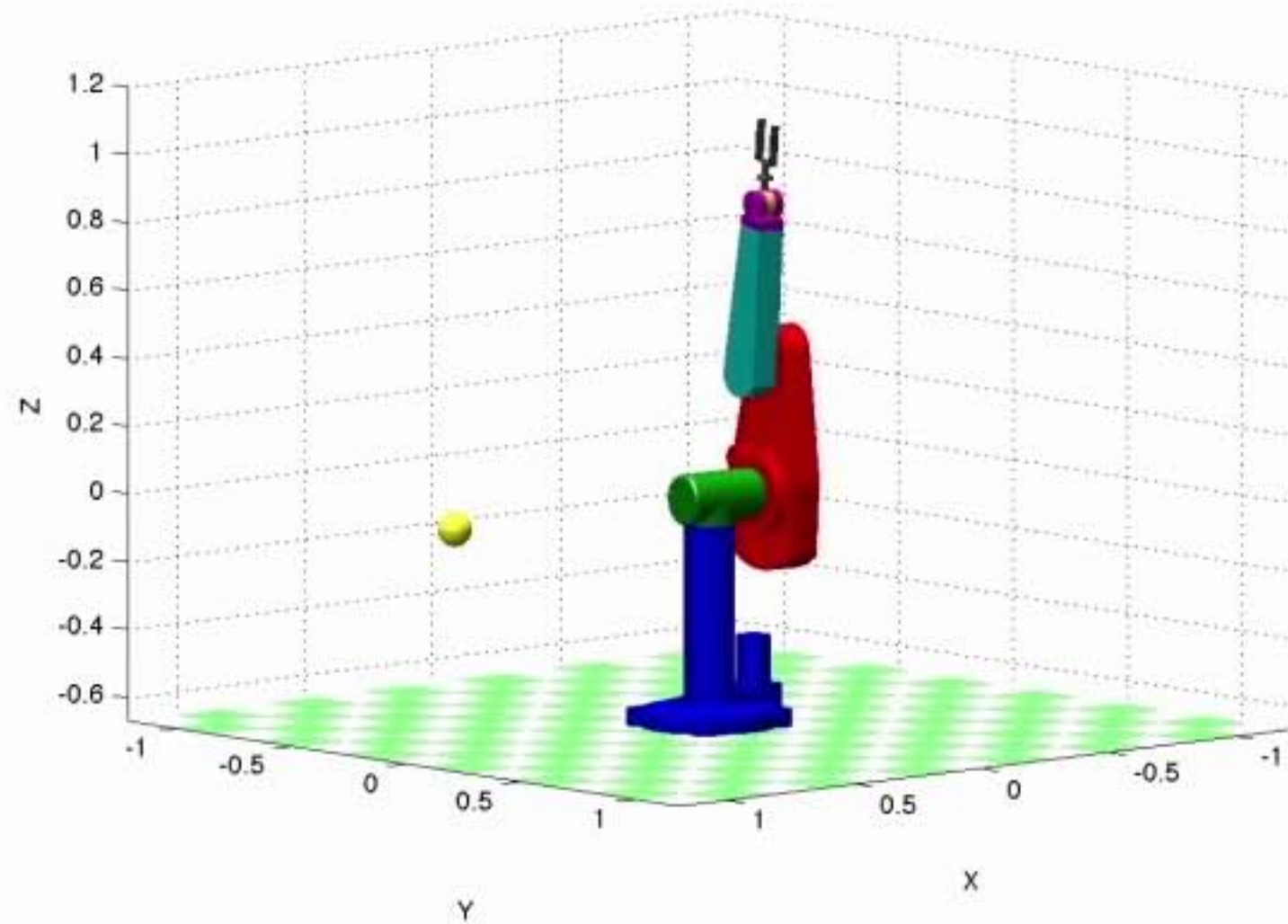
Two solutions



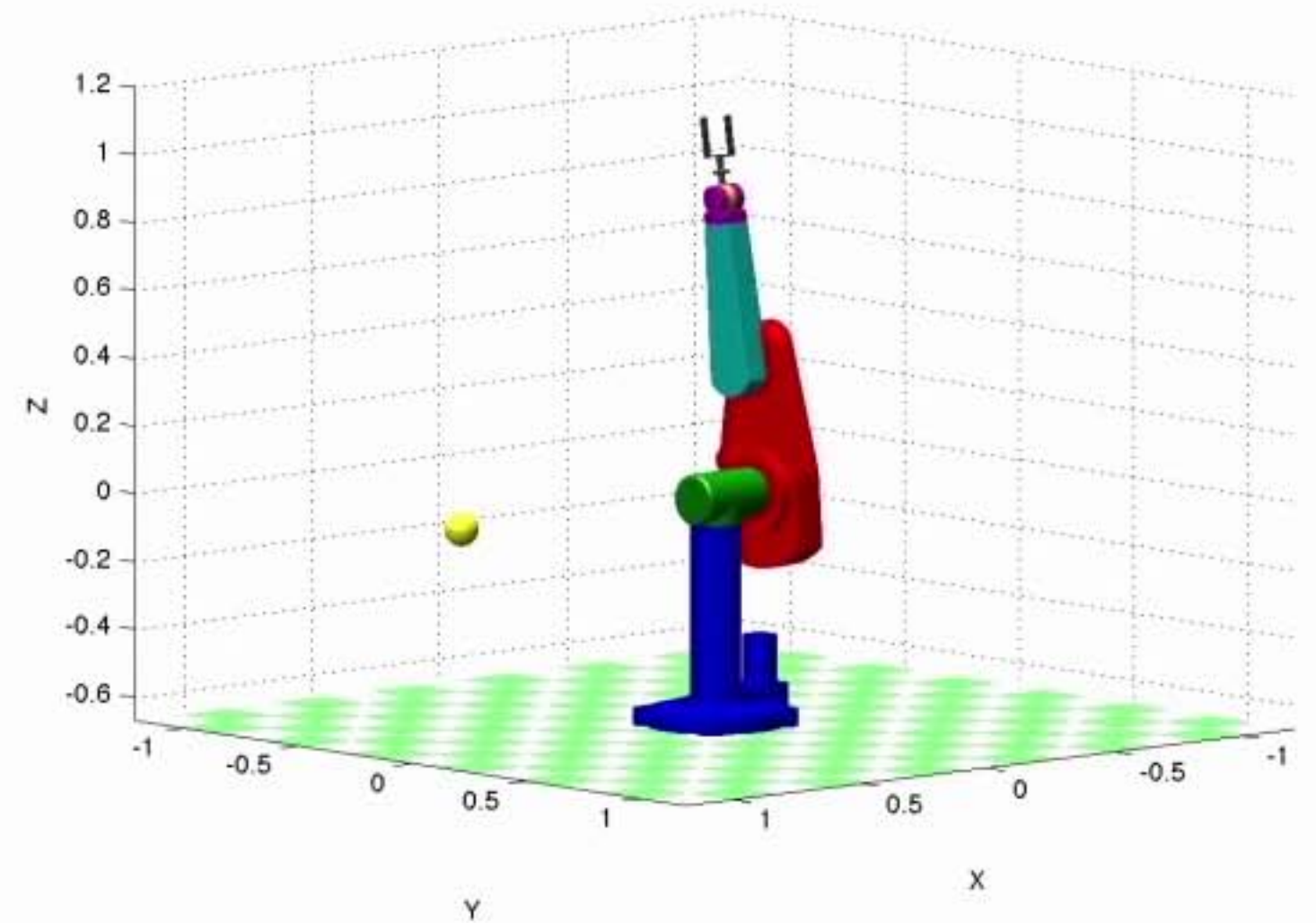
Two solutions



Two solutions

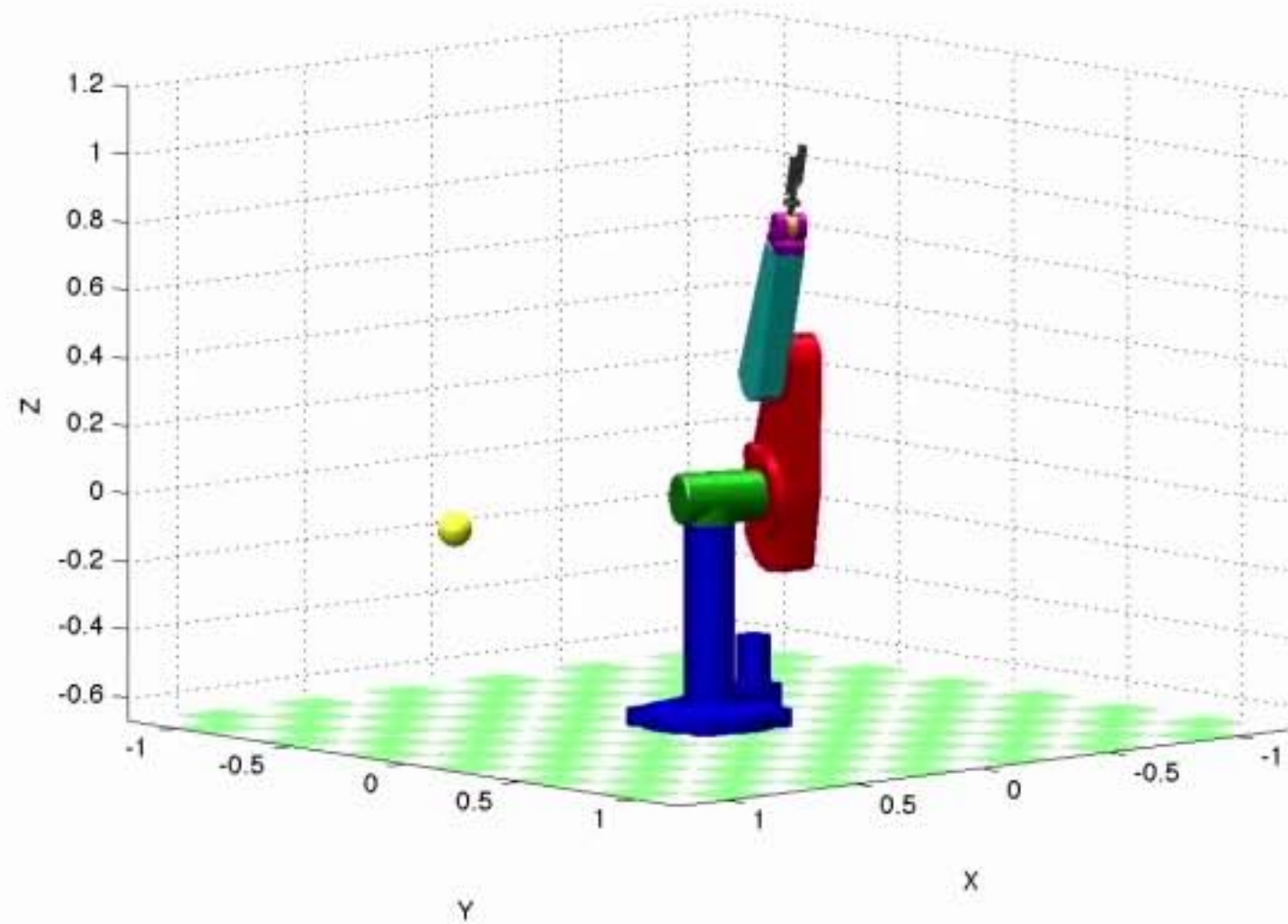


■ left-handed

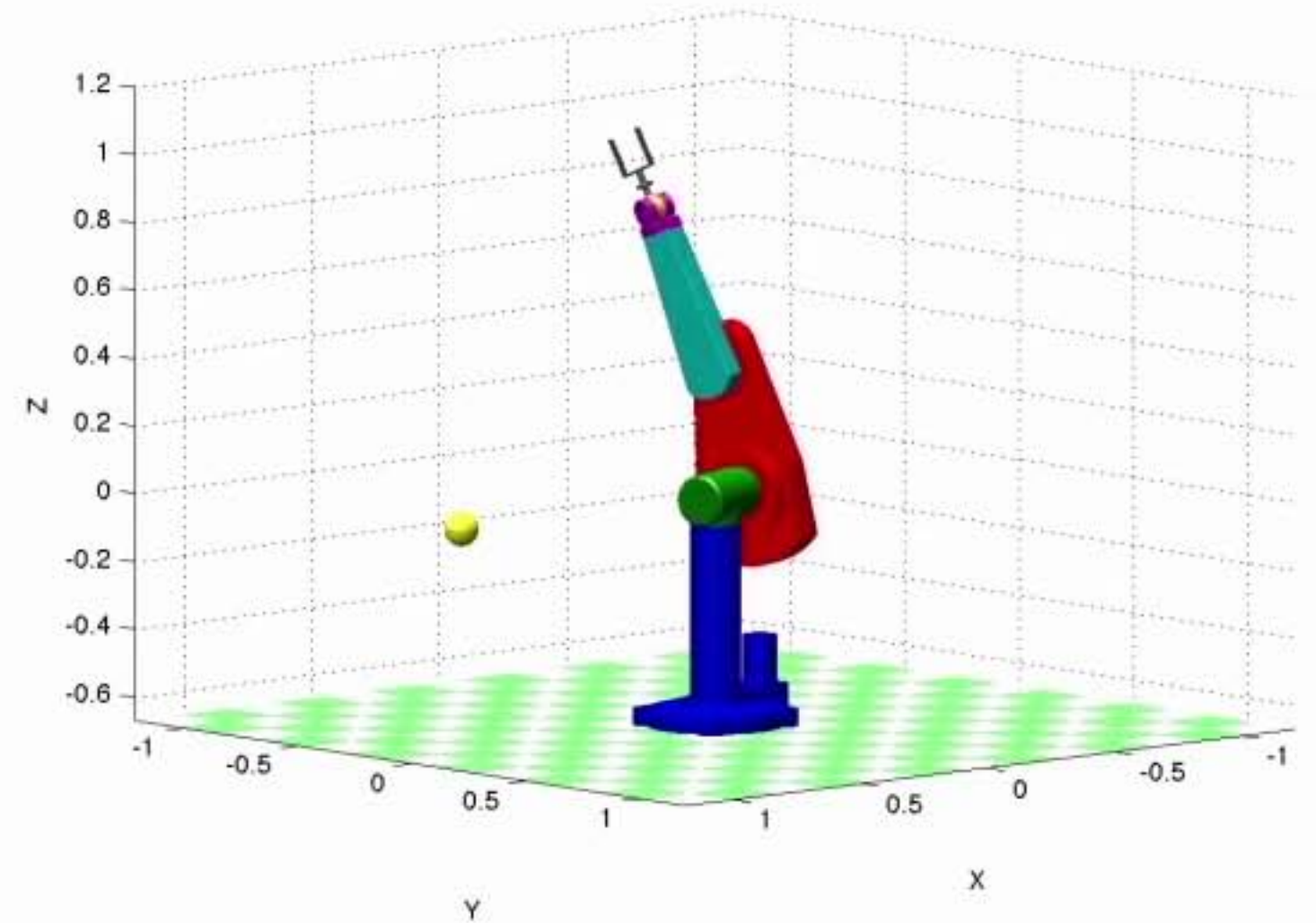


■ right-handed

Two solutions

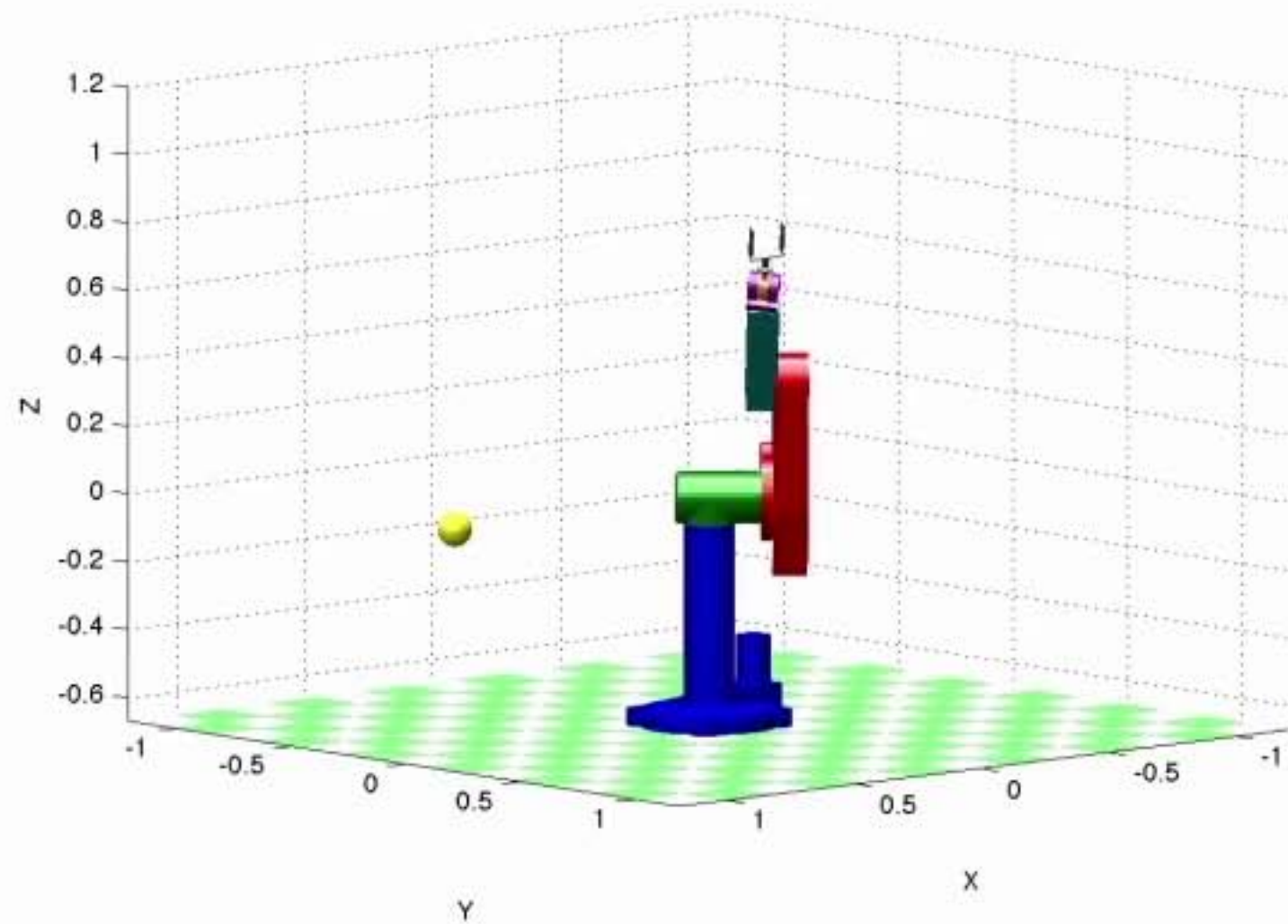


■ left-handed

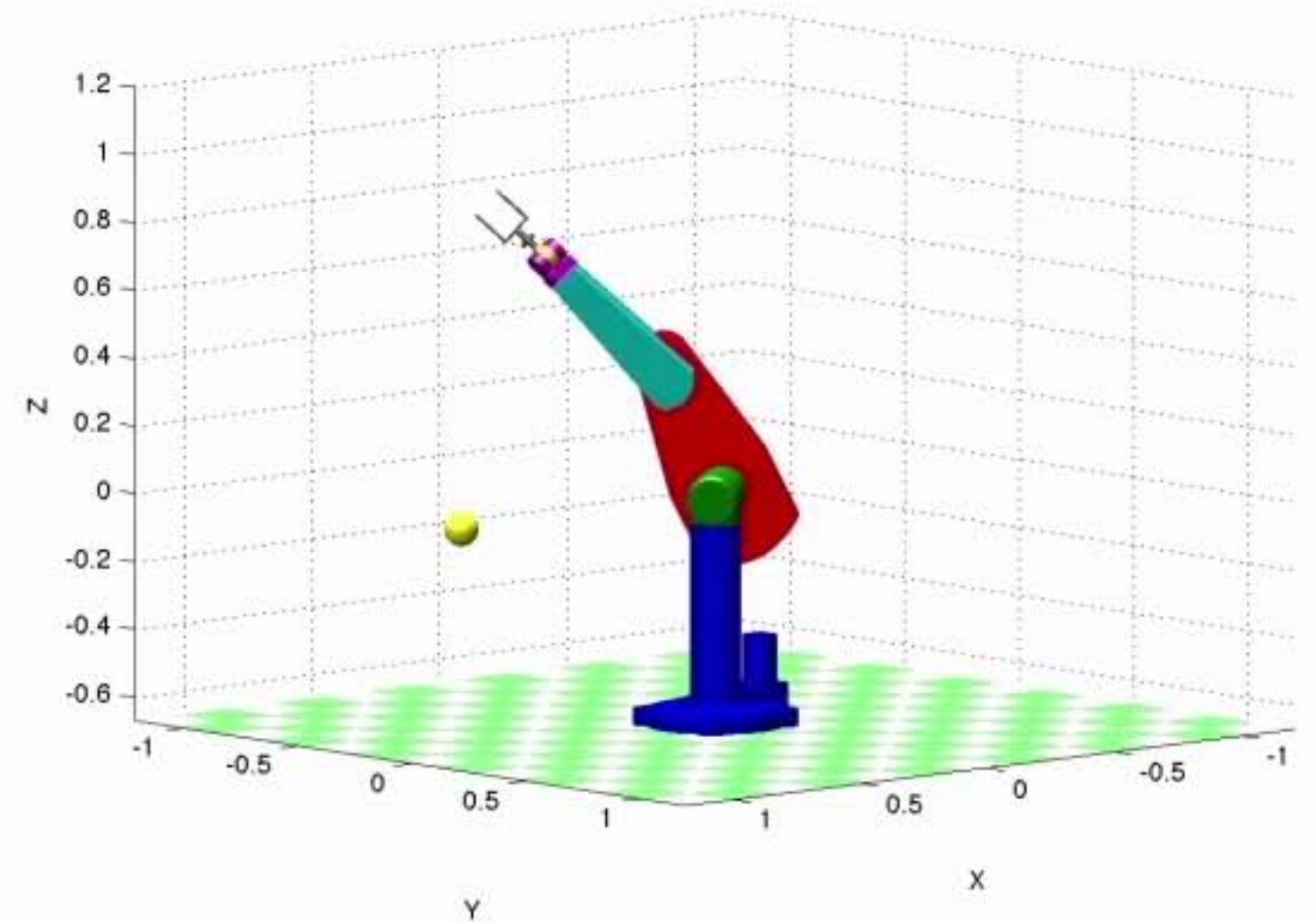


■ right-handed

Two solutions

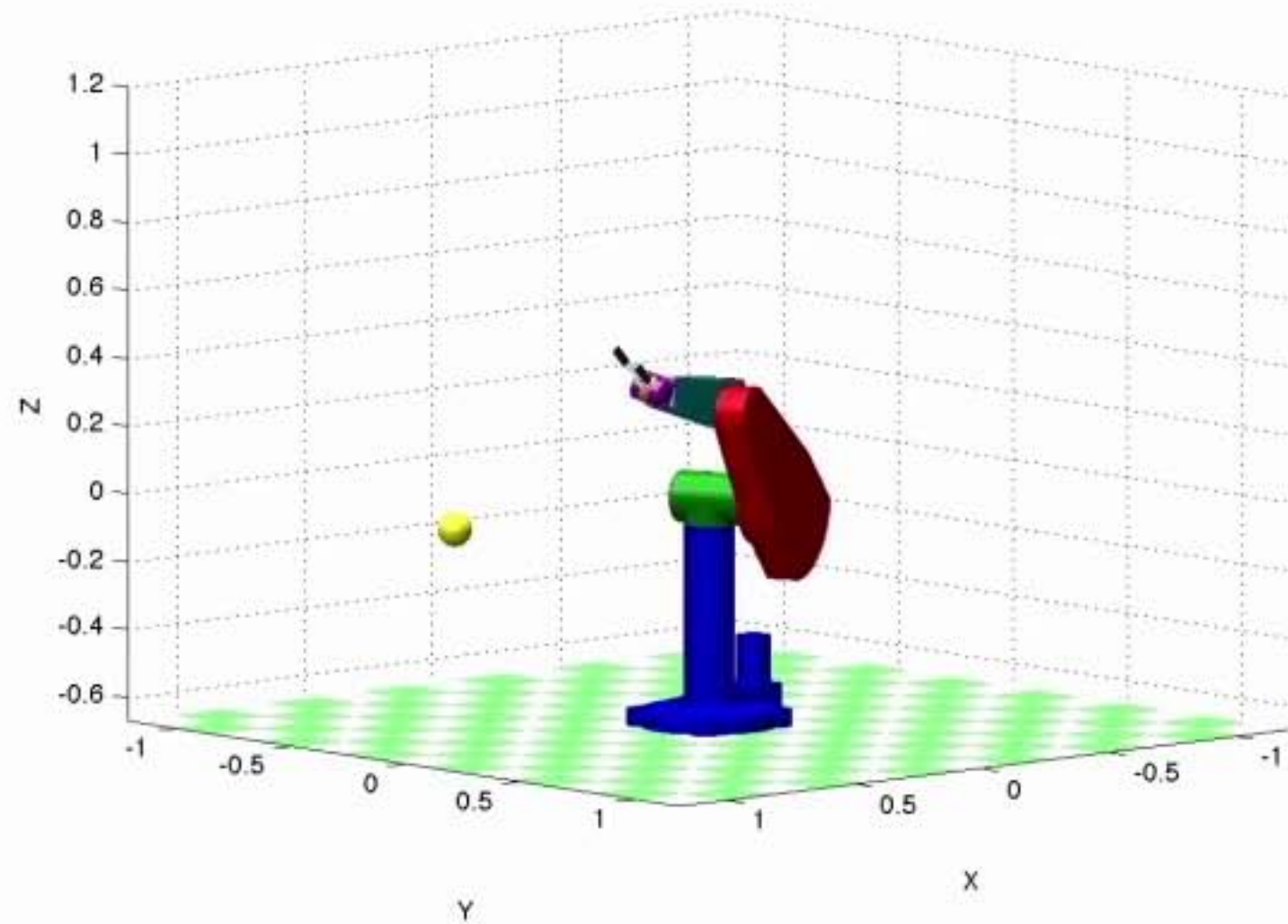


■ left-handed

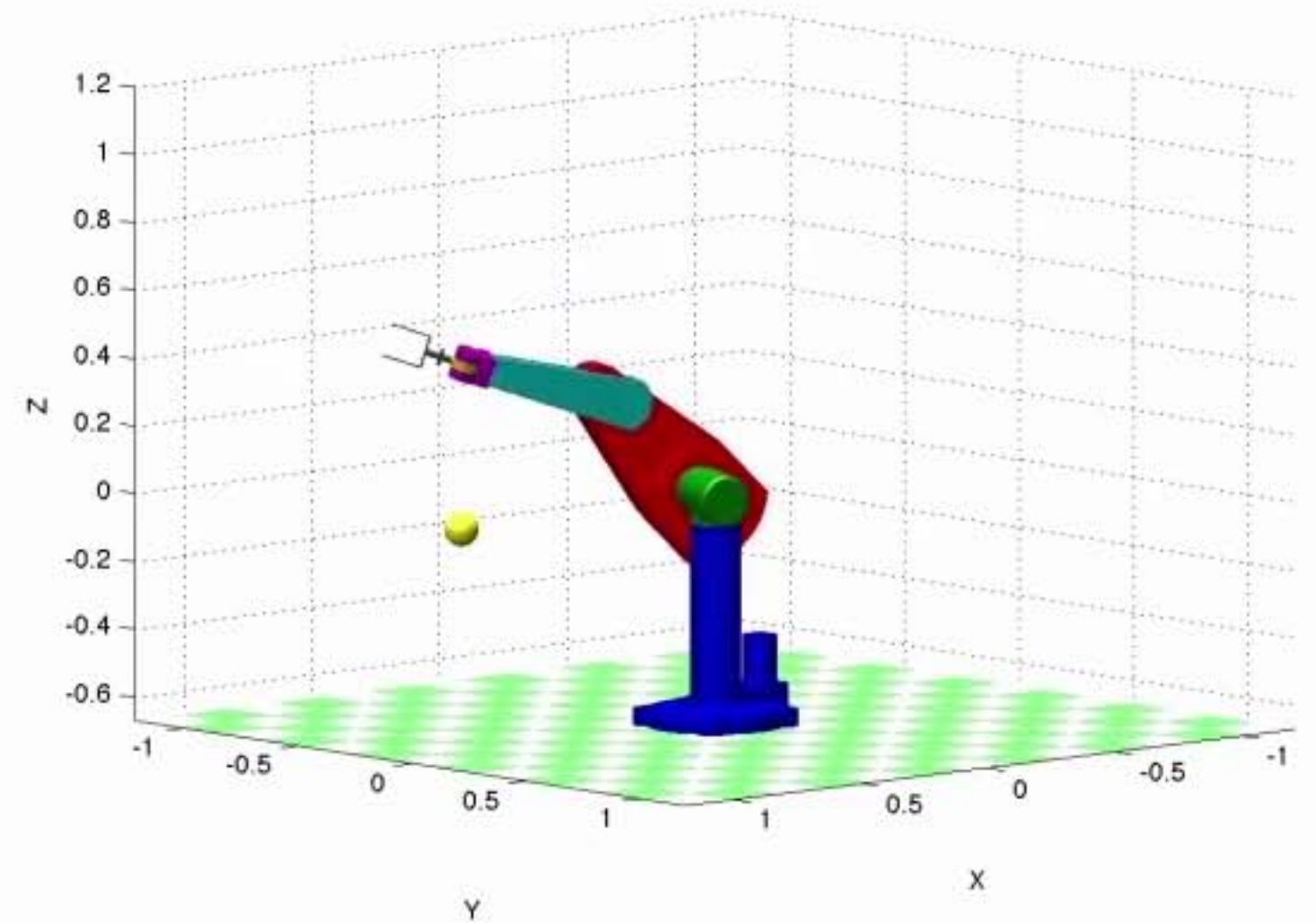


■ right-handed

Two solutions

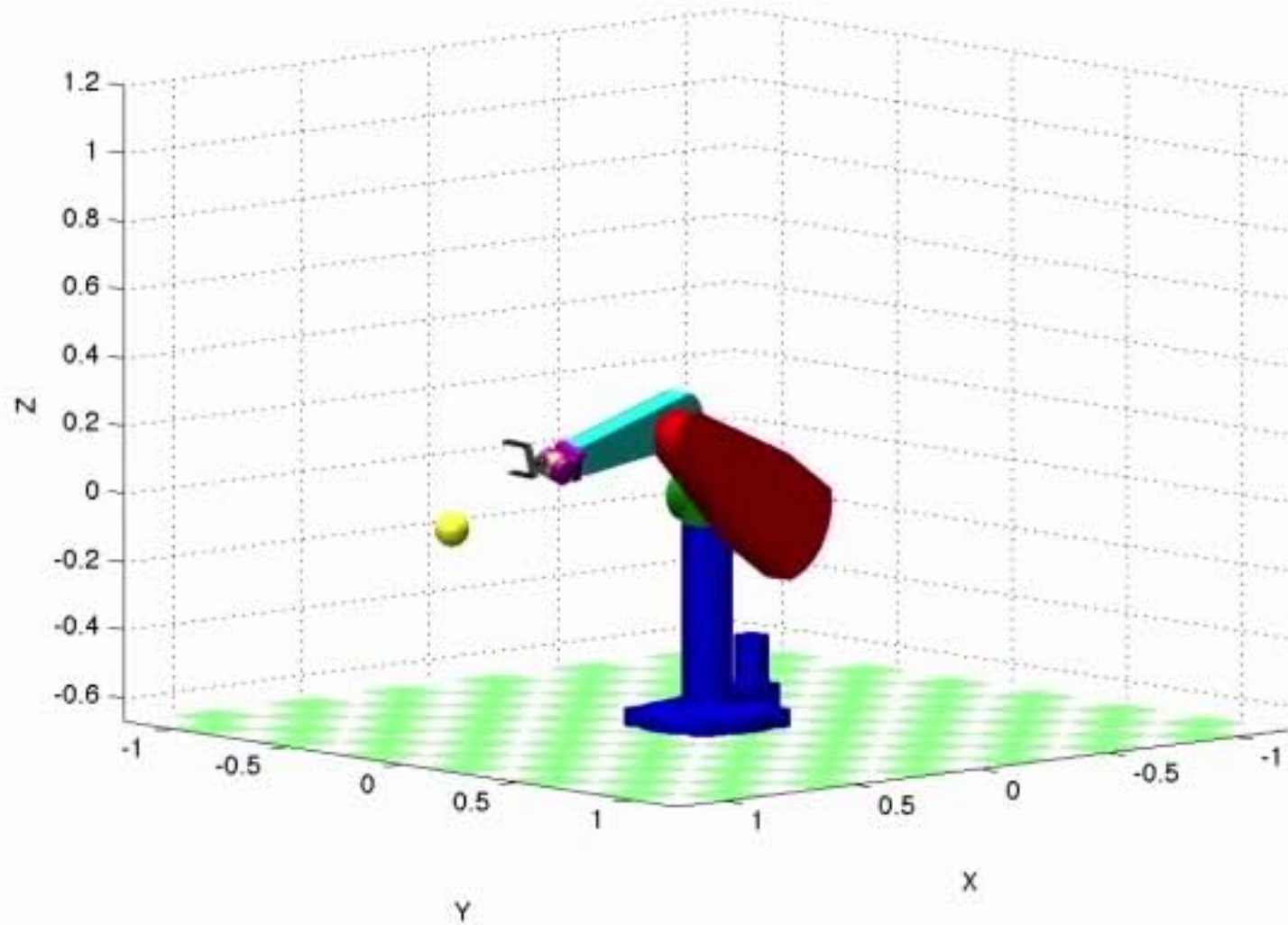


■ left-handed

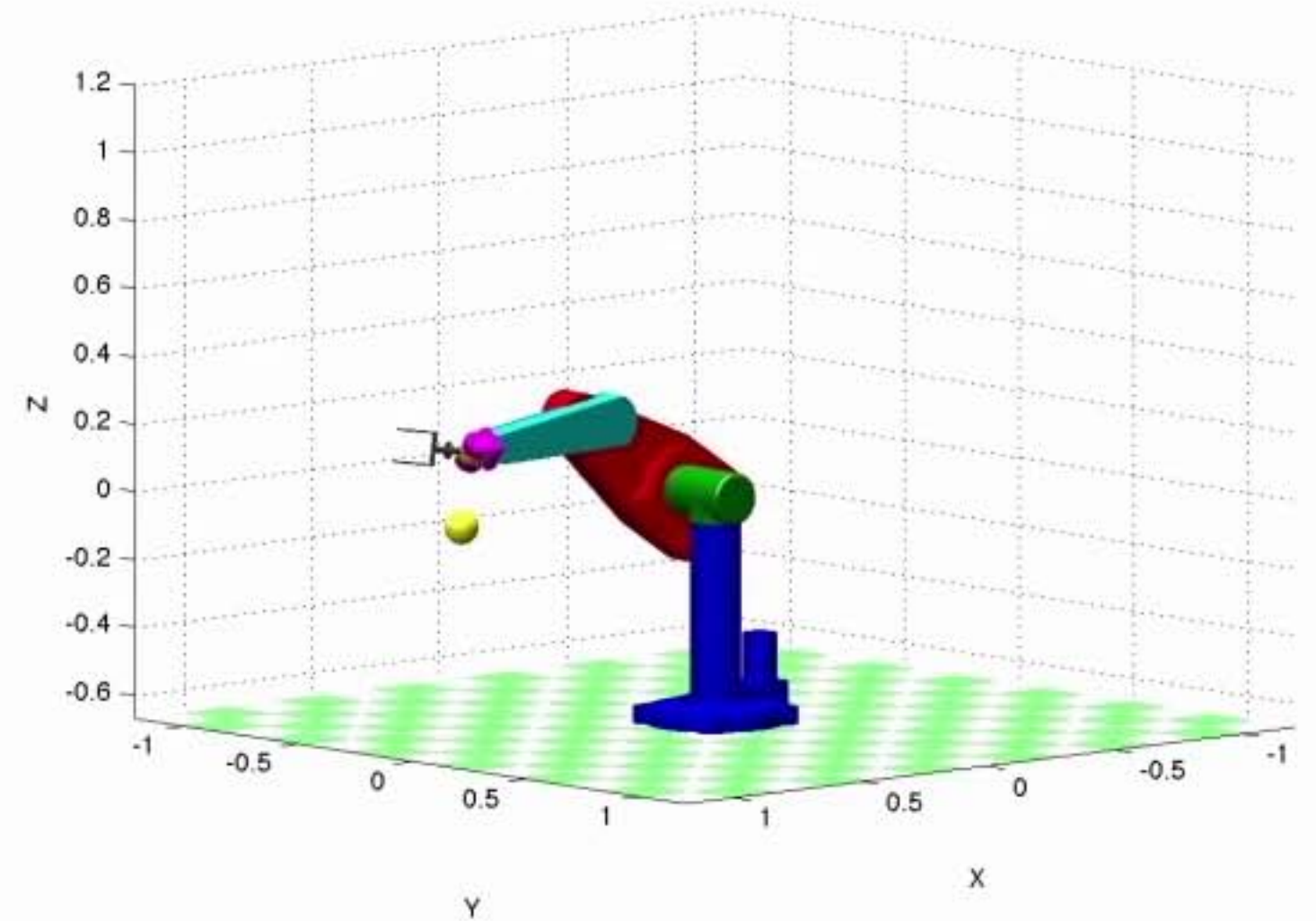


■ right-handed

Two solutions

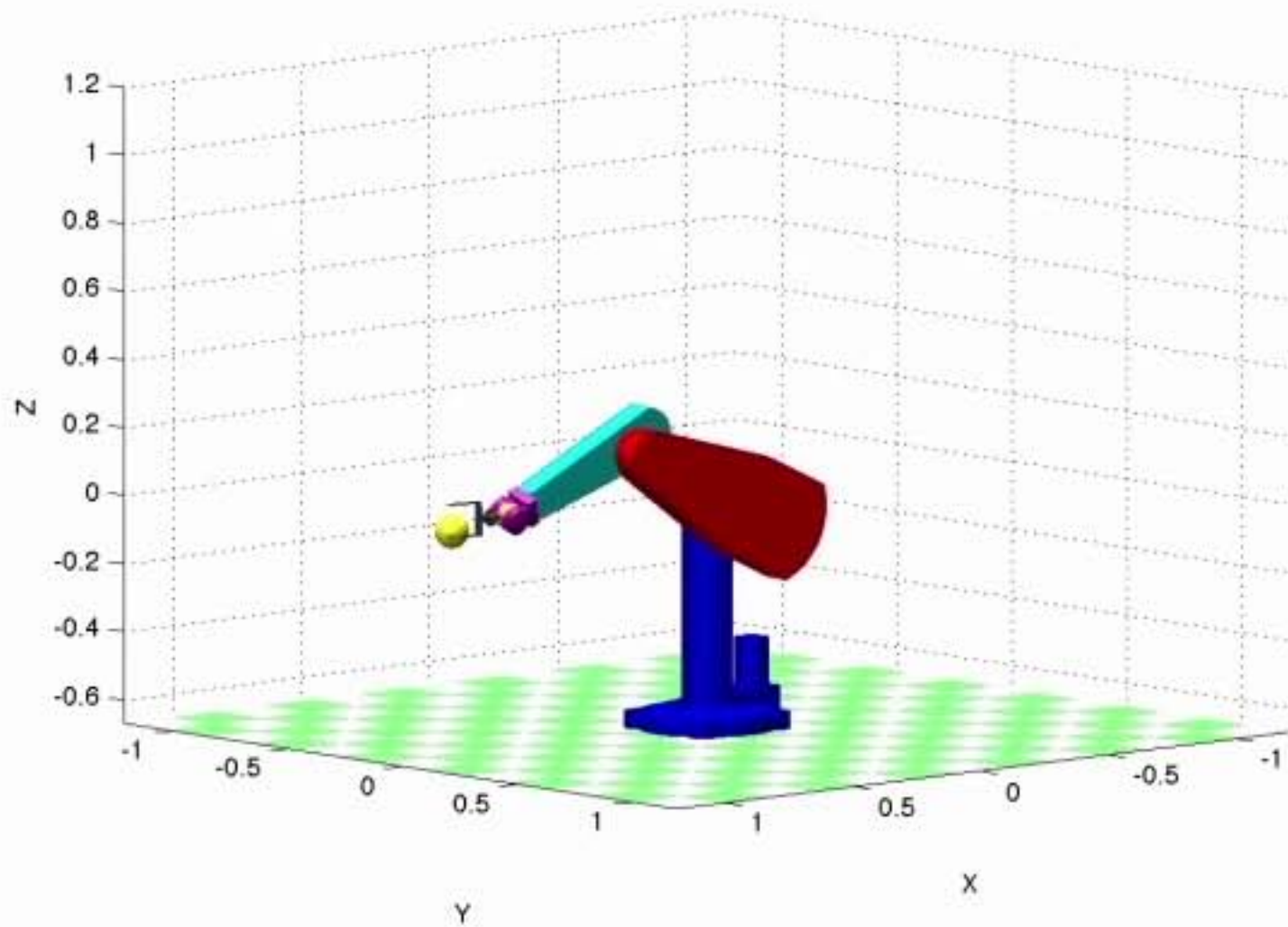


■ left-handed

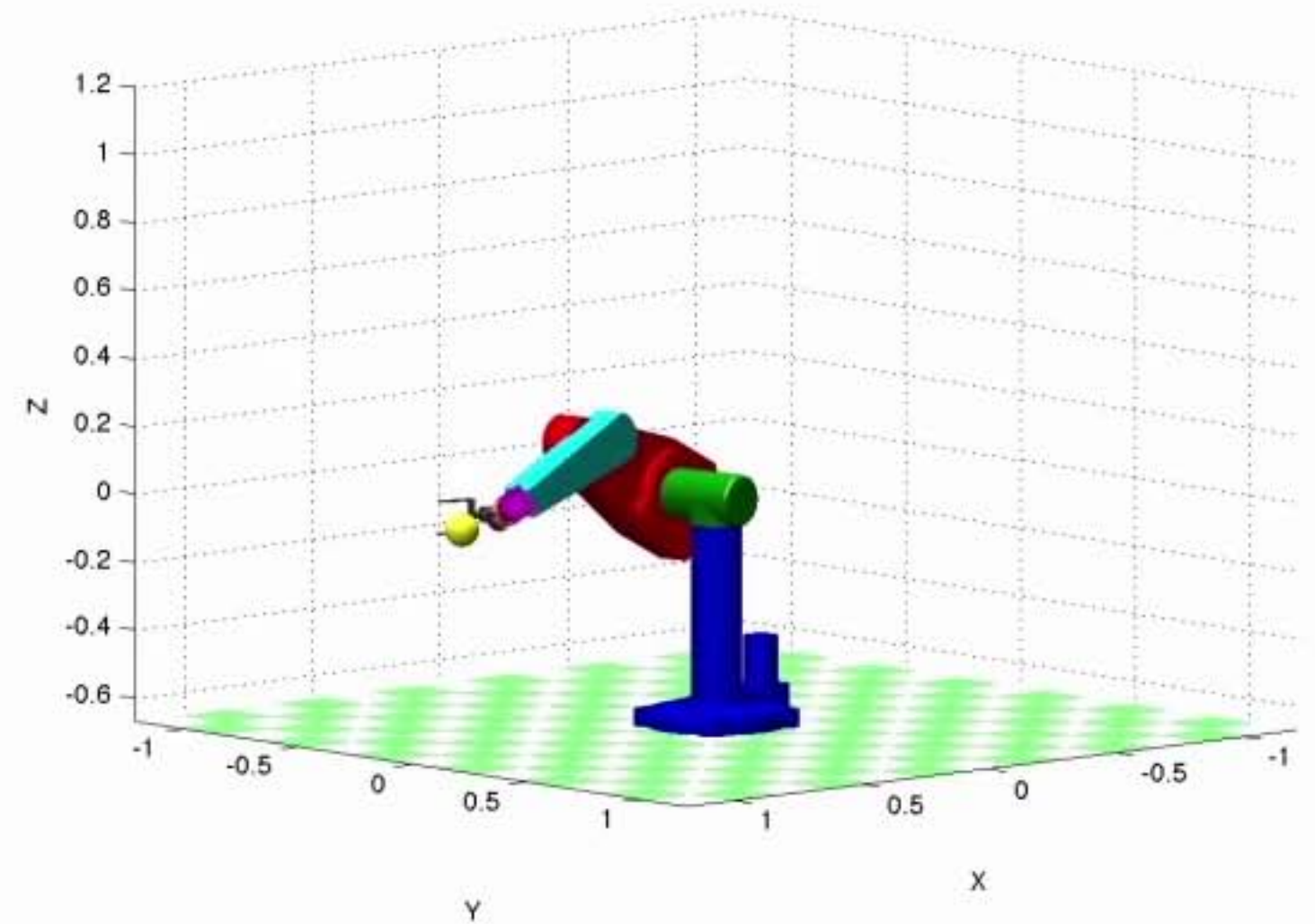


■ right-handed

Two solutions



■ left-handed



■ right-handed

Command Window

```

>> mdl_puma560
fx >> |
  
```

Workspace

Name ▲	Value	Class
deg	0.0175	double
p560	<1x1 SerialLink>	SerialLink
qn	[0,0.7854,3.1416,0,0.78...	double
qr	[0,1.5708,-1.5708,0,0,0]	double
qs	[0,0,-1.5708,0,0,0]	double
qz	[0,0,0,0,0,0]	double

Command Window

```
>> mdl_puma560
>> T = transl(0.6, 0.1, 0) * rpy2tr(0, 180, 0, 'deg')
```

T =

```
-1.0000    0    0.0000    0.6000
         0    1.0000    0    0.1000
-0.0000    0   -1.0000    0
         0    0         0    1.0000
```

fx >> q1 =

Workspace

Name	Value	Class
T	<4x4 double>	double
deg	0.0175	double
p560	<1x1 SerialLink>	SerialLink
qn	[0,0.7854,3.1416,0,0.78...	double
qr	[0,1.5708,-1.5708,0,0,0]	double
qs	[0,0,-1.5708,0,0,0]	double
qz	[0,0,0,0,0,0]	double

Command Window

```
>> mdl_puma560
>> T = transl(0.6, 0.1, 0) * rpy2tr(0, 180, 0, 'deg')

T =

    -1.0000         0         0.0000         0.6000
         0         1.0000         0         0.1000
   -0.0000         0    -1.0000         0
         0         0         0         1.0000

>> ql = p560.ikine6s(T, 'l')

ql =

    3.0575    2.3210    0.1162   -0.0000    0.7044    3.0575

fx >> ql = p560.ikine6s(T, 'l')
```

Workspace

Name	Value	Class
T	<4x4 double>	double
deg	0.0175	double
p560	<1x1 SerialLink>	SerialLink
ql	[3.0575,2.3210,0.1162,-...	double
qn	[0,0.7854,3.1416,0,0.78...	double
qr	[0,1.5708,-1.5708,0,0,0]	double
qs	[0,0,-1.5708,0,0,0]	double
qz	[0,0,0,0,0,0]	double

Command Window

```
>> mdl_puma560
>> T = transl(0.6, 0.1, 0) * rpy2tr(0, 180, 0, 'deg')

T =

    -1.0000         0     0.0000     0.6000
         0     1.0000         0     0.1000
   -0.0000         0    -1.0000         0
         0         0         0     1.0000

>> ql = p560.ikine6s(T, 'l')

ql =

    3.0575    2.3210    0.1162   -0.0000    0.7044    3.0575

>> qr = p560.ikine6s(T, 'r')

qr =

    0.4144    0.8206    3.1193   -3.1416    0.7983   -2.7272
```

fx >> |

Workspace

Name ▲	Value	Class
T	<4x4 double>	double
deg	0.0175	double
p560	<1x1 SerialLink>	SerialLink
ql	[3.0575,2.3210,0.1162,-...	double
qn	[0,0.7854,3.1416,0,0.78...	double
qr	[0.4144,0.8206,3.1193,-...	double
qs	[0,0,-1.5708,0,0,0]	double
qz	[0,0,0,0,0,0]	double

Command Window

```
>> q1 = p560.ikine6s(T, 'l')

q1 =

    3.0575    2.3210    0.1162   -0.0000    0.7044    3.0575

>> qr = p560.ikine6s(T, 'r')

qr =

    0.4144    0.8206    3.1193   -3.1416    0.7983   -2.7272

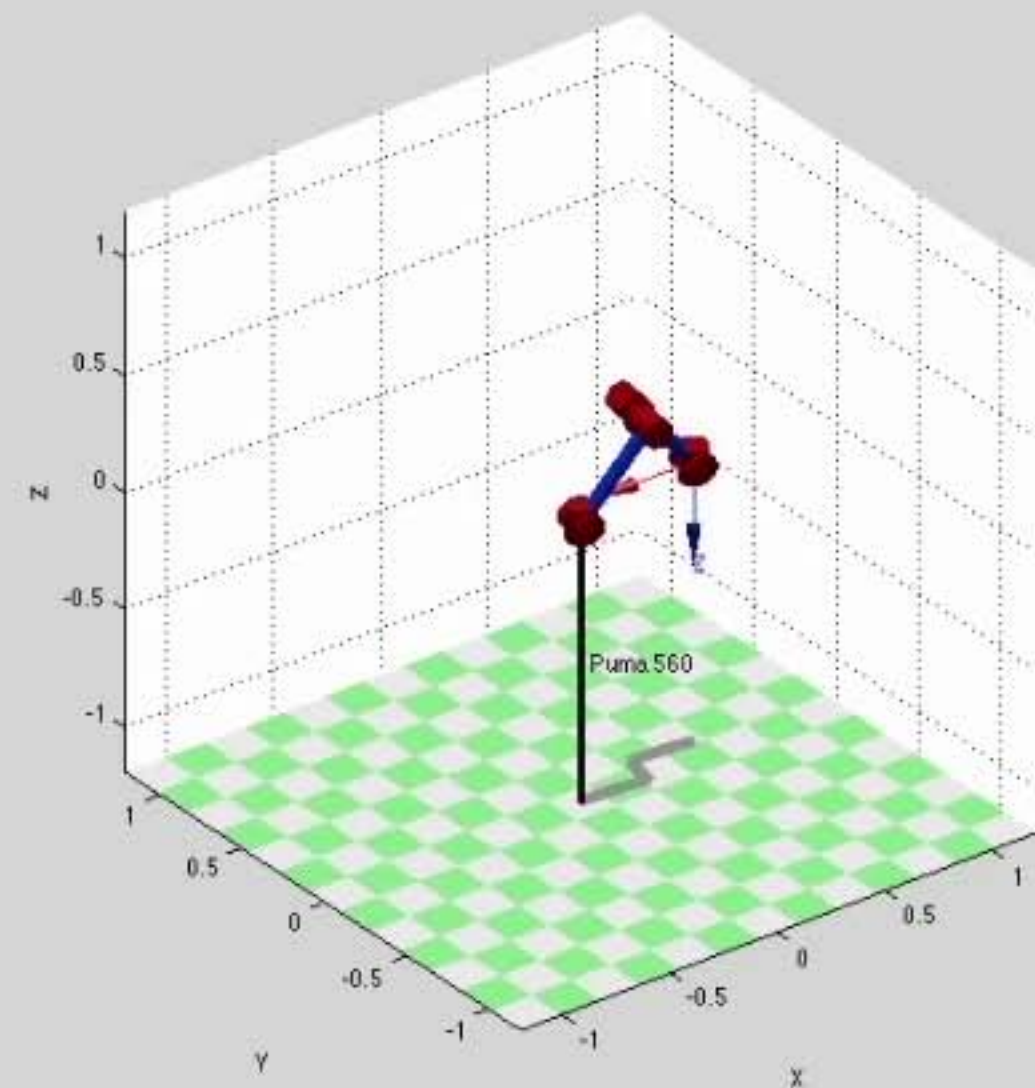
>> p560.plot(q1)
fx >>
```

Workspace

Name	Value	Class
T	<4x4 double>	double
deg	0.0175	double
p560	<1x1 SerialLink>	SerialLink
q1	[3.0575,2.3210,0.1162,-...	double
qn	[0,0.7854,3.1416,0,0.78...	double
qr	[0.4144,0.8206,3.1193,-...	double
qs	[0,0,-1.5708,0,0,0]	double
qz	[0,0,0,0,0,0]	double

Figures

Figure 1



Command Window

```
>> q1 = p560.ikine6s(T, 'l')

q1 =

    3.0575    2.3210    0.1162   -0.0000    0.7044    3.0575

>> qr = p560.ikine6s(T, 'r')

qr =

    0.4144    0.8206    3.1193   -3.1416    0.7983   -2.7272

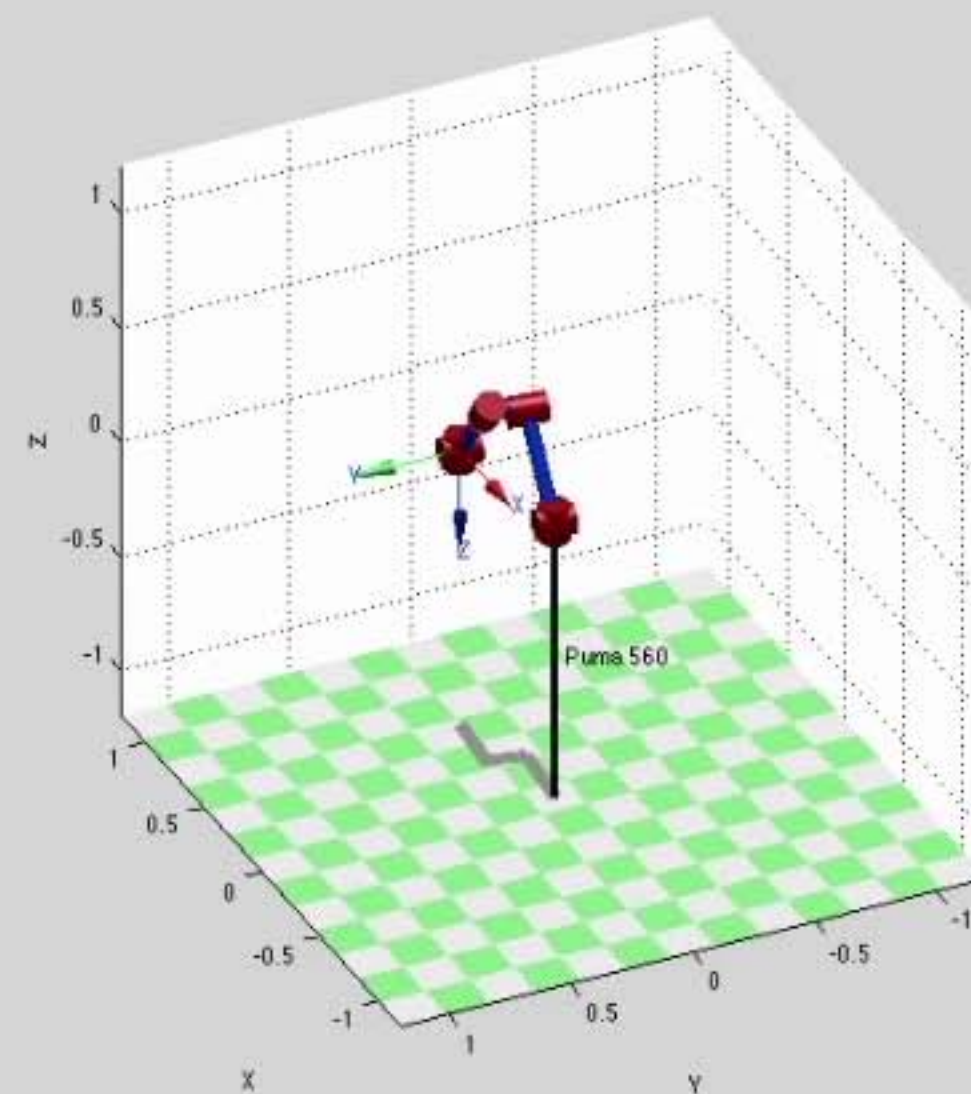
>> p560.plot(q1)
fx >> |
```

Workspace

Name	Value	Class
T	<4x4 double>	double
deg	0.0175	double
p560	<1x1 SerialLink>	SerialLink
q1	[3.0575,2.3210,0.1162,-...	double
qn	[0,0.7854,3.1416,0,0.78...	double
qr	[0.4144,0.8206,3.1193,-...	double
qs	[0,0,-1.5708,0,0,0]	double
qz	[0,0,0,0,0,0]	double

Figures

Figure 1



Command Window

```
>> q1 = p560.ikine6s(T, 'l')

q1 =

    3.0575    2.3210    0.1162   -0.0000    0.7044    3.0575

>> qr = p560.ikine6s(T, 'r')

qr =

    0.4144    0.8206    3.1193   -3.1416    0.7983   -2.7272

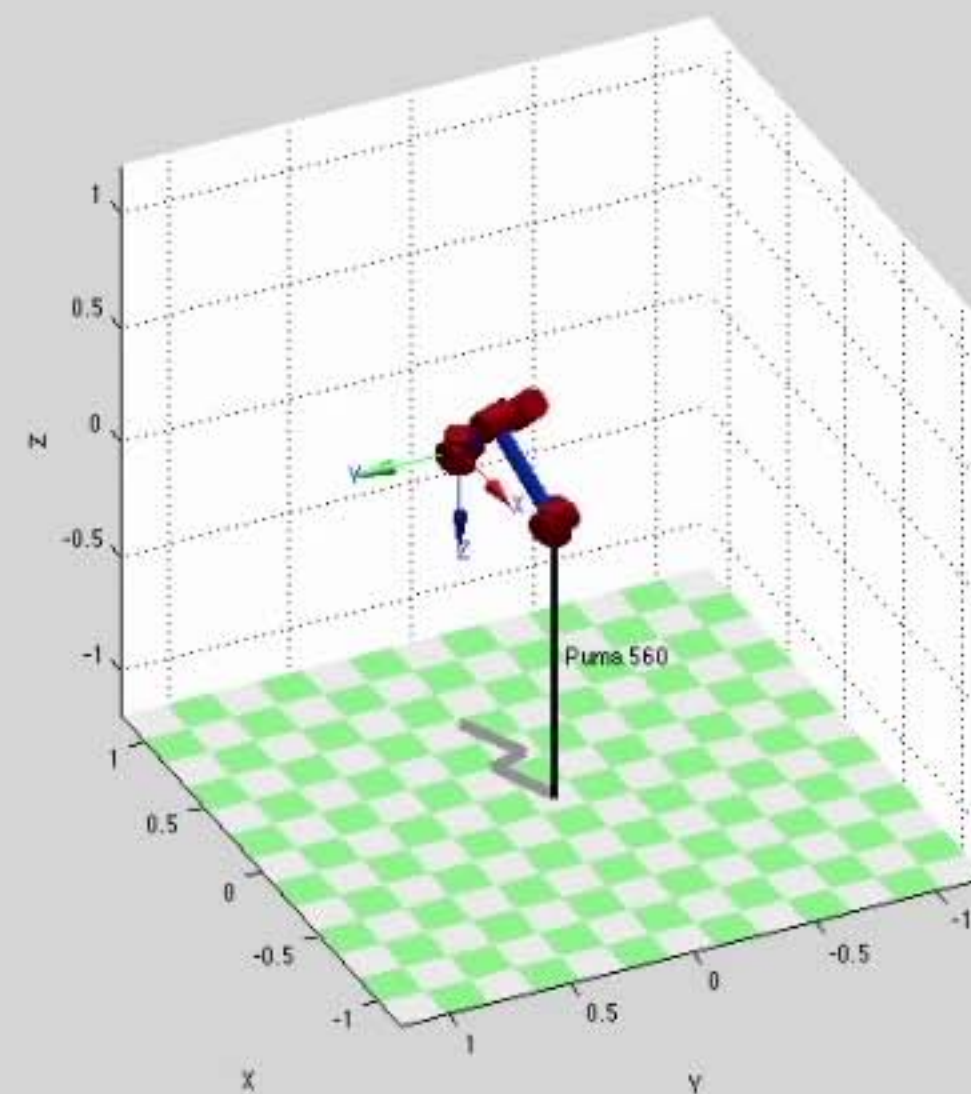
>> p560.plot(q1)
>> p560.plot(qr)
fx >> |
```

Workspace

Name	Value	Class
T	<4x4 double>	double
deg	0.0175	double
p560	<1x1 SerialLink>	SerialLink
q1	[3.0575,2.3210,0.1162,-...	double
qn	[0,0.7854,3.1416,0,0.78...	double
qr	[0.4144,0.8206,3.1193,-...	double
qs	[0,0,-1.5708,0,0,0]	double
qz	[0,0,0,0,0,0]	double

Figures

Figure 1



Command Window

```
>> q1 = p560.ikine6s(1, 1)

q1 =

    3.0575    2.3210    0.1162   -0.0000    0.7044    3.0575

>> qr = p560.ikine6s(T, 'r')

qr =

    0.4144    0.8206    3.1193   -3.1416    0.7983   -2.7372

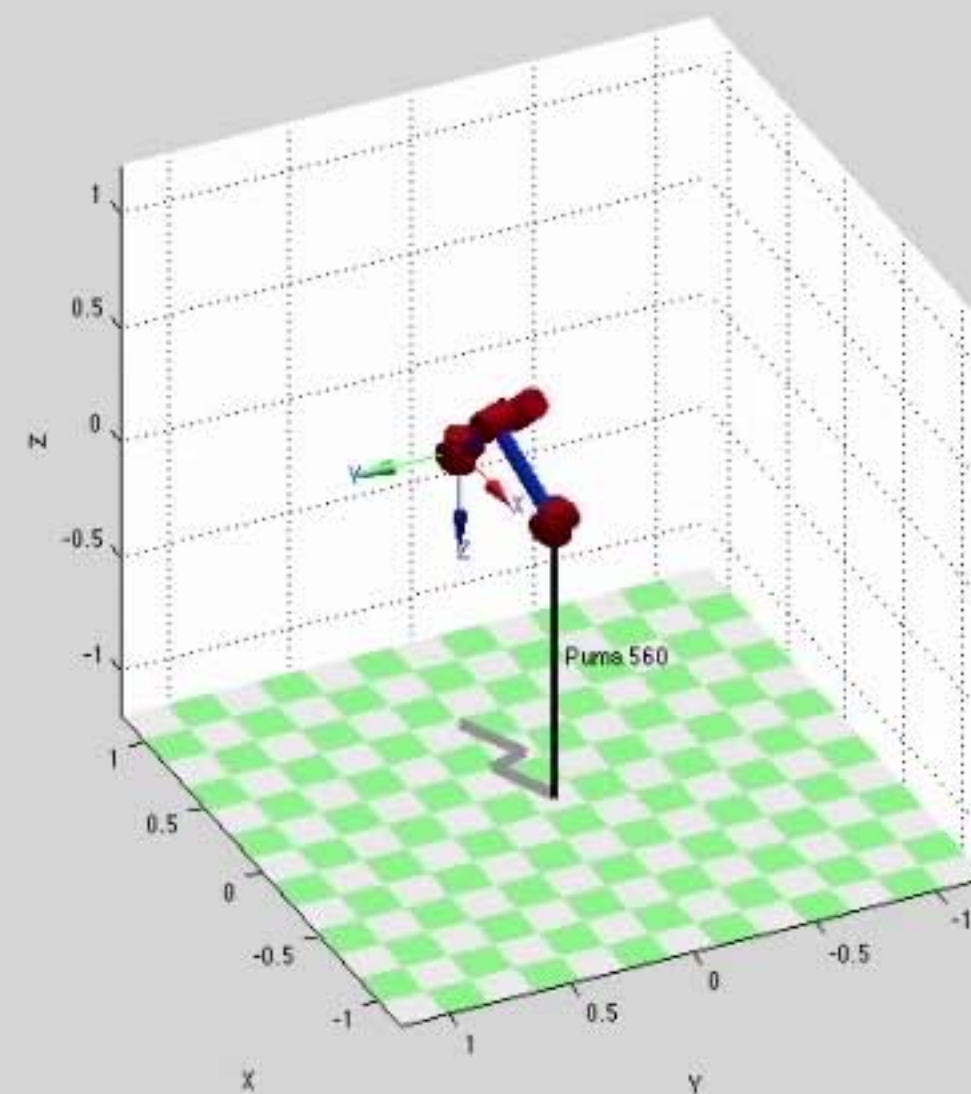
>> p560.plot(q1)
>> p560.plot(qr)
>> tg = jtraj(qz, q1, 50);
fx >>
```

Workspace

Name	Value	Class
T	<4x4 double>	double
deg	0.0175	double
p560	<1x1 SerialLink>	SerialLink
q1	[3.0575,2.3210,0.1162,-...	double
qn	[0,0.7854,3.1416,0,0.78...	double
qr	[0.4144,0.8206,3.1193,-...	double
qs	[0,0,-1.5708,0,0,0]	double
qz	[0,0,0,0,0,0]	double
tg	<50x6 double>	double

Figures

Figure 1



Command Window

```
ql =
    3.0575    2.3210    0.1162   -0.0000    0.7044    3.0575

>> qr = p560.ikine6s(T, 'r')

qr =
    0.4144    0.8206    3.1193   -3.1416    0.7983   -2.7272

>> p560.plot(ql)
>> p560.plot(qr)
>> tg = jtraj(qz, ql, 50);
>> p560.plot(tg)
```

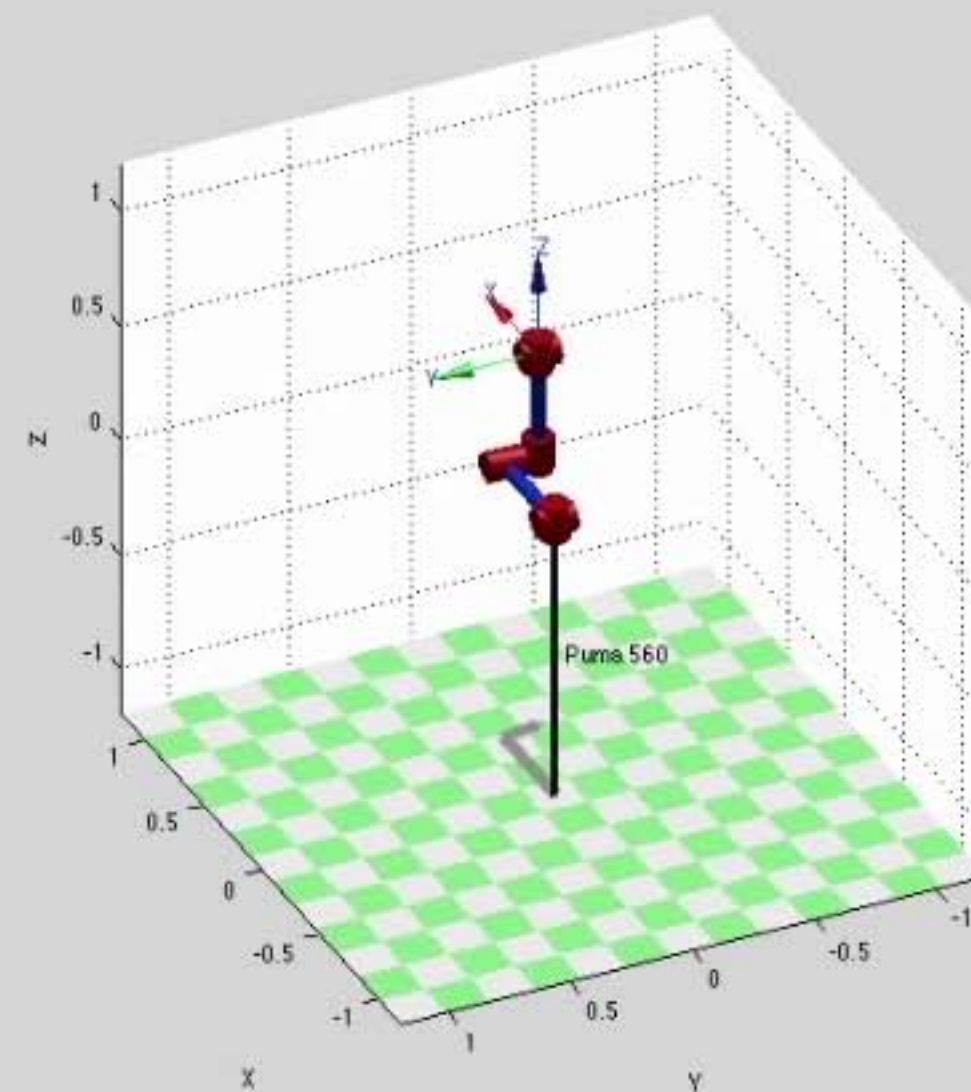
fx

Workspace

Name	Value	Class
T	<4x4 double>	double
deg	0.0175	double
p560	<1x1 SerialLink>	SerialLink
ql	[3.0575,2.3210,0.1162,-...	double
qn	[0,0.7854,3.1416,0,0.78...	double
qr	[0.4144,0.8206,3.1193,-...	double
qs	[0,0,-1.5708,0,0,0]	double
qz	[0,0,0,0,0,0]	double
tg	<50x6 double>	double

Figures

Figure 1



Command Window

```
ql =
    3.0575    2.3210    0.1162   -0.0000    0.7044    3.0575

>> qr = p560.ikine6s(T, 'r')

qr =
    0.4144    0.8206    3.1193   -3.1416    0.7983   -2.7272

>> p560.plot(ql)
>> p560.plot(qr)
>> tg = jtraj(qz, ql, 50);
>> p560.plot(tg)
```

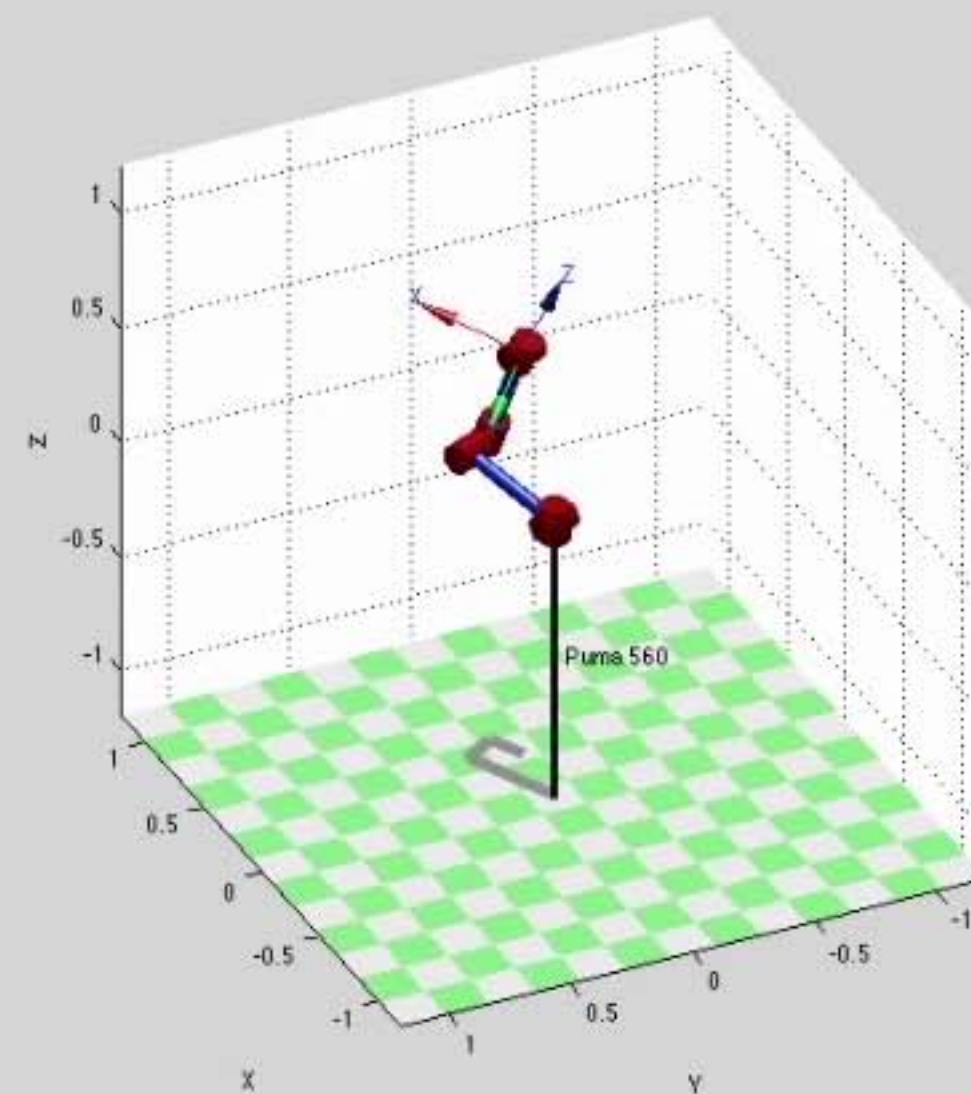
fx

Workspace

Name	Value	Class
T	<4x4 double>	double
deg	0.0175	double
p560	<1x1 SerialLink>	SerialLink
ql	[3.0575,2.3210,0.1162,-...	double
qn	[0,0.7854,3.1416,0,0.78...	double
qr	[0.4144,0.8206,3.1193,-...	double
qs	[0,0,-1.5708,0,0,0]	double
qz	[0,0,0,0,0,0]	double
tg	<50x6 double>	double

Figures

Figure 1



Command Window

```
ql =
    3.0575    2.3210    0.1162   -0.0000    0.7044    3.0575

>> qr = p560.ikine6s(T, 'r')

qr =
    0.4144    0.8206    3.1193   -3.1416    0.7983   -2.7272

>> p560.plot(ql)
>> p560.plot(qr)
>> tg = jtraj(qz, ql, 50);
>> p560.plot(tg)
```

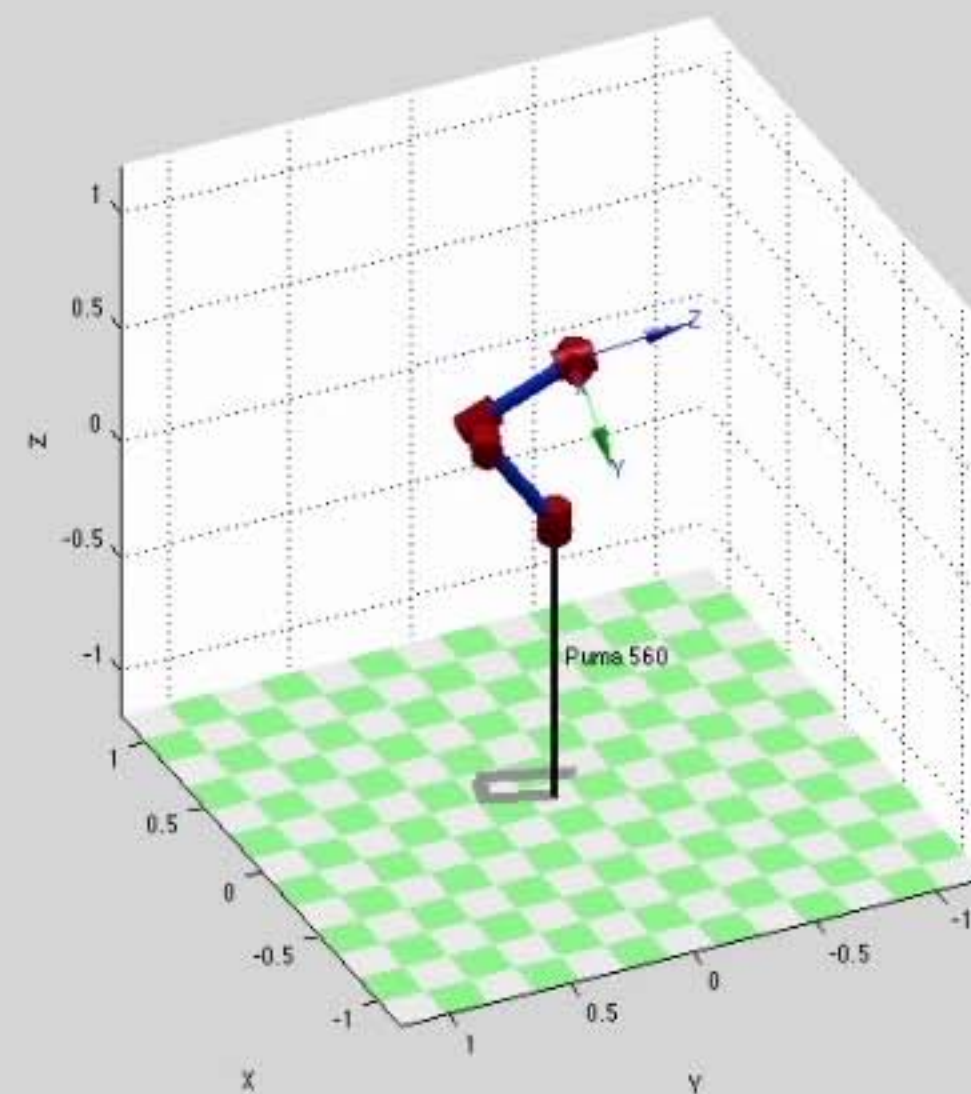
fx

Workspace

Name	Value	Class
T	<4x4 double>	double
deg	0.0175	double
p560	<1x1 SerialLink>	SerialLink
ql	[3.0575,2.3210,0.1162,-...	double
qn	[0,0.7854,3.1416,0,0.78...	double
qr	[0.4144,0.8206,3.1193,-...	double
qs	[0,0,-1.5708,0,0,0]	double
qz	[0,0,0,0,0,0]	double
tg	<50x6 double>	double

Figures

Figure 1



Command Window

```
ql =
    3.0575    2.3210    0.1162   -0.0000    0.7044    3.0575

>> qr = p560.ikine6s(T, 'r')

qr =
    0.4144    0.8206    3.1193   -3.1416    0.7983   -2.7272

>> p560.plot(ql)
>> p560.plot(qr)
>> tg = jtraj(qz, ql, 50);
>> p560.plot(tg)
```

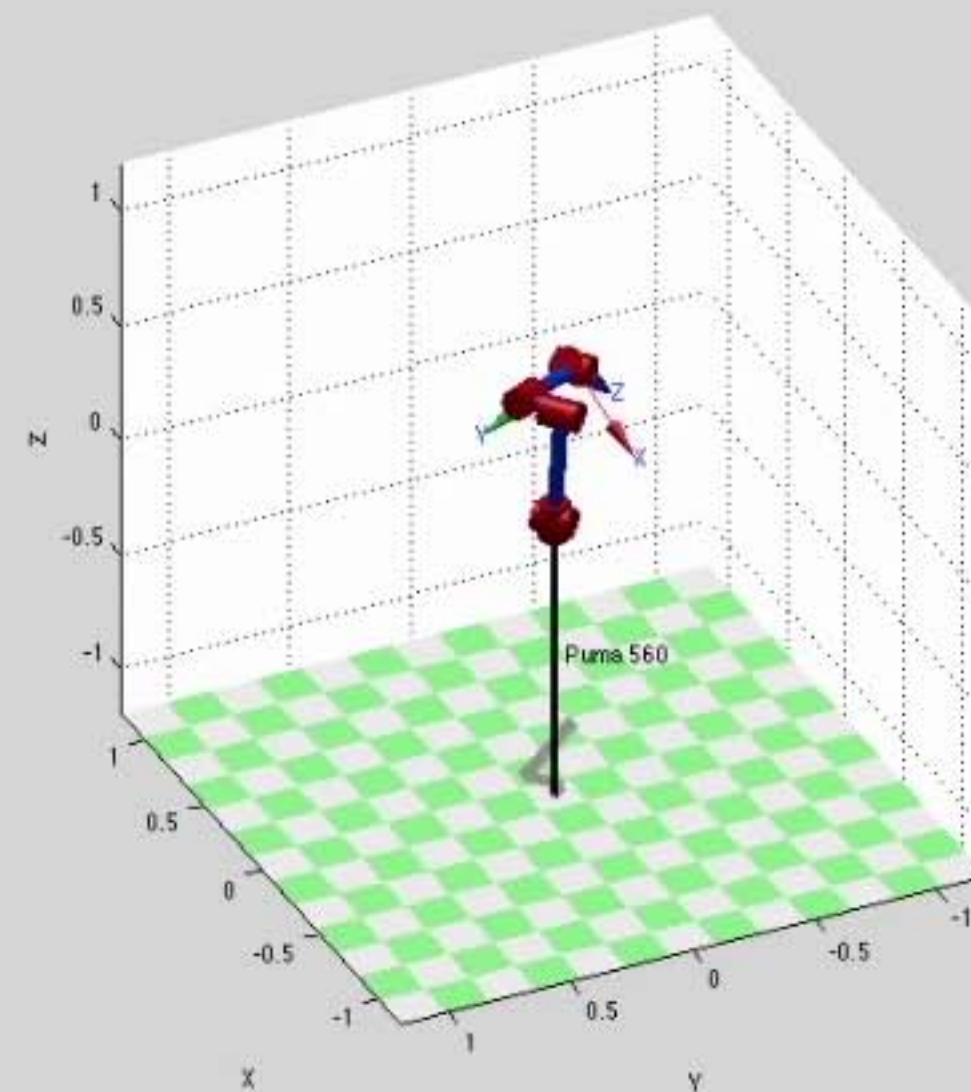
fx

Workspace

Name	Value	Class
T	<4x4 double>	double
deg	0.0175	double
p560	<1x1 SerialLink>	SerialLink
ql	[3.0575,2.3210,0.1162,-...	double
qn	[0,0.7854,3.1416,0,0.78...	double
qr	[0.4144,0.8206,3.1193,-...	double
qs	[0,0,-1.5708,0,0,0]	double
qz	[0,0,0,0,0,0]	double
tg	<50x6 double>	double

Figures

Figure 1



Command Window

```
ql =
    3.0575    2.3210    0.1162   -0.0000    0.7044    3.0575

>> qr = p560.ikine6s(T, 'r')

qr =
    0.4144    0.8206    3.1193   -3.1416    0.7983   -2.7272

>> p560.plot(ql)
>> p560.plot(qr)
>> tg = jtraj(qz, ql, 50);
>> p560.plot(tg)
```

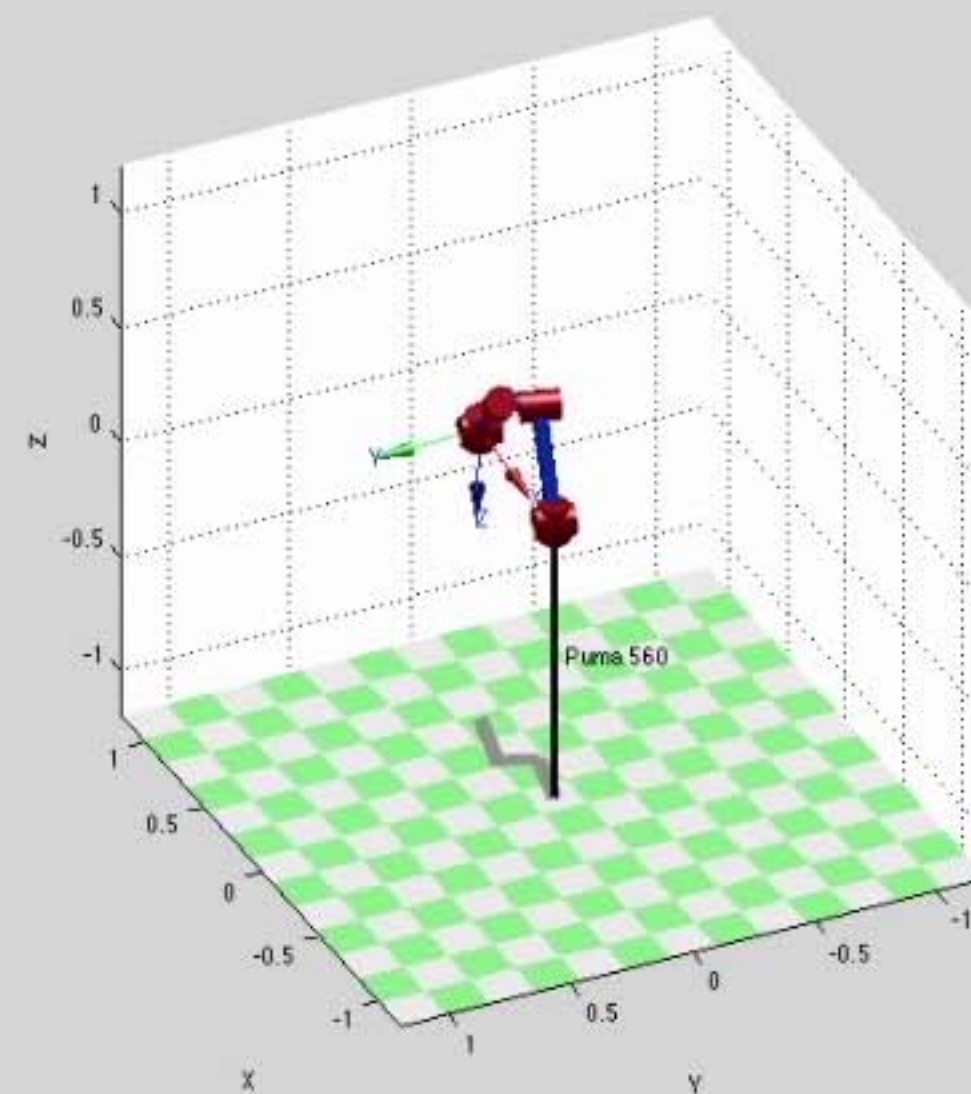
fx

Workspace

Name	Value	Class
T	<4x4 double>	double
deg	0.0175	double
p560	<1x1 SerialLink>	SerialLink
ql	[3.0575,2.3210,0.1162,-...	double
qn	[0,0.7854,3.1416,0,0.78...	double
qr	[0.4144,0.8206,3.1193,-...	double
qs	[0,0,-1.5708,0,0,0]	double
qz	[0,0,0,0,0,0]	double
tg	<50x6 double>	double

Figures

Figure 1



Command Window

```

3.0575    2.3210    0.1162   -0.0000    0.7044    3.0575
>> qr = p560.ikine6s(T, 'r')
qr =
    0.4144    0.8206    3.1193   -3.1416    0.7983   -2.7272
>> p560.plot(q1)
>> p560.plot(qr)
>> tg = jtraj(qz, q1, 50);
>> p560.plot(tg)
>> tg = jtraj(qz, qr, 50);
>> p560.plot(tg)

```

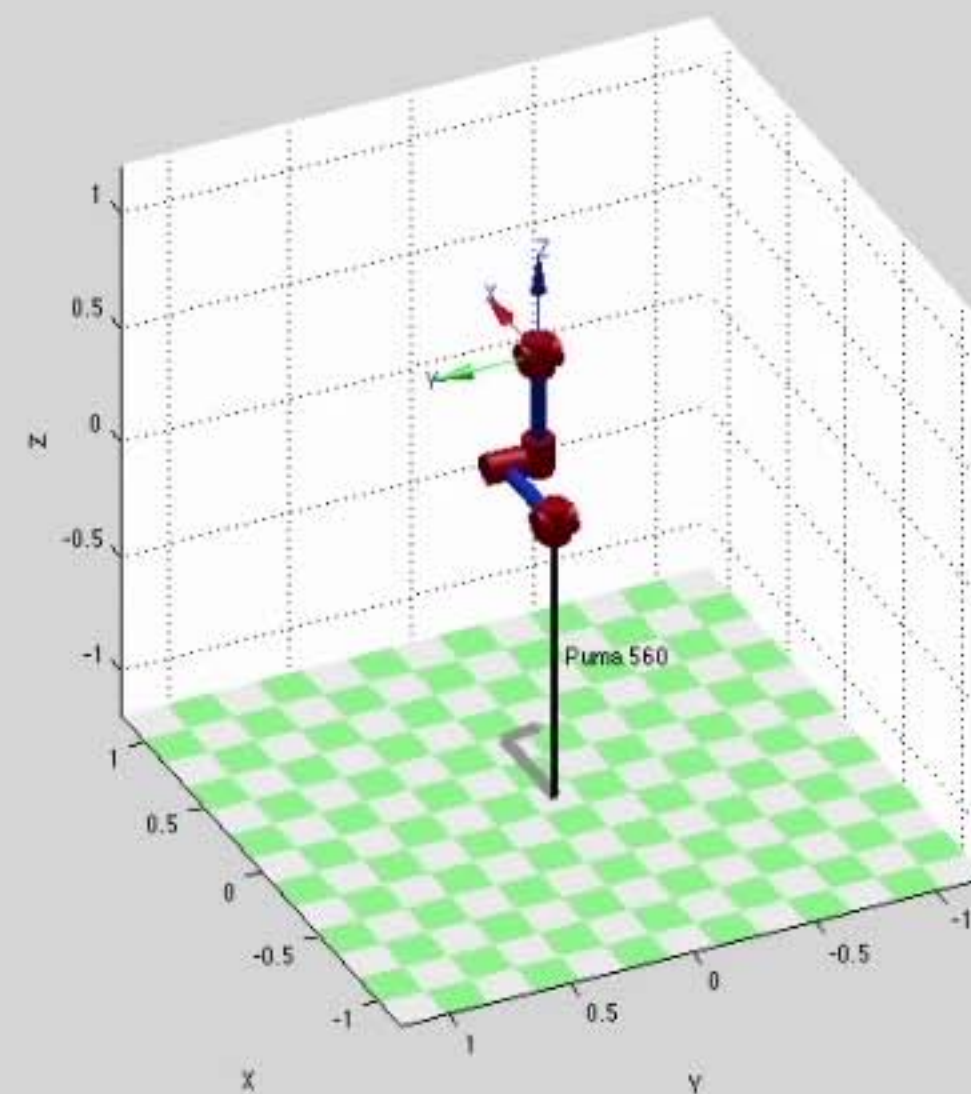
fx

Workspace

Name	Value	Class
T	<4x4 double>	double
deg	0.0175	double
p560	<1x1 SerialLink>	SerialLink
q1	[3.0575,2.3210,0.1162,-...	double
qn	[0,0.7854,3.1416,0,0.78...	double
qr	[0.4144,0.8206,3.1193,-...	double
qs	[0,0,-1.5708,0,0,0]	double
qz	[0,0,0,0,0,0]	double
tg	<50x6 double>	double

Figures

Figure 1



Command Window

```

3.0575    2.3210    0.1162   -0.0000    0.7044    3.0575

>> qr = p560.ikine6s(T, 'r')

qr =

    0.4144    0.8206    3.1193   -3.1416    0.7983   -2.7272

>> p560.plot(q1)
>> p560.plot(qr)
>> tg = jtraj(qz, q1, 50);
>> p560.plot(tg)
>> tg = jtraj(qz, qr, 50);
>> p560.plot(tg)

```

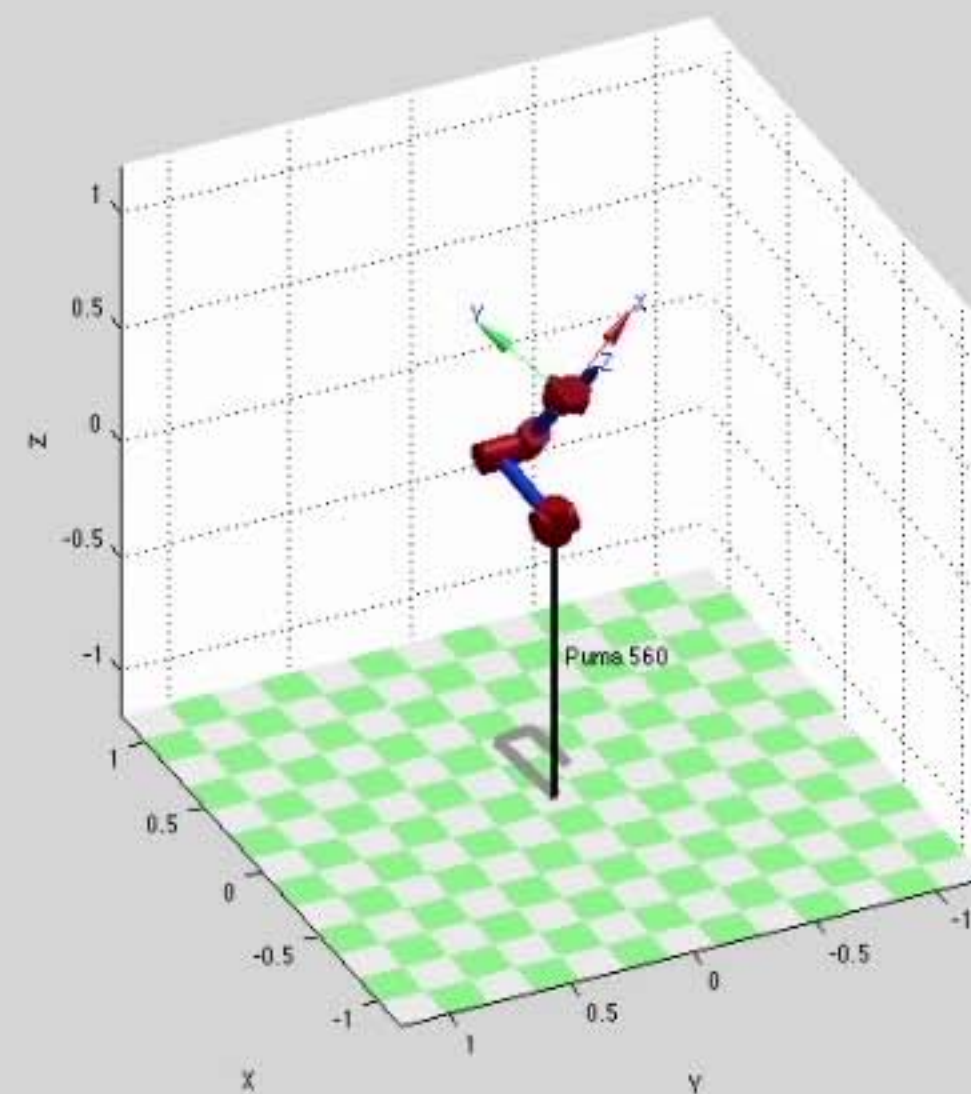
fx

Workspace

Name	Value	Class
T	<4x4 double>	double
deg	0.0175	double
p560	<1x1 SerialLink>	SerialLink
q1	[3.0575,2.3210,0.1162,-...	double
qn	[0,0.7854,3.1416,0,0.78...	double
qr	[0.4144,0.8206,3.1193,-...	double
qs	[0,0,-1.5708,0,0,0]	double
qz	[0,0,0,0,0,0]	double
tg	<50x6 double>	double

Figures

Figure 1



Command Window

```

3.0575    2.3210    0.1162   -0.0000    0.7044    3.0575

>> qr = p560.ikine6s(T, 'r')

qr =

    0.4144    0.8206    3.1193   -3.1416    0.7983   -2.7272

>> p560.plot(q1)
>> p560.plot(qr)
>> tg = jtraj(qz, q1, 50);
>> p560.plot(tg)
>> tg = jtraj(qz, qr, 50);
>> p560.plot(tg)

```

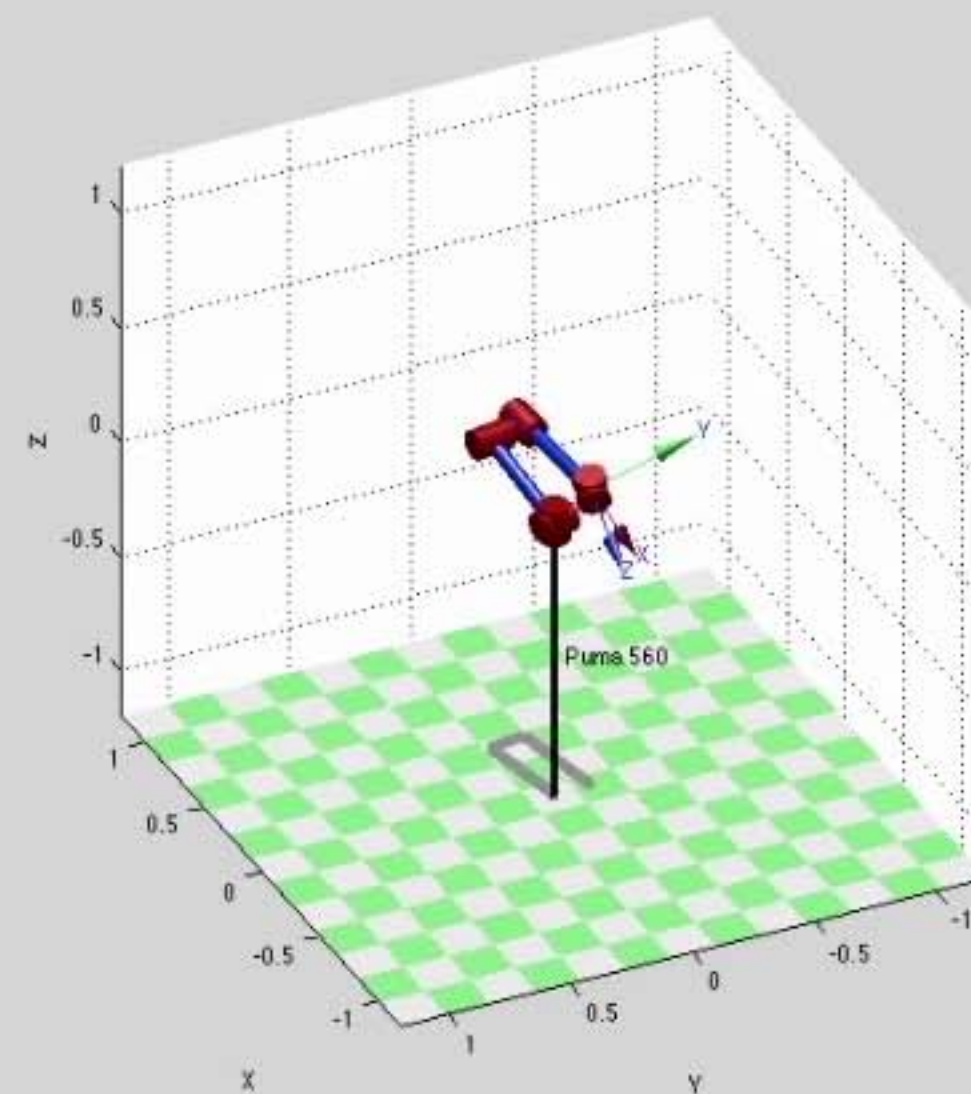
fx

Workspace

Name	Value	Class
T	<4x4 double>	double
deg	0.0175	double
p560	<1x1 SerialLink>	SerialLink
q1	[3.0575,2.3210,0.1162,-...	double
qn	[0,0.7854,3.1416,0,0.78...	double
qr	[0.4144,0.8206,3.1193,-...	double
qs	[0,0,-1.5708,0,0,0]	double
qz	[0,0,0,0,0,0]	double
tg	<50x6 double>	double

Figures

Figure 1



Command Window

```

3.0575    2.3210    0.1162   -0.0000    0.7044    3.0575

>> qr = p560.ikine6s(T, 'r')

qr =

    0.4144    0.8206    3.1193   -3.1416    0.7983   -2.7272

>> p560.plot(q1)
>> p560.plot(qr)
>> tg = jtraj(qz, q1, 50);
>> p560.plot(tg)
>> tg = jtraj(qz, qr, 50);
>> p560.plot(tg)

```

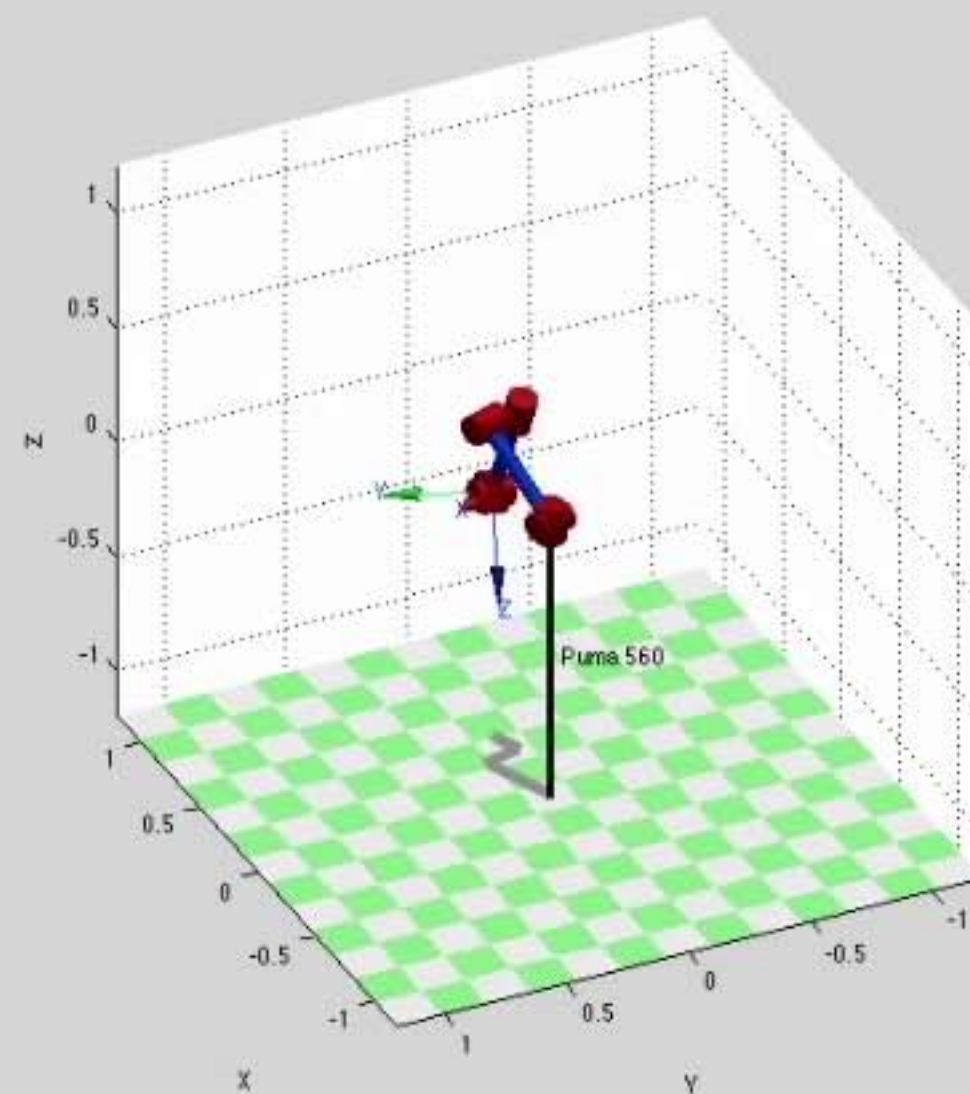
fx

Workspace

Name	Value	Class
T	<4x4 double>	double
deg	0.0175	double
p560	<1x1 SerialLink>	SerialLink
q1	[3.0575,2.3210,0.1162,-...	double
qn	[0,0.7854,3.1416,0,0.78...	double
qr	[0.4144,0.8206,3.1193,-...	double
qs	[0,0,-1.5708,0,0,0]	double
qz	[0,0,0,0,0,0]	double
tg	<50x6 double>	double

Figures

Figure 1



Command Window

```

3.0575    2.3210    0.1162   -0.0000    0.7044    3.0575

>> qr = p560.ikine6s(T, 'r')

qr =

    0.4144    0.8206    3.1193   -3.1416    0.7983   -2.7272

>> p560.plot(q1)
>> p560.plot(qr)
>> tg = jtraj(qz, q1, 50);
>> p560.plot(tg)
>> tg = jtraj(qz, qr, 50);
>> p560.plot(tg)
fx>> tg = jtraj(q1, qr, 50);

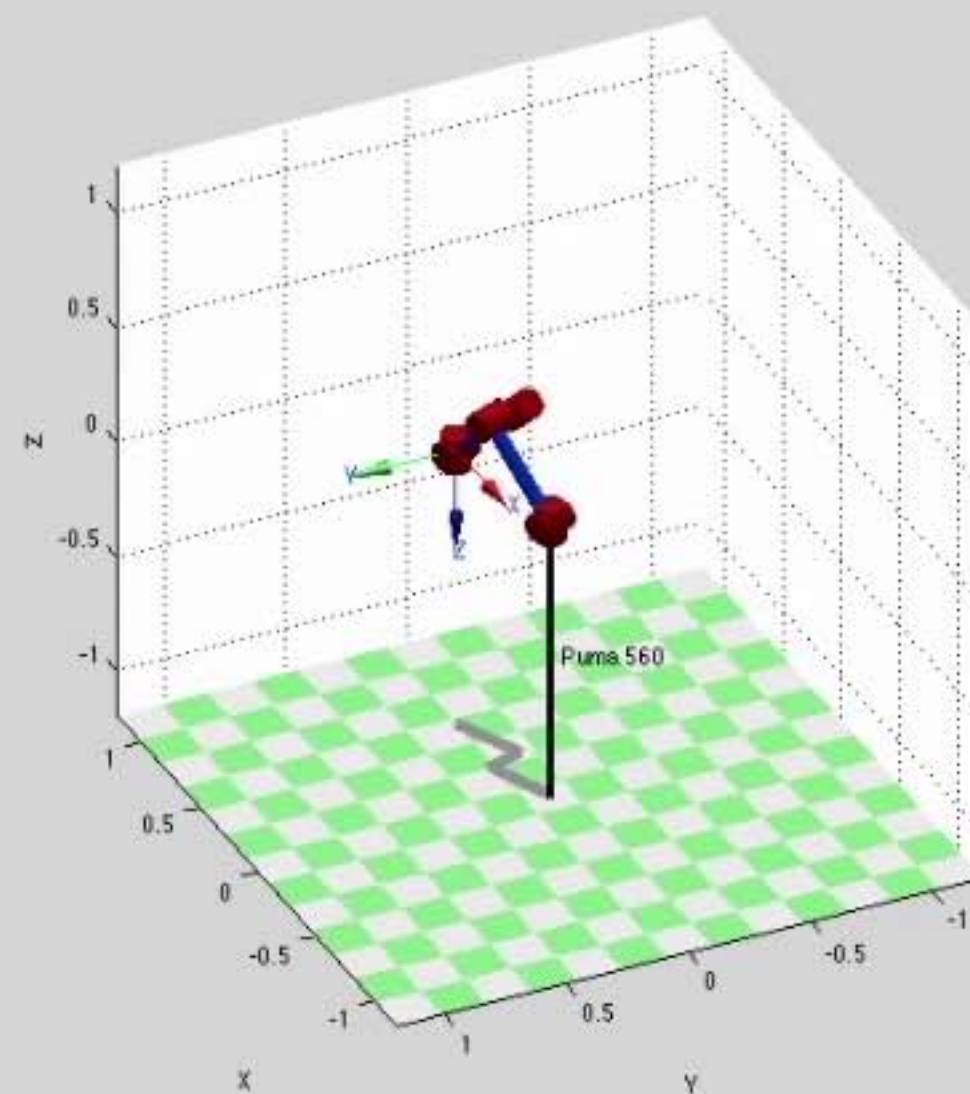
```

Workspace

Name	Value	Class
T	<4x4 double>	double
deg	0.0175	double
p560	<1x1 SerialLink>	SerialLink
q1	[3.0575,2.3210,0.1162,-...	double
qn	[0,0.7854,3.1416,0,0.78...	double
qr	[0.4144,0.8206,3.1193,-...	double
qs	[0,0,-1.5708,0,0,0]	double
qz	[0,0,0,0,0,0]	double
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Figures

Figure 1



Command Window

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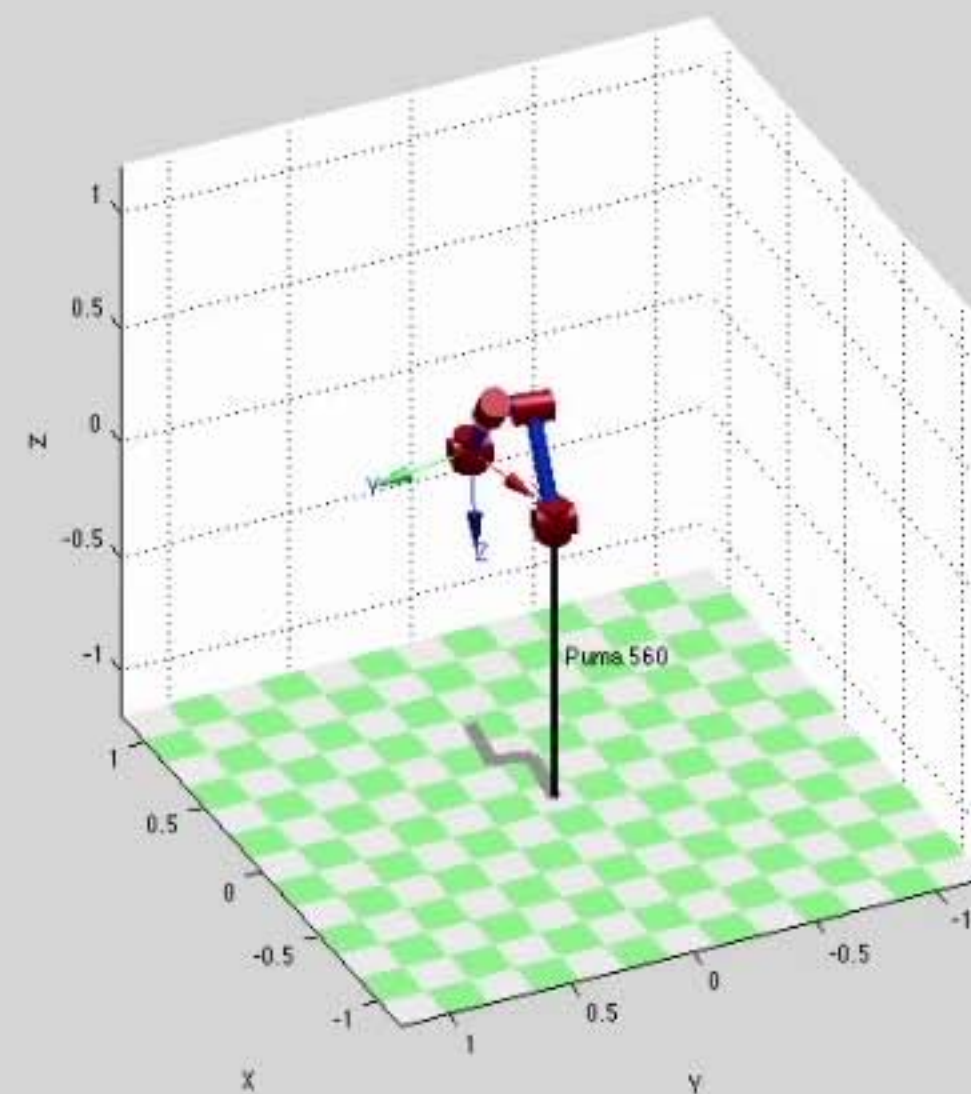
fx

Workspace

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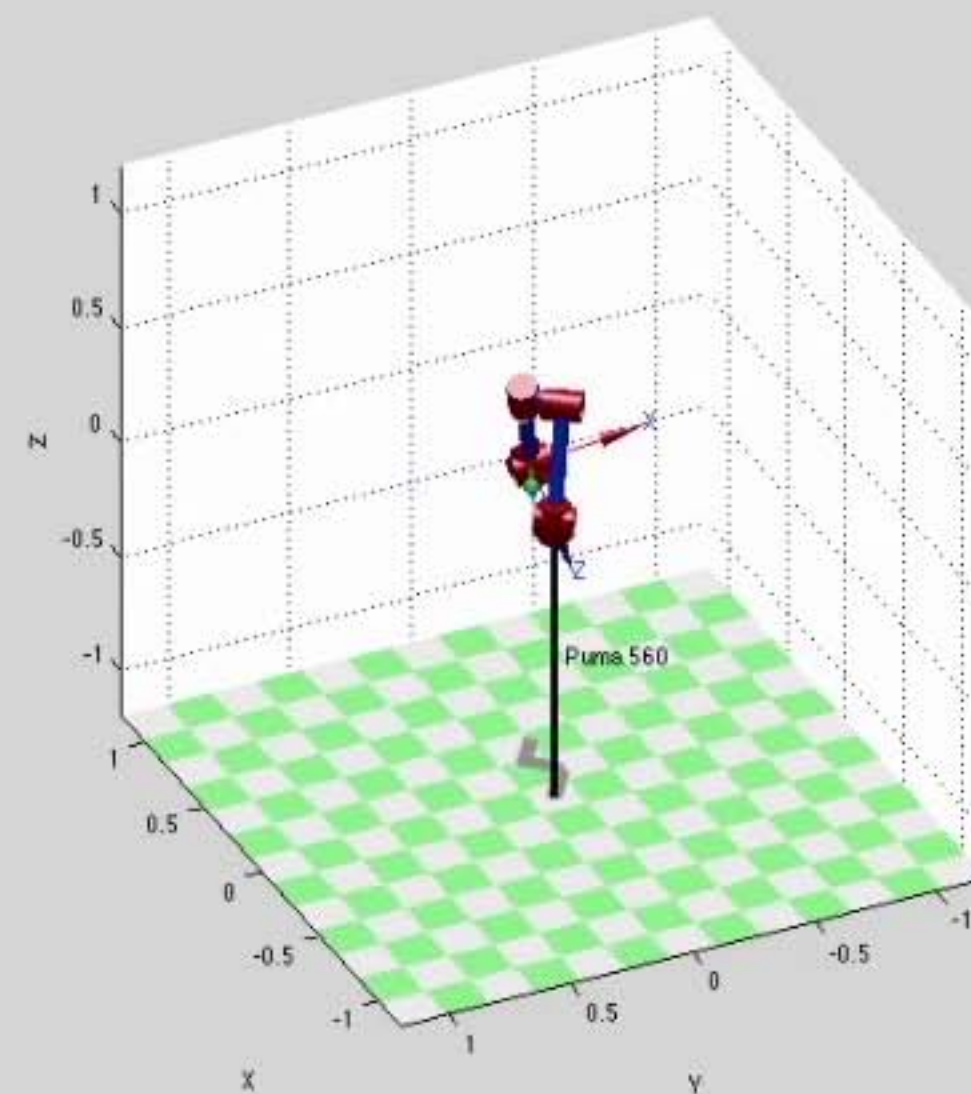
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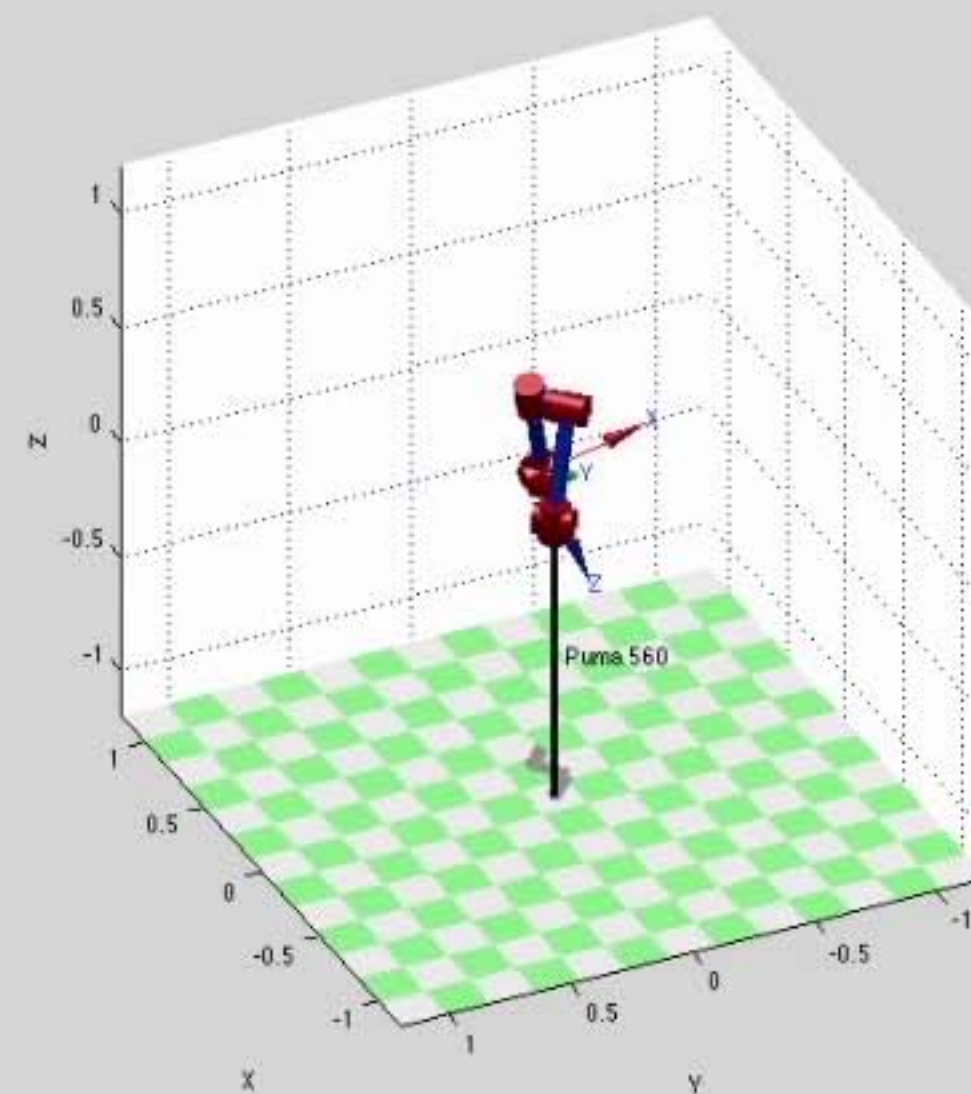
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Workspace

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Figures

Figure 1



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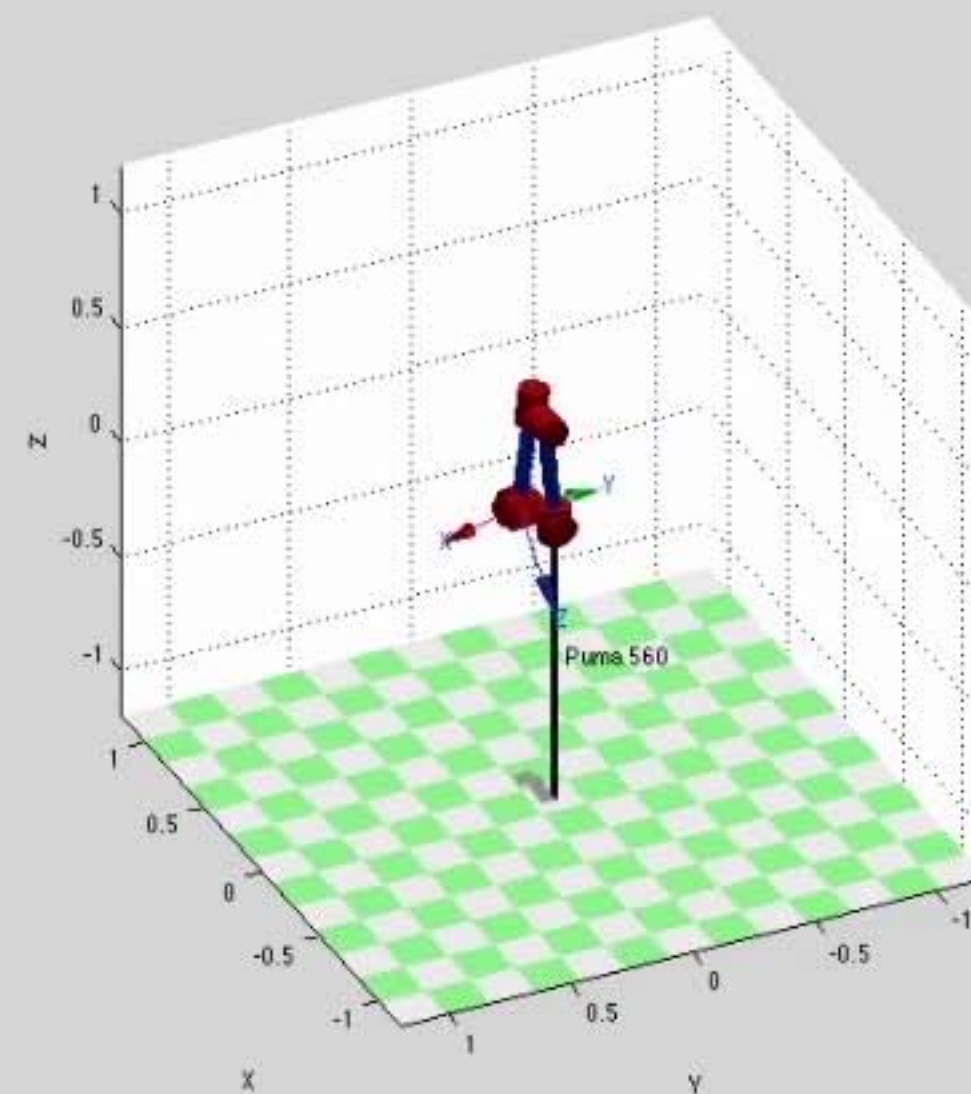
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Workspace

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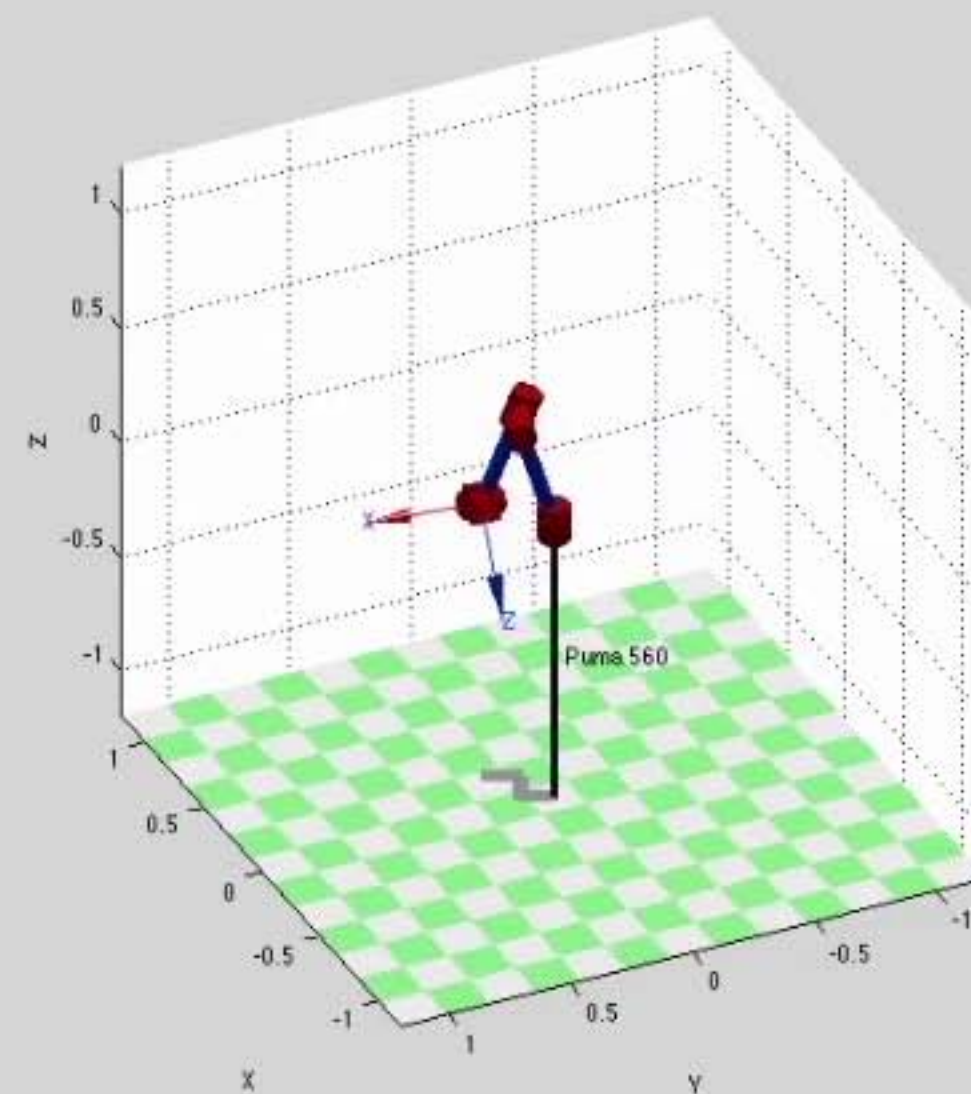
fx

Workspace

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Figures

Figure 1



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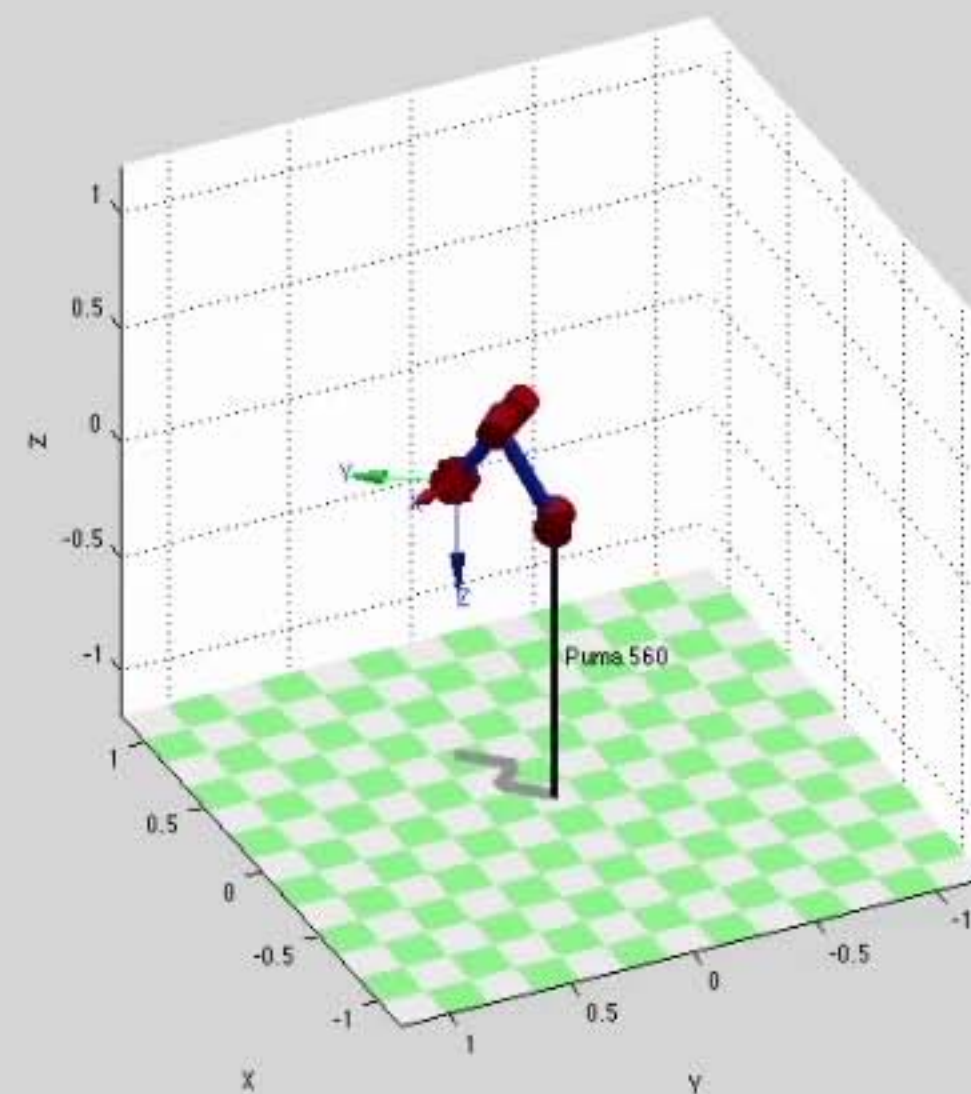
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Name	Value	Class
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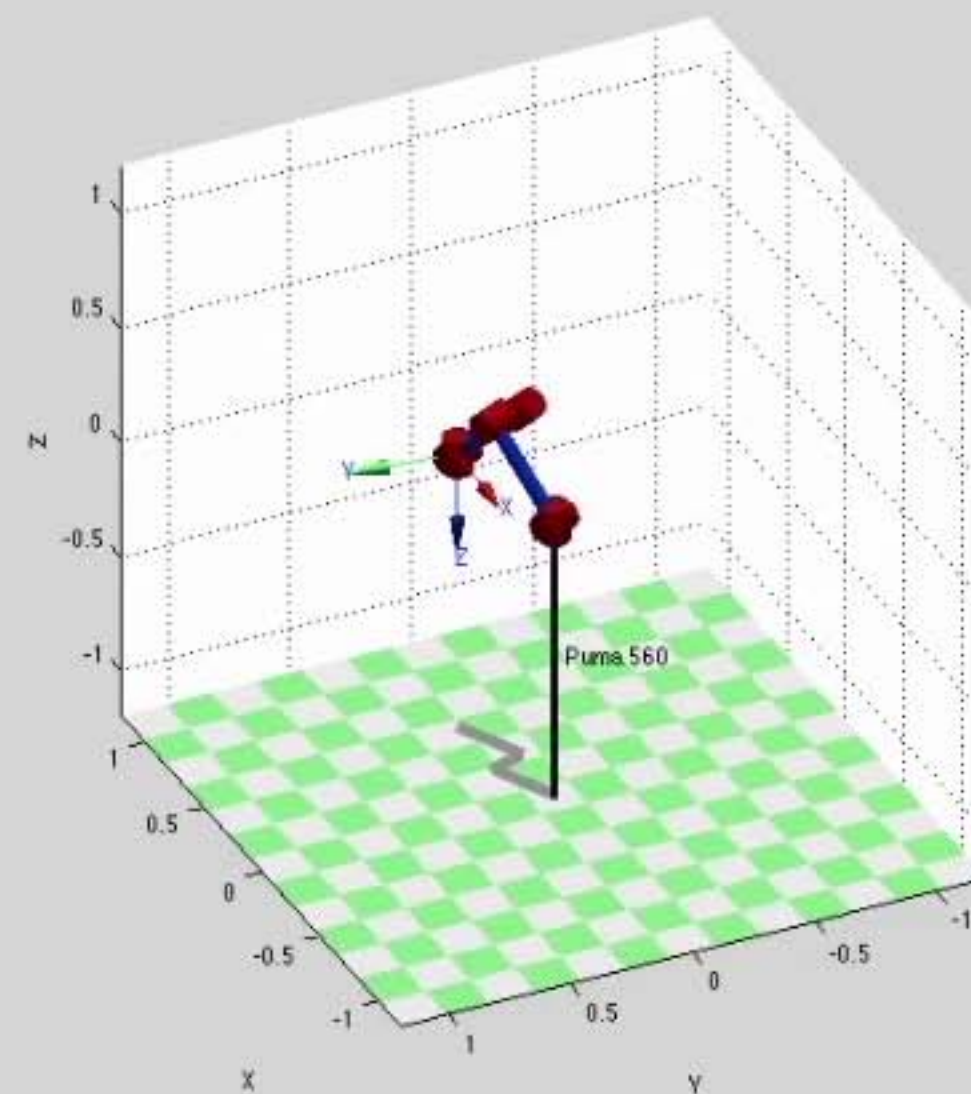
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Workspace

Name	Value	Class
T	<4x4 double>	double
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Figures

Figure 1



Section 6 Approaches to inverse kinematics

Analytic solution

- create a model of the robots joints and links

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- write the forward kinematics

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- write down lots of equations and attempt to solve for the unknown joint variables
 - ➔ needs some human insight
 - ➔ can be automated

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Analytic solution

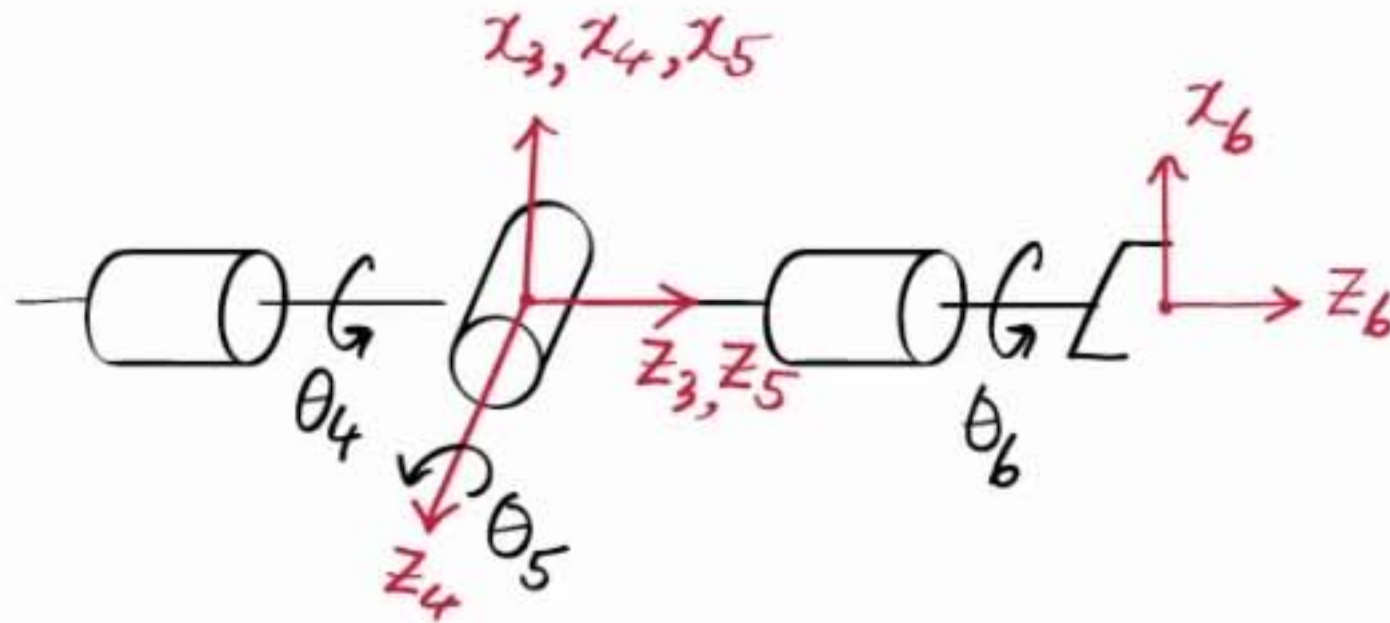
- create a model of the robots joints and links
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- explicitly shows the multiple configurations
- solution are compact and fast to execute

Analytic solution

- create a model of the robots joints and links
- write the forward kinematics
- write down lots of equations and attempt to solve for the unknown joint variables
 - ➡ needs some human insight
 - ➡ can be automated
- explicitly shows the multiple configurations
- solution are compact and fast to execute
- increasingly difficult as the number of joints increases
 - ➡ however many solutions have been derived and published

Spherical wrist

- A critical factor for 6-DOF robots is having a spherical wrist
- All 3 wrist joint axes intersect at a single point



Numerical solution

- What if the analytic solution
 - ➔ does not exist ?
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- Just have to adjust $\mathbf{q}$ until $\xi \rightarrow \xi^*$

Numerical solution

- What if the analytic solution
 - ➔ does not exist ?
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- We can **always** figure out the forward kinematics $\xi = \mathcal{K}(\mathbf{q})$
- We know the pose we want ξ^*
- Just have to adjust $\mathbf{q}$ until $\xi \rightarrow \xi^*$
- This can be posed as an **optimization problem**

$$\mathbf{q}^* = \arg \min_{\mathbf{q}} |\mathcal{K}(\mathbf{q}) \ominus \xi^*|$$

Serial and parallel **link robots**



Used with permission of ABB



Used with permission of ABB

Forward kinematics	Easy, analytic form exists	Complex, analytic form may not exist
Inverse kinematics	Complex, analytic form may not exist	Easy, analytic form exists

Section 7 Numerical inverse kinematics

Numerical Inverse Kinematics in MATLAB

Numerical IKs take 6 steps:

- Initialization
- Forward Kinematics
- Error Calculation
- Jacobian Matrix
- Update the Joint Angles
- Iterate

1. Initialization

Start with an initial guess of joint angles.

- Typically the current robot configuration q .

Serves as the **seed** for iterative refinement.

2. Forward Kinematics

Compute the desired and current end-effector positions and orientation from the current joint angles.

$$\boldsymbol{\varepsilon}_{curr} = K(\boldsymbol{q}_{curr})$$

$$\boldsymbol{\varepsilon}_{des} = K(\boldsymbol{q}_{des})$$

3. Error Calculation

Compute the error between the desired and current end-effector pose.

$$e = \varepsilon_{des} - \varepsilon_{curr}$$

The error will be used to select the direction in which to update the joint updates.

4. Jacobian Computation

Calculate the Jacobian matrix: relates joint velocities to end-effector velocities.

Used to determine how joint changes affect end-effector position.

$$\begin{bmatrix} \Delta x \\ \Delta y \\ \Delta z \end{bmatrix} = \begin{bmatrix} \frac{\partial x}{\partial \theta_1} & \frac{\partial x}{\partial \theta_2} & \frac{\partial x}{\partial \theta_3} & \frac{\partial x}{\partial \theta_4} & \frac{\partial x}{\partial \theta_5} & \frac{\partial x}{\partial \theta_6} \\ \frac{\partial y}{\partial \theta_1} & \frac{\partial y}{\partial \theta_2} & \frac{\partial y}{\partial \theta_3} & \frac{\partial y}{\partial \theta_4} & \frac{\partial y}{\partial \theta_5} & \frac{\partial y}{\partial \theta_6} \\ \frac{\partial z}{\partial \theta_1} & \frac{\partial z}{\partial \theta_2} & \frac{\partial z}{\partial \theta_3} & \frac{\partial z}{\partial \theta_4} & \frac{\partial z}{\partial \theta_5} & \frac{\partial z}{\partial \theta_6} \end{bmatrix} \begin{bmatrix} \Delta \theta_1 \\ \Delta \theta_2 \\ \Delta \theta_3 \\ \Delta \theta_4 \\ \Delta \theta_5 \\ \Delta \theta_6 \end{bmatrix}$$

$$\Delta x = J \Delta \theta$$

5. Joint Angle Update

Update joint angles using optimization (e.g., Jacobian pseudoinverse or damped least squares).

Iterate until the error is below a threshold.

$$\delta\theta = J^\dagger e$$

where,

$$e = x_{\text{des}} - x_{\text{curr}}$$

$$J^\dagger = \text{inv}(J) \text{ <<pseudo-inverse>>}$$

6. Pseudocode

```
function jointAngles = numericalIK(targetPosition, initialGuess)
    jointAngles = initialGuess
    while not converged:
        currentPos = forwardKinematics(jointAngles)
        error = targetPosition - currentPos
        J = computeJacobian(jointAngles)
        deltaTheta = pinv(J) * error
        jointAngles = jointAngles + deltaTheta
    return jointAngles
```

Command Window

>> mdl_planar2
fx >> |

Workspace

Name	Value	Class
a1	1	double
a2	1	double
p2	<1x1 SerialLink>	SerialLink
qz	[0,0]	double

Command Window

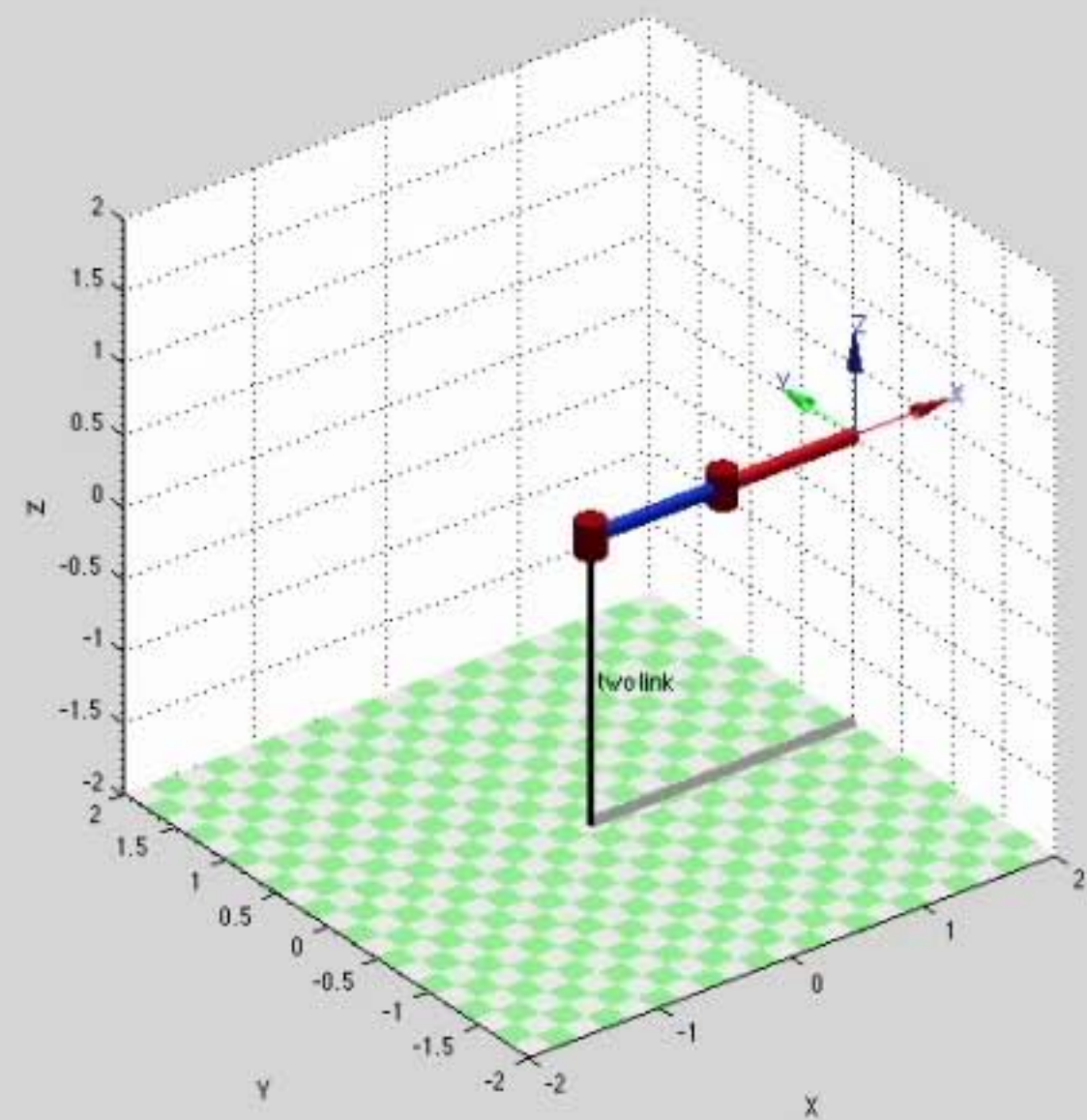
```
>> mdl_planar2
>> p2.plot(qz)
fx >>
```

Workspace

Name	Value	Class
a1	1	double
a2	1	double
p2	<1x1 SerialLink>	SerialLink
qz	[0,0]	double

Figures

Figure 1



Command Window

```
>> mdl_planar2
>> p2.plot(qz)
>> T = transl(1, 1, 0)
```

T =

```
1 0 0 1
0 1 0 1
0 0 1 0
0 0 0 1
```

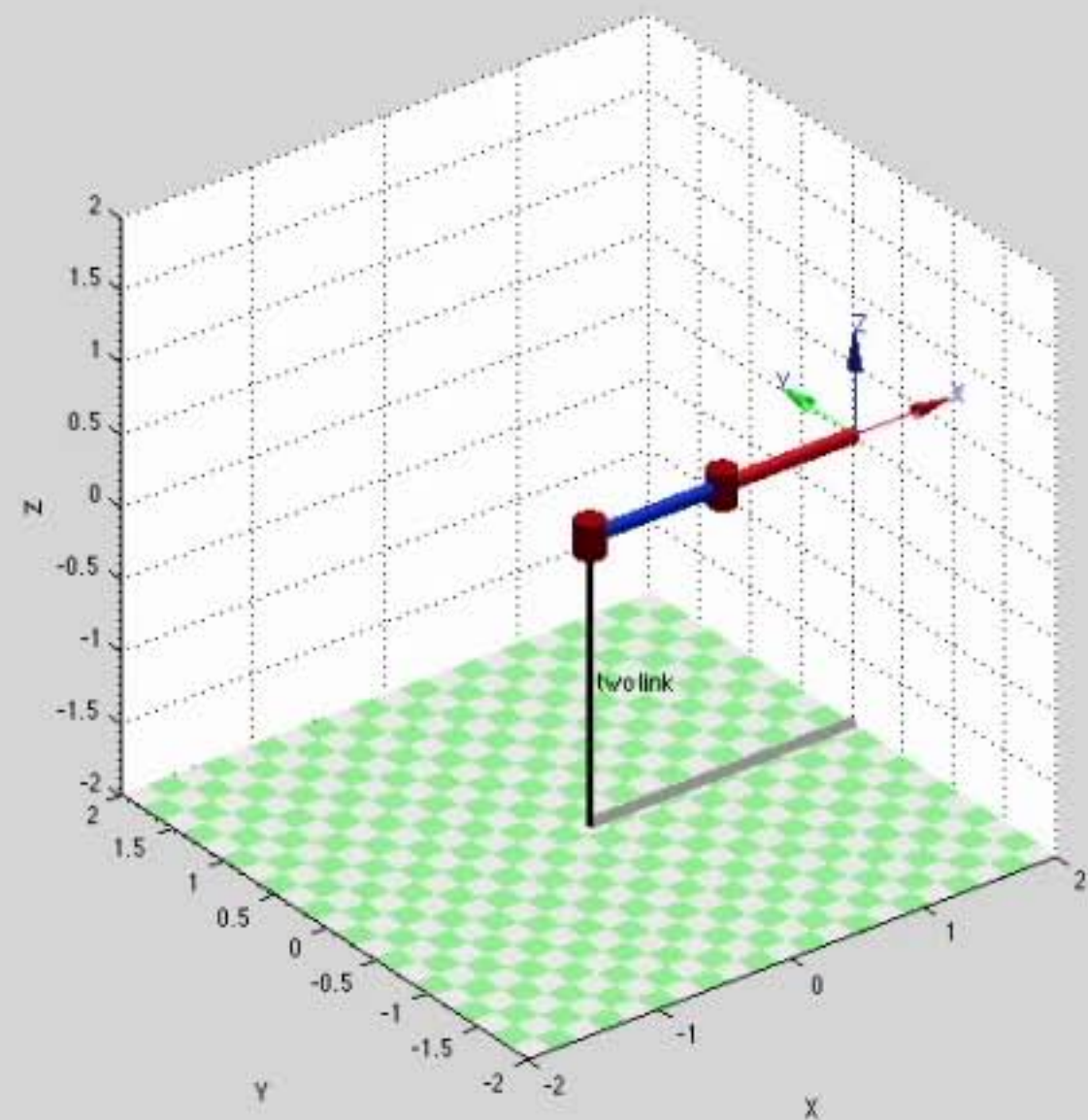
fx >>

Workspace

Name	Value	Class
T	<4x4 double>	double
a1	1	double
a2	1	double
p2	<1x1 SerialLink>	SerialLink
qz	[0,0]	double

Figures

Figure 1



Command Window

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```

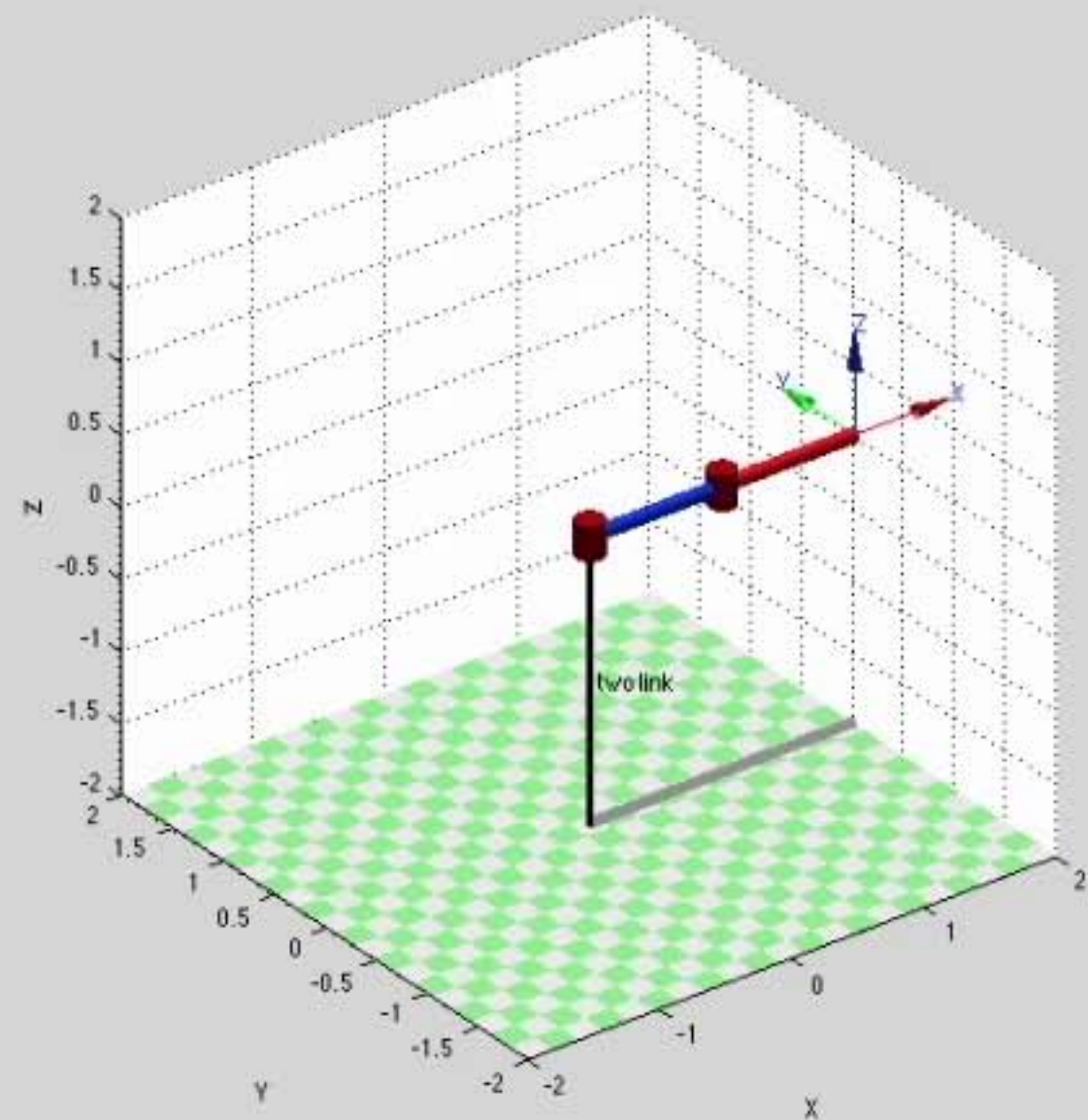
```
fx >> q = p2.ikine(T, [0 0], [1 1 0 0 0 0])
```

Workspace

Name	Value	Class
T	<4x4 double>	double
a1	1	double
a2	1	double
p2	<1x1 SerialLink>	SerialLink
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Figures

Figure 1



Command Window

```
1 0 0 1
0 1 0 1
0 0 1 0
0 0 0 1
```

```
>> q = p2.ikine(T, [0 0], [1 1 0 0 0 0])
```

Warning: Initial joint angles results in near-singular configuration, this may slow convergence

> In SerialLink.ikine at 152

Warning: solution diverging at step 3, try reducing alpha

> In SerialLink.ikine at 257

```
q =
```

```
1.5708 -1.5708
```

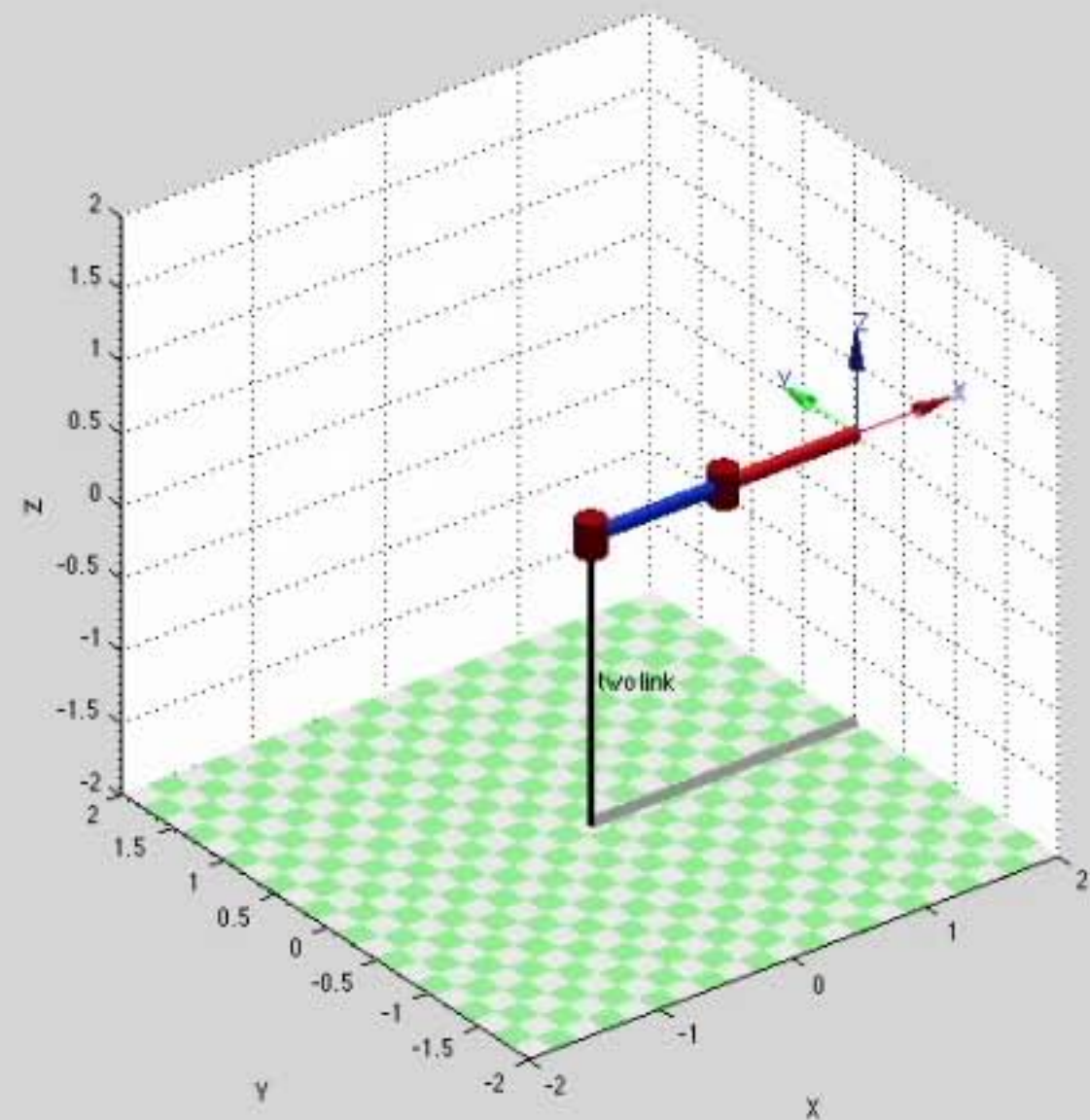
```
f_x >> p
```

Workspace

Name	Value	Class
T	<4x4 double>	double
a1	1	double
a2	1	double
p2	<1x1 SerialLink>	SerialLink
q	[1.5708,-1.5708]	double
qz	[0,0]	double

Figures

Figure 1



Command Window

```

0 1 0 1
0 0 1 0
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```

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> In SerialLink.ikine at 257

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q =
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1.5708 -1.5708
```

```
>> p2.plot(q)
```

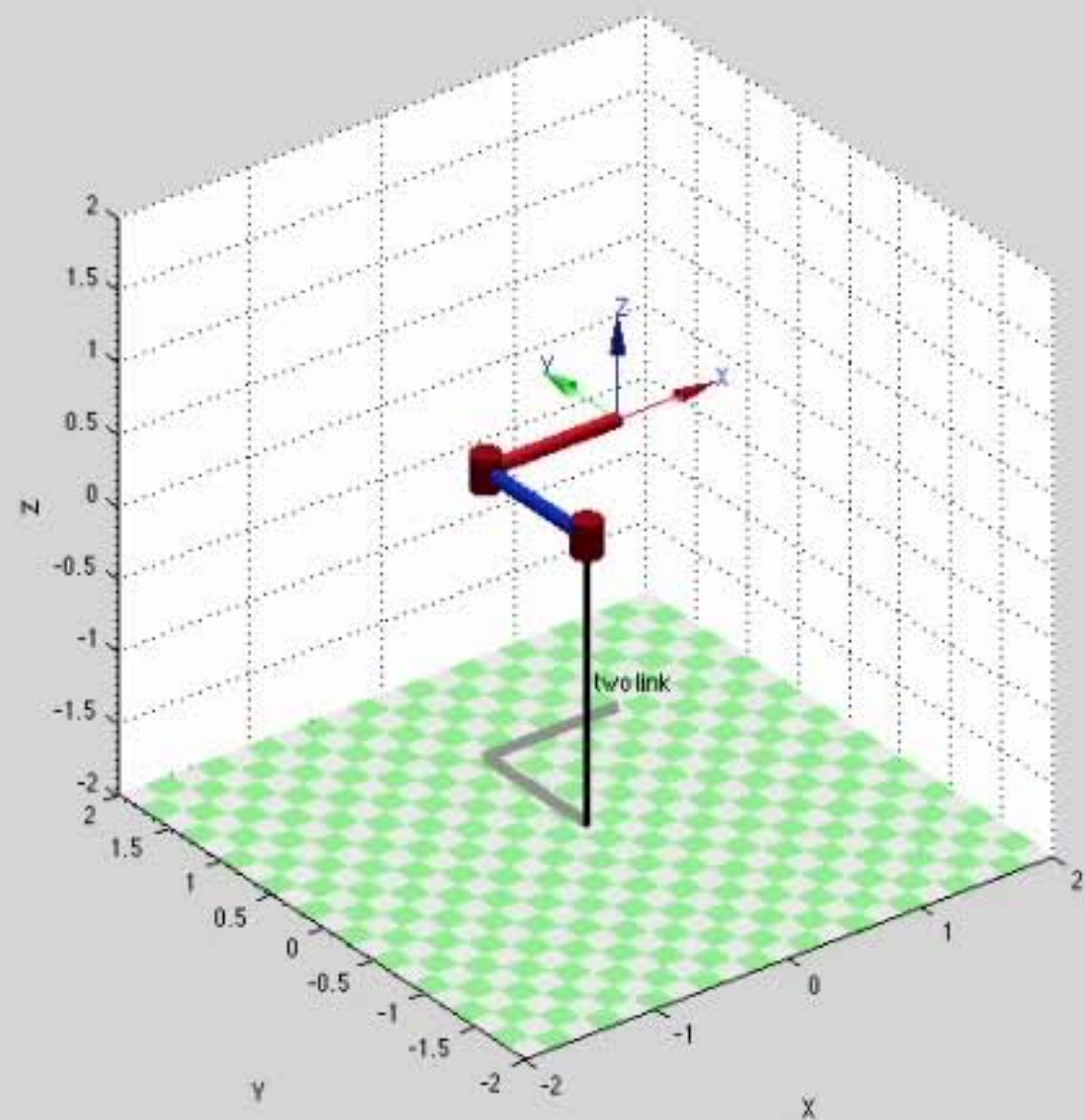
```
fx >>
```

Workspace

Name	Value	Class
T	<4x4 double>	double
a1	1	double
a2	1	double
p2	<1x1 SerialLink>	SerialLink
q	[1.5708,-1.5708]	double
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Figures

Figure 1



Command Window

```
0 1 0 1
0 0 1 0
0 0 0 1
```

```
>> q = p2.ikine(T, [0 0], [1 1 0 0 0 0])
```

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> In SerialLink.ikine at 257

```
q =
```

```
1.5708 -1.5708
```

```
>> p2.plot(q)
```

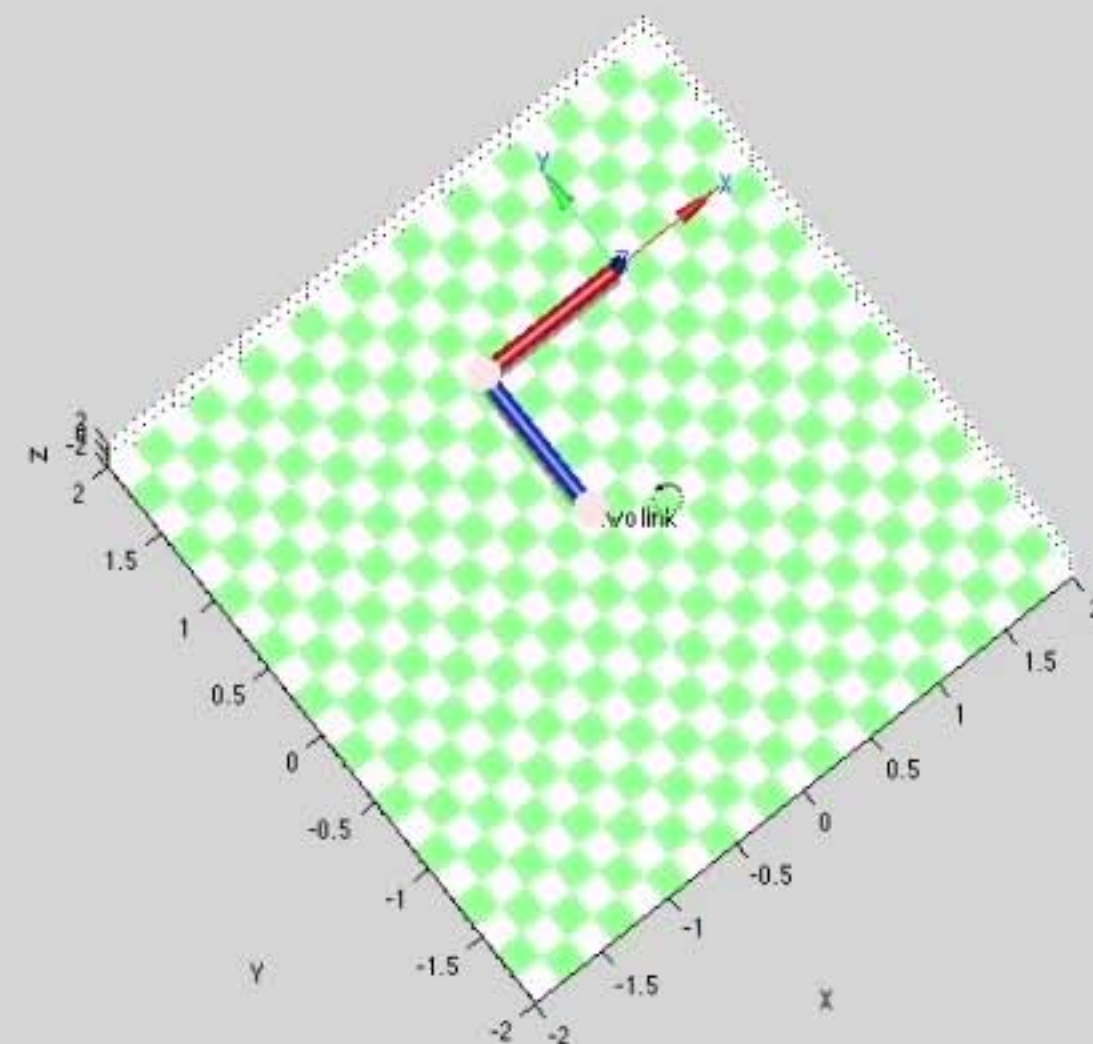
```
fx >>
```

Workspace

Name	Value	Class
T	<4x4 double>	double
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Figures

Figure 1



Command Window

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> In SerialLink.ikine at 257

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1.5708 -1.5708
```

```
>> p2.plot(q)
```

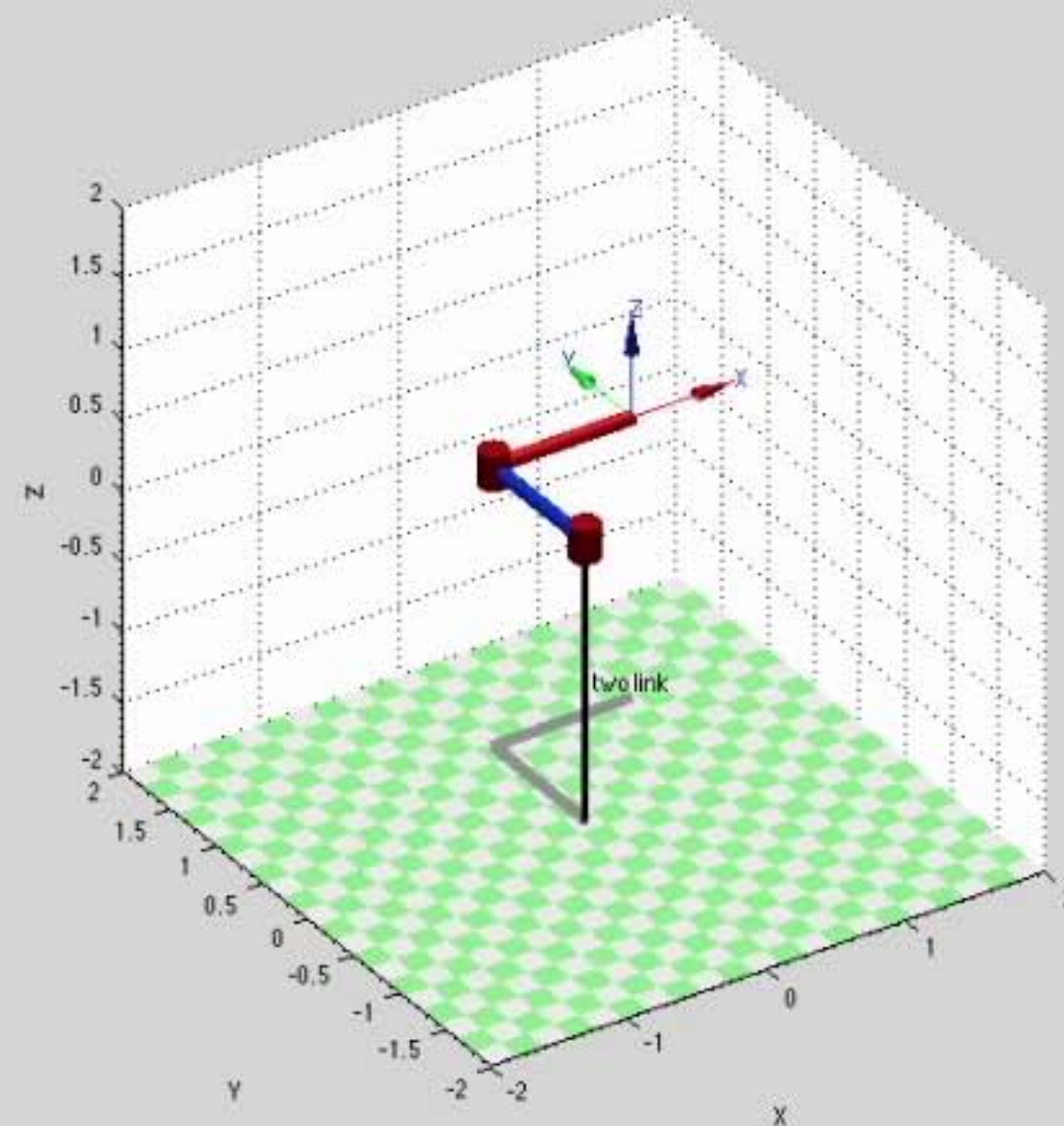
```
fx>> q = p2.ikine(T, [-1 0], [1 1 0 0 0 0])
```

Workspace

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T	<4x4 double>	double
a1	1	double
a2	1	double
p2	<1x1 SerialLink>	SerialLink
q	[1.5708,-1.5708]	double
qz	[0,0]	double

Figures

Figure 1



Command Window

> In SerialLink.ikine at 237

q =

1.5708 -1.5708

>> p2.plot(q)

>> q = p2.ikine(T, [-1 0], [1 1 0 0 0 0])

Warning: Initial joint angles results in near-singular configuration, this may slow convergence

> In SerialLink.ikine at 152

q =

1.5708 -1.5708

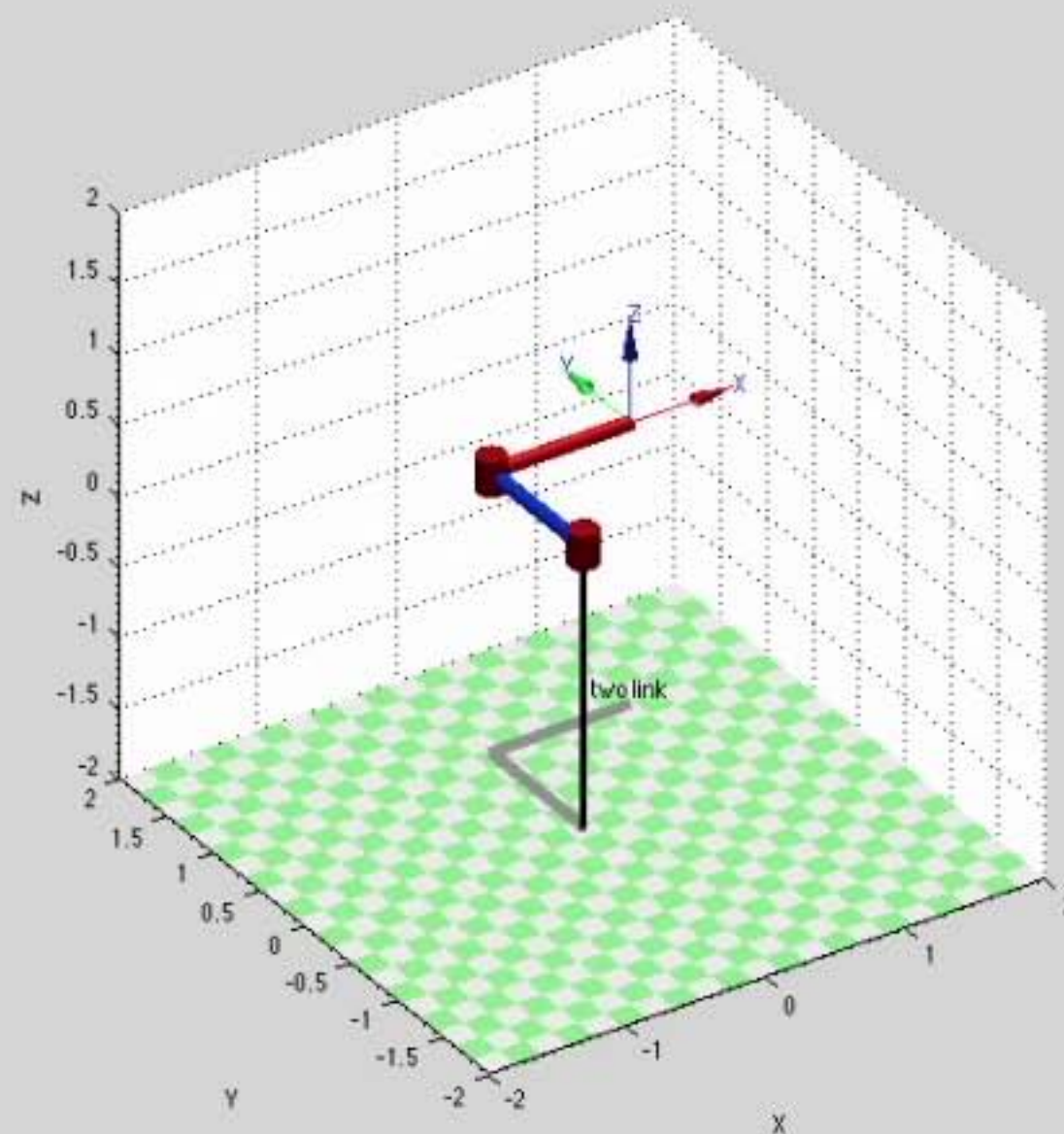
fx >>

Workspace

Name	Value	Class
T	<4x4 double>	double
a1	1	double
a2	1	double
p2	<1x1 SerialLink>	SerialLink
q	[1.5708,-1.5708]	double
qz	[0,0]	double

Figures

Figure 1



Command Window

Warning: initial joint angles results in near-singular configuration, this may slow convergence

> In SerialLink.ikine at 152

q =

1.5708 -1.5708

>> q = p2.ikine(T, [-1 -1], [1 1 0 0 0 0])

Warning: solution diverging at step 4, try reducing alpha

> In SerialLink.ikine at 257

q =

-0.0000 1.5708

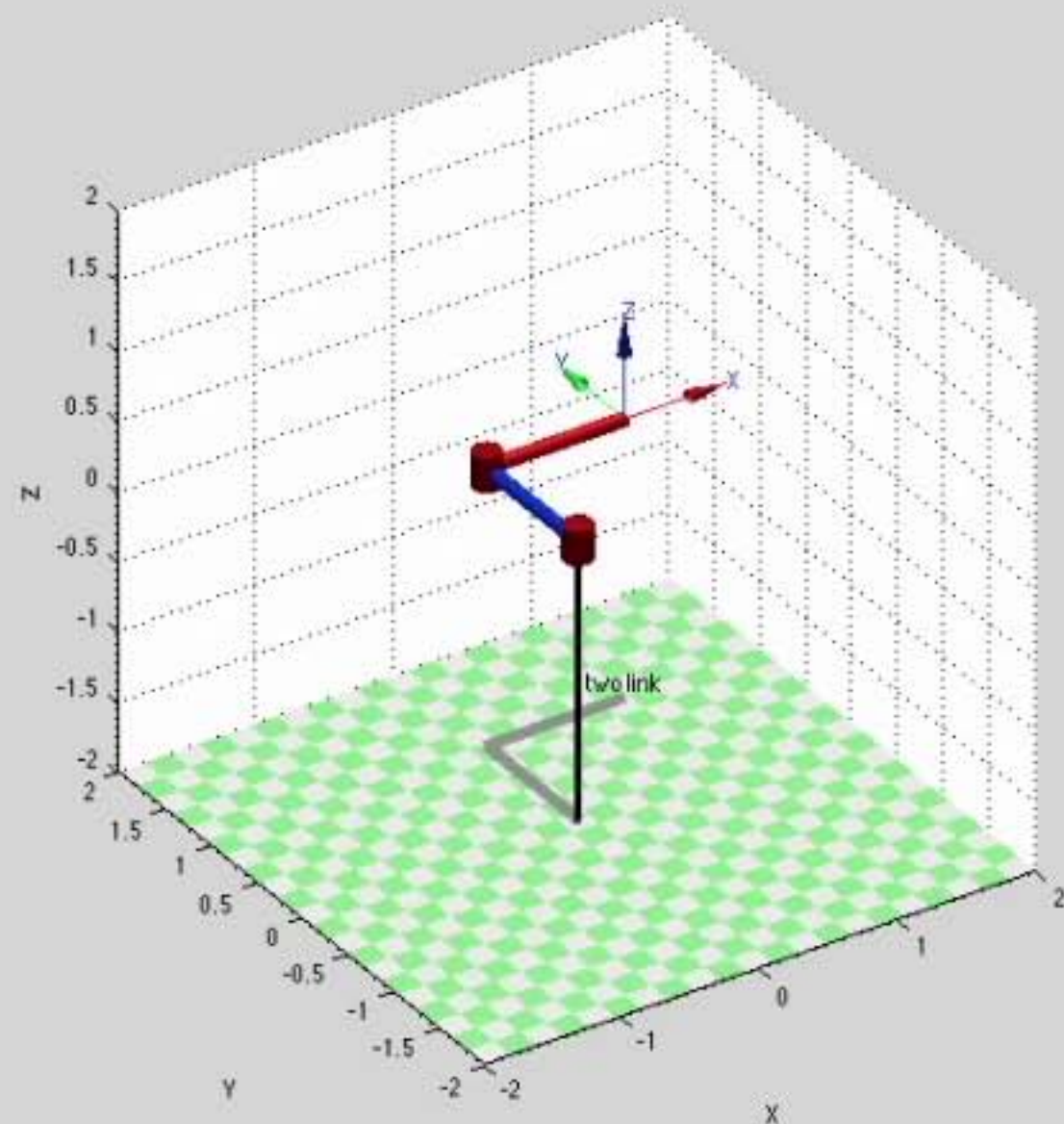
fx >> |

Workspace

Name	Value	Class
T	<4x4 double>	double
a1	1	double
a2	1	double
p2	<1x1 SerialLink>	SerialLink
q	[-1.2218e-17,1.5708]	double
qz	[0,0]	double

Figures

Figure 1



Command Window

```
> In SerialLink.ikine at 152

q =

    1.5708   -1.5708

>> q = p2.ikine(T, [-1 -1], [1 1 0 0 0 0])
Warning: solution diverging at step 4, try reducing alpha
> In SerialLink.ikine at 257

q =

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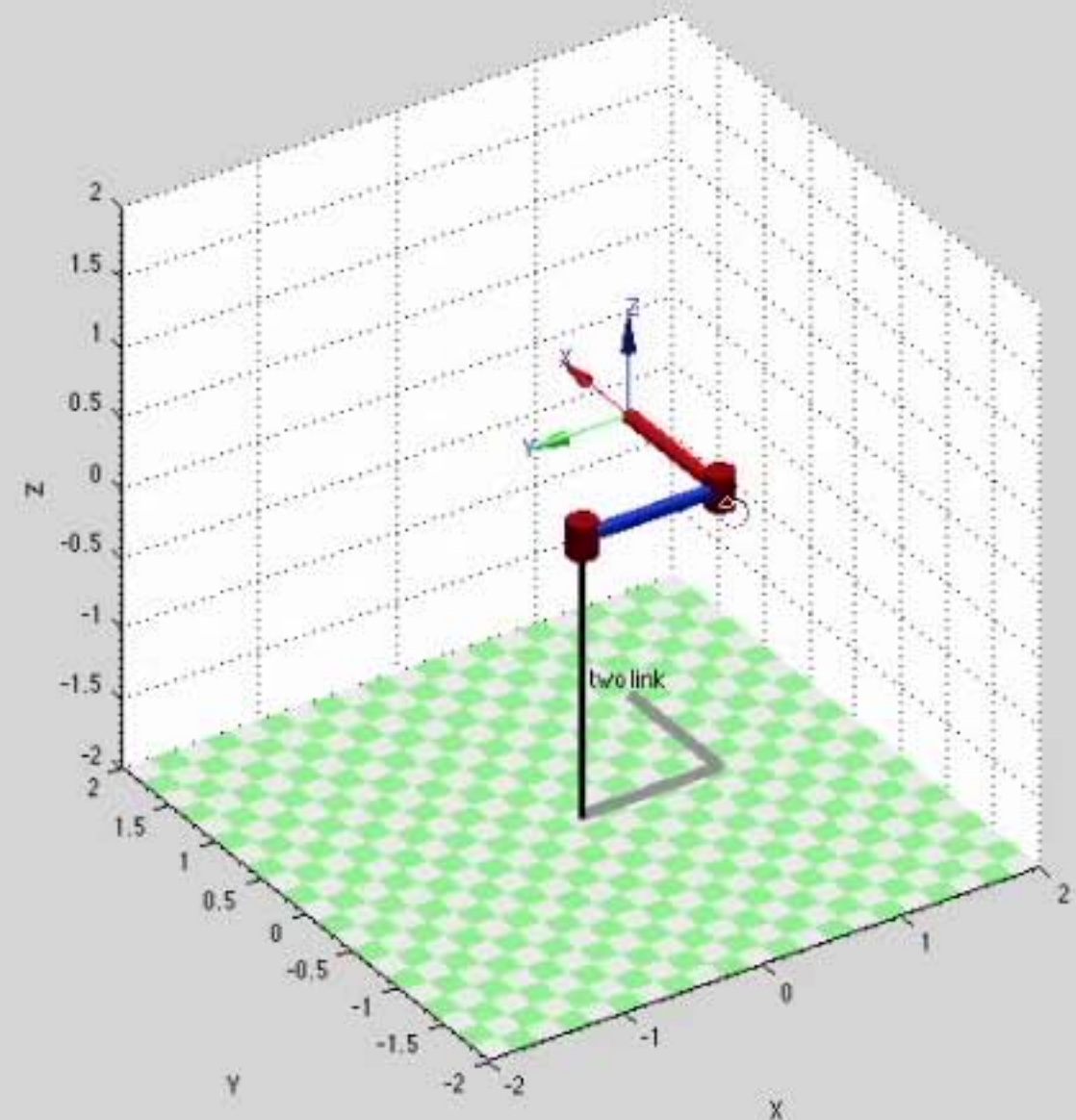
>> p2.plot(q)
fx >>
```

Workspace

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T	<4x4 double>	double
a1	1	double
a2	1	double
p2	<1x1 SerialLink>	SerialLink
q	[-1.2218e-17,1.5708]	double
qz	[0,0]	double

Figures

Figure 1





Unimate Puma 500
Oussama Khatib | Used with permission

Command Window

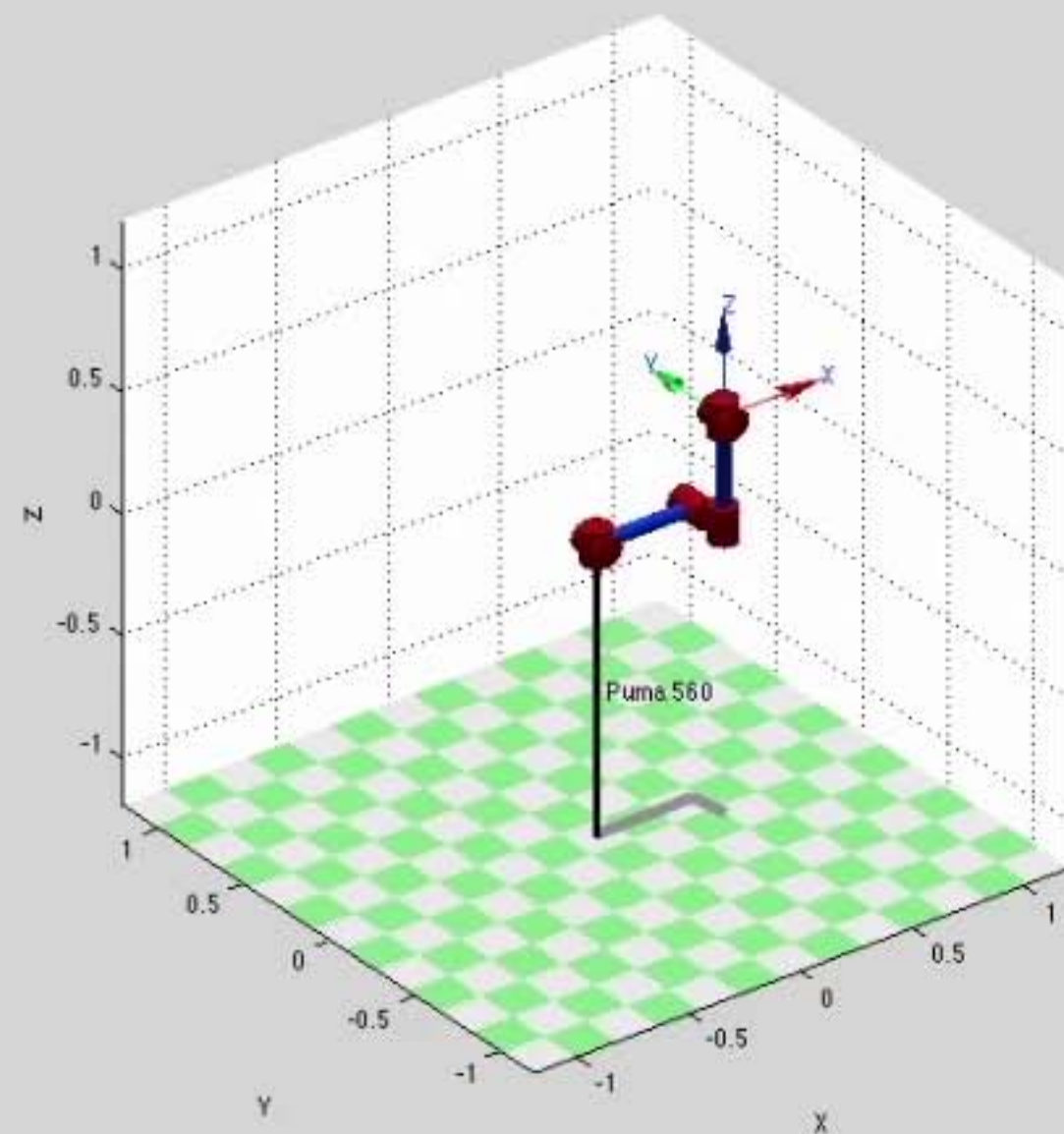
```
>> mdl_puma560
>> p560.plot(qz)
fx >>
```

Workspace

Name	Value	Class
deg	0.0175	double
p560	<1x1 SerialLink>	SerialLink
qn	[0,0.7854,3.1416,0,0.78...	double
qr	[0,1.5708,-1.5708,0,0,0]	double
qs	[0,0,-1.5708,0,0,0]	double
qz	[0,0,0,0,0,0]	double

Figures

Figure 1



Command Window

```
>> mdl_puma560
>> p560.plot(qz)
>> T2 = transl(0.6, 0.2, 0) * rpy2tr(0, 180, 180, 'deg')
```

T2 =

```
1.0000    0.0000    0.0000    0.6000
0.0000   -1.0000         0    0.2000
0.0000    0.0000   -1.0000         0
         0         0         0    1.0000
```

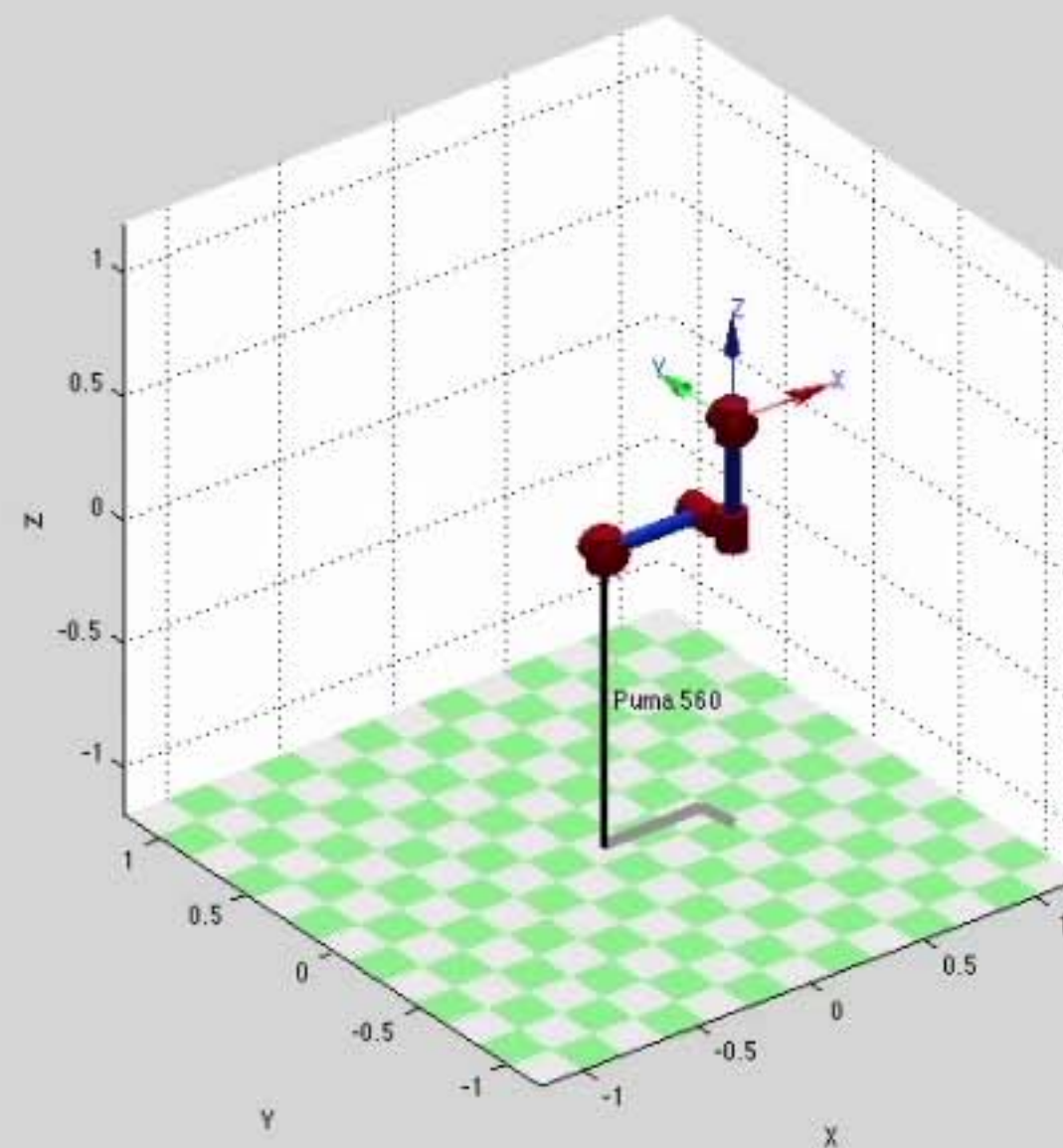
fx >> |

Workspace

Name	Value	Class
T2	<4x4 double>	double
deg	0.0175	double
p560	<1x1 SerialLink>	SerialLink
qn	[0,0.7854,3.1416,0,0.78...	double
qr	[0,1.5708,-1.5708,0,0,0]	double
qs	[0,0,-1.5708,0,0,0]	double
qz	[0,0,0,0,0,0]	double

Figures

Figure 1



Command Window

```
>> mdl_puma560
>> p560.plot(qz)
>> T2 = transl(0.6, 0.2, 0) * rpy2tr(0, 180, 180, 'deg')

T2 =

    1.0000    0.0000    0.0000    0.6000
    0.0000   -1.0000         0    0.2000
    0.0000    0.0000   -1.0000         0
         0         0         0    1.0000

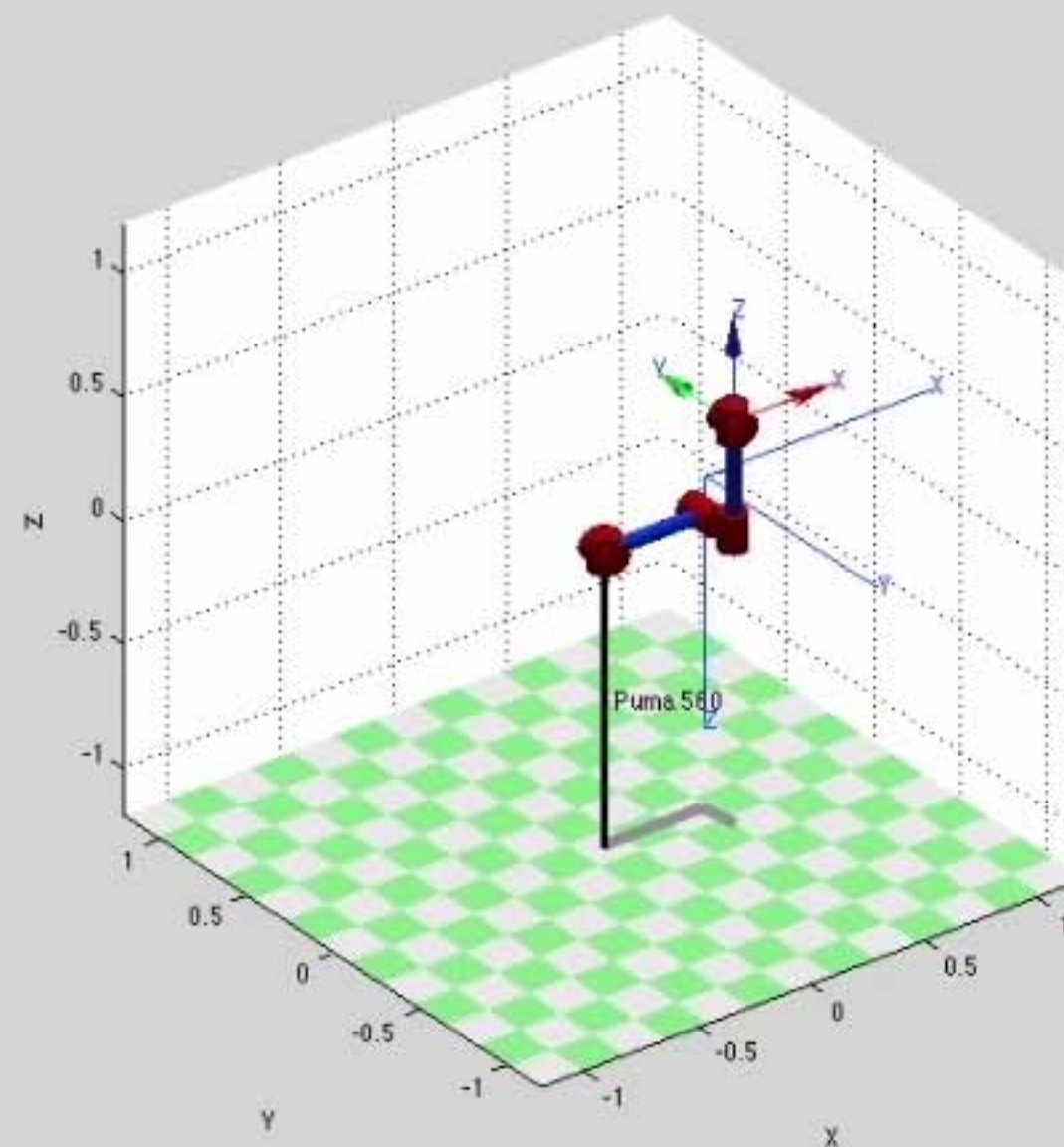
>> hold on
>> trplot(T2)
fx >> q = p560.ikine(T2, [0 0 0 0 0 0])
```

Workspace

Name	Value	Class
T2	<4x4 double>	double
deg	0.0175	double
p560	<1x1 SerialLink>	SerialLink
qn	[0,0.7854,3.1416,0,0.78...	double
qr	[0,1.5708,-1.5708,0,0,0]	double
qs	[0,0,-1.5708,0,0,0]	double
qz	[0,0,0,0,0,0]	double

Figures

Figure 1



Command Window

```
1.0000 0.0000 0.0000 0.0000
0.0000 -1.0000 0 0.2000
0.0000 0.0000 -1.0000 0
0 0 0 1.0000
```

```
>> hold on
>> trplot(T2)
>> q = p560.ikine(T2, [0 0 0 0 0 0])
```

Warning: Initial joint configuration results in a (near-)singular configuration, this may slow convergence
> In SerialLink.ikine at 152

q =

```
0.5613 -0.7804 0.0358 -3.1416 2.3970 0.5613
```

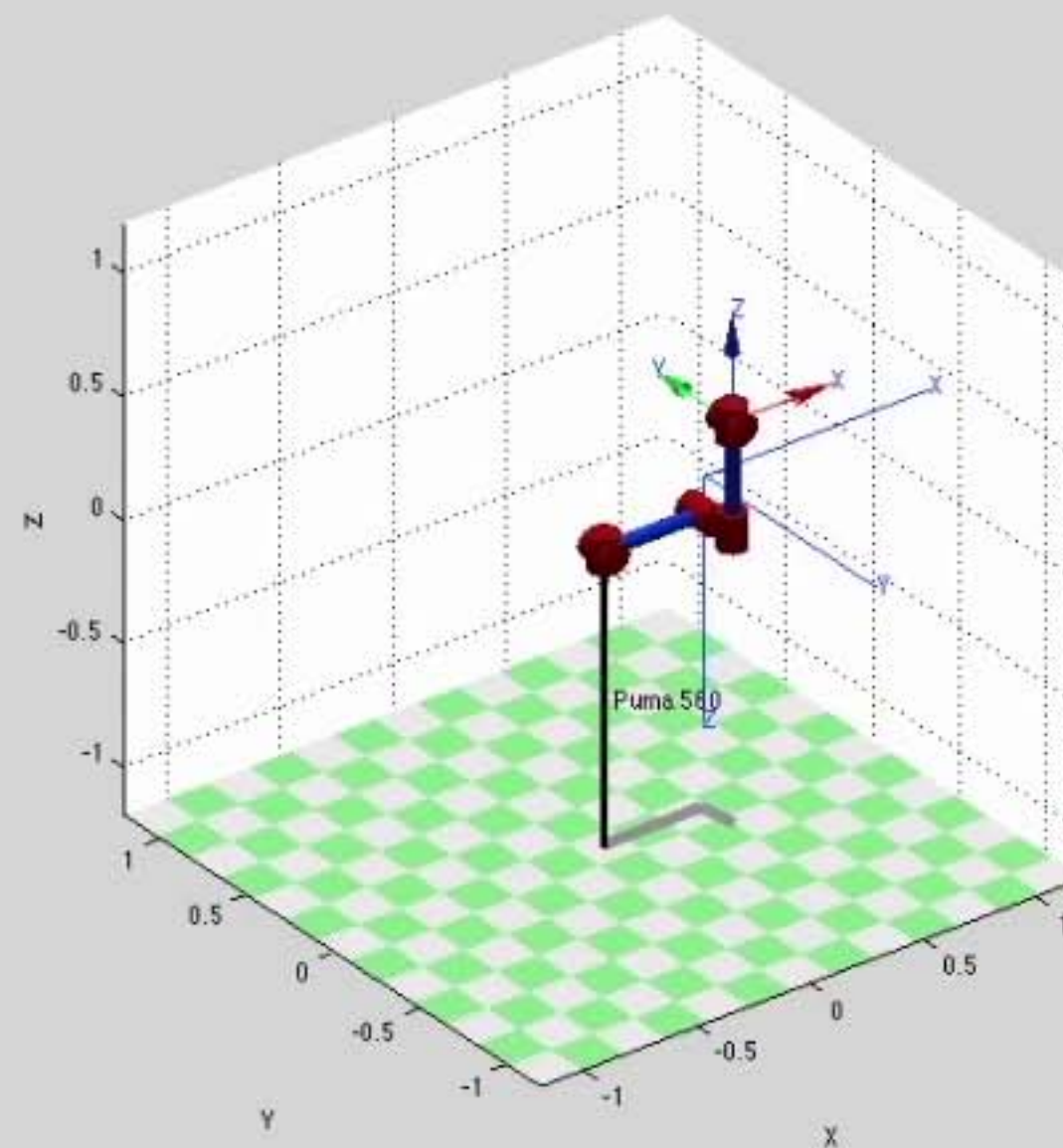
fx >> |

Workspace

Name	Value	Class
T2	<4x4 double>	double
deg	0.0175	double
p560	<1x1 SerialLink>	SerialLink
q	[0.5613,-0.7804,0.0358,...	double
qn	[0,0.7854,3.1416,0,0.78...	double
qr	[0,1.5708,-1.5708,0,0,0]	double
qs	[0,0,-1.5708,0,0,0]	double
qz	[0,0,0,0,0,0]	double

Figures

Figure 1



Command Window

```

0.0000    -1.0000     0.0000     0.2000
0.0000     0.0000    -1.0000     0.0000
0.0000     0.0000     0.0000     1.0000

>> hold on
>> trplot(T2)
>> q = p560.ikine(T2, [0 0 0 0 0 0])
Warning: Initial joint configuration results in a (near-)singular configuration, this may slow convergence.
> In SerialLink.ikine at 152

q =

    0.5613   -0.7804    0.0358   -3.1416    2.3970    0.5613

>> p560.plot(q)
fx >>

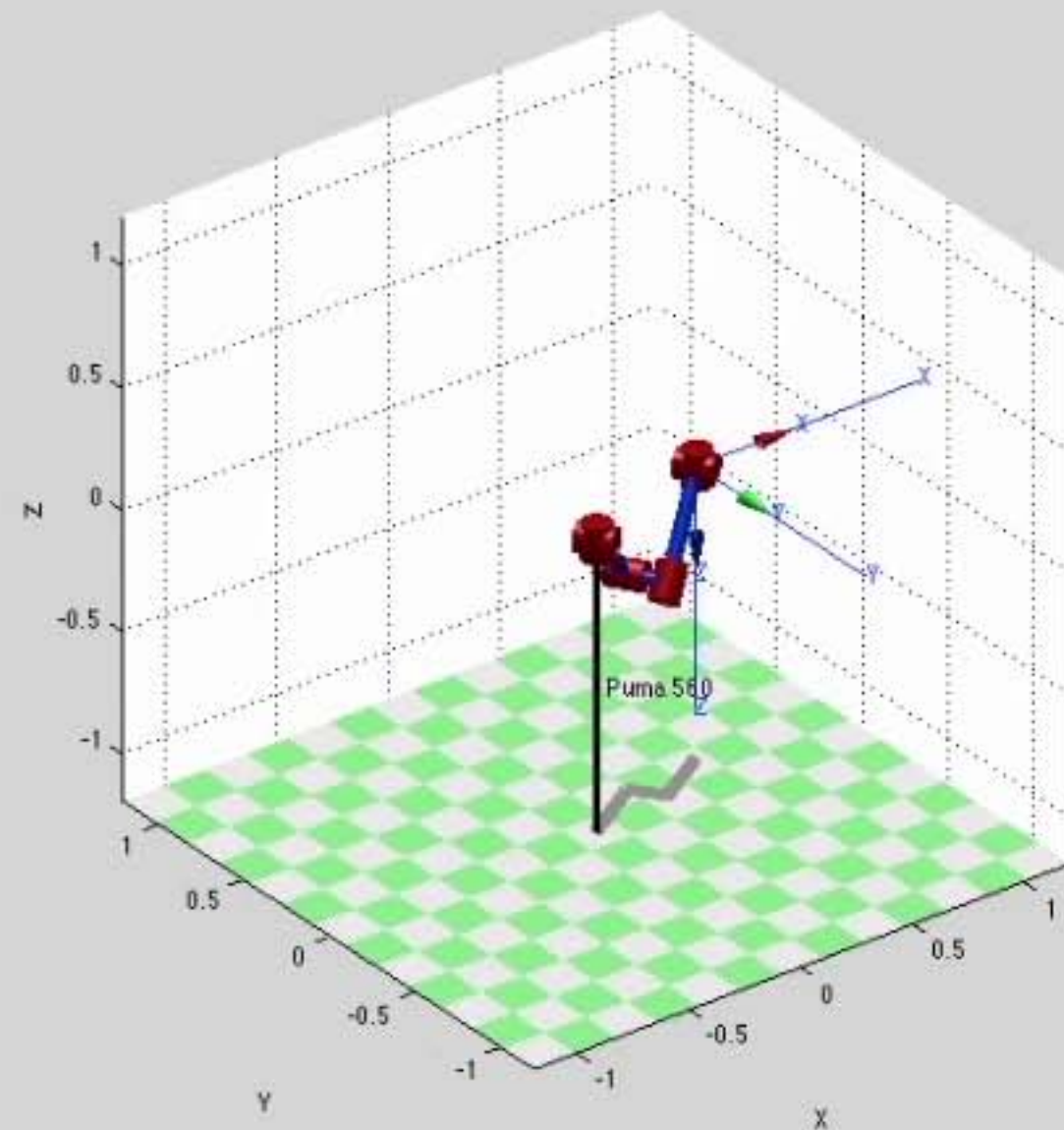
```

Workspace

Name	Value	Class
T2	<4x4 double>	double
deg	0.0175	double
p560	<1x1 SerialLink>	SerialLink
q	[0.5613,-0.7804,0.0358,...	double
qn	[0,0.7854,3.1416,0,0.78...	double
qr	[0,1.5708,-1.5708,0,0,0]	double
qs	[0,0,-1.5708,0,0,0]	double
qz	[0,0,0,0,0,0]	double

Figures

Figure 1



Issues with **numerical solution**

- Choosing good initial joint coordinates

Issues with **numerical solution**

- Choosing good initial joint coordinates
- Can't guarantee a particular configuration

Issues with **numerical solution**

- Choosing good initial joint coordinates
- Can't guarantee a particular configuration
- Can be computationally expensive

Numerical Inverse Kinematics in MATLAB

Numerical IKs take 6 steps:

- Initialization
- Forward Kinematics
- Error Calculation
- Jacobian Matrix
- Update the Joint Angles
- Iterate

1. Initialization

Start with an initial guess of joint angles.

- Typically the current robot configuration q .

Serves as the **seed** for iterative refinement.

2. Forward Kinematics

Compute the desired and current end-effector positions and orientation from the current joint angles.

$$\boldsymbol{\varepsilon}_{curr} = K(\boldsymbol{q}_{curr})$$

$$\boldsymbol{\varepsilon}_{des} = K(\boldsymbol{q}_{des})$$

3. Error Calculation

Compute the error between the desired and current end-effector pose.

$$e = \varepsilon_{des} - \varepsilon_{curr}$$

The error will be used to select the direction in which to update the joint updates.

4. Jacobian Computation

Calculate the Jacobian matrix: relates joint velocities to end-effector velocities.

Used to determine how joint changes affect end-effector position.

$$\begin{bmatrix} \Delta x \\ \Delta y \\ \Delta z \end{bmatrix} = \begin{bmatrix} \frac{\partial x}{\partial \theta_1} & \frac{\partial x}{\partial \theta_2} & \frac{\partial x}{\partial \theta_3} & \frac{\partial x}{\partial \theta_4} & \frac{\partial x}{\partial \theta_5} & \frac{\partial x}{\partial \theta_6} \\ \frac{\partial y}{\partial \theta_1} & \frac{\partial y}{\partial \theta_2} & \frac{\partial y}{\partial \theta_3} & \frac{\partial y}{\partial \theta_4} & \frac{\partial y}{\partial \theta_5} & \frac{\partial y}{\partial \theta_6} \\ \frac{\partial z}{\partial \theta_1} & \frac{\partial z}{\partial \theta_2} & \frac{\partial z}{\partial \theta_3} & \frac{\partial z}{\partial \theta_4} & \frac{\partial z}{\partial \theta_5} & \frac{\partial z}{\partial \theta_6} \end{bmatrix} \begin{bmatrix} \Delta \theta_1 \\ \Delta \theta_2 \\ \Delta \theta_3 \\ \Delta \theta_4 \\ \Delta \theta_5 \\ \Delta \theta_6 \end{bmatrix}$$

$$\Delta x = J \Delta \theta$$

5. Joint Angle Update

Update joint angles using optimization (e.g., Jacobian pseudoinverse or damped least squares).

Iterate until the error is below a threshold.

$$\delta\theta = J^\dagger e$$

where,

$$e = x_{\text{des}} - x_{\text{curr}}$$

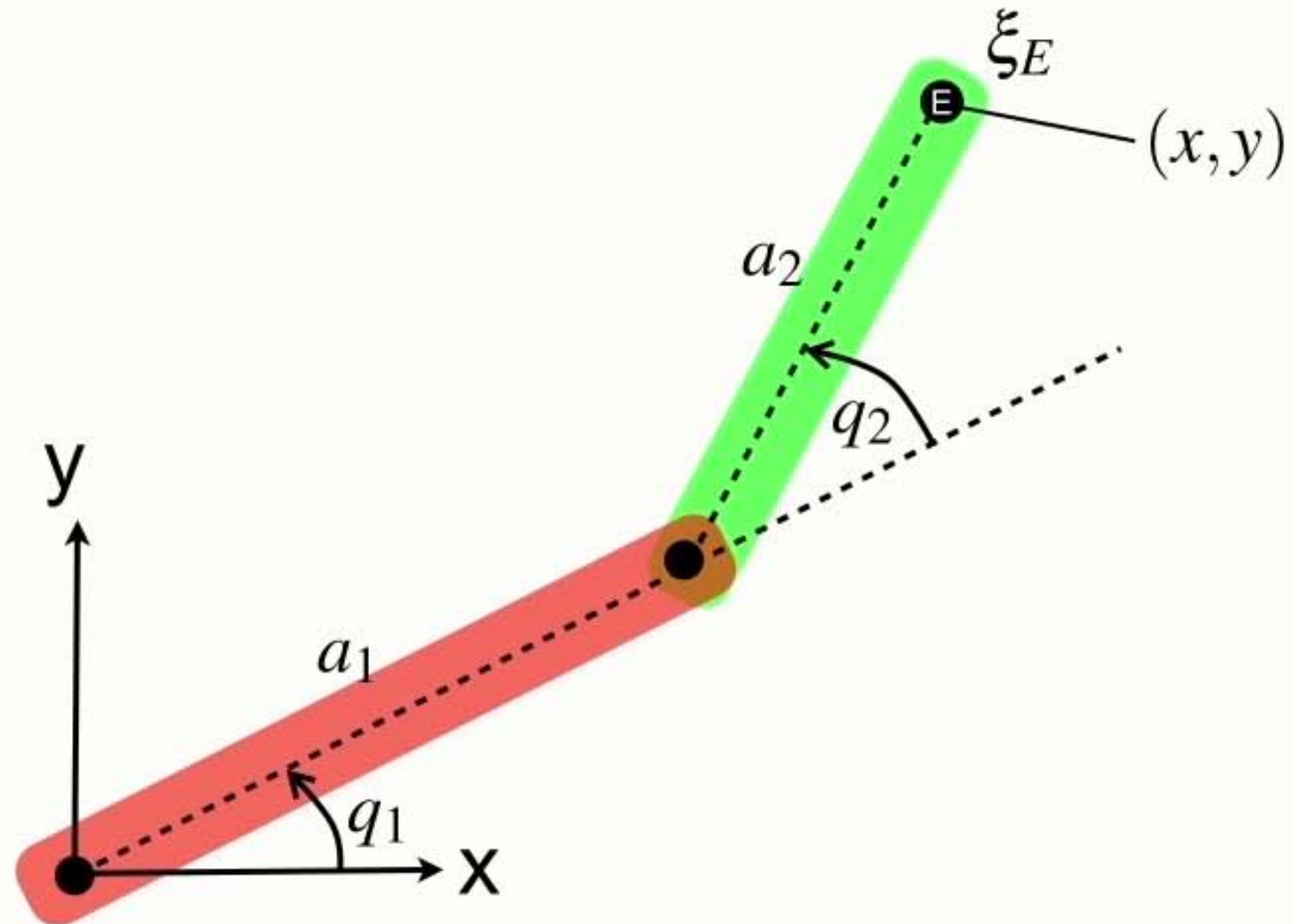
$$J^\dagger = \text{inv}(J) \text{ <<pseudo-inverse>>}$$

6. Pseudocode

```
function jointAngles = numericalIK(targetPosition, initialGuess)
    jointAngles = initialGuess
    while not converged:
        currentPos = forwardKinematics(jointAngles)
        error = targetPosition - currentPos
        J = computeJacobian(jointAngles)
        deltaTheta = pinv(J) * error
        jointAngles = jointAngles + deltaTheta
    return jointAngles
```

Section 8 Practical considerations

Reachability

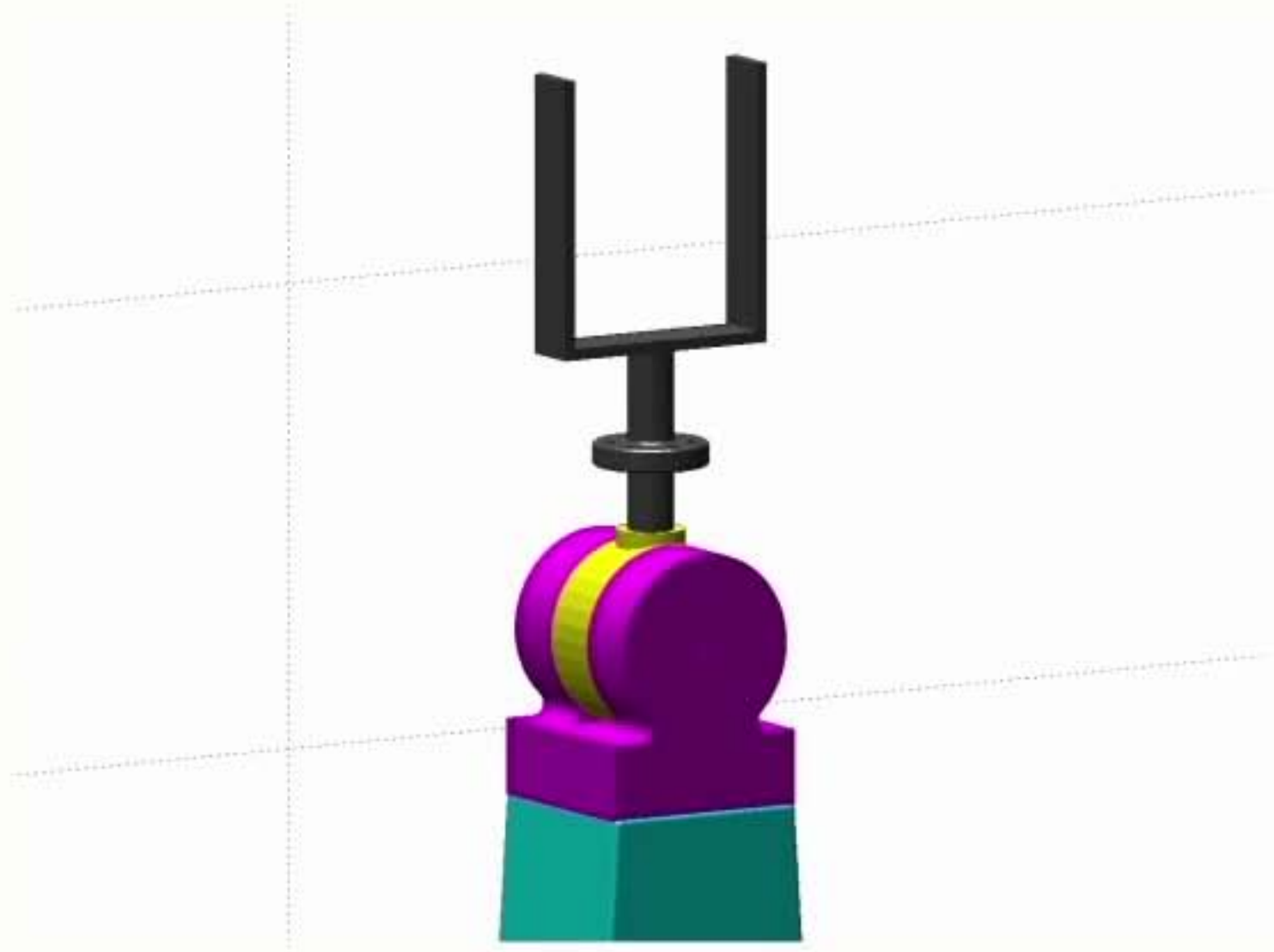


Ⓒ

- The robot may not be able to reach the pose that we request

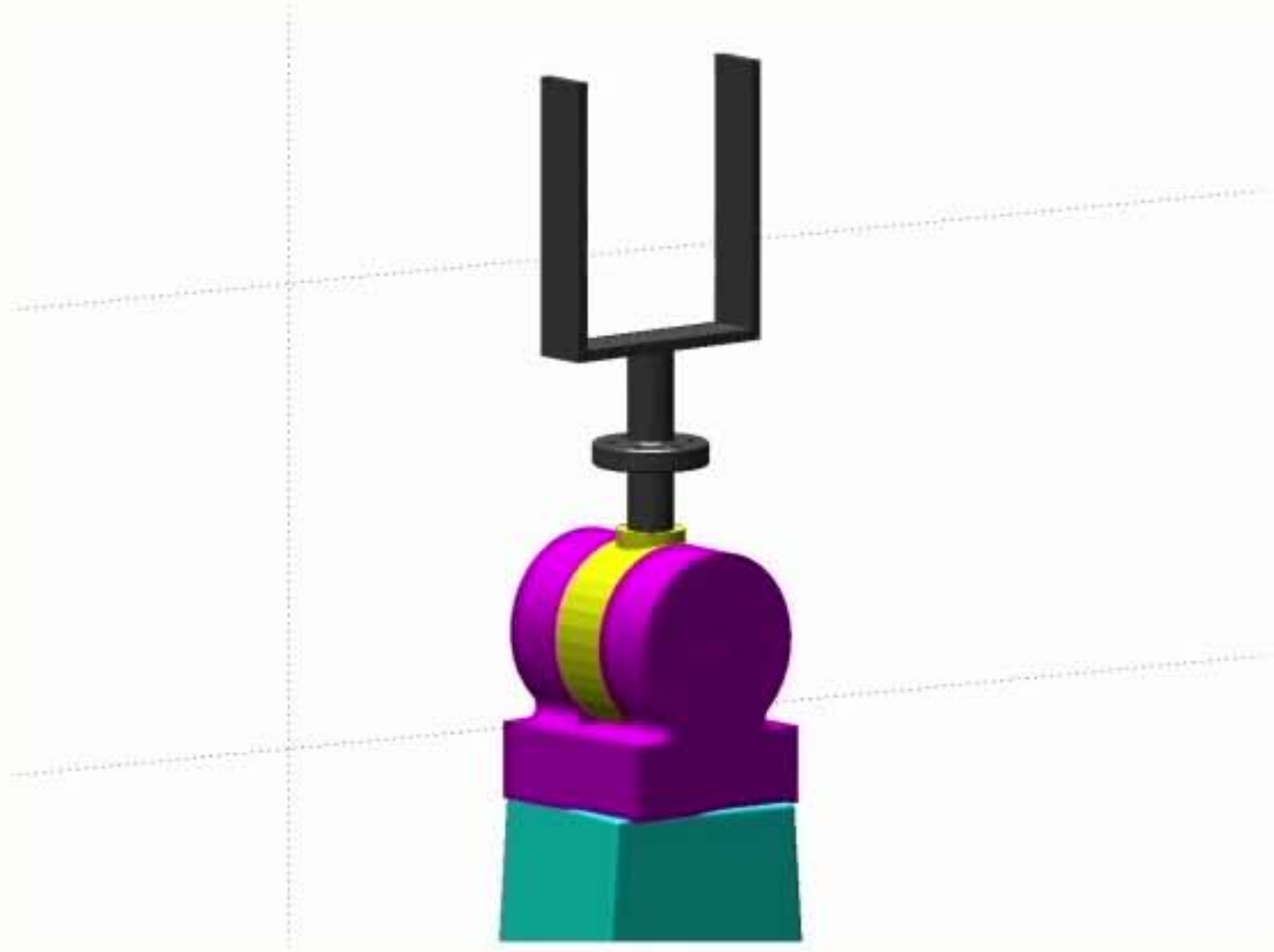
Singularity

- This is related to the gimbal lock problem we looked at earlier



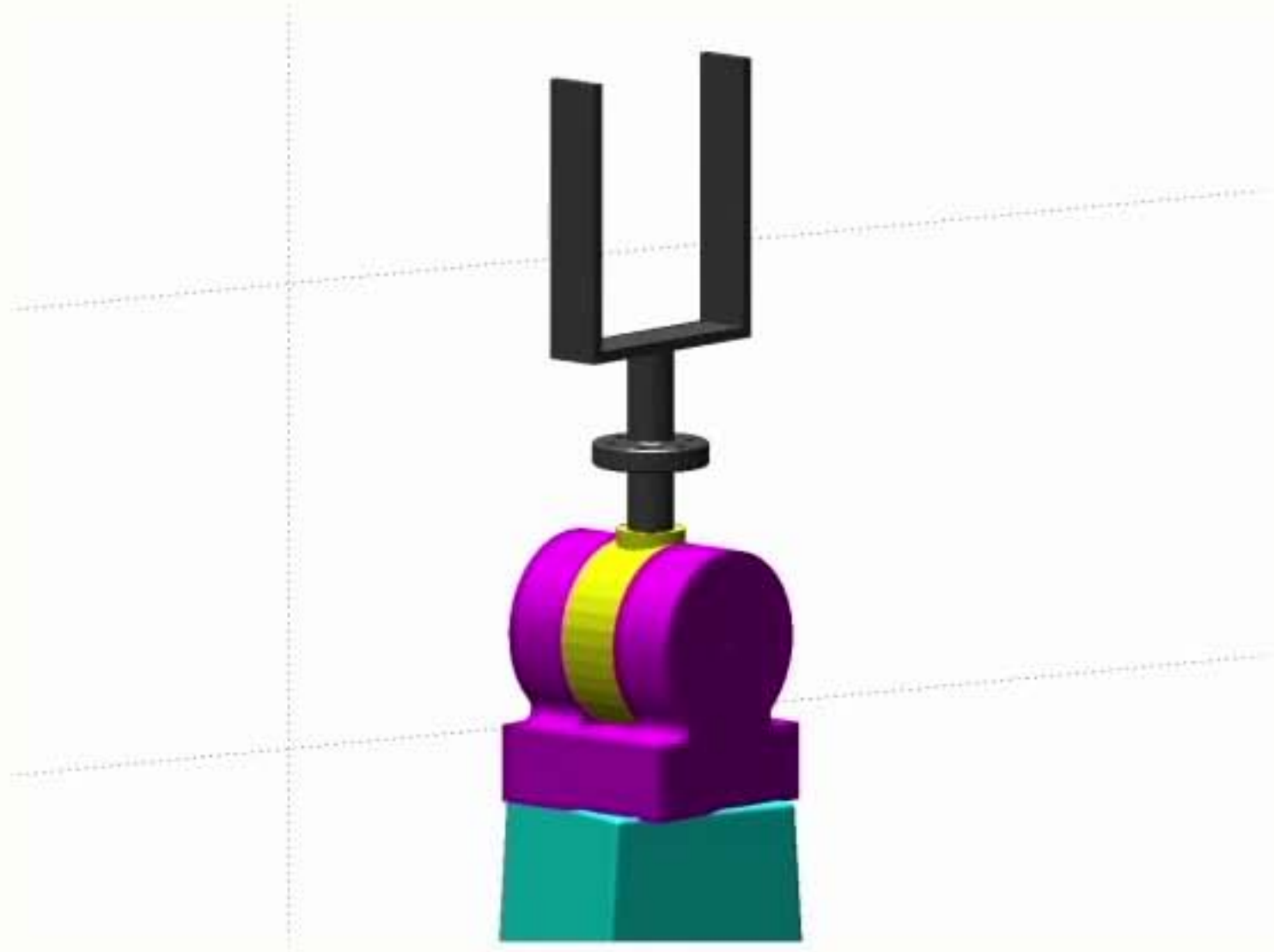
Singularity

- This is related to the gimbal lock problem we looked at earlier



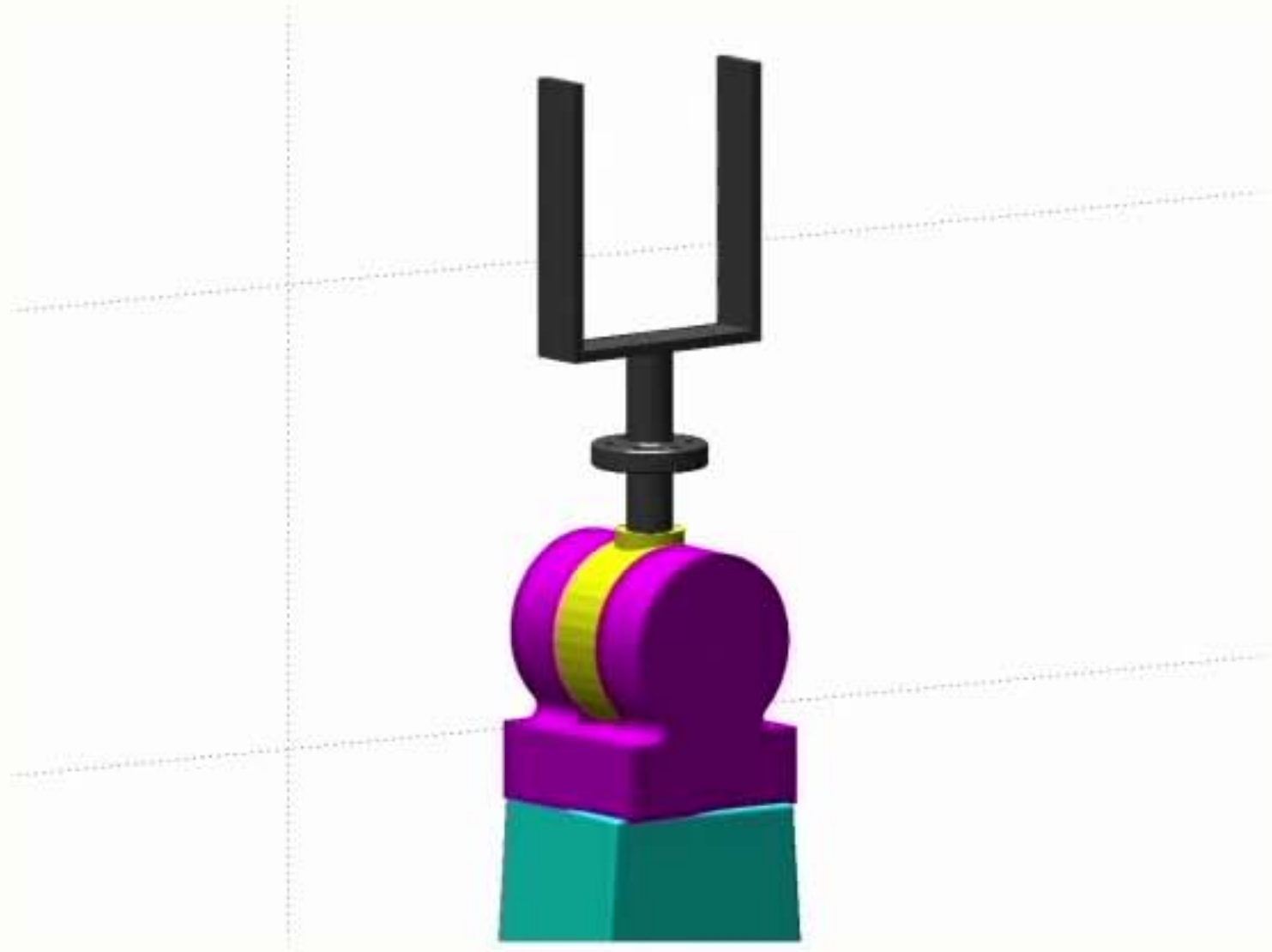
Singularity

- This is related to the gimbal lock problem we looked at earlier



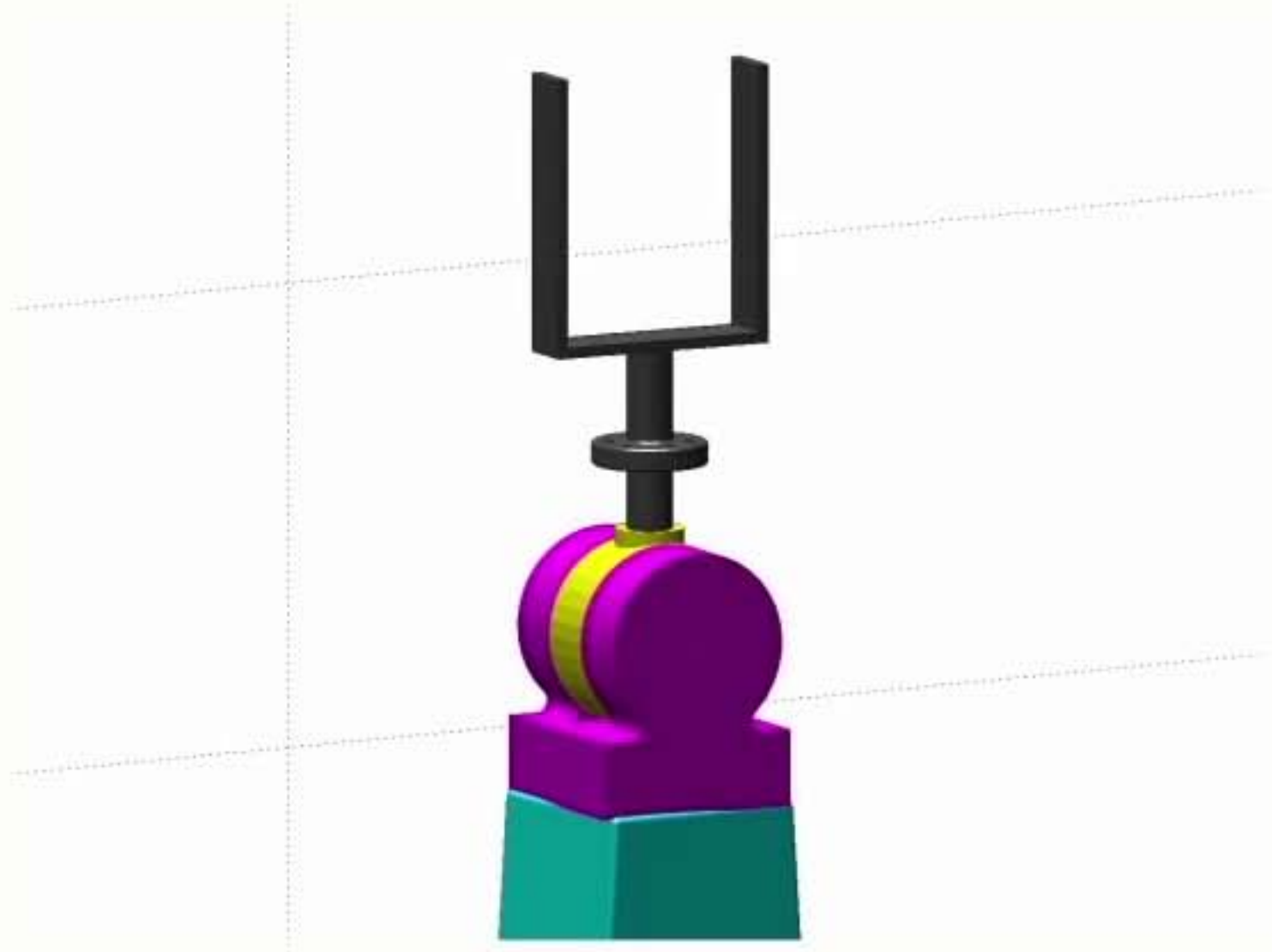
Singularity

- This is related to the gimbal lock problem we looked at earlier



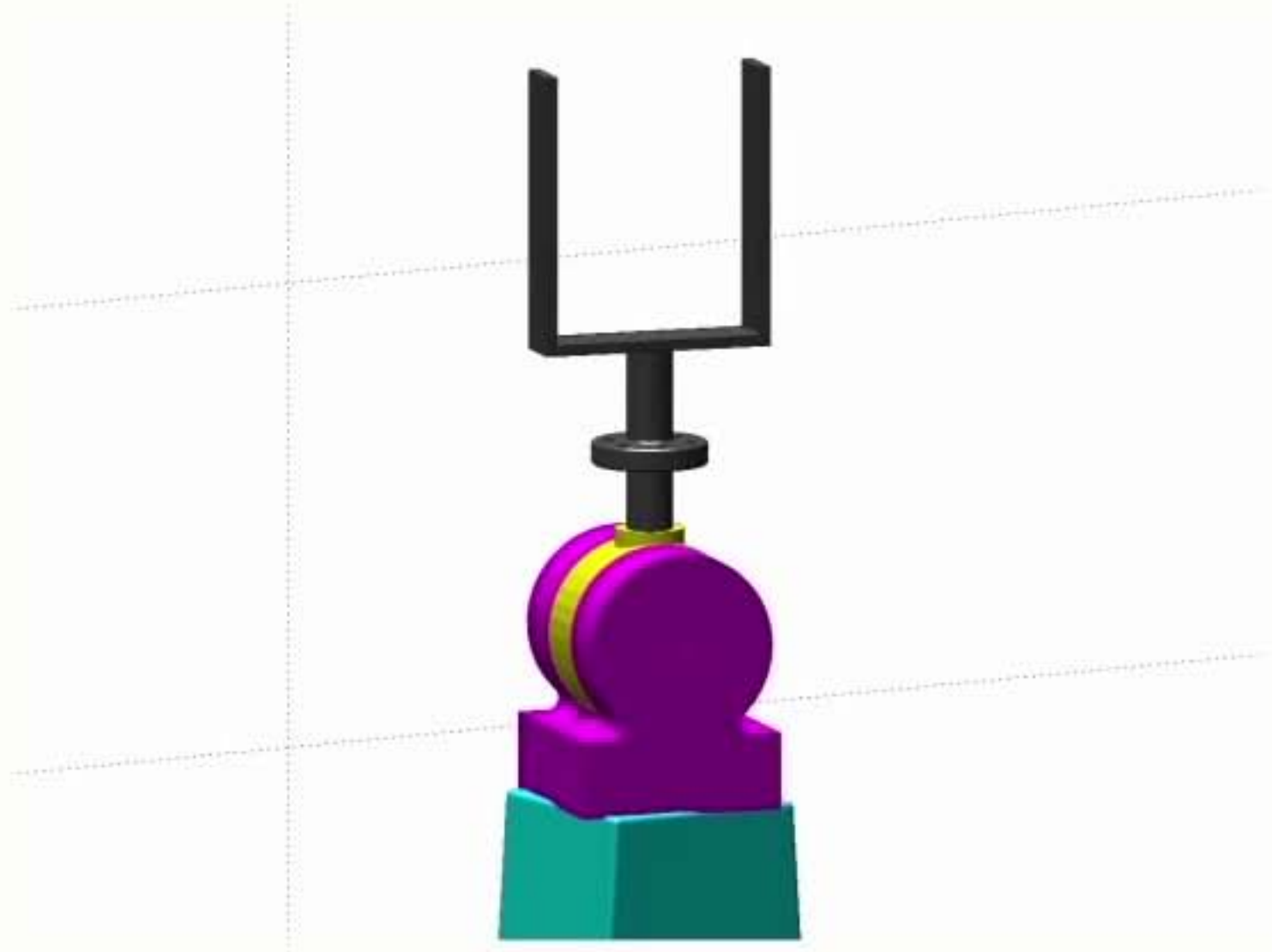
Singularity

- This is related to the gimbal lock problem we looked at earlier



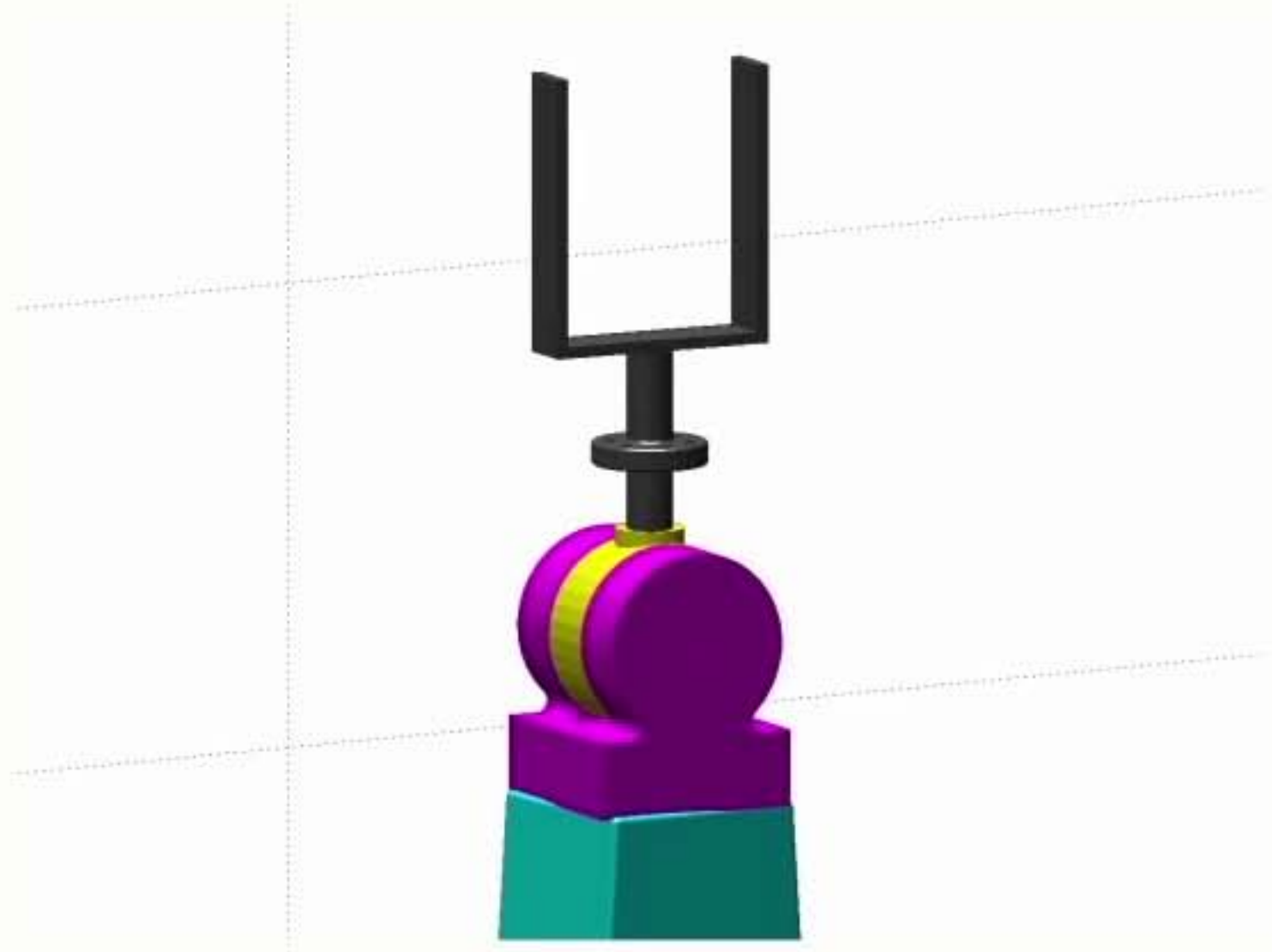
Singularity

- This is related to the gimbal lock problem we looked at earlier



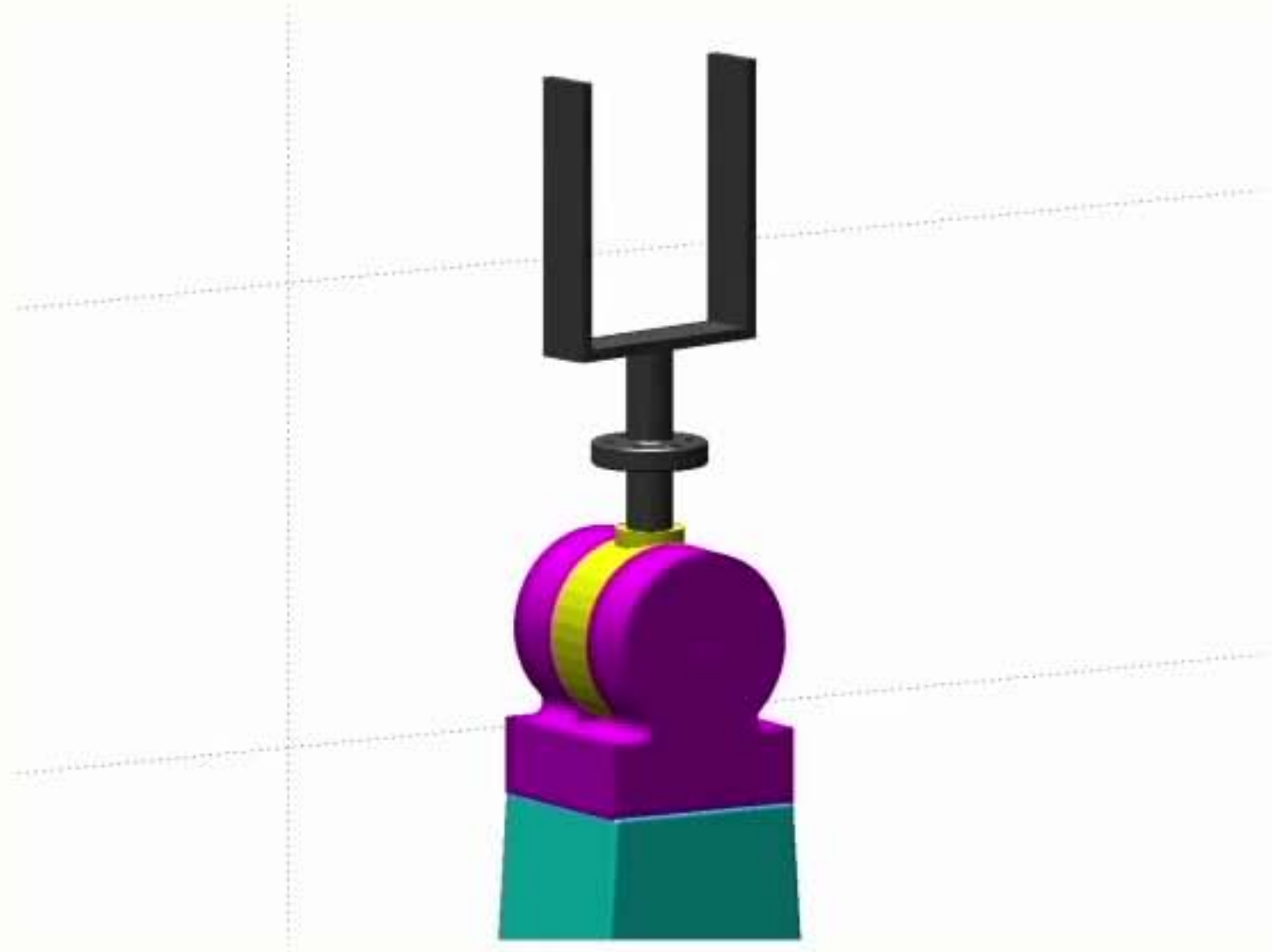
Singularity

- This is related to the gimbal lock problem we looked at earlier



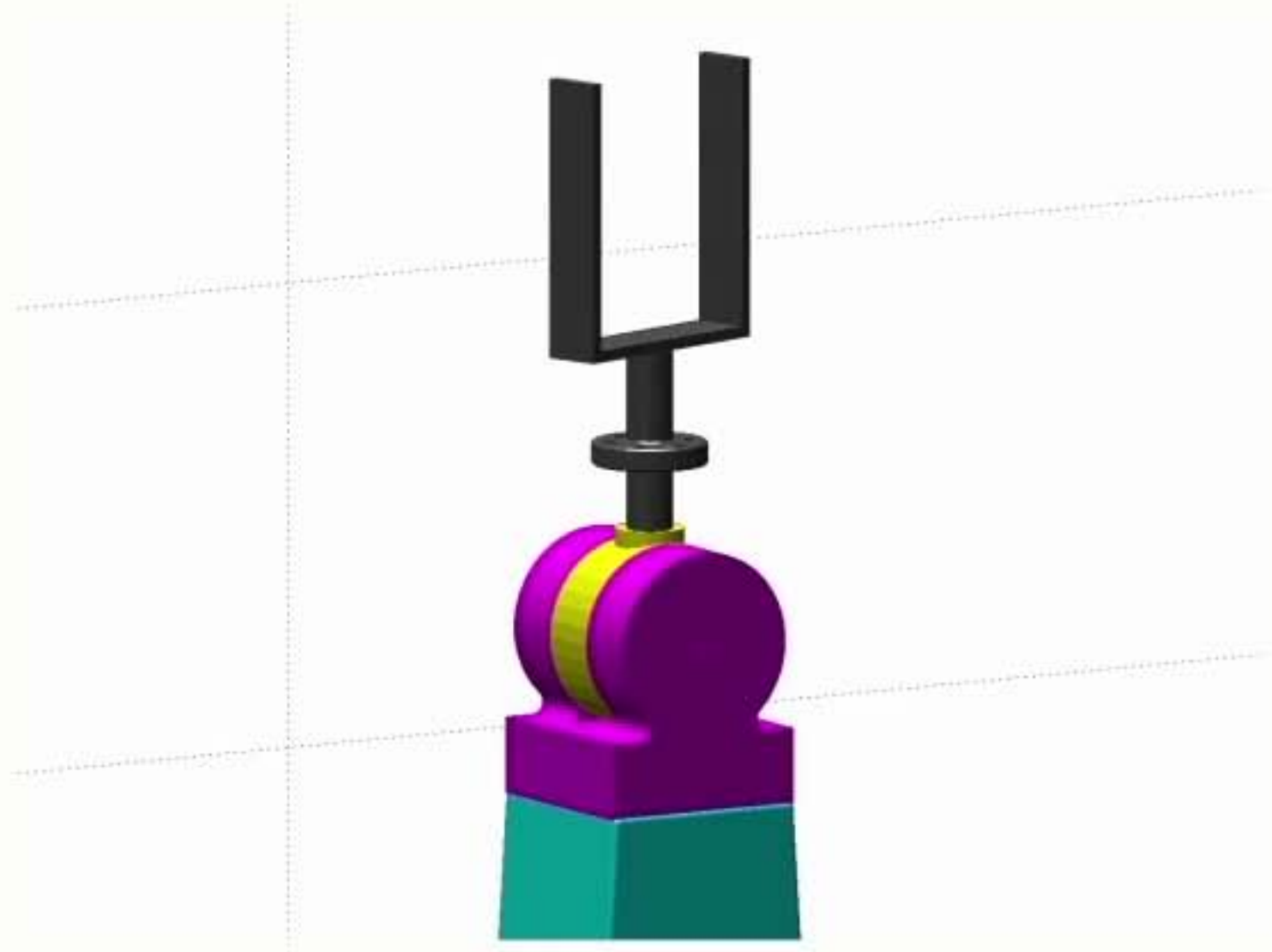
Singularity

- This is related to the gimbal lock problem we looked at earlier



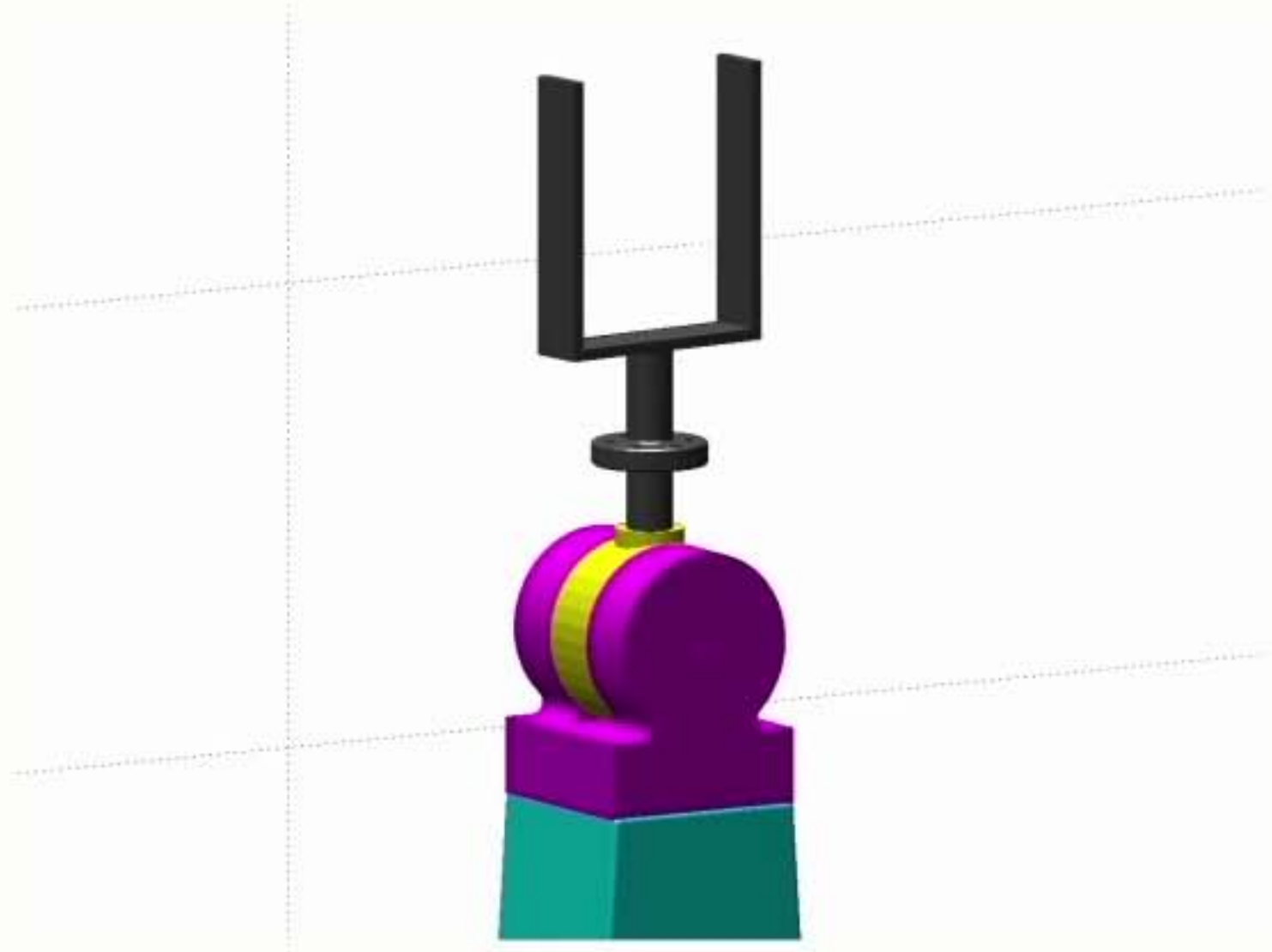
Singularity

- This is related to the gimbal lock problem we looked at earlier



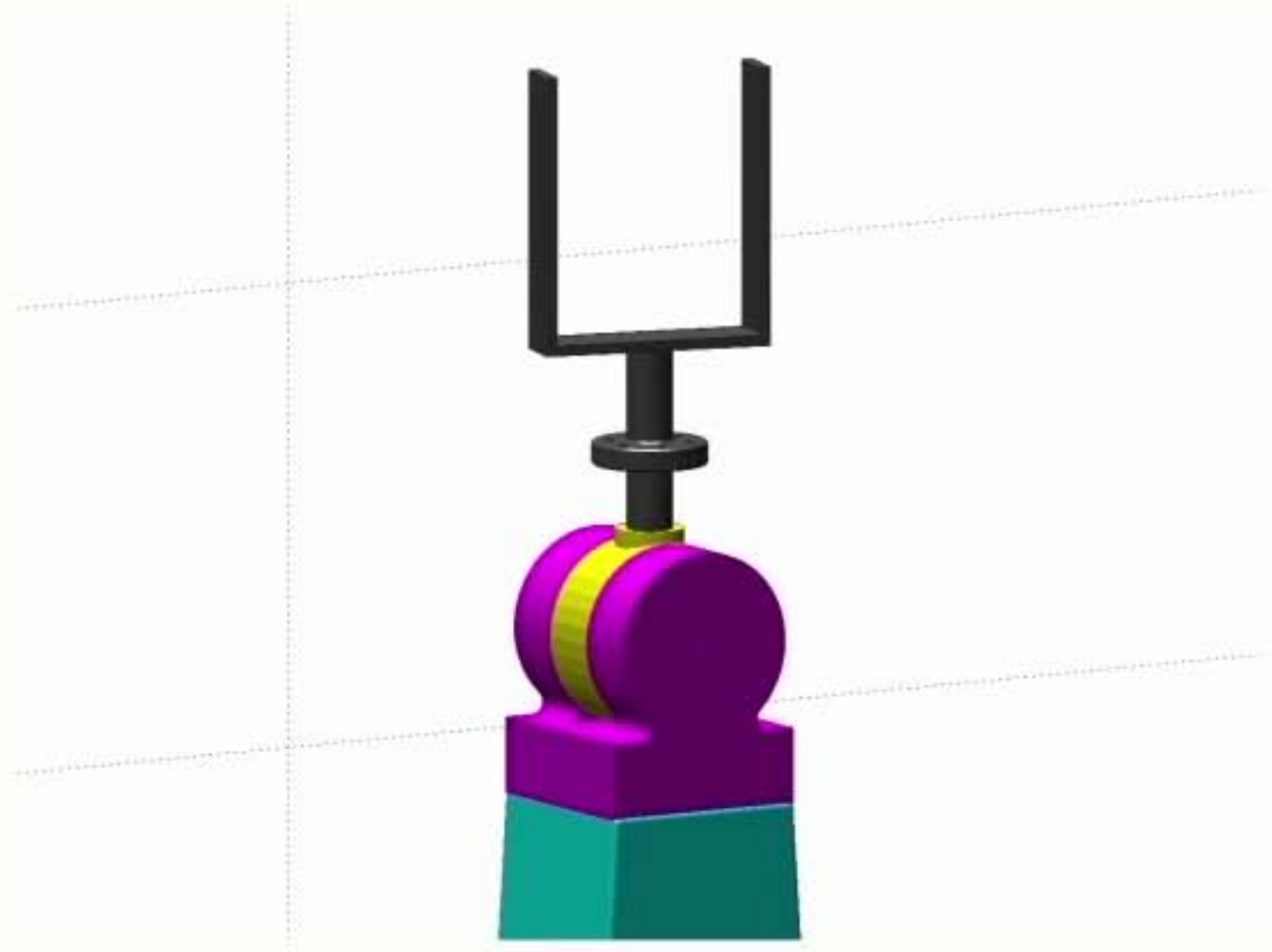
Singularity

- This is related to the gimbal lock problem we looked at earlier



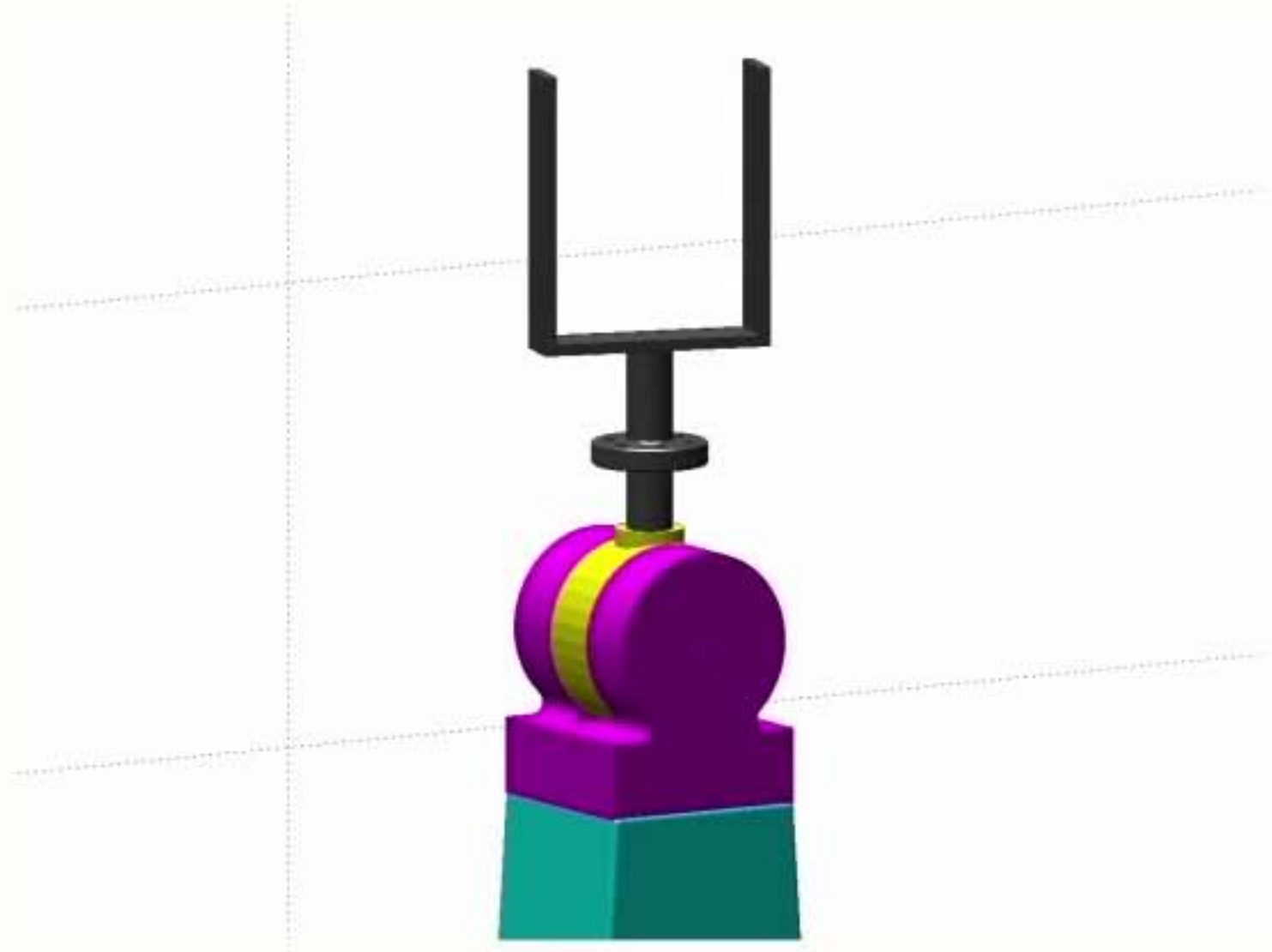
Singularity

- This is related to the gimbal lock problem we looked at earlier



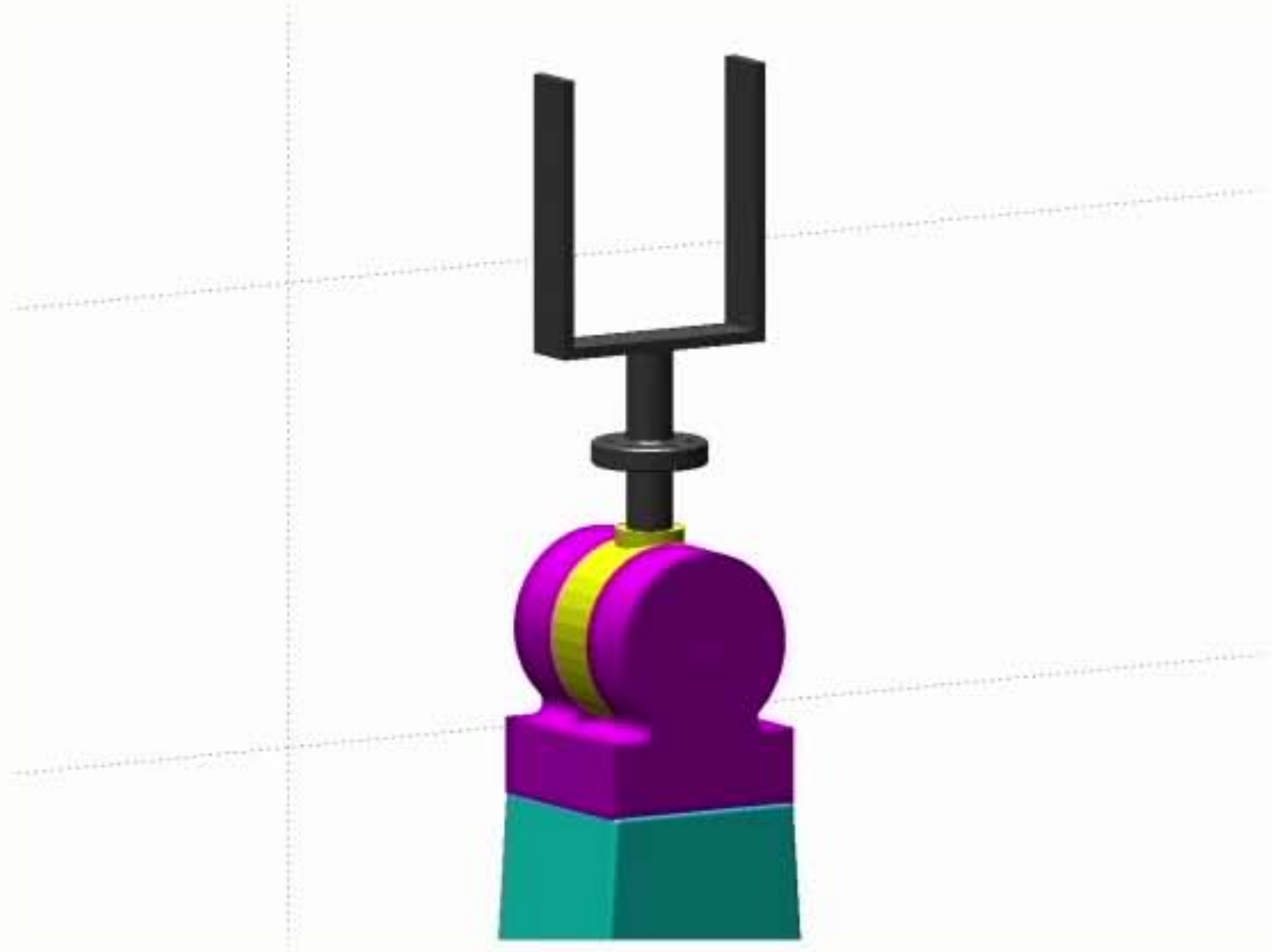
Singularity

- This is related to the gimbal lock problem we looked at earlier



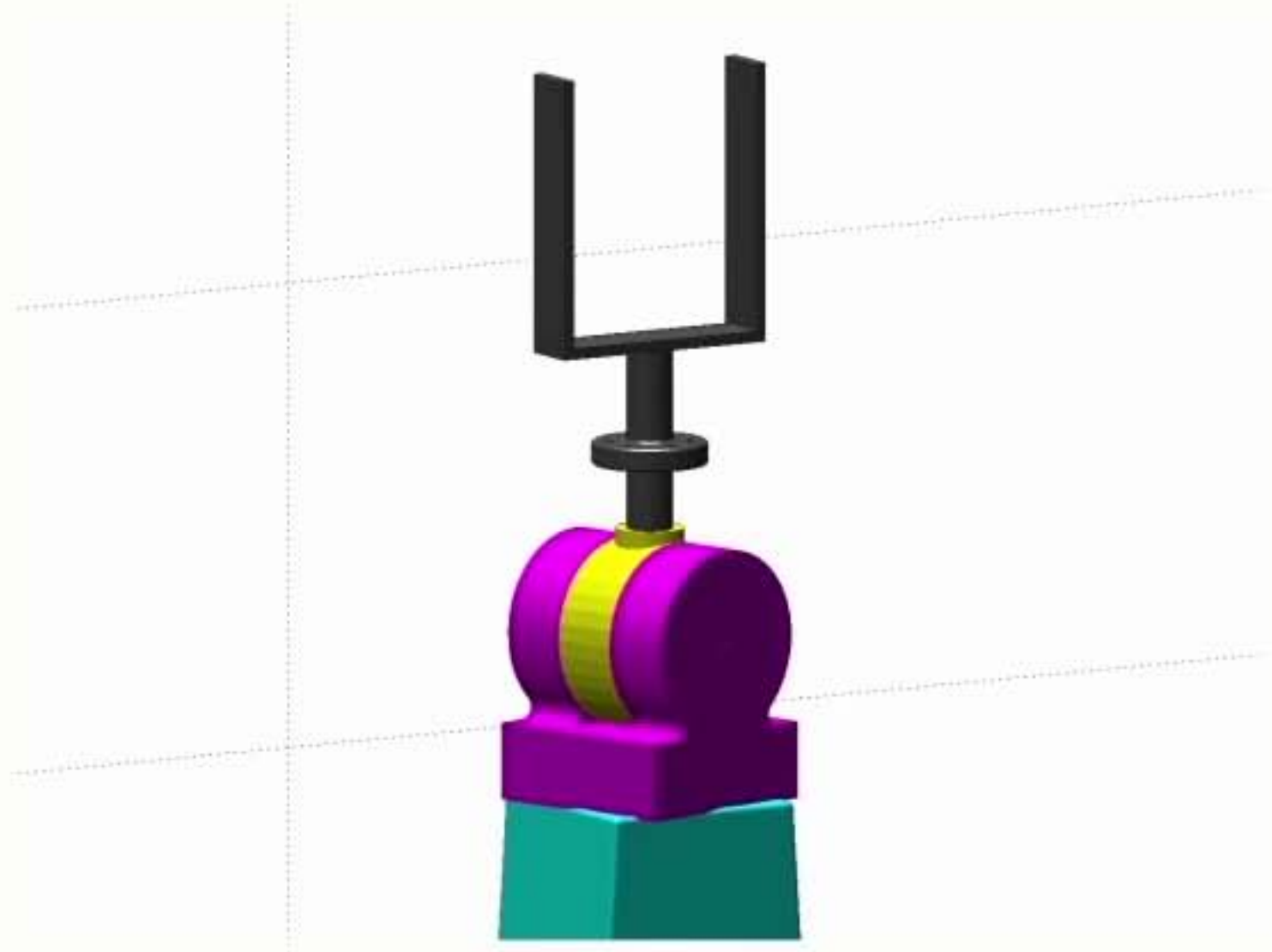
Singularity

- This is related to the gimbal lock problem we looked at earlier



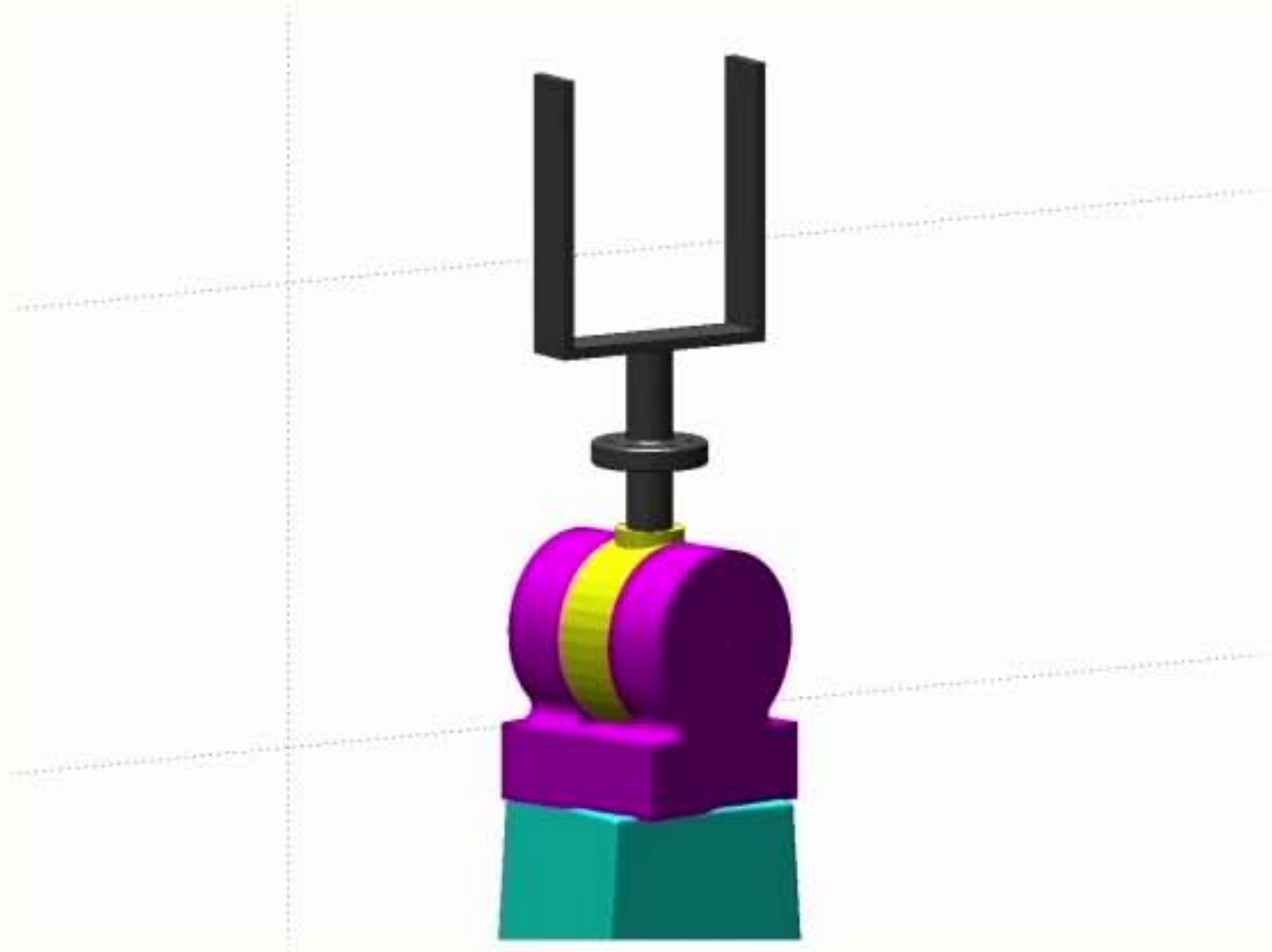
Singularity

- This is related to the gimbal lock problem we looked at earlier



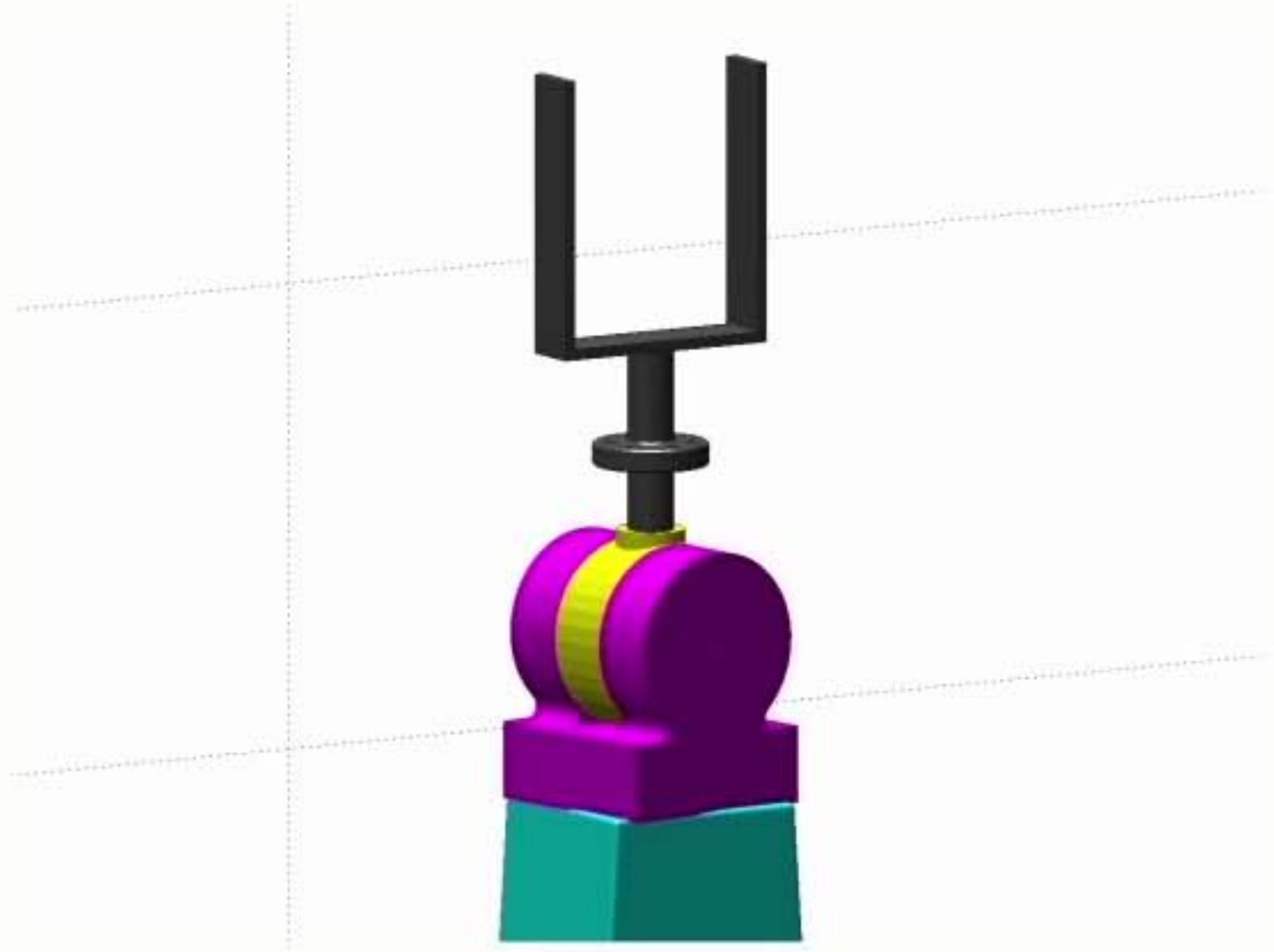
Singularity

- This is related to the gimbal lock problem we looked at earlier



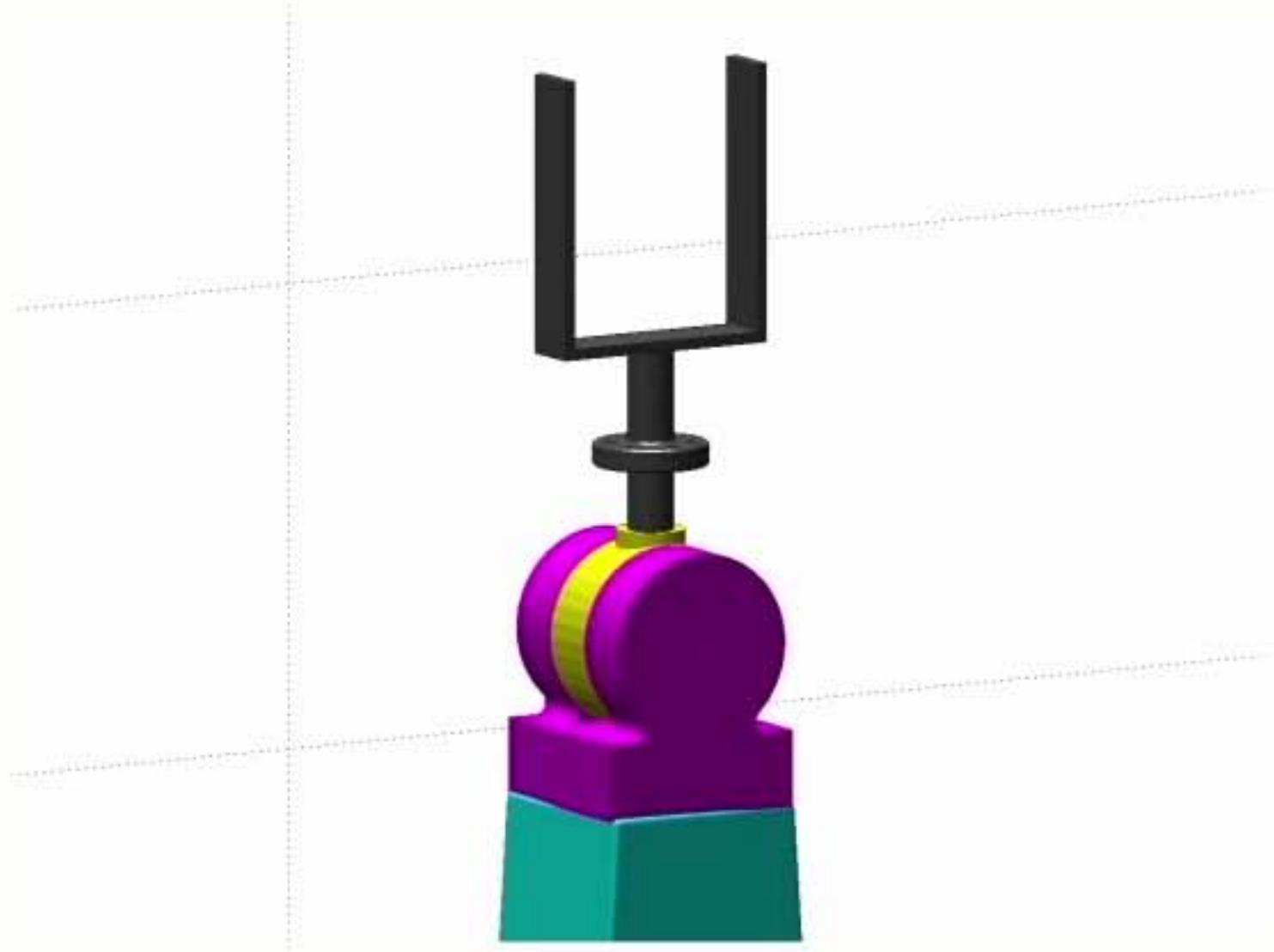
Singularity

- This is related to the gimbal lock problem we looked at earlier



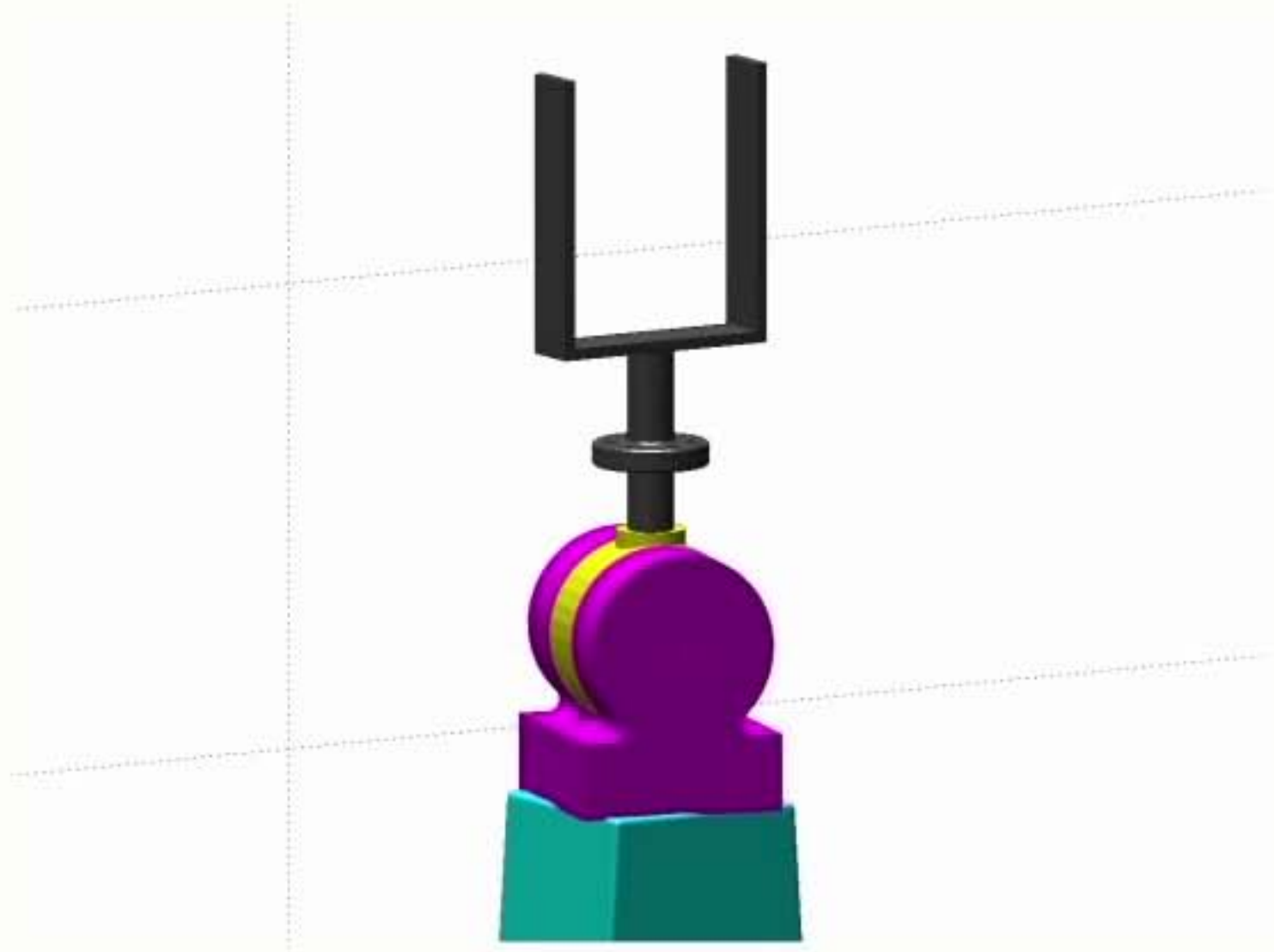
Singularity

- This is related to the gimbal lock problem we looked at earlier



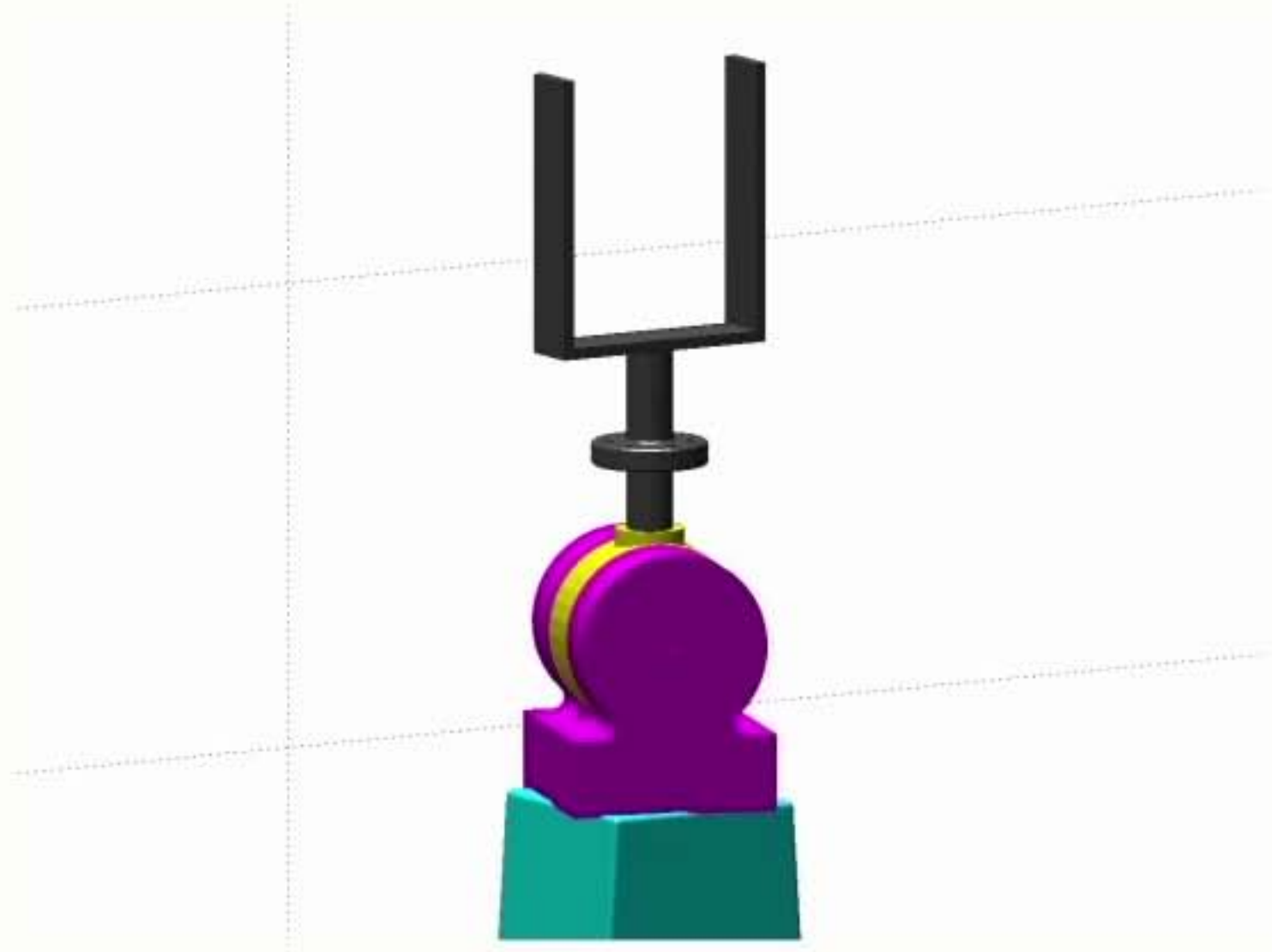
Singularity

- This is related to the gimbal lock problem we looked at earlier



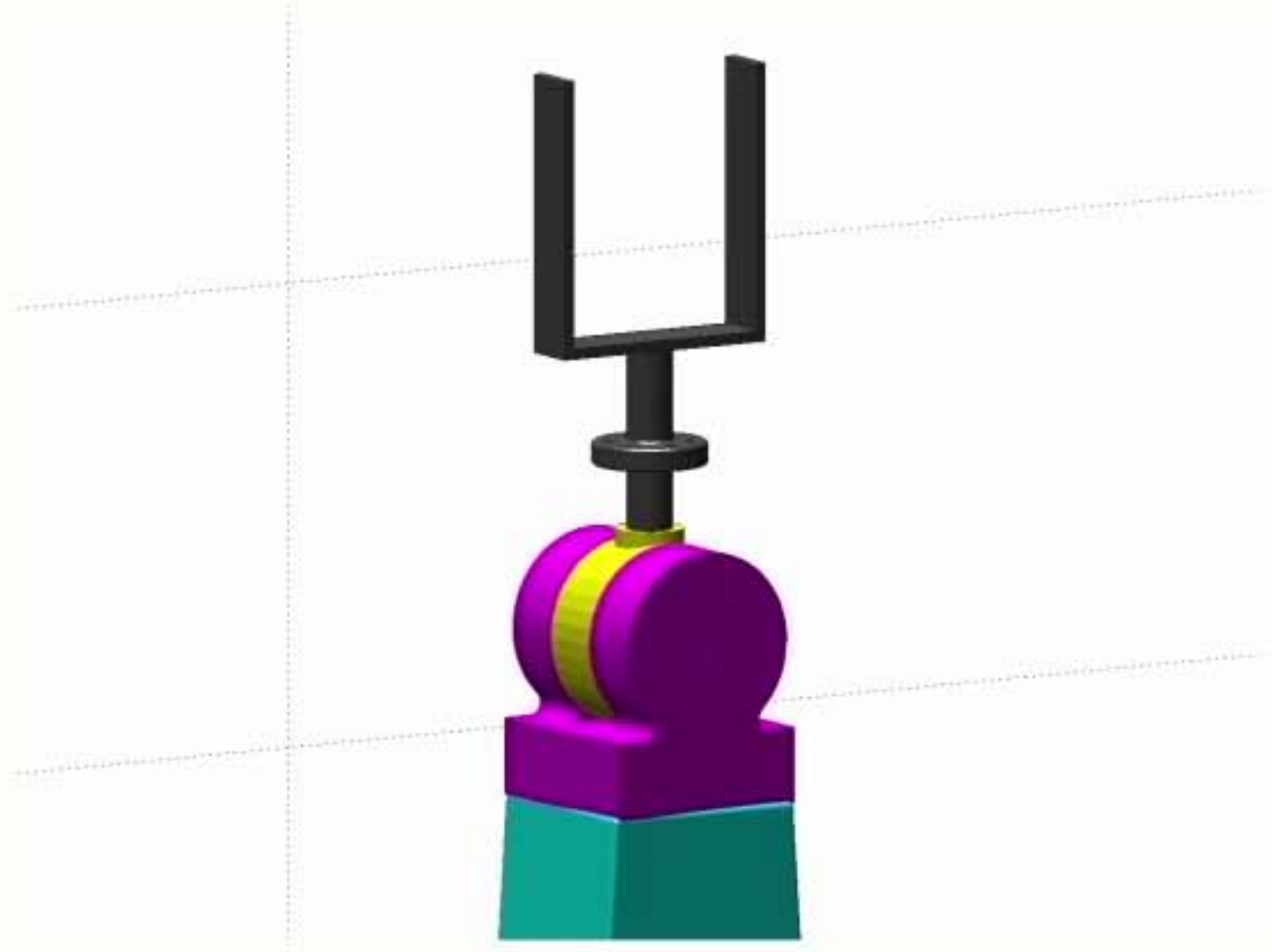
Singularity

- This is related to the gimbal lock problem we looked at earlier



Singularity

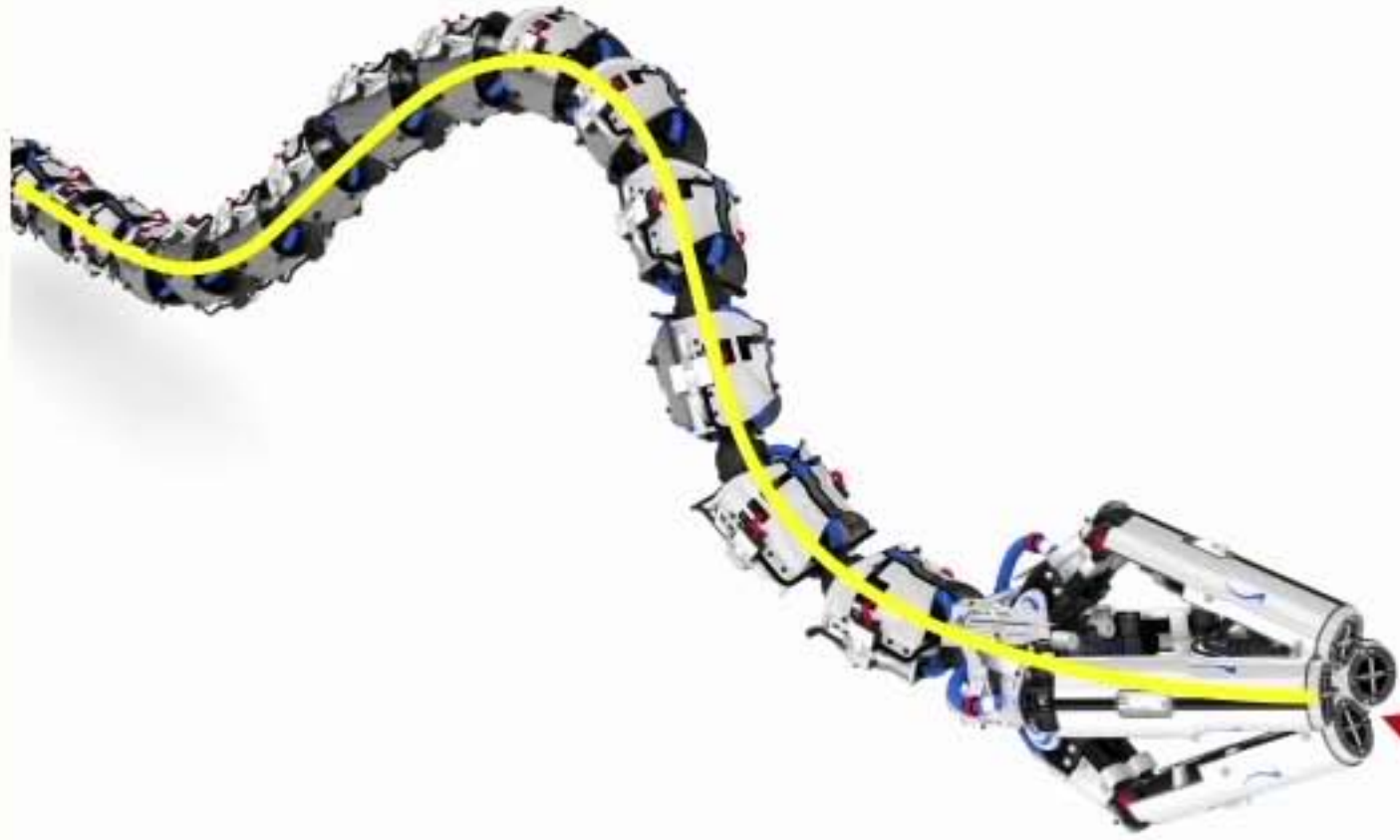
- This is related to the gimbal lock problem we looked at earlier



Section 9

Redundant robots

Very high **DOF** robot



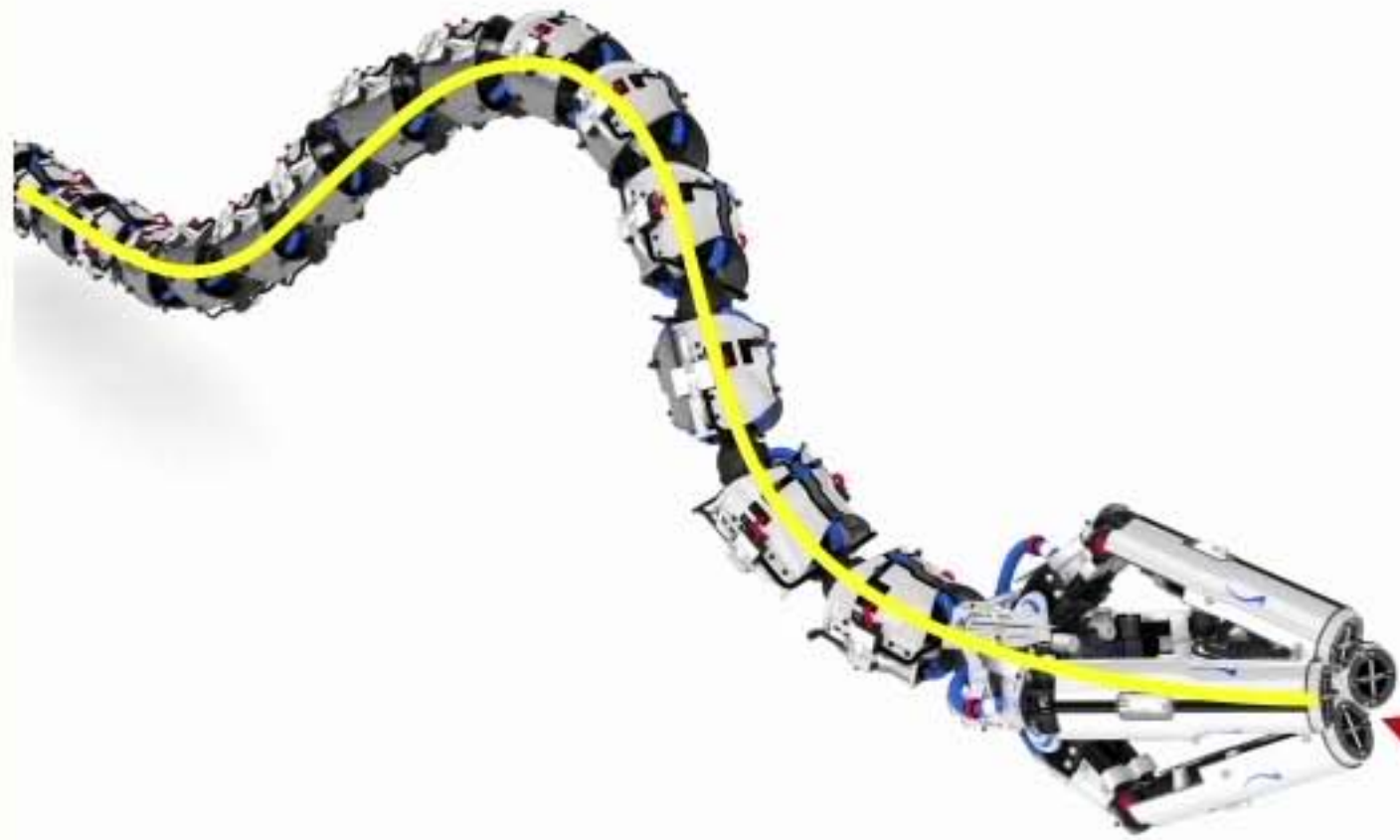
$$\mathcal{C} \subset \mathbb{R}^{100}$$

$$\mathcal{T} \subset \text{SE}(3)$$

$$\xi_E \sim (x, y, z, \theta_r, \theta_p, \theta_y)$$

■ Redundant robot

Very high **DOF** robot



configuration space
is much bigger

$$\mathcal{C} \subset \mathbb{R}^{100}$$

$$\mathcal{T} \subset \text{SE}(3)$$

task space is the same
as the Puma robot

$\xi_E \sim (x, y, z, \theta_r, \theta_p, \theta_y)$

■ Redundant robot

Current Folder

Command Window

```
>> T2 = transl(0.6, 0.2, 0) * rpy2tr(0, 180, 180, 'deg')
```

```
T2 =
```

```

    1.0000    0.0000    0.0000    0.6000
    0.0000   -1.0000         0    0.2000
    0.0000    0.0000   -1.0000         0
         0         0         0    1.0000

```

```
fx >> mdl_hyper3d
```

Workspace

Name ▲	Value	Class
T2	<4x4 double>	double

Command Window

```
>> T2 = transl(0.6, 0.2, 0) * rpy2tr(0, 180, 180, 'deg')
```

```
T2 =
```

```
    1.0000    0.0000    0.0000    0.6000
    0.0000   -1.0000         0    0.2000
    0.0000    0.0000   -1.0000         0
         0         0         0    1.0000
```

```
>> mdl_hyper3d(20)
```

```
/x>>
```

Workspace

Name *	Value	Class
T2	<4x4 double>	double
h3d	<1x1 SerialLink>	SerialLink
qz	<1x20 double>	double

Command Window

```
>> mdl_hyper3d(20)
```

```
>> h3d
```

```
h3d =
```

```
hyper3d (20 axis, RRRRRRRRRRRRRRRRRRRRRR, stdDH, fastRNE)
```

j	theta	d	a	alpha
1	q1	0	0.05	1.571
2	q2	0	0.05	0
3	q3	0	0.05	1.571
4	q4	0	0.05	0
5	q5	0	0.05	1.571
6	q6	0	0.05	0
7	q7	0	0.05	1.571
8	q8	0	0.05	0
9	q9	0	0.05	1.571
10	q10	0	0.05	0
11	q11	0	0.05	1.571
12	q12	0	0.05	0
13	q13	0	0.05	1.571
14	q14	0	0.05	0
15	q15	0	0.05	1.571
16	q16	0	0.05	0
17	q17	0	0.05	1.571
18	q18	0	0.05	0
19	q19	0	0.05	1.571
20	q20	0	0.05	0

```
grav = 0 base = 1 0 0 0 tool = 1 0 0 0
        0         0 1 0 0         0 1 0 0
        9.81      0 0 1 0         0 0 1 0
                0 0 0 1         0 0 0 1
```

f_x >> |

Workspace

Name	Value	Class
T2	<4x4 double>	double
h3d	<1x1 SerialLink>	SerialLink
qz	<1x20 double>	double

Command Window

```
>> mdl_hyper3d(20)
>> h3d
```

```
h3d =
```

```
hyper3d (20 axis, RRRRRRRRRRRRRRRRRRRR, stdDH, fastHNE)
```

j	theta	d	a	alpha
1	q1	0	0.05	1.571
2	q2	0	0.05	0
3	q3	0	0.05	1.571
4	q4	0	0.05	0
5	q5	0	0.05	1.571
6	q6	0	0.05	0
7	q7	0	0.05	1.571
8	q8	0	0.05	0
9	q9	0	0.05	1.571
10	q10	0	0.05	0
11	q11	0	0.05	1.571
12	q12	0	0.05	0
13	q13	0	0.05	1.571
14	q14	0	0.05	0
15	q15	0	0.05	1.571
16	q16	0	0.05	0
17	q17	0	0.05	1.571
18	q18	0	0.05	0
19	q19	0	0.05	1.571
20	q20	0	0.05	0

```
grav = 0 base = 1 0 0 0 tool = 1 0 0 0
0 0 1 0 0 0 1 0 0
9.81 0 0 1 0 0 0 1 0
0 0 0 1 0 0 0 0 1
```

```
/s >>
```

Workspace

Name *	Value	Class
T2	<4x4 double>	double
h3d	<1x1 SerialLink>	SerialLink
qz	<1x20 double>	double

Command Window

```

h3d =

hyper3d (20 axis, RRRRRRRRRRRRRRRRRRRR, stdDH, fastRNE)
    
```

j	theta	d	a	alpha
1	q1	0	0.05	1.571
2	q2	0	0.05	0
3	q3	0	0.05	1.571
4	q4	0	0.05	0
5	q5	0	0.05	1.571
6	q6	0	0.05	0
7	q7	0	0.05	1.571
8	q8	0	0.05	0
9	q9	0	0.05	1.571
10	q10	0	0.05	0
11	q11	0	0.05	1.571
12	q12	0	0.05	0
13	q13	0	0.05	1.571
14	q14	0	0.05	0
15	q15	0	0.05	1.571
16	q16	0	0.05	0
17	q17	0	0.05	1.571
18	q18	0	0.05	0
19	q19	0	0.05	1.571
20	q20	0	0.05	0

```

grav =      0  base = 1  0  0  0  tool = 1  0  0  0
           0          0  1  0  0         0  1  0  0
          9.81        0  0  1  0         0  0  1  0
                   0  0  0  1         0  0  0  1
    
```

```

fx >> q = h3d.ikine(T2, qz)
    
```

Workspace

Name	Value	Class
T2	<4x4 double>	double
h3d	<1x1 SerialLink>	SerialLink
qz	<1x20 double>	double

Command Window

h3d =

hyper3d (20 axis, RRRRRRRRRRRRRRRRRRRRRRR, stdDH, fastRNE)

j	theta	d	a	alpha
1	q1	0	0.05	1.571
2	q2	0	0.05	0
3	q3	0	0.05	1.571
4	q4	0	0.05	0
5	q5	0	0.05	1.571
6	q6	0	0.05	0
7	q7	0	0.05	1.571
8	q8	0	0.05	0
9	q9	0	0.05	1.571
10	q10	0	0.05	0
11	q11	0	0.05	1.571
12	q12	0	0.05	0
13	q13	0	0.05	1.571
14	q14	0	0.05	0
15	q15	0	0.05	1.571
16	q16	0	0.05	0
17	q17	0	0.05	1.571
18	q18	0	0.05	0
19	q19	0	0.05	1.571
20	q20	0	0.05	0

```
grav = 0 base = 1 0 0 0 tool = 1 0 0 0
0 0 1 0 0 0 1 0 0
9.81 0 0 1 0 0 0 1 0
0 0 0 1 0 0 0 0 1
```

fz >> q = h3d.ikine(T2, qz)

Workspace

Name *	Value	Class
T2	<4x4 double>	double
h3d	<1x1 SerialLink>	SerialLink
qz	<1x20 double>	double

Command Window

```

4|      q4|      0|      0.05|      0|
5|      q5|      0|      0.05|     1.571|
6|      q6|      0|      0.05|      0|
7|      q7|      0|      0.05|     1.571|
8|      q8|      0|      0.05|      0|
9|      q9|      0|      0.05|     1.571|
10|     q10|      0|      0.05|      0|
11|     q11|      0|      0.05|     1.571|
12|     q12|      0|      0.05|      0|
13|     q13|      0|      0.05|     1.571|
14|     q14|      0|      0.05|      0|
15|     q15|      0|      0.05|     1.571|
16|     q16|      0|      0.05|      0|
17|     q17|      0|      0.05|     1.571|
18|     q18|      0|      0.05|      0|
19|     q19|      0|      0.05|     1.571|
20|     q20|      0|      0.05|      0|

```

```

grav =      0  base = 1  0  0  0  tool = 1  0  0  0
           0          0  1  0  0      0  1  0  0
          9.81        0  0  1  0      0  0  1  0
                   0  0  0  1      0  0  0  1

```

```
>> q = h3d.ikine(T2, qz)
```

Warning: Initial joint configuration results in a (near-)singular configuration, this may slow convergence

> In SerialLink.ikine at 152

```
q =
```

Columns 1 through 17

```
-0.0446 -0.0000 -0.0000 -2.7354  2.0571  0.0000  0.0000 -2.4222  2.4302 -0.0000  0.0000
```

Columns 18 through 20

```
0.0000  0.0000 -0.0446
```

Workspace

Name	Value	Class
T2	<4x4 double>	double
h3d	<1x1 SerialLink>	SerialLink
q	<1x20 double>	double
qz	<1x20 double>	double

Command Window

4	q4	0	0.05	0
5	q5	0	0.05	1.571
6	q6	0	0.05	0
7	q7	0	0.05	1.571
8	q8	0	0.05	0
9	q9	0	0.05	1.571
10	q10	0	0.05	0
11	q11	0	0.05	1.571
12	q12	0	0.05	0
13	q13	0	0.05	1.571
14	q14	0	0.05	0
15	q15	0	0.05	1.571
16	q16	0	0.05	0
17	q17	0	0.05	1.571
18	q18	0	0.05	0
19	q19	0	0.05	1.571
20	q20	0	0.05	0

```

grav =    0  base = 1  0  0  0  tool = 1  0  0  0
          0          0  1  0  0          0  1  0  0
        9.81        0  0  1  0          0  0  1  0
                  0  0  0  1          0  0  0  1

```

```
>> q = h3d.ikine(T2, qz)
```

Warning: Initial joint configuration results in a (near-)singular configuration, this may slow convergence
> In SerialLink.ikine at 152

```
q =
```

Columns 1 through 17

```
-0.0446 -0.0000 -0.0000 -2.7354  2.0571  0.0000  0.0000 -2.4222  2.4302 -0.0000  0.0000
```

Columns 18 through 20

```
0.0000  0.0000 -0.0446
```

```
>> h3d.fkine(q)
```

Workspace

Name	Value	Class
T2	<4x4 double>	double
h3d	<1x1 SerialLink>	SerialLink
q	<1x20 double>	double
qz	<1x20 double>	double

Command Window

```
grav =    0   base = 1   0   0   0   tool = 1   0   0   0
          0         0   1   0   0         0   1   0   0
        9.81       0   0   1   0         0   0   1   0
              0   0   0   1         0   0   0   1
```

```
>> q = h3d.ikine(T2, qz)
```

Warning: Initial joint configuration results in a (near-)singular configuration, this may slow convergence
> In SerialLink.ikine at 152

```
q =
```

Columns 1 through 17

```
-0.0446   -0.0000   -0.0000   -2.7354    2.0571    0.0000    0.0000   -2.4222    2.4302   -0.0000    0.0000
```

Columns 18 through 20

```
0.0000    0.0000   -0.0446
```

```
>> h3d.fkine(q)
```

```
ans =
```

```
1.0000    0.0000    0.0000    0.6000
0.0000   -1.0000    0.0000    0.2000
0.0000   -0.0000   -1.0000   -0.0000
      0         0         0         1.0000
```

```
>> T2
```

```
T2 =
```

```
1.0000    0.0000    0.0000    0.6000
0.0000   -1.0000         0    0.2000
0.0000    0.0000   -1.0000         0
      0         0         0         1.0000
```

fx >>

Workspace

Name	Value	Class
T2	<4x4 double>	double
ans	<4x4 double>	double
h3d	<1x1 SerialLink>	SerialLink
q	<1x20 double>	double
qz	<1x20 double>	double

Command Window

```
1.0000 0.0000 0.0000 0.0000
0.0000 -1.0000 0.0000 0.2000
0.0000 -0.0000 -1.0000 -0.0000
0 0 0 1.0000
```

```
>> T2
```

```
T2 =
```

```
1.0000 0.0000 0.0000 0.6000
0.0000 -1.0000 0 0.2000
0.0000 0.0000 -1.0000 0
0 0 0 1.0000
```

```
>> h3d.plot(q)
```

```
fx >>
```

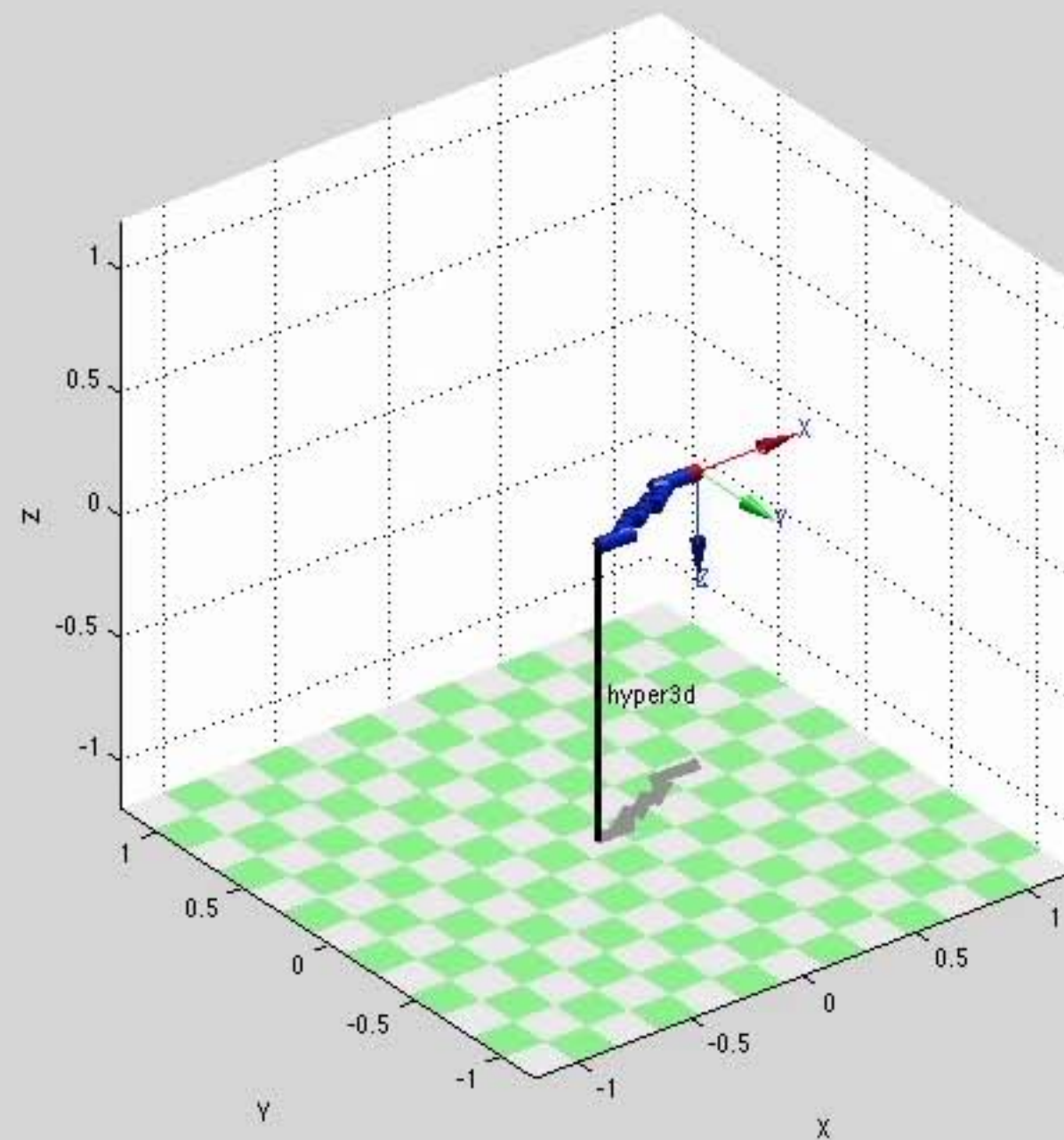
Workspace

Name	Value	Class
T2	<4x4 double>	double
ans	<4x4 double>	double
h3d	<1x1 SerialLink>	SerialLink
q	<1x20 double>	double
qz	<1x20 double>	double

Current Folder

Figures

Figure 1



Command Window

```
0.0000 -0.0000 -1.0000 -0.0000
0 0 0 1.0000
```

```
>> T2
```

```
T2 =
```

```
1.0000 0.0000 0.0000 0.6000
0.0000 -1.0000 0 0.2000
0.0000 0.0000 -1.0000 0
0 0 0 1.0000
```

```
>> h3d.plot(q)
```

```
>> hold on
```

```
>> trplot(T2)
```

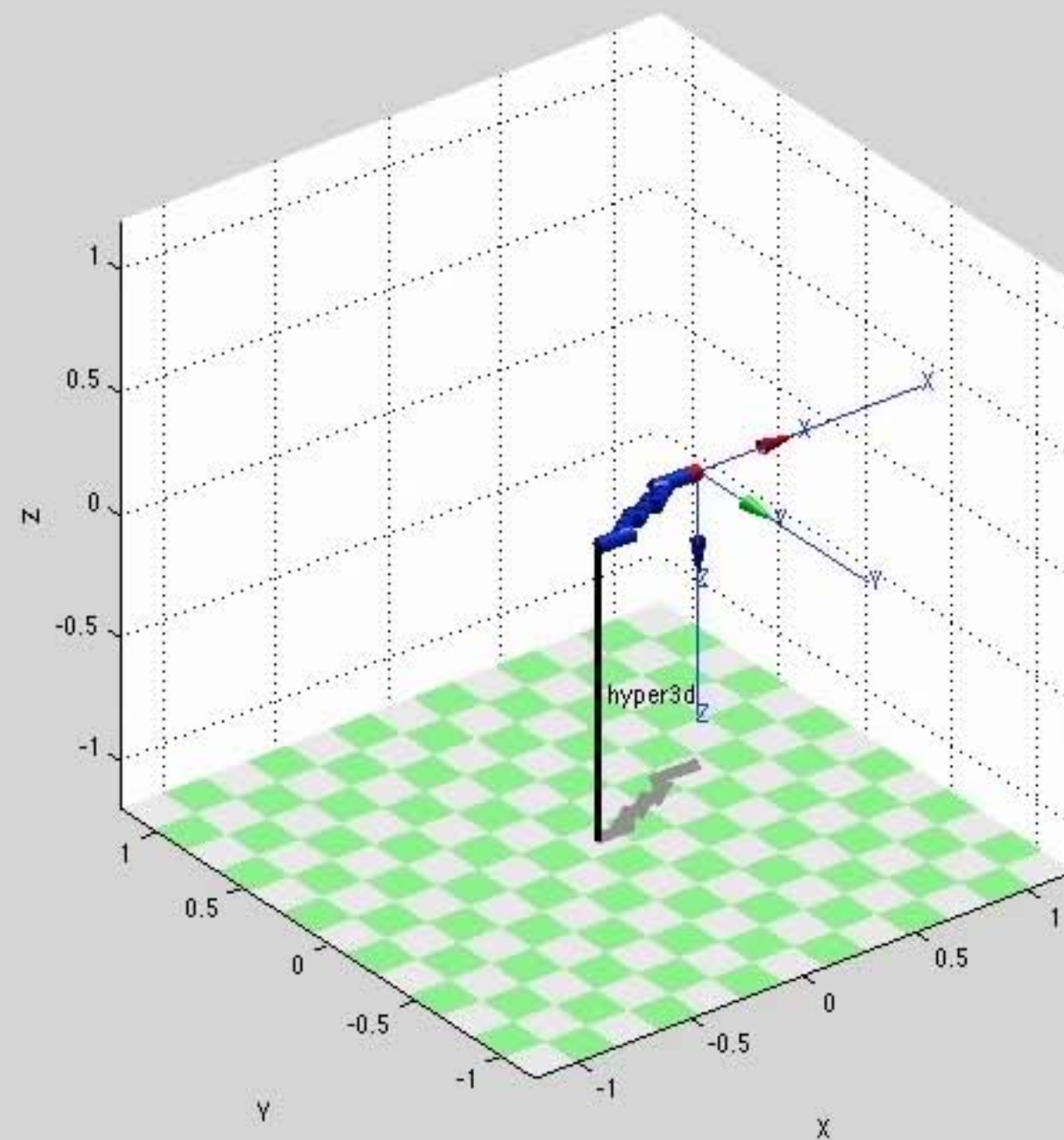
```
>> |
```

Workspace

Name	Value	Class
T2	<4x4 double>	double
ans	<4x4 double>	double
h3d	<1x1 SerialLink>	SerialLink
q	<1x20 double>	double
qz	<1x20 double>	double

Figures

Figure 1



Section10 Joint interpolated motion

Making smooth paths

- We wish to move from pose A to pose B $\xi_A \rightarrow \xi_B$

Making **smooth paths**

- We wish to move from pose A to pose B $\xi_A \rightarrow \xi_B$
- We can determine the joint angles at each pose

$$\mathbf{q}_A = \mathcal{K}^{-1}(\xi_A)$$

$$\mathbf{q}_B = \mathcal{K}^{-1}(\xi_B)$$

Making **smooth** paths

- We wish to move from pose A to pose B $\xi_A \rightarrow \xi_B$
- We can determine the joint angles at each pose

$$\mathbf{q}_A = \mathcal{K}^{-1}(\xi_A)$$

$$\mathbf{q}_B = \mathcal{K}^{-1}(\xi_B)$$

- Now we need to move **smoothly** $\mathbf{q}_A \rightarrow \mathbf{q}_B$ $\mathbf{q}_A, \mathbf{q}_B \in \mathbb{R}^6$
- We've previously discussed how to do this

Command Window

```
>> mdl_puma560
>> TA = transl(0.4, 0.2, 0) * trotx(pi)
```

TA =

```

    1.0000         0         0    0.4000
         0   -1.0000   -0.0000    0.2000
         0    0.0000   -1.0000         0
         0         0         0    1.0000
```

fx >> |

I

Workspace

Name ▲	Value	Class
TA	<4x4 double>	double
deg	0.0175	double
p560	<1x1 SerialLink>	SerialLink
qn	[0,0.7854,3.1416,0,0.78...	double
qr	[0,1.5708,-1.5708,0,0,0]	double
qs	[0,0,-1.5708,0,0,0]	double
qz	[0,0,0,0,0,0]	double

Command Window

```
>> mdl_puma560
>> TA = transl(0.4, 0.2, 0) * trotx(pi)
```

TA =

```
    1.0000    0    0    0.4000
         0   -1.0000   -0.0000    0.2000
         0    0.0000   -1.0000    0
         0    0    0    1.0000
```

```
>> TB = transl(0.4, -0.2, 0) * trotx(pi/2)
```

TB =

```
    1.0000    0    0    0.4000
         0    0.0000   -1.0000   -0.2000
         0    1.0000    0.0000    0
         0    0    0    1.0000
```

f_x >> |

Workspace

Name ▲	Value	Class
TA	<4x4 double>	double
TB	<4x4 double>	double
deg	0.0175	double
p560	<1x1 SerialLink>	SerialLink
qn	[0,0.7854,3.1416,0,0.78...	double
qr	[0,1.5708,-1.5708,0,0,0]	double
qs	[0,0,-1.5708,0,0,0]	double
qz	[0,0,0,0,0,0]	double

Command Window

```
>> mdl_puma560
>> TA = transl(0.4, 0.2, 0) * trotx(pi)

TA =

    1.0000         0         0    0.4000
         0   -1.0000   -0.0000    0.2000
         0    0.0000   -1.0000         0
         0         0         0    1.0000

>> TB = transl(0.4, -0.2, 0) * trotx(pi/2)

TB =

    1.0000         0         0    0.4000
         0    0.0000   -1.0000   -0.2000
         0    1.0000    0.0000         0
         0         0         0    1.0000

>> qA = p560.ikine6s(TA)

qA =

    3.2631    2.0791    0.5992   -0.0000    0.4633    0.1215
```

fx >> |

Workspace

Name ▲	Value	Class
TA	<4x4 double>	double
TB	<4x4 double>	double
deg	0.0175	double
p560	<1x1 SerialLink>	SerialLink
qA	[3.2631,2.0791,0.5992,-...	double
qn	[0,0.7854,3.1416,0,0.78...	double
qr	[0,1.5708,-1.5708,0,0,0]	double
qs	[0,0,-1.5708,0,0,0]	double
qz	[0,0,0,0,0,0]	double

Command Window

```
>> mdl_puma560
>> TA = transl(0.4, 0.2, 0) * trotx(pi)

TA =

    1.0000    0    0    0.4000
         0   -1.0000   -0.0000    0.2000
         0    0.0000   -1.0000    0
         0    0    0    1.0000

>> TB = transl(0.4, -0.2, 0) * trotx(pi/2)

TB =

    1.0000    0    0    0.4000
         0    0.0000   -1.0000   -0.2000
         0    1.0000    0.0000    0
         0    0    0    1.0000

>> qA = p560.ikine6s(TA)

qA =

    3.2631    2.0791    0.5992   -0.0000    0.4633    0.1215

>> qB = p560.ikine6s(TB)

qB =

    2.3358    2.0791    0.5992   -2.3209    1.2425    1.2377

fx >> |
```

Workspace

Name	Value	Class
TA	<4x4 double>	double
TB	<4x4 double>	double
deg	0.0175	double
p560	<1x1 SerialLink>	SerialLink
qA	[3.2631,2.0791,0.5992,-...	double
qB	[2.3358,2.0791,0.5992,-...	double
qn	[0,0.7854,3.1416,0,0.78...	double
qr	[0,1.5708,-1.5708,0,0,0]	double
qs	[0,0,-1.5708,0,0,0]	double
qz	[0,0,0,0,0,0]	double

Command Window

```
>> mdl_puma560
>> TA = transl(0.4, 0.2, 0) * trotx(pi)

TA =

    1.0000    0    0    0.4000
         0   -1.0000   -0.0000    0.2000
         0    0.0000   -1.0000    0
         0    0    0    1.0000

>> TB = transl(0.4, -0.2, 0) * trotx(pi/2)

TB =

    1.0000    0    0    0.4000
         0    0.0000   -1.0000   -0.2000
         0    1.0000    0.0000    0
         0    0    0    1.0000

>> qA = p560.ikine6s(TA)

qA =

    3.2631    2.0791    0.5992   -0.0000    0.4633    0.1215

>> qB = p560.ikine6s(TB)

qB =

    2.3358    2.0791    0.5992   -2.3209    1.2425    1.2377

fx >> tg = jtraj(qA, qB, 50);
```

Workspace

Name	Value	Class
TA	<4x4 double>	double
TB	<4x4 double>	double
deg	0.0175	double
p560	<1x1 SerialLink>	SerialLink
qA	[3.2631,2.0791,0.5992,-...	double
qB	[2.3358,2.0791,0.5992,-...	double
qn	[0,0.7854,3.1416,0,0.78...	double
qr	[0,1.5708,-1.5708,0,0,0]	double
qs	[0,0,-1.5708,0,0,0]	double
qz	[0,0,0,0,0,0]	double

Command Window

```
>> qA = p560.ikine6s(TA)

qA =

    3.2631    2.0791    0.5992   -0.0000    0.4633    0.1215

>> qB = p560.ikine6s(TB)

qB =

    2.3358    2.0791    0.5992   -2.3209    1.2425    1.2377

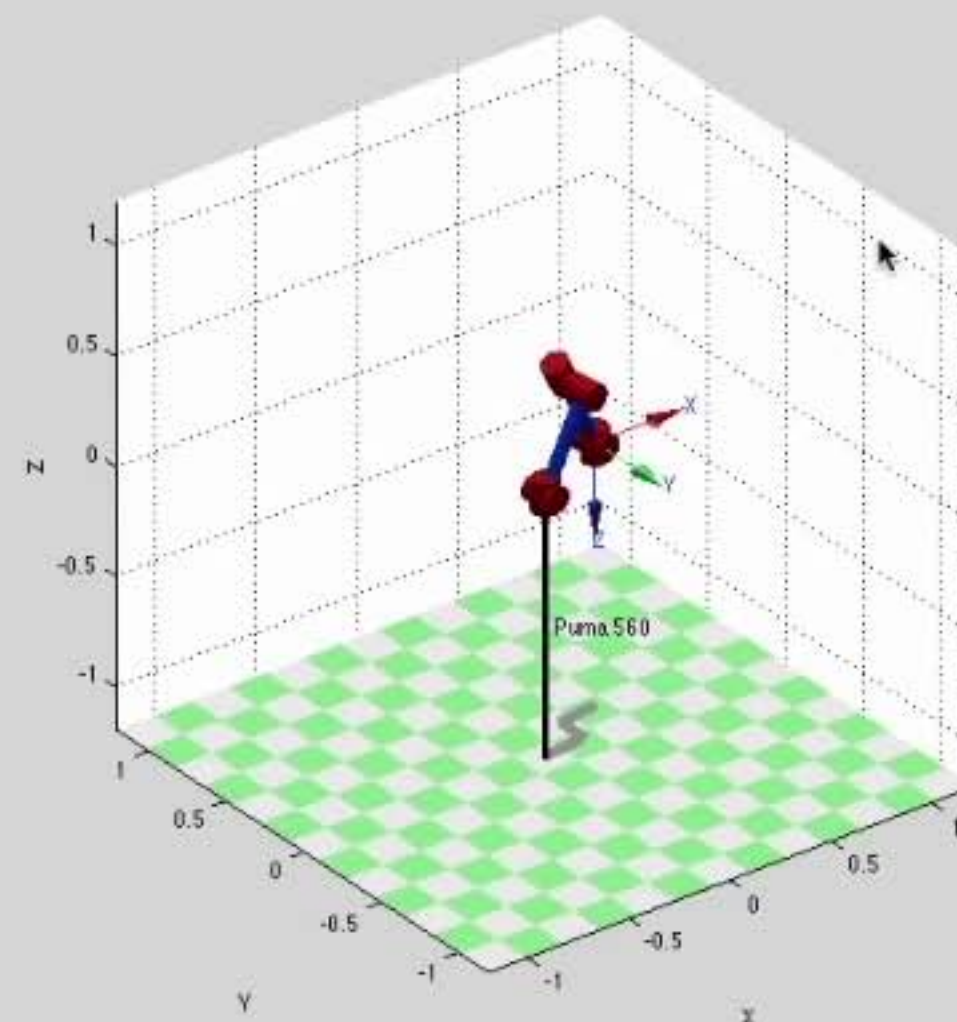
>> tg = jtraj(qA, qB, 50);
>> p560.plot(tg)
```

fx

Workspace

Name	Value	Class
TA	<4x4 double>	double
TB	<4x4 double>	double
deg	0.0175	double
p560	<1x1 SerialLink>	SerialLink
qA	[3.2631,2.0791,0.5992,-...	double
qB	[2.3358,2.0791,0.5992,-...	double
qn	[0,0.7854,3.1416,0,0.78...	double
qr	[0,1.5708,-1.5708,0,0,0]	double
qs	[0,0,-1.5708,0,0,0]	double
qz	[0,0,0,0,0,0]	double
tg	<50x6 double>	double

Figure 1



Command Window

```
>> qA = p560.ikine6s(TA)

qA =

    3.2631    2.0791    0.5992   -0.0000    0.4633    0.1215

>> qB = p560.ikine6s(TB)

qB =

    2.3358    2.0791    0.5992   -2.3209    1.2425    1.2377

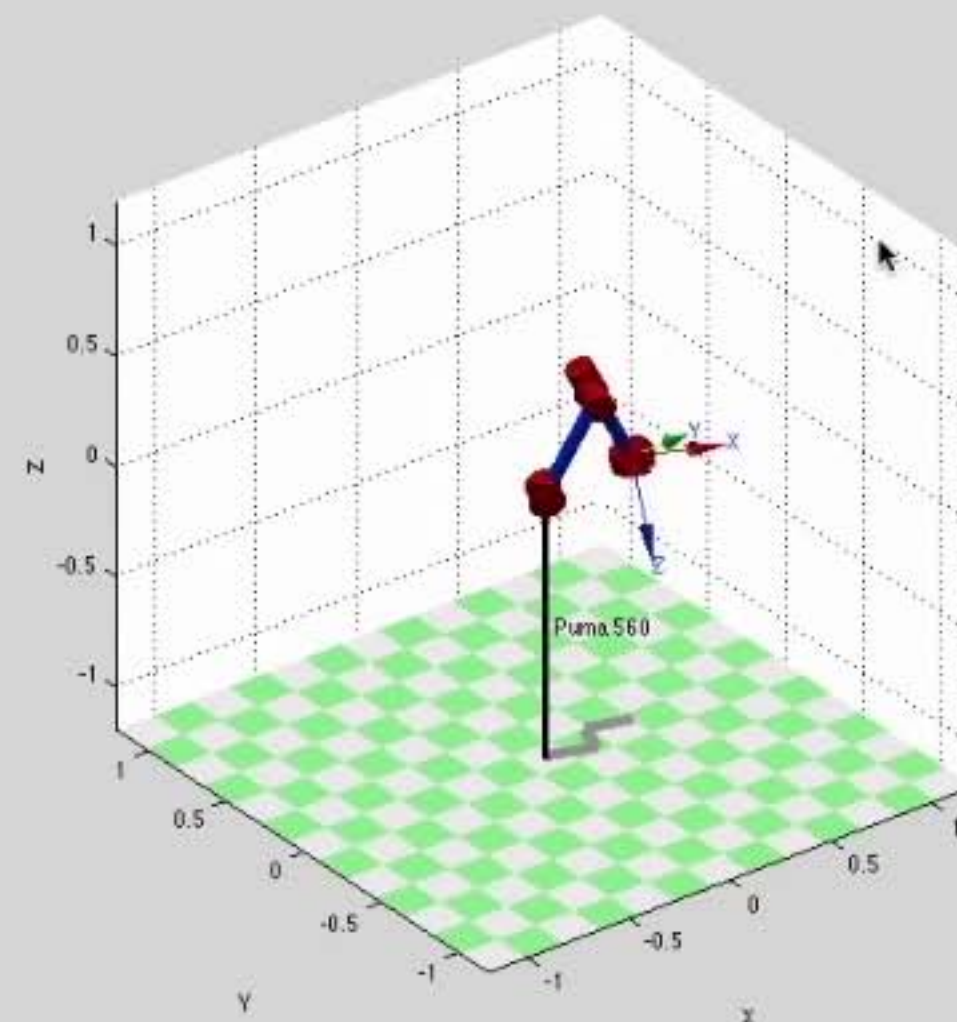
>> tg = jtraj(qA, qB, 50);
>> p560.plot(tg)
```

fx

Workspace

Name	Value	Class
TA	<4x4 double>	double
TB	<4x4 double>	double
deg	0.0175	double
p560	<1x1 SerialLink>	SerialLink
qA	[3.2631,2.0791,0.5992,-...	double
qB	[2.3358,2.0791,0.5992,-...	double
qn	[0,0.7854,3.1416,0,0.78...	double
qr	[0,1.5708,-1.5708,0,0,0]	double
qs	[0,0,-1.5708,0,0,0]	double
qz	[0,0,0,0,0,0]	double
tg	<50x6 double>	double

Figure 1



Command Window

```
>> qA = p560.ikine6s(TA)

qA =

    3.2631    2.0791    0.5992   -0.0000    0.4633    0.1215

>> qB = p560.ikine6s(TB)

qB =

    2.3358    2.0791    0.5992   -2.3209    1.2425    1.2377

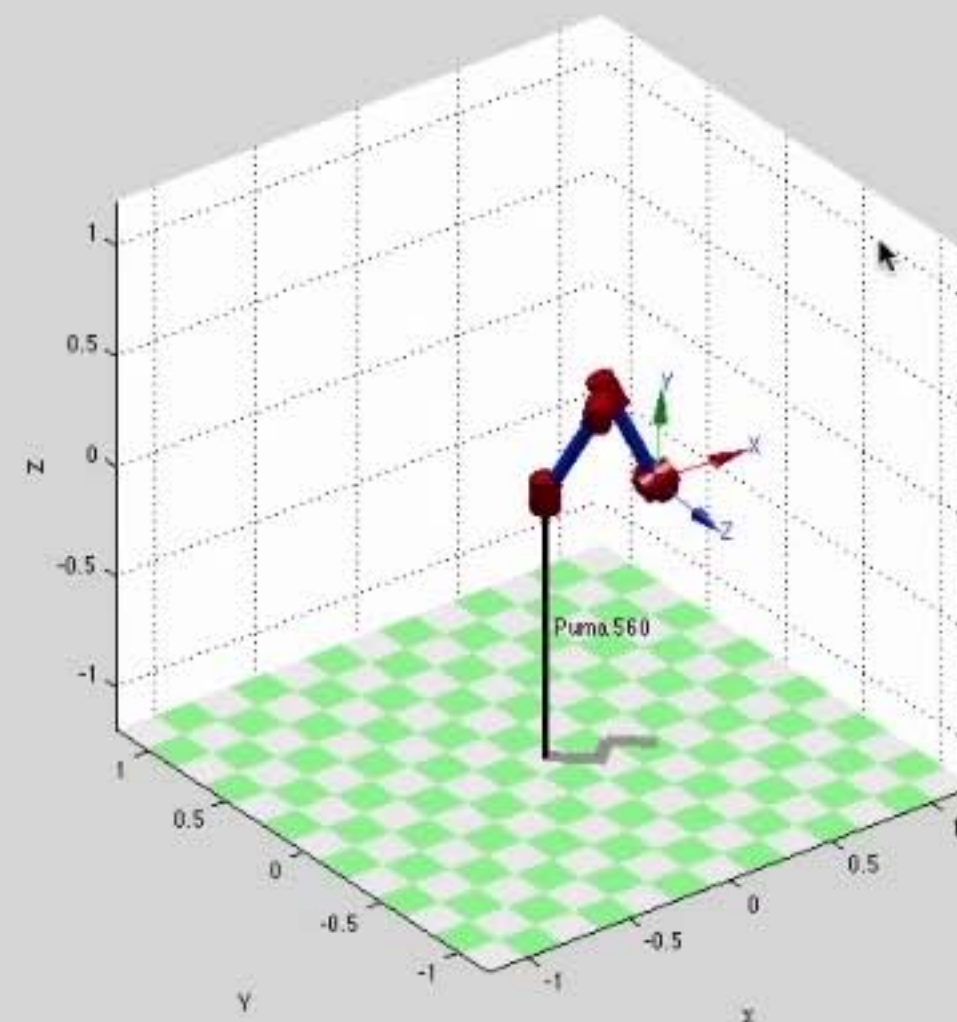
>> tg = jtraj(qA, qB, 50);
>> p560.plot(tg)
```

fx

Workspace

Name	Value	Class
TA	<4x4 double>	double
TB	<4x4 double>	double
deg	0.0175	double
p560	<1x1 SerialLink>	SerialLink
qA	[3.2631,2.0791,0.5992,-...	double
qB	[2.3358,2.0791,0.5992,-...	double
qn	[0,0.7854,3.1416,0,0.78...	double
qr	[0,1.5708,-1.5708,0,0,0]	double
qs	[0,0,-1.5708,0,0,0]	double
qz	[0,0,0,0,0,0]	double
tg	<50x6 double>	double

Figure 1



Command Window

```
>> qA = p560.ikine6s(TA)

qA =

    3.2631    2.0791    0.5992   -0.0000    0.4633    0.1215

>> qB = p560.ikine6s(TB)

qB =

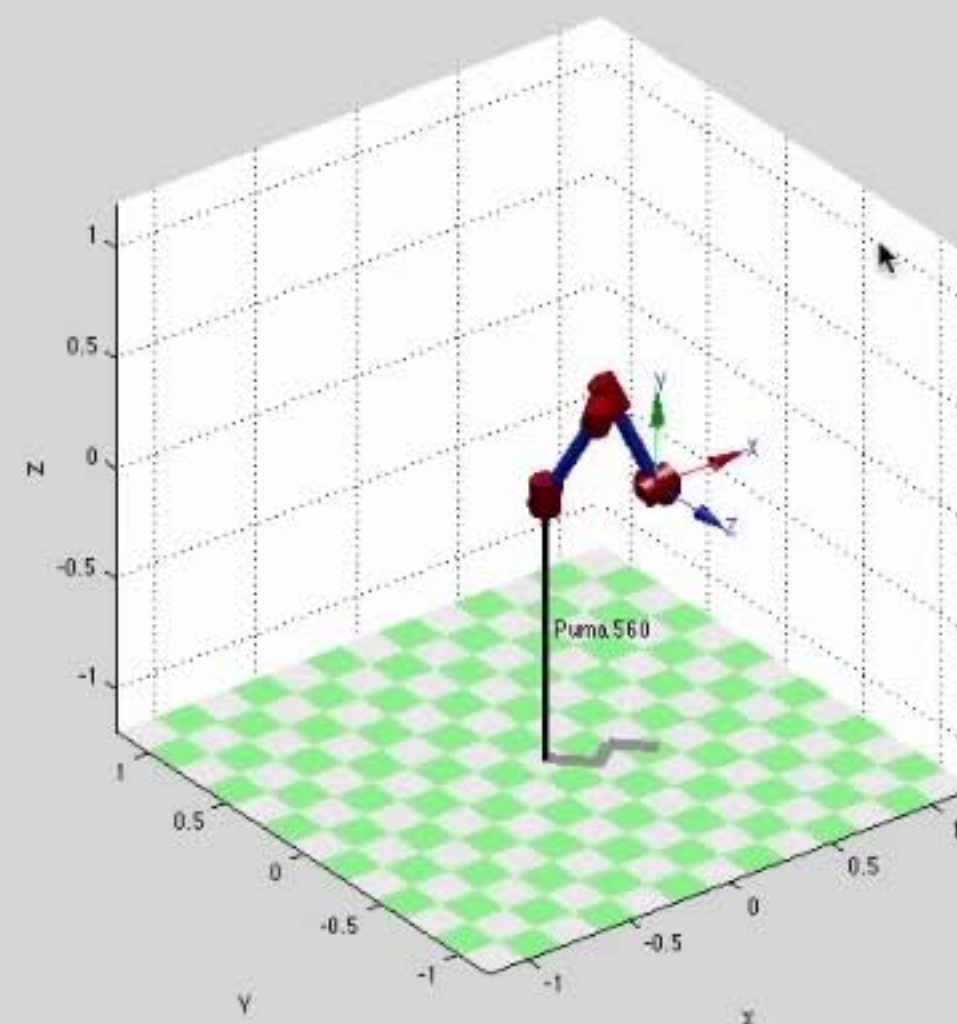
    2.3358    2.0791    0.5992   -2.3209    1.2425    1.2377

>> tg = jtraj(qA, qB, 50);
>> p560.plot(tg)
fx >>
```

Workspace

Name	Value	Class
TA	<4x4 double>	double
TB	<4x4 double>	double
deg	0.0175	double
p560	<1x1 SerialLink>	SerialLink
qA	[3.2631,2.0791,0.5992,-...	double
qB	[2.3358,2.0791,0.5992,-...	double
qn	[0,0.7854,3.1416,0,0.78...	double
qr	[0,1.5708,-1.5708,0,0,0]	double
qs	[0,0,-1.5708,0,0,0]	double
qz	[0,0,0,0,0,0]	double
tg	<50x6 double>	double

Figure 1



Command Window

```
qA =
    3.2631    2.0791    0.5992   -0.0000    0.4633    0.1215

>> qB = p560.ikine6s(TB)

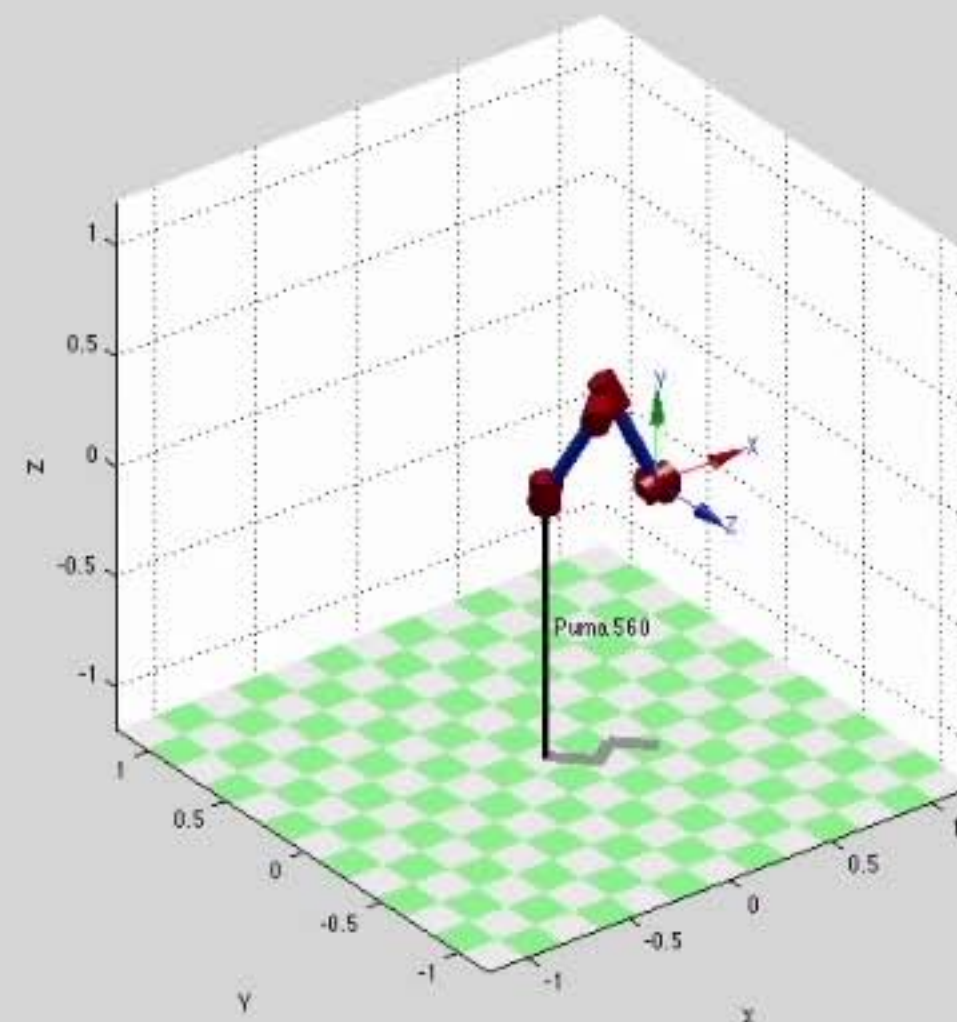
qB =
    2.3358    2.0791    0.5992   -2.3209    1.2425    1.2377

>> tg = jtraj(qA, qB, 50);
>> p560.plot(tg)
>> figure
>> qplot(tg)
fx >> |
```

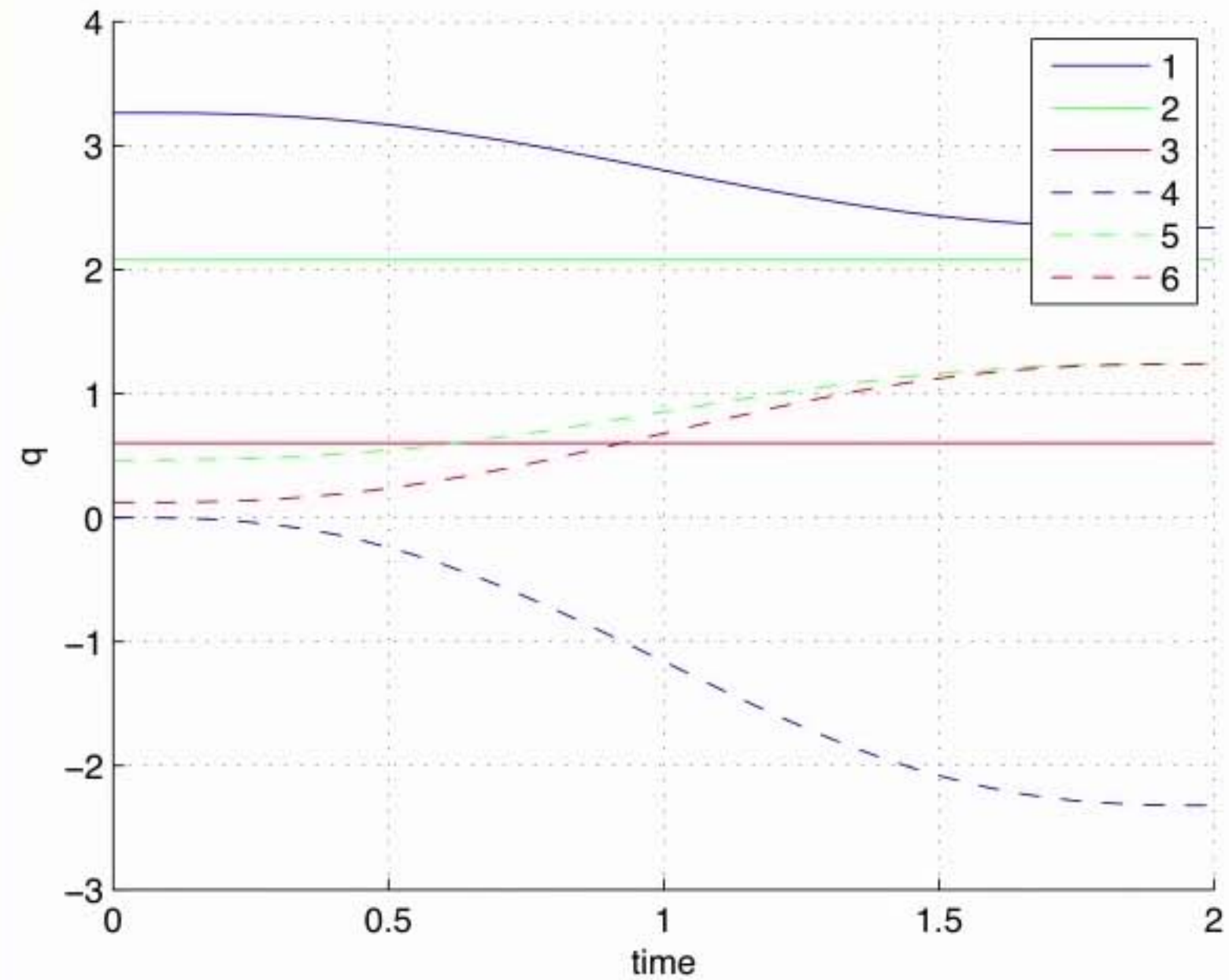
Workspace

Name	Value	Class
TA	<4x4 double>	double
TB	<4x4 double>	double
deg	0.0175	double
p560	<1x1 SerialLink>	SerialLink
qA	[3.2631,2.0791,0.5992,-...	double
qB	[2.3358,2.0791,0.5992,-...	double
qn	[0,0.7854,3.1416,0,0.78...	double
qr	[0,1.5708,-1.5708,0,0,0]	double
qs	[0,0,-1.5708,0,0,0]	double
qz	[0,0,0,0,0,0]	double
tg	<50x6 double>	double

Figure 1

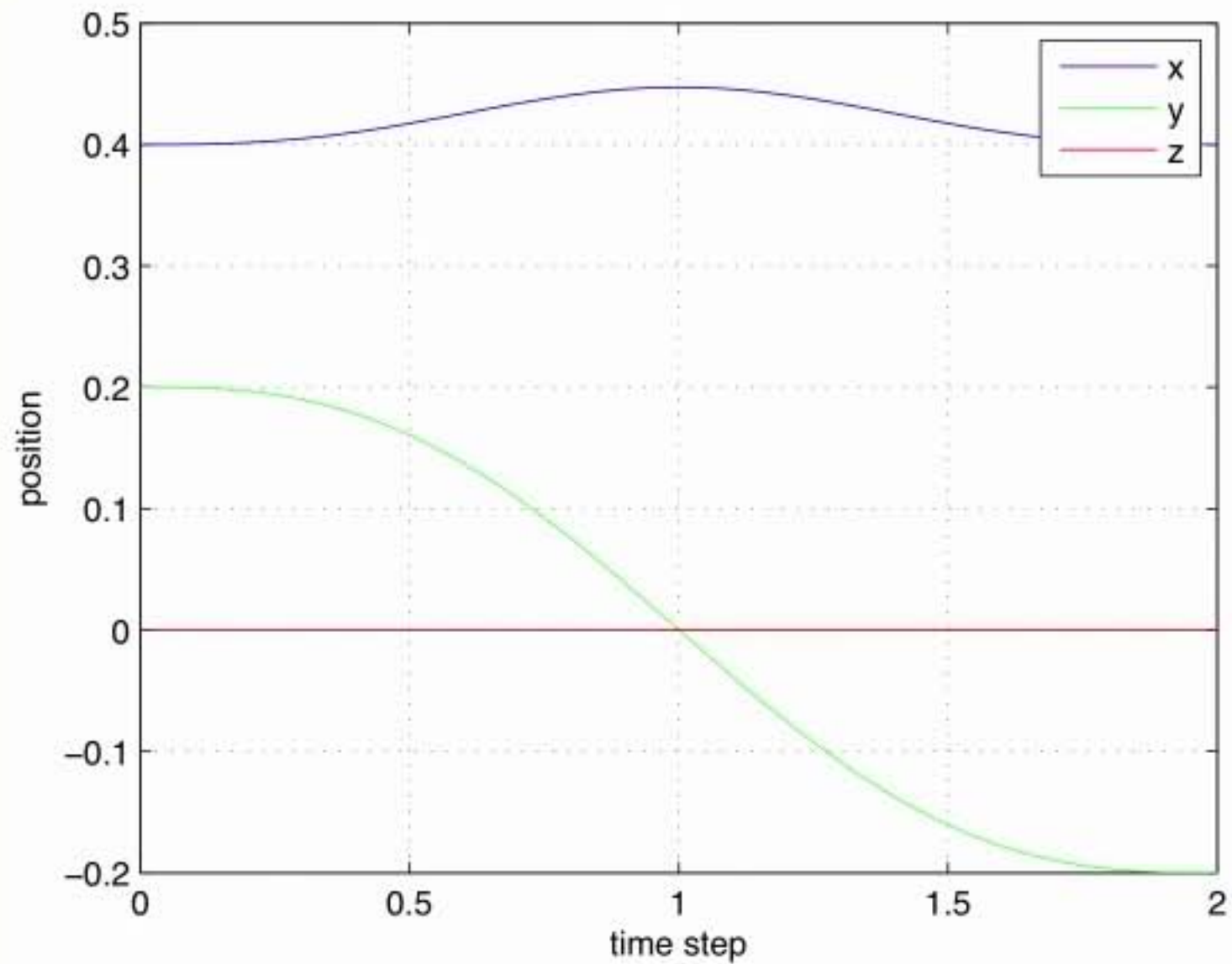


Joint angle trajectories



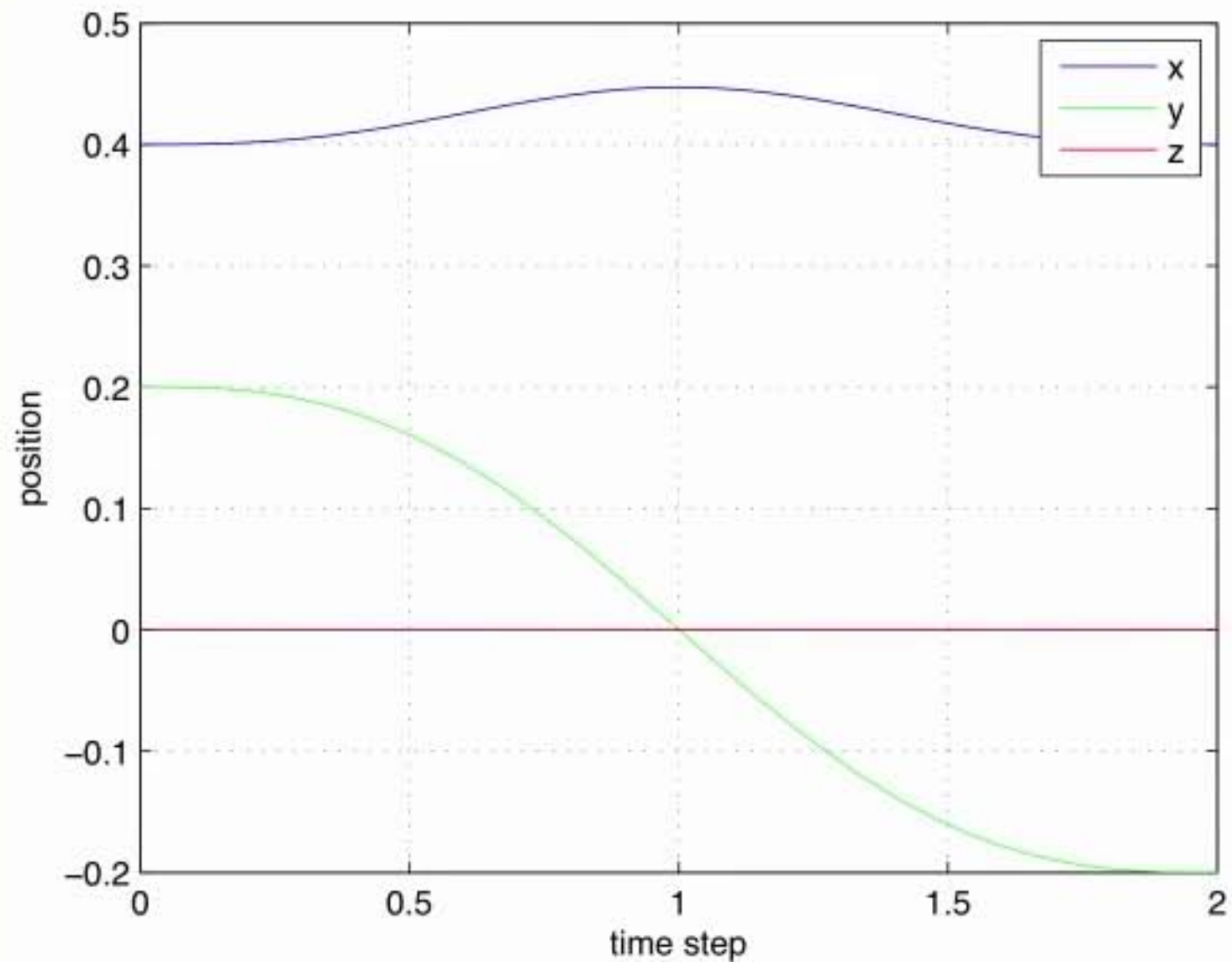
With kind permission of Springer Science+Business Media

Cartesian position

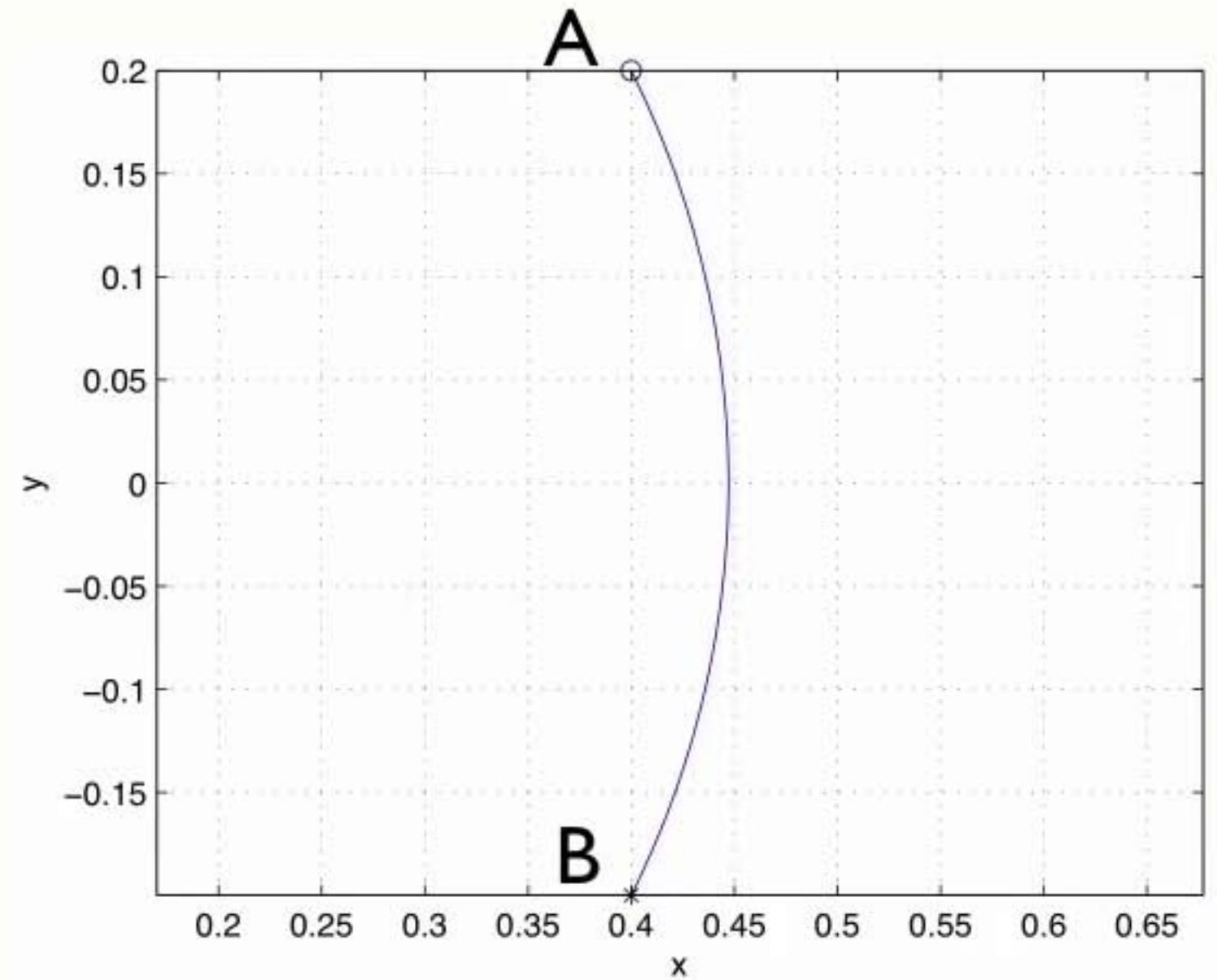


With kind permission of Springer Science+Business Media

Cartesian position



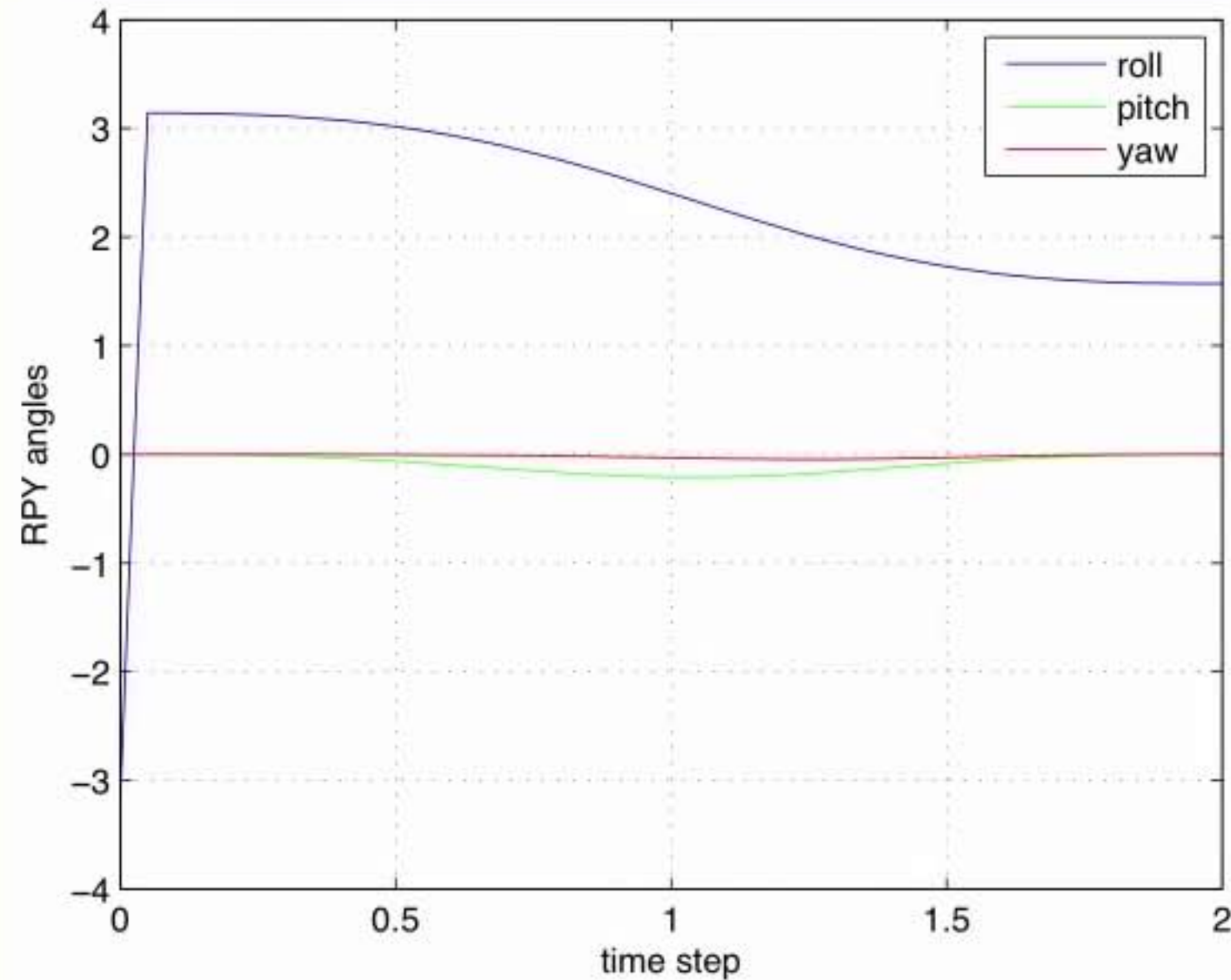
With kind permission of Springer Science+Business Media



With kind permission of Springer Science+Business Media

■ The path in the x-y plane is not straight

Cartesian **orientation**



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- The tool orientation changes along the path

Section 11 Cartesian interpolated motion

Making smooth paths

- We wish to move from pose A to pose B $\xi_A \rightarrow \xi_B$

Making **smooth paths**

- We wish to move from pose A to pose B $\xi_A \rightarrow \xi_B$
- We determine the pose at each intermediate point

$$\xi_s = f(\xi_A, \xi_B, s), \quad 0 \leq s \leq 1$$



s
varies smoothly
from 0 to 1

Making **smooth paths**

- We wish to move from pose A to pose B $\xi_A \rightarrow \xi_B$
- We determine the pose at each intermediate point

$$\xi_s = f(\xi_A, \xi_B, s), \quad 0 \leq s \leq 1$$



s
varies smoothly
from 0 to 1

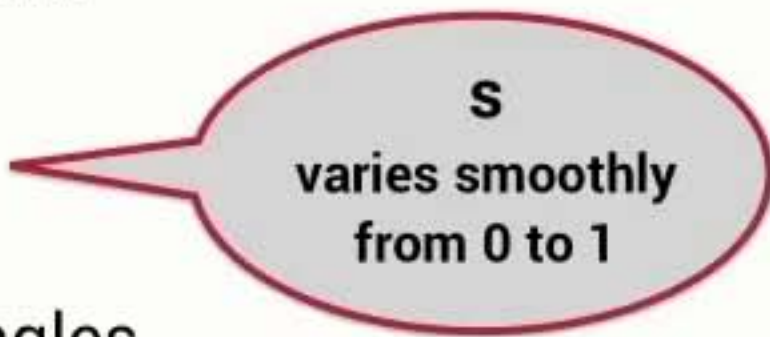
- Then use inverse kinematics to find the joint angles

$$\mathbf{q}_s = \mathcal{K}^{-1}(\xi_s)$$

Making **smooth paths**

- We wish to move from pose A to pose B $\xi_A \rightarrow \xi_B$
- We determine the pose at each intermediate point

$$\xi_s = f(\xi_A, \xi_B, s), \quad 0 \leq s \leq 1$$

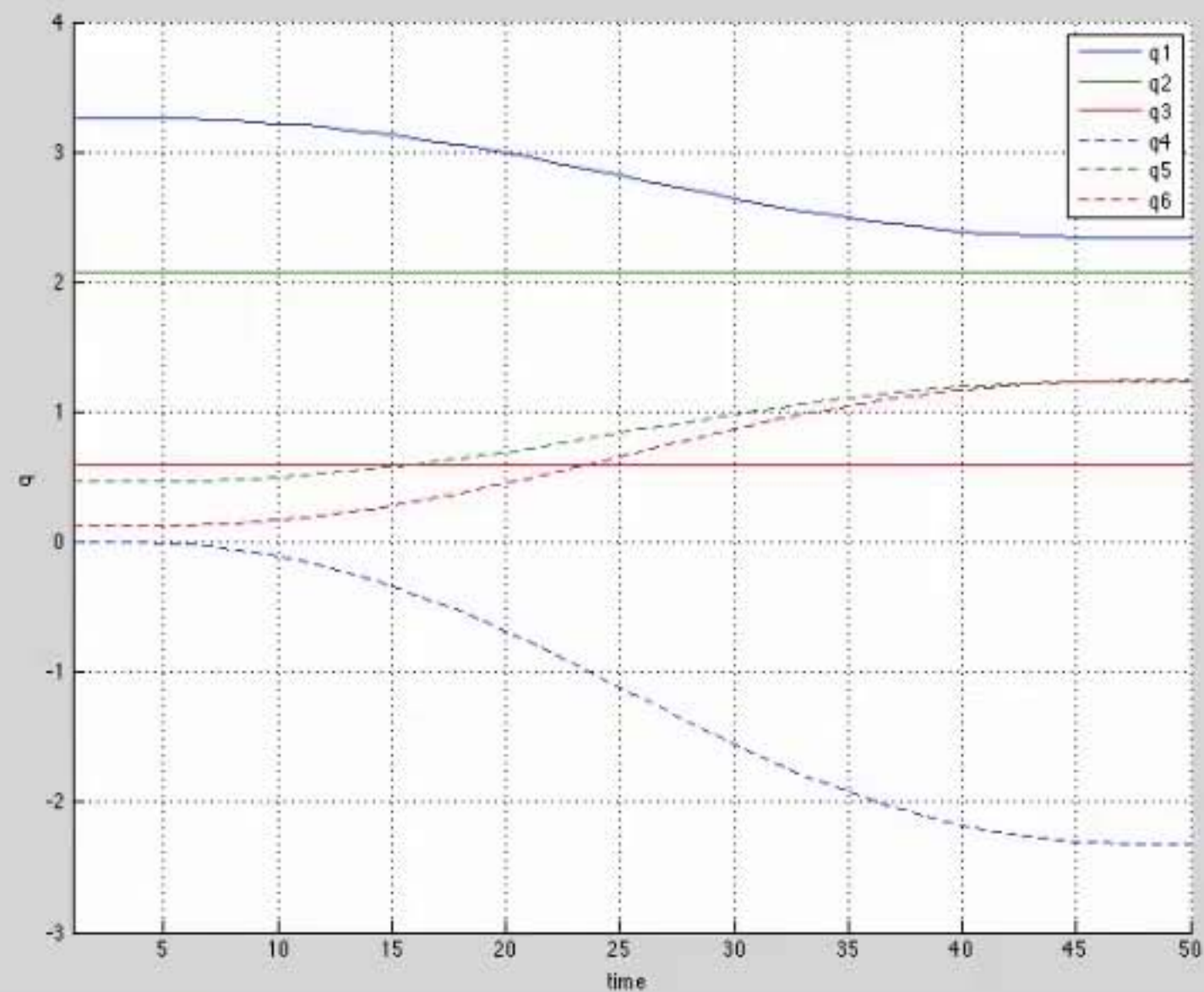
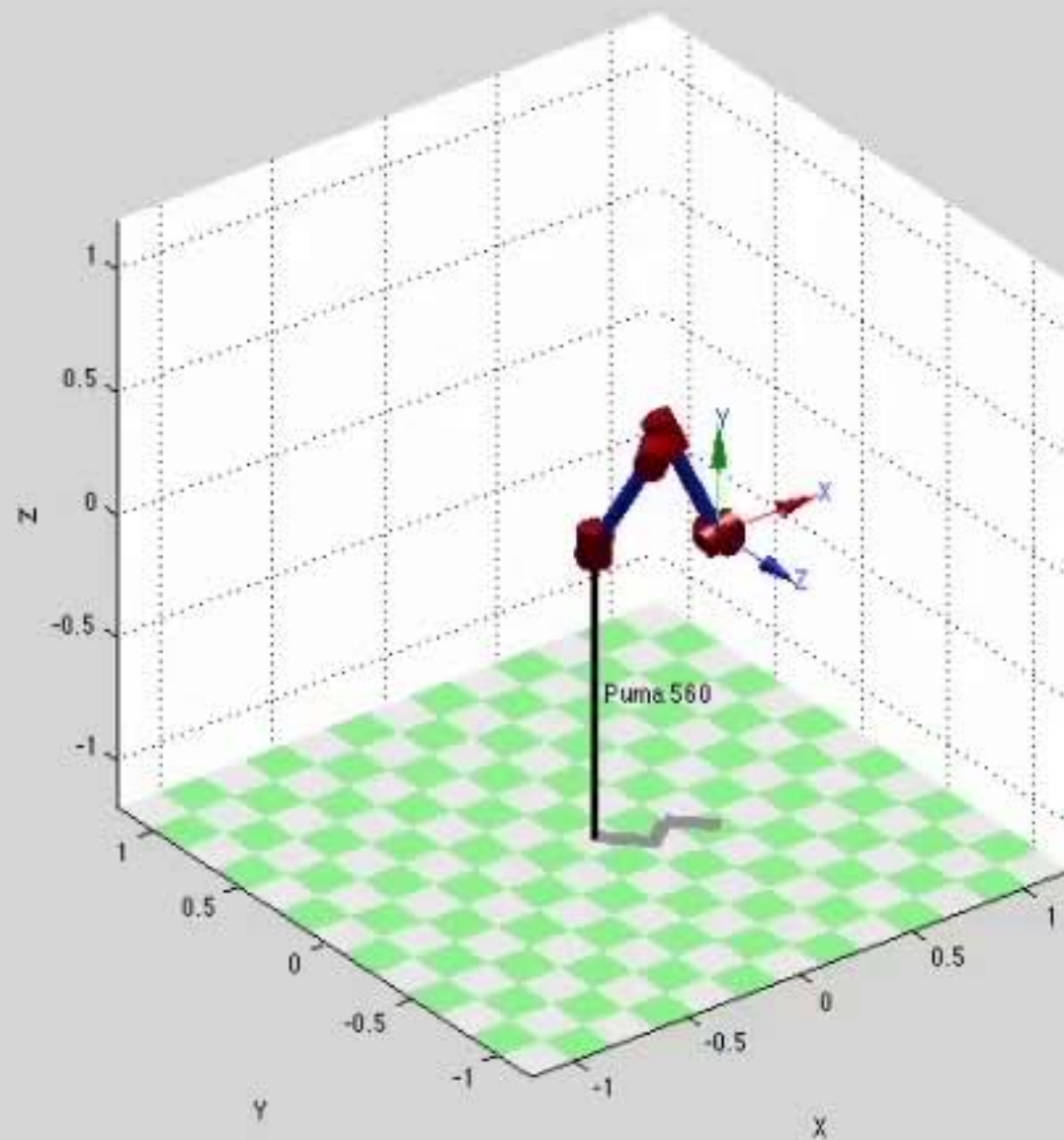


s
varies smoothly
from 0 to 1

- Then use inverse kinematics to find the joint angles

$$q_s = \mathcal{K}^{-1}(\xi_s)$$

- At every time step we
 - ➡ interpolate the pose
 - ➡ perform inverse kinematics



Command Window

```

qA =
    3.2631    2.0791    0.5992   -0.0000    0.4633    0.1215

>> qB = p560.ikine6s(TB)

qB =
    2.3358    2.0791    0.5992   -2.3209    1.2425    1.2377

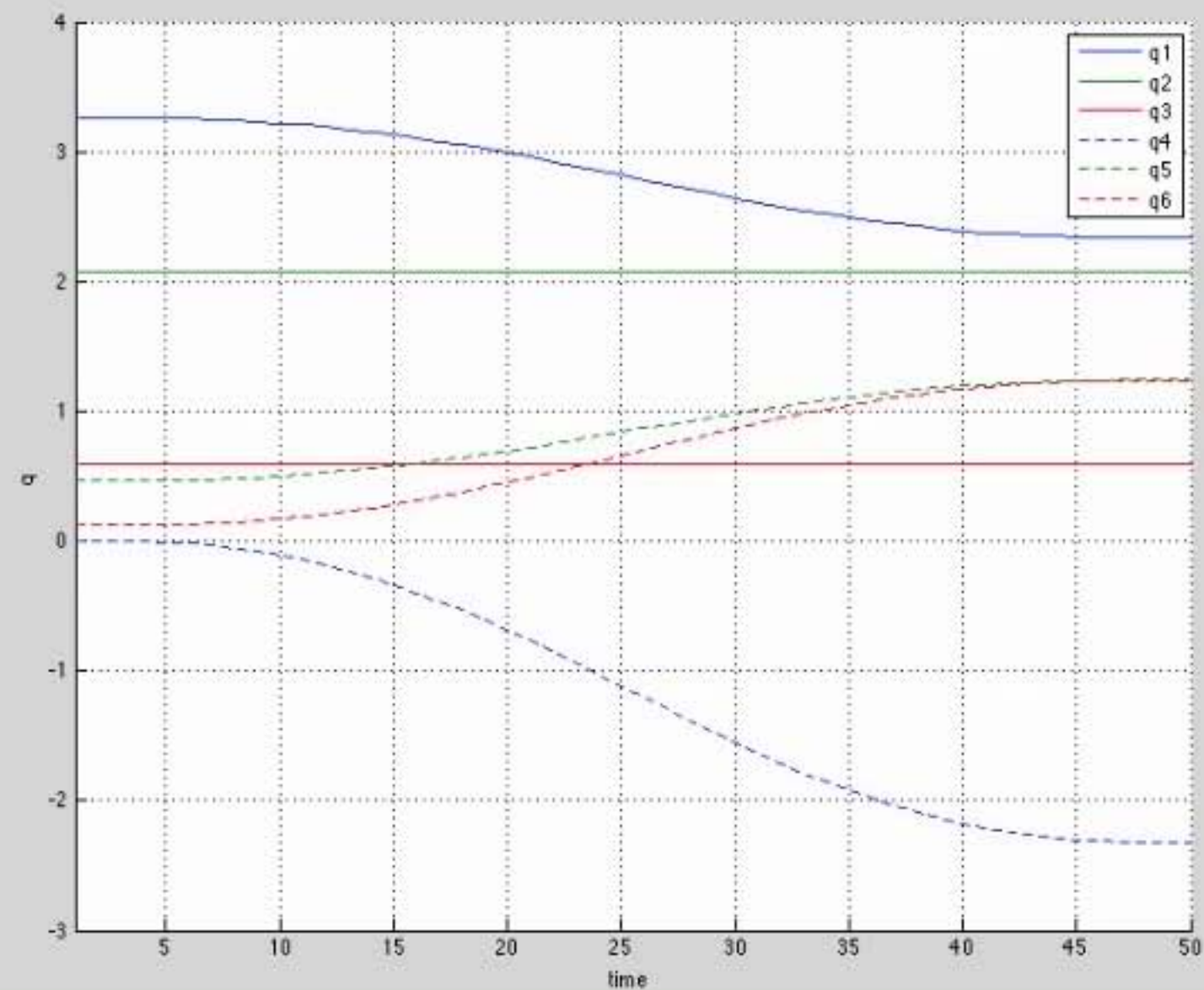
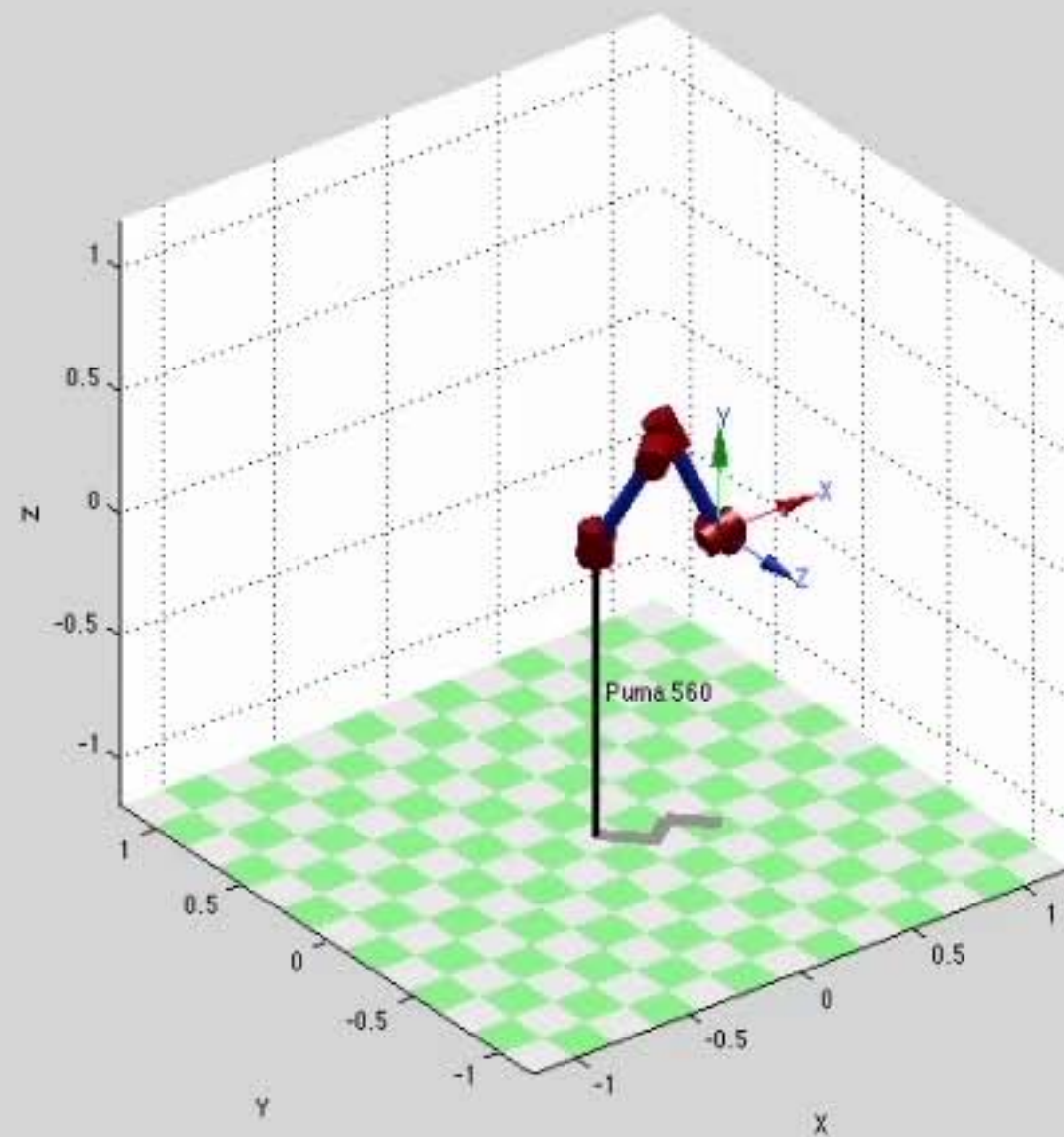
>> tg = jtraj(qA, qB, 50);
>> p560.plot(tg)
>> figure
>> qplot(tg)

```

fx >>

Workspace

Name	Value	Class
TA	<4x4 double>	double
TB	<4x4 double>	double
deg	0.0175	double
p560	<1x1 SerialLink>	SerialLink
qA	[3.2631,2.0791,0.5992,-...	double
qB	[2.3358,2.0791,0.5992,-...	double
qn	[0,0.7854,3.1416,0,0.78...	double
qr	[0,1.5708,-1.5708,0,0,0]	double
qs	[0,0,-1.5708,0,0,0]	double
qz	[0,0,0,0,0,0]	double
tg	<50x6 double>	double



Command Window

```

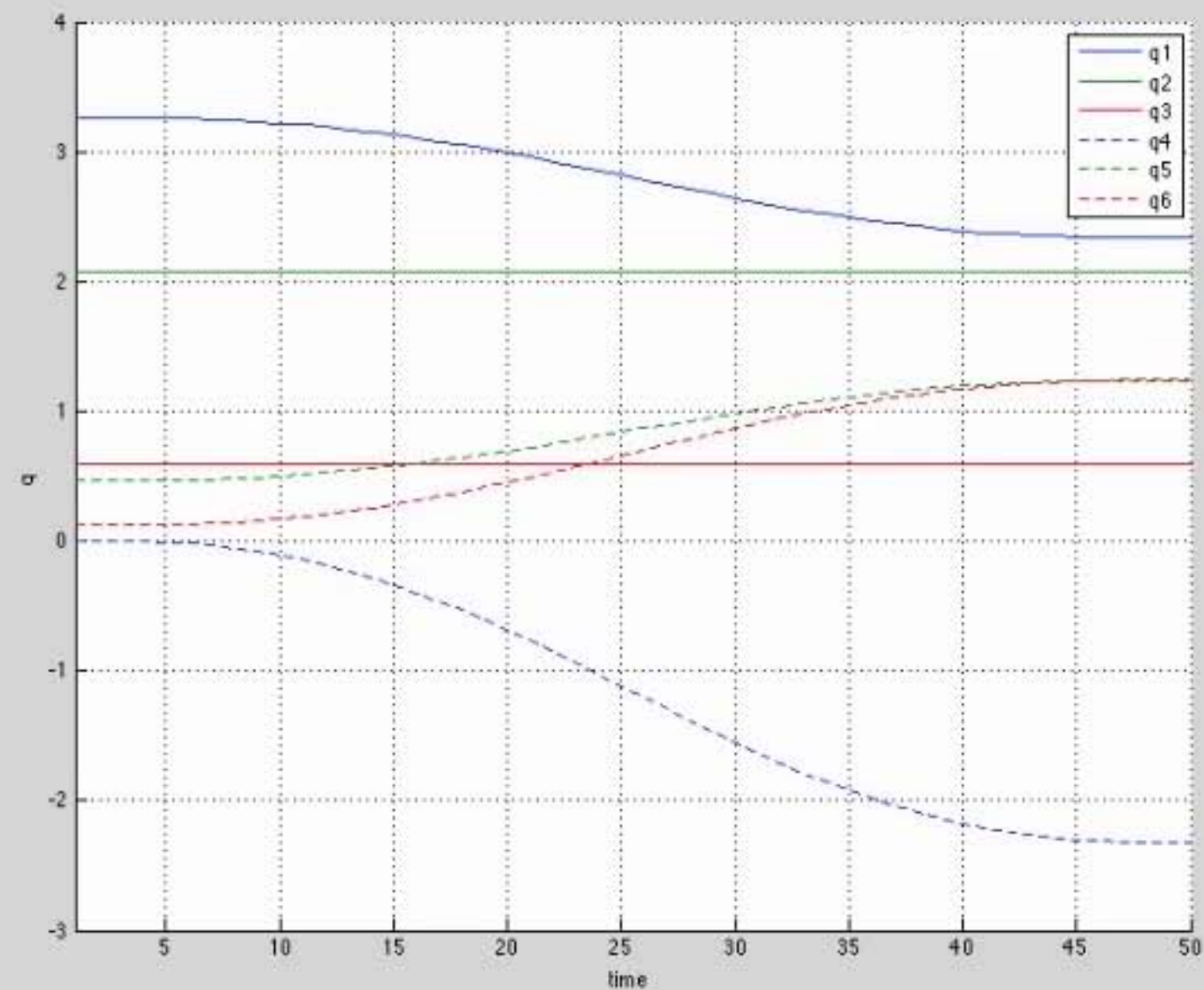
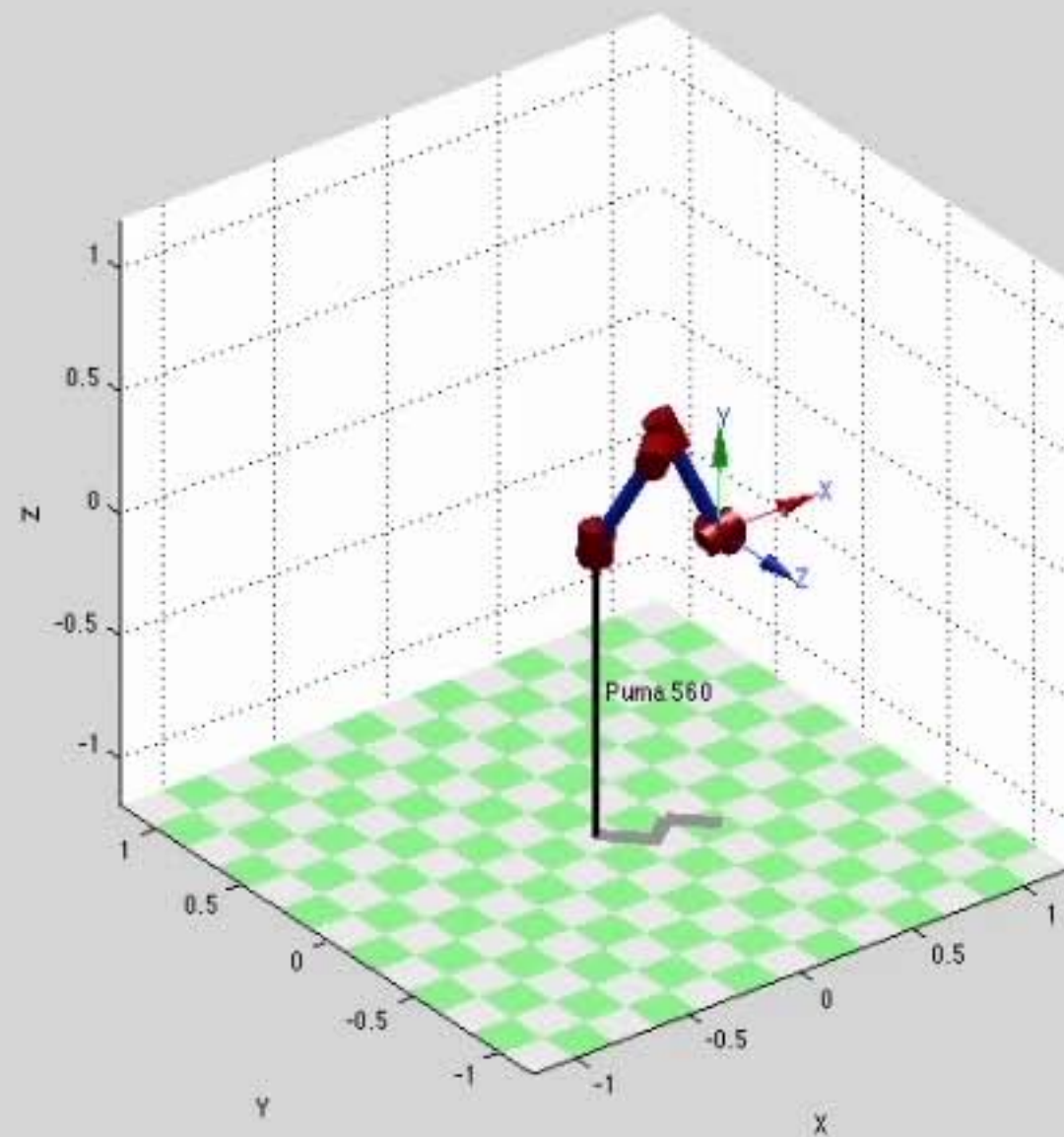
3.2631    2.0791    0.5992   -0.0000    0.4633    0.1215
>> qB = p560.ikine6s(TB)
qB =
    2.3358    2.0791    0.5992   -2.3209    1.2425    1.2377

>> tg = jtraj(qA, qB, 50);
>> p560.plot(tg)
>> figure
>> qplot(tg)
>>
>> Ts = ctraj(TA, TB, 50);
fx >>

```

Workspace

Name	Value	Class
TA	<4x4 double>	double
TB	<4x4 double>	double
Ts	<4x4x50 double>	double
deg	0.0175	double
p560	<1x1 SerialLink>	SerialLink
qA	[3.2631,2.0791,0.5992,-...	double
qB	[2.3358,2.0791,0.5992,-...	double
qn	[0,0.7854,3.1416,0,0.78...	double
qr	[0,1.5708,-1.5708,0,0,0]	double
qs	[0,0,-1.5708,0,0,0]	double
qz	[0,0,0,0,0,0]	double
tg	<50x6 double>	double



Command Window

```
>> tg = jtraj(qA, qB, 50);
>> p560.plot(tg)
>> figure
>> qplot(tg)
>>
>> Ts = ctraj(TA, TB, 50);
>> Ts(:, :, 1)
```

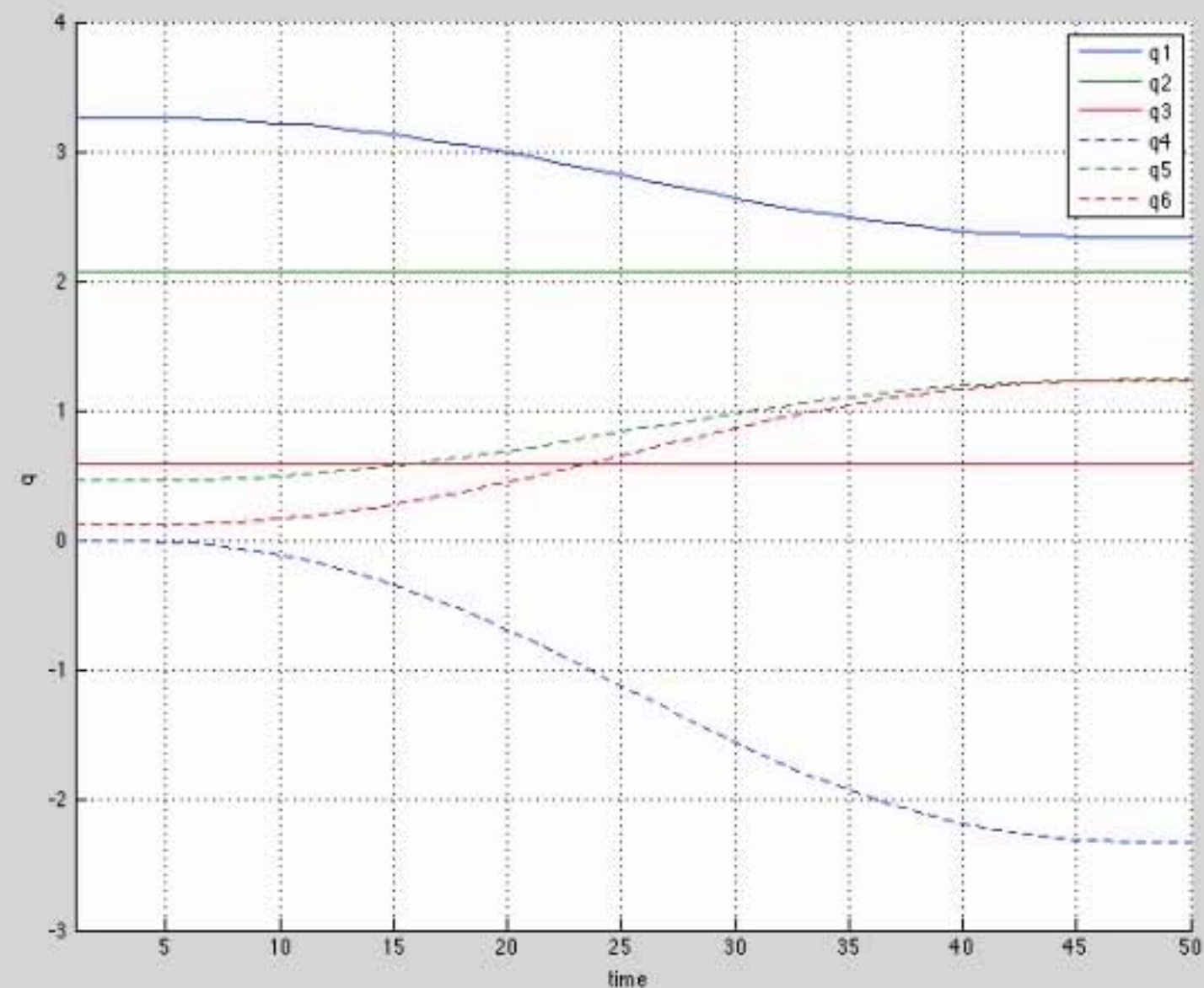
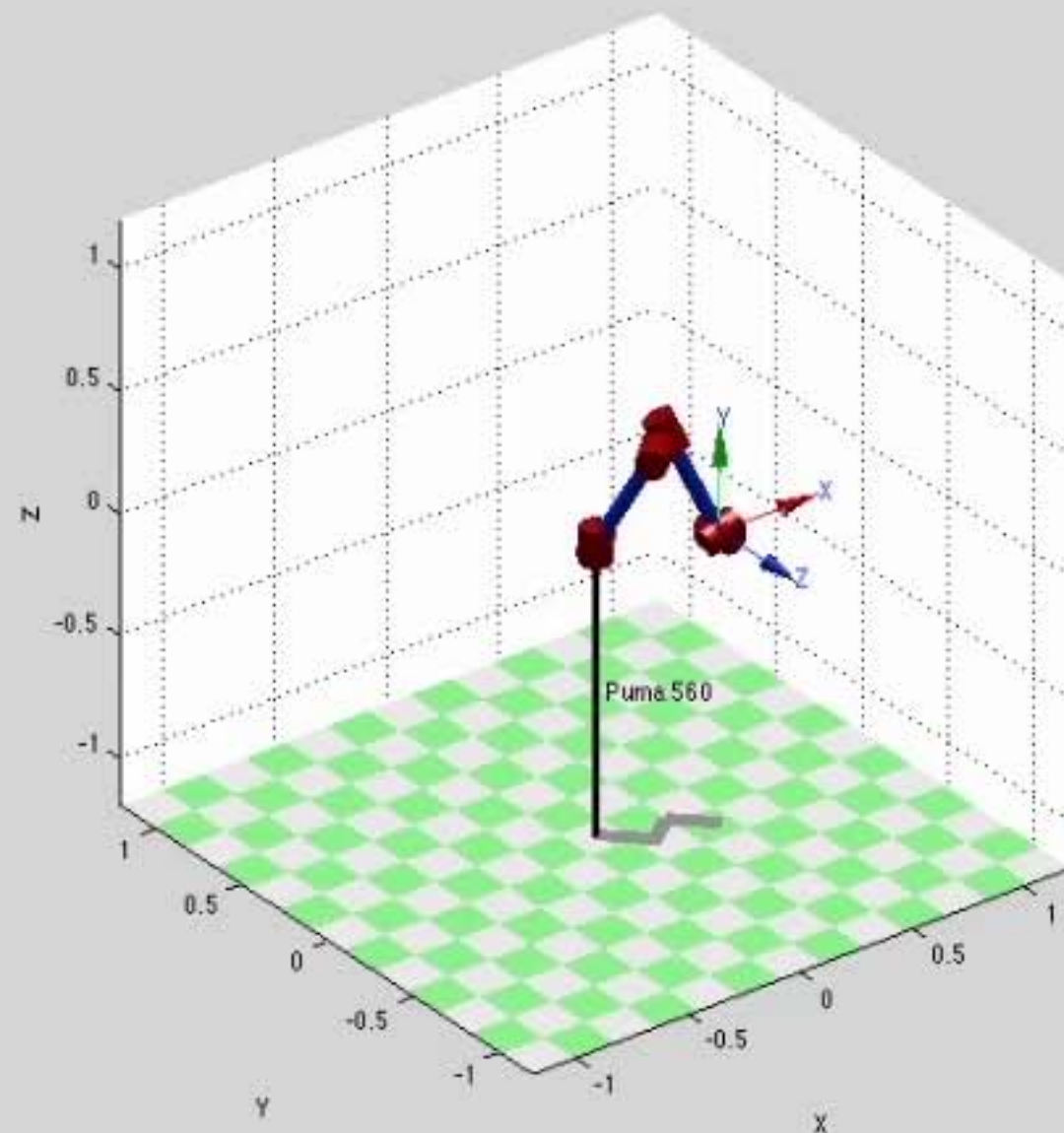
ans =

```
1.0000    0    0    0.4000
0   -1.0000    0    0.2000
0    0   -1.0000    0
0    0    0    1.0000
```

f_g >>

Workspace

Name	Value	Class
TA	<4x4 double>	double
TB	<4x4 double>	double
Ts	<4x4x50 double>	double
ans	<4x4 double>	double
deg	0.0175	double
p560	<1x1 SerialLink>	SerialLink
qA	[3.2631,2.0791,0.5992,-...	double
qB	[2.3358,2.0791,0.5992,-...	double
qn	[0,0.7854,3.1416,0,0.78...	double
qr	[0,1.5708,-1.5708,0,0,0]	double
qs	[0,0,-1.5708,0,0,0]	double
qz	[0,0,0,0,0,0]	double
tg	<50x6 double>	double



Command Window

```

1.0000    0    0    0.4000
    0   -1.0000   -0.0015    0.1996
    0    0.0015   -1.0000    0
    0    0    0    1.0000

```

```
>> Ts(:, :, 3)
```

```
ans =
```

```

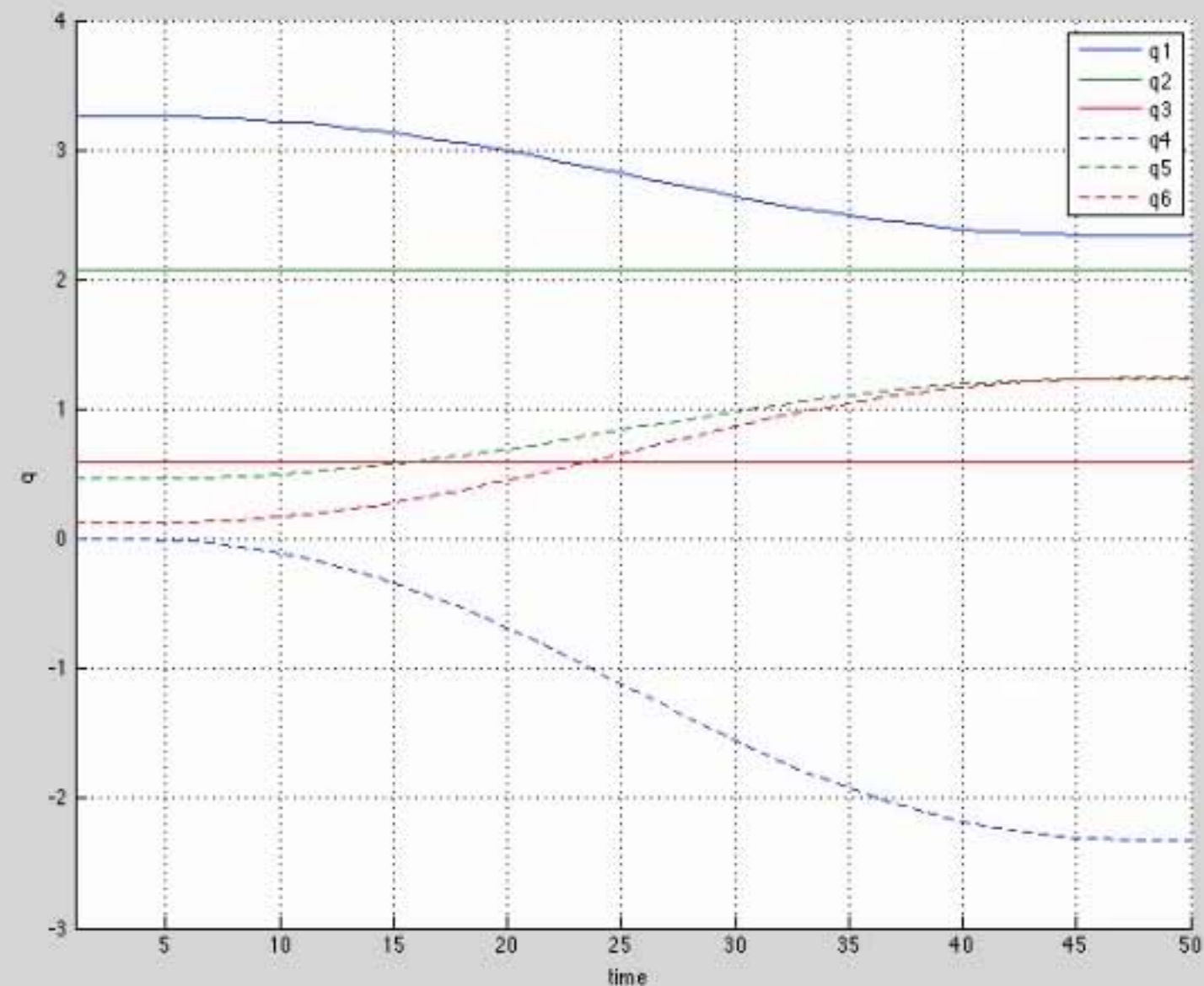
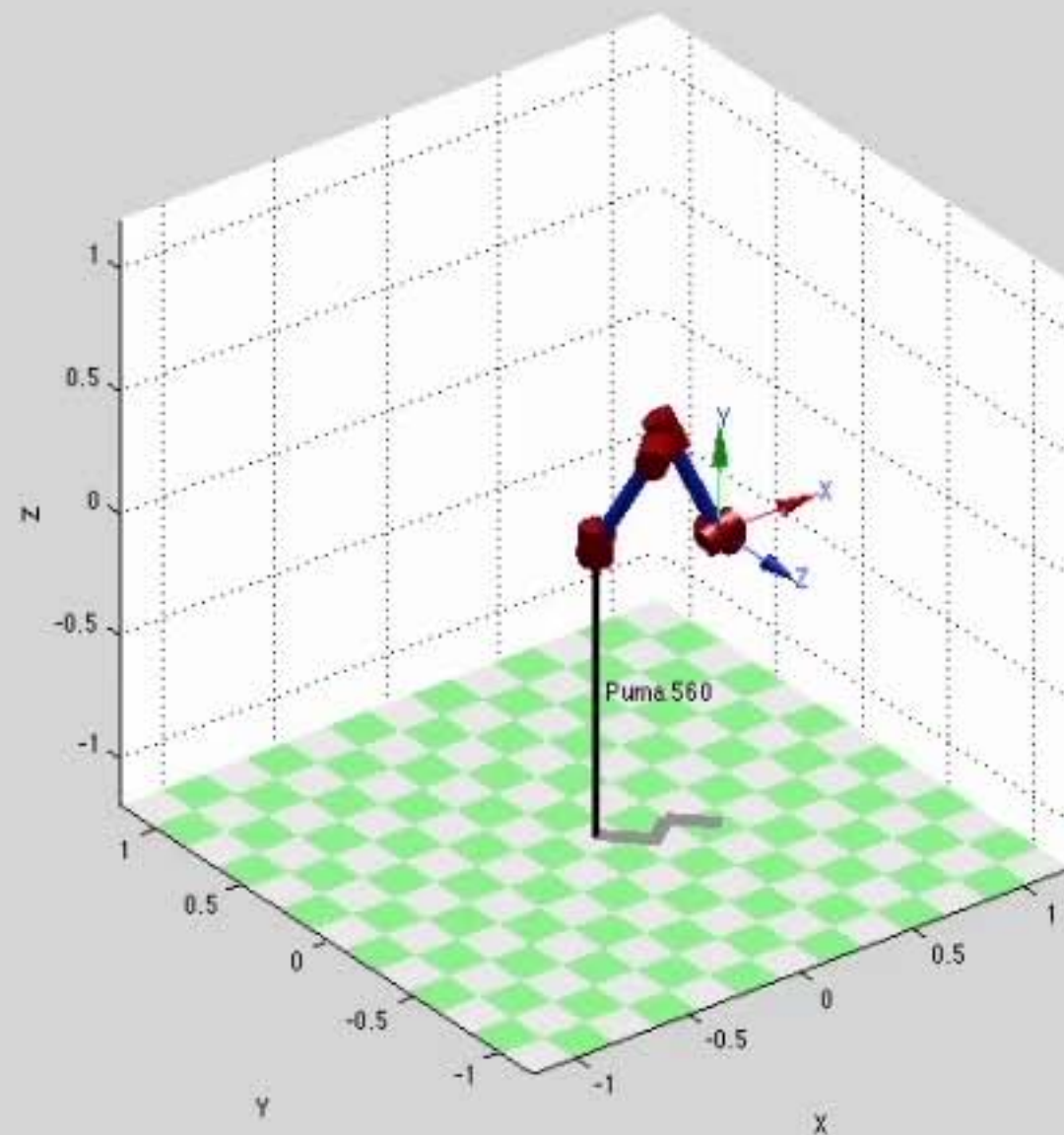
1.0000    0    0    0.4000
    0   -1.0000   -0.0059    0.1985
    0    0.0059   -1.0000    0
    0    0    0    1.0000

```

```
fz >>
```

Workspace

Name	Value	Class
TA	<4x4 double>	double
TB	<4x4 double>	double
Ts	<4x4x50 double>	double
ans	<4x4 double>	double
deg	0.0175	double
p560	<1x1 SerialLink>	SerialLink
qA	[3.2631,2.0791,0.5992,-...	double
qB	[2.3358,2.0791,0.5992,-...	double
qn	[0,0.7854,3.1416,0,0.78...	double
qr	[0,1.5708,-1.5708,0,0,0]	double
qs	[0,0,-1.5708,0,0,0]	double
qz	[0,0,0,0,0,0]	double
tg	<50x6 double>	double



Command Window

```
1.0000    0    0    0.4000
    0   -1.0000   -0.0015    0.1996
    0    0.0015   -1.0000    0
    0    0    0    1.0000
```

```
>> Ts(:, :, 3)
```

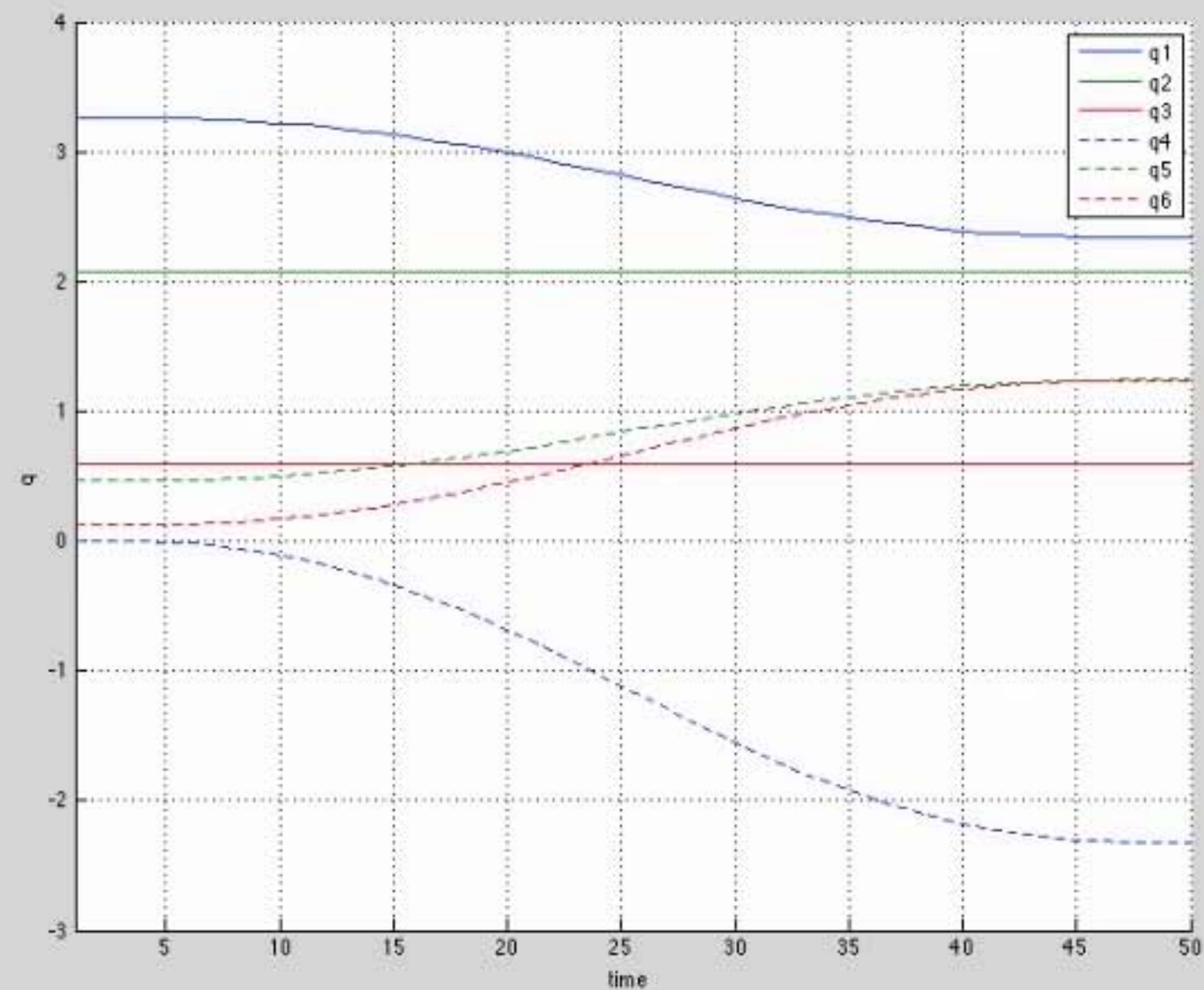
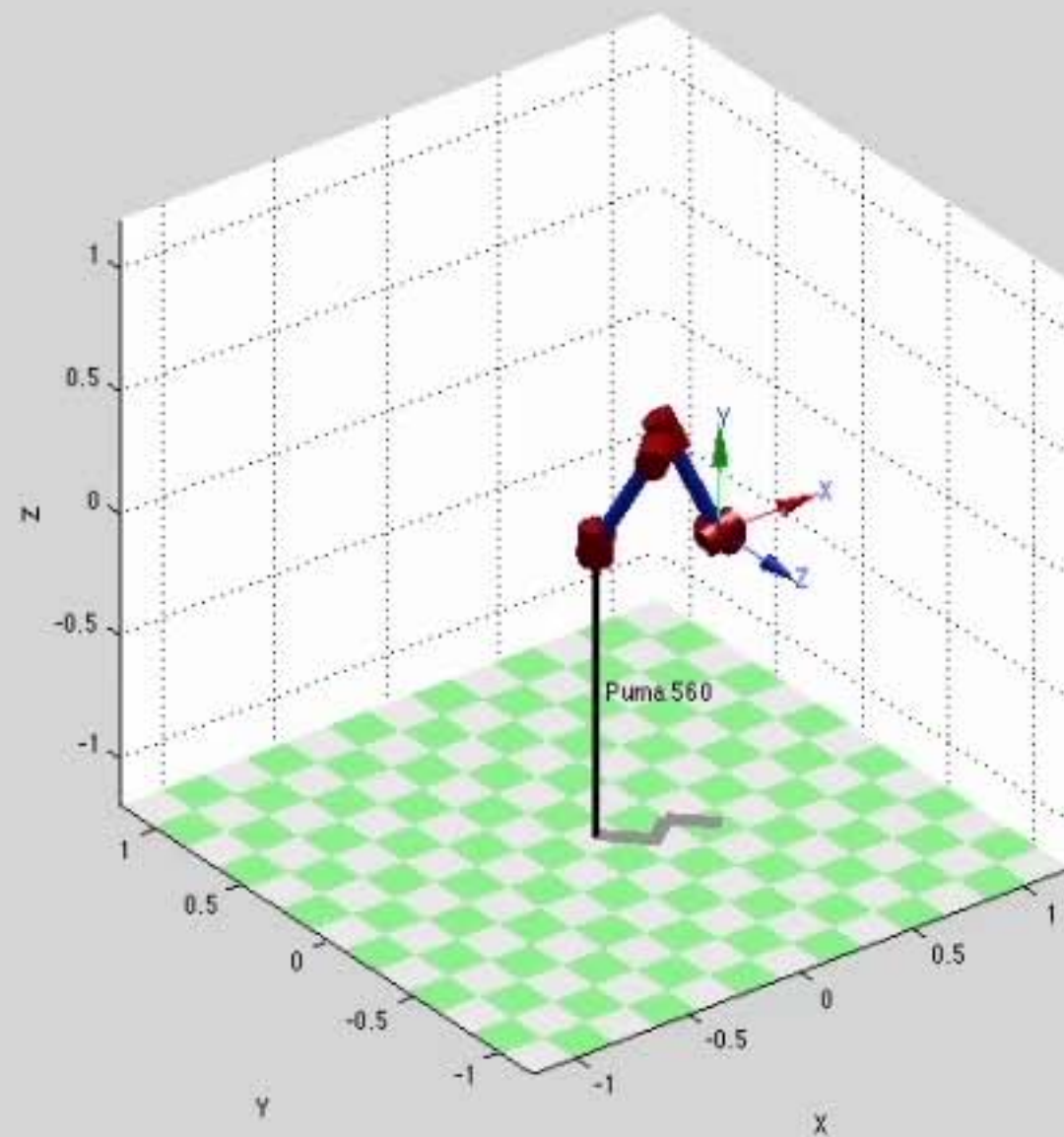
```
ans =
```

```
1.0000    0    0    0.4000
    0   -1.0000   -0.0059    0.1985
    0    0.0059   -1.0000    0
    0    0    0    1.0000
```

```
f_x >> qs = p560.ikine6s(Ts)
```

Workspace

Name	Value	Class
TA	<4x4 double>	double
TB	<4x4 double>	double
Ts	<4x4x50 double>	double
ans	<4x4 double>	double
deg	0.0175	double
p560	<1x1 SerialLink>	SerialLink
qA	[3.2631,2.0791,0.5992,-...	double
qB	[2.3358,2.0791,0.5992,-...	double
qn	[0,0.7854,3.1416,0,0.78...	double
qr	[0,1.5708,-1.5708,0,0,0]	double
qs	[0,0,-1.5708,0,0,0]	double
qz	[0,0,0,0,0,0]	double
tg	<50x6 double>	double



Command Window

```

1.0000    0    0    0.4000
    0   -1.0000   -0.0015    0.1996
    0    0.0015   -1.0000    0
    0    0    0    1.0000

>> Ts(:, :, 3)

ans =

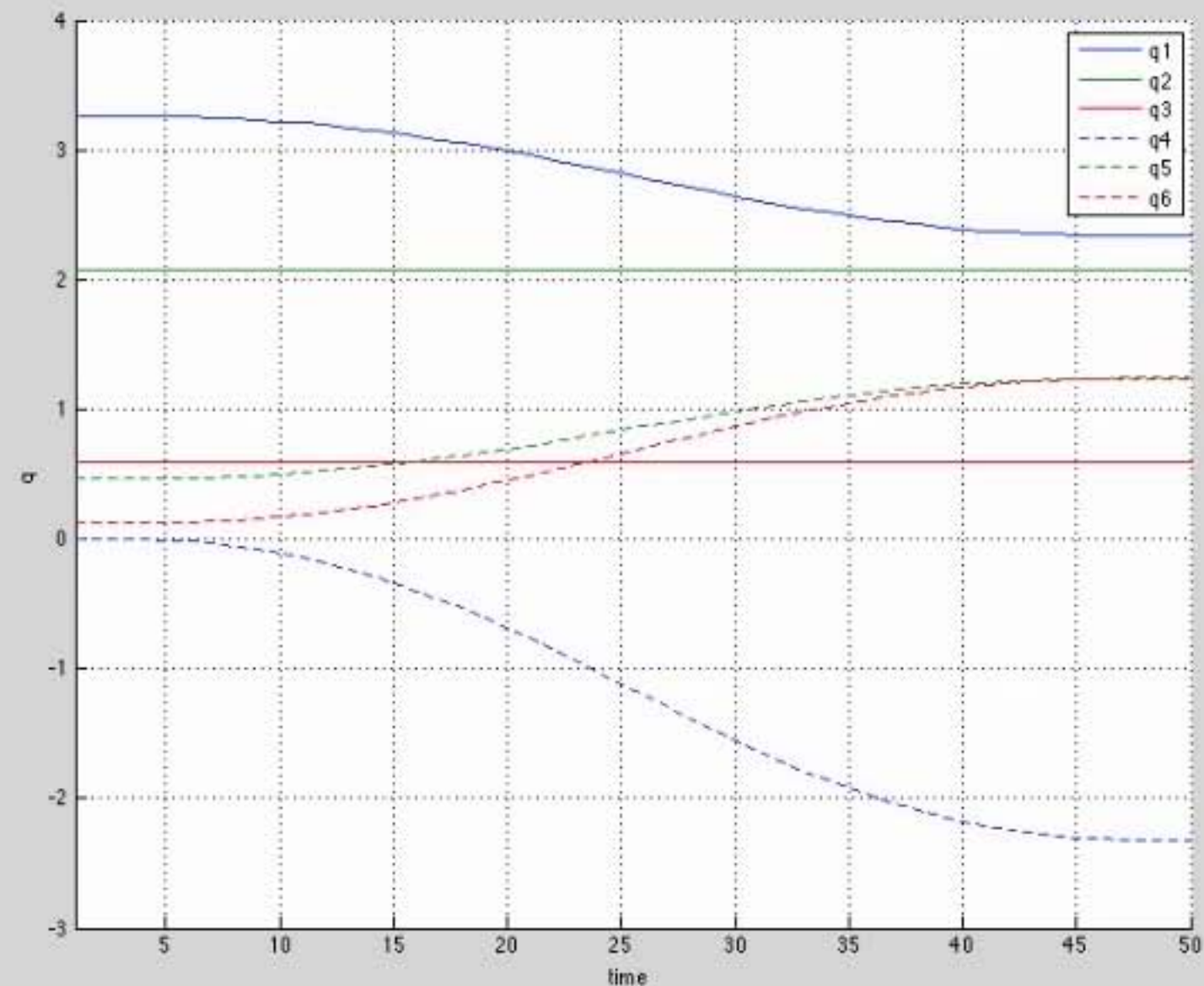
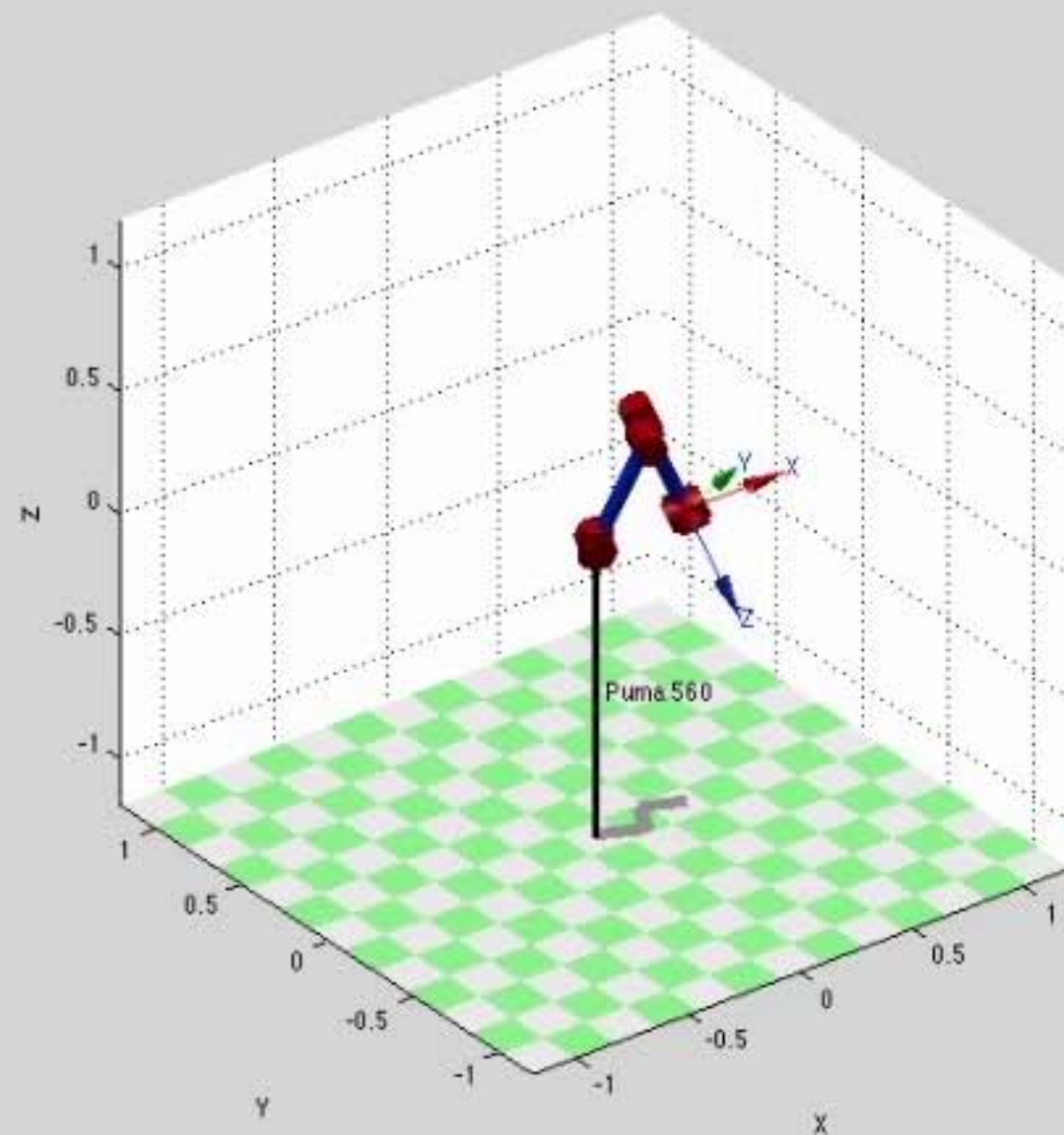
    1.0000    0    0    0.4000
    0   -1.0000   -0.0059    0.1985
    0    0.0059   -1.0000    0
    0    0    0    1.0000

>> qs = p560.ikine6s(Ts);
fx >>

```

Workspace

Name	Value	Class
TA	<4x4 double>	double
TB	<4x4 double>	double
Ts	<4x4x50 double>	double
ans	<4x4 double>	double
deg	0.0175	double
p560	<1x1 SerialLink>	SerialLink
qA	[3.2631,2.0791,0.5992,-...	double
qB	[2.3358,2.0791,0.5992,-...	double
qn	[0,0.7854,3.1416,0,0.78...	double
qr	[0,1.5708,-1.5708,0,0,0]	double
qs	<50x6 double>	double
qz	[0,0,0,0,0,0]	double
tg	<50x6 double>	double



Command Window

```

0 0.0015 -1.0000 0
0 0 0 1.0000

>> Ts(:, :, 3)

ans =

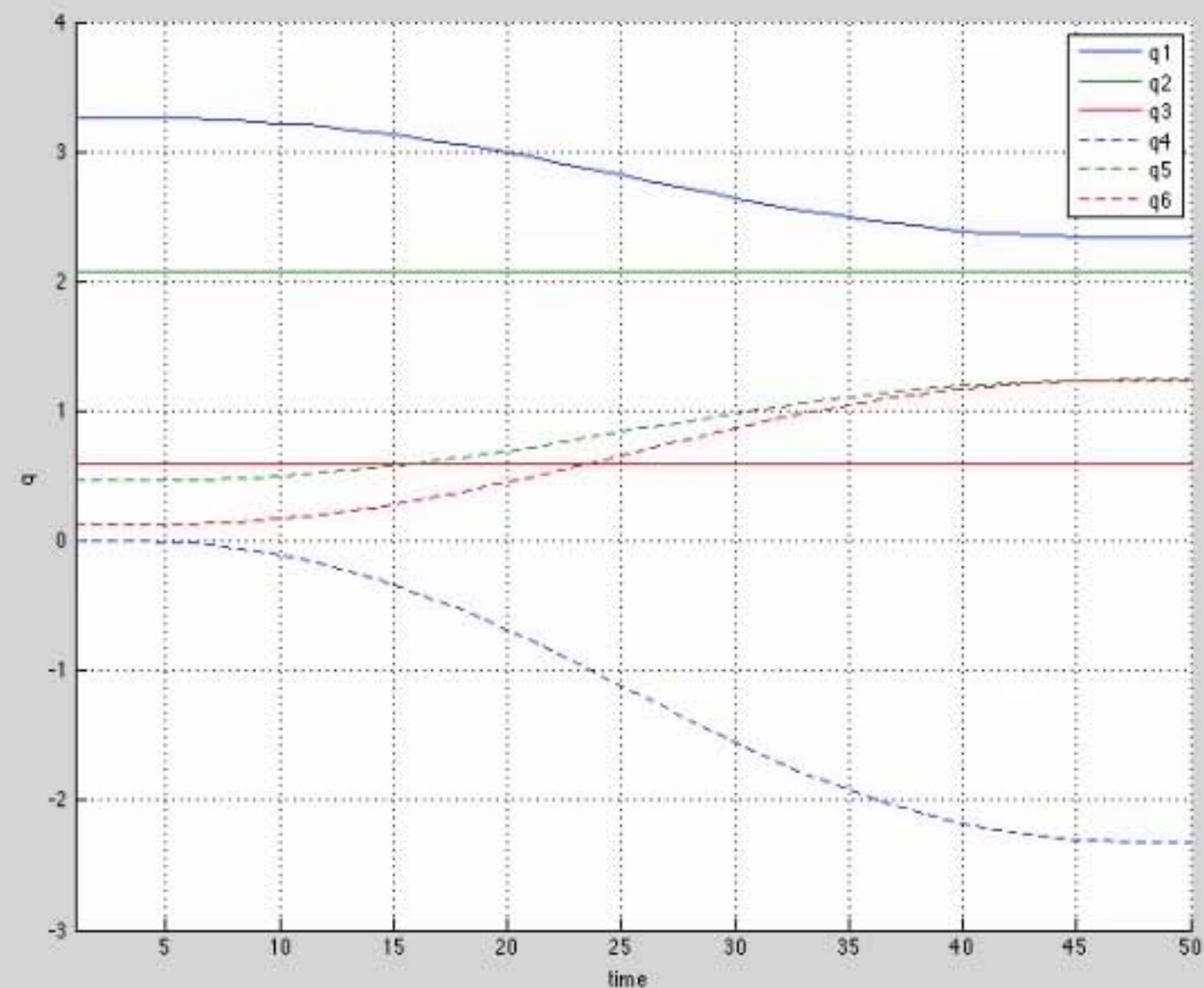
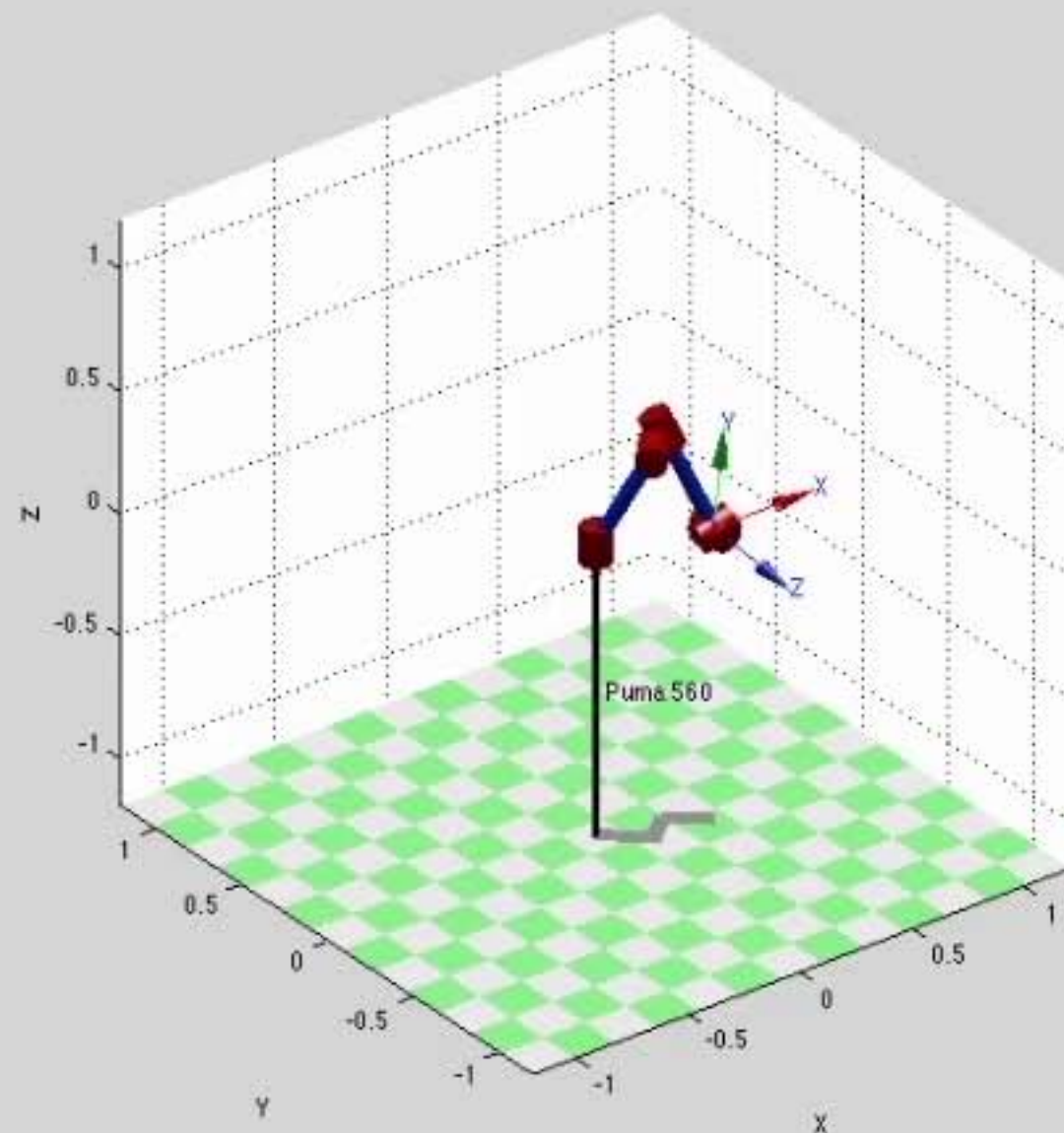
    1.0000         0         0    0.4000
         0    -1.0000    -0.0059    0.1985
         0     0.0059    -1.0000         0
         0         0         0    1.0000

>> qs = p560.ikine6s(Ts);
>> figure(1)
>> p560.plot(qs)

```

Workspace

Name	Value	Class
TA	<4x4 double>	double
TB	<4x4 double>	double
Ts	<4x4x50 double>	double
ans	<4x4 double>	double
deg	0.0175	double
p560	<1x1 SerialLink>	SerialLink
qA	[3.2631,2.0791,0.5992,-...	double
qB	[2.3358,2.0791,0.5992,-...	double
qn	[0,0.7854,3.1416,0,0.78...	double
qr	[0,1.5708,-1.5708,0,0,0]	double
qs	<50x6 double>	double
qz	[0,0,0,0,0,0]	double
tg	<50x6 double>	double



Command Window

```

>> Ts(:, :, 3)

ans =

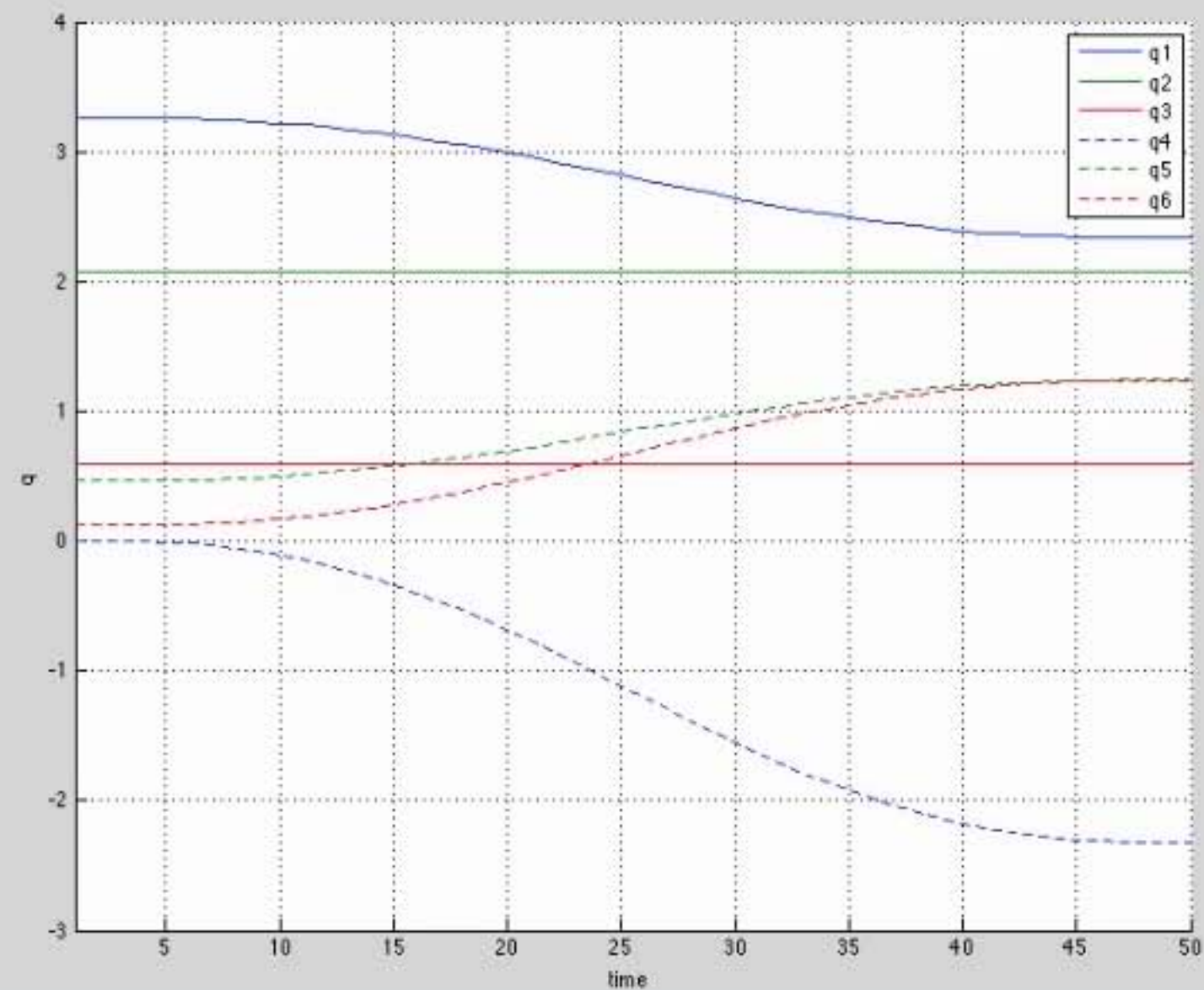
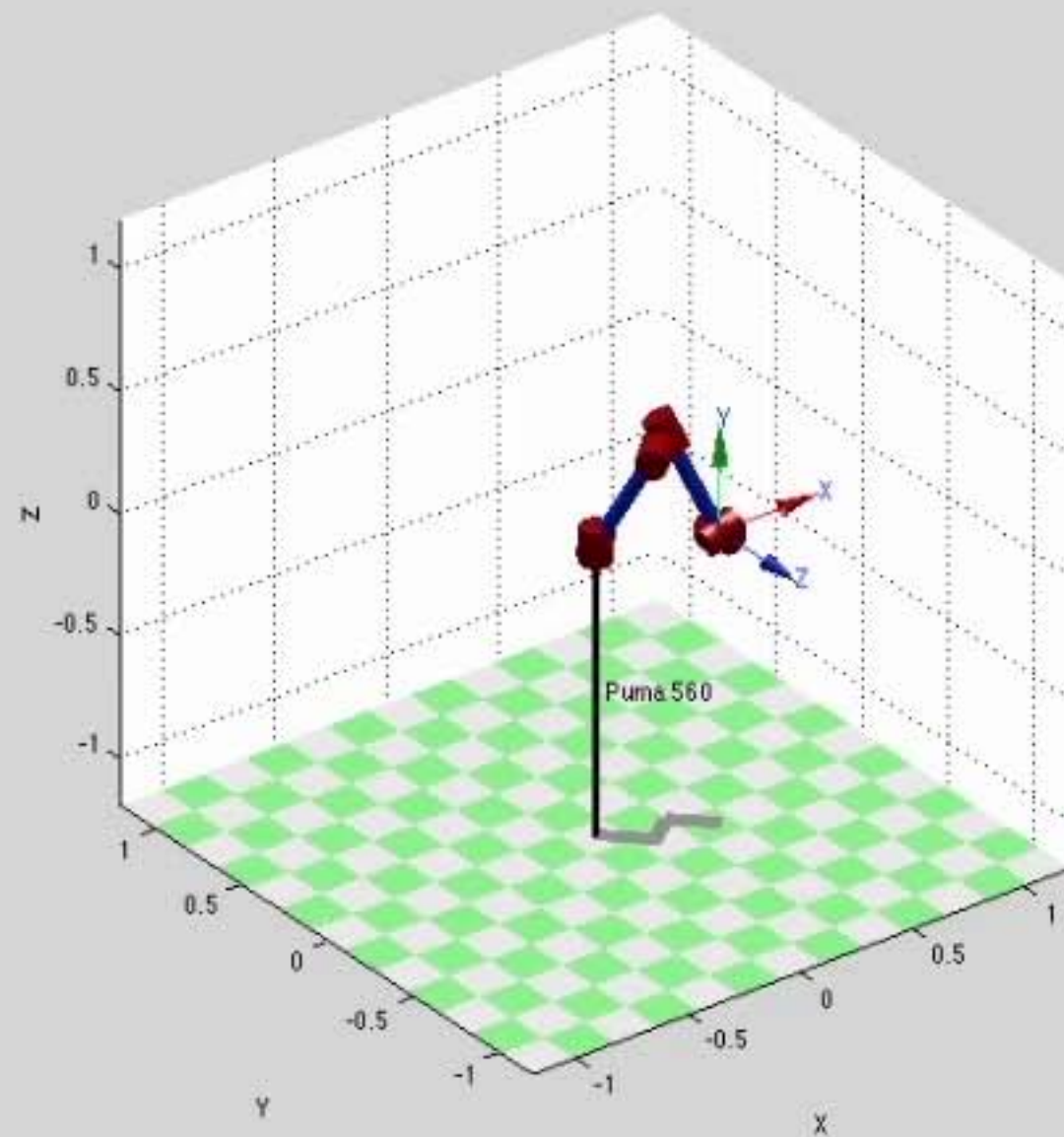
    1.0000    0    0    0.4000
         0   -1.0000   -0.0059    0.1985
         0    0.0059   -1.0000    0
         0    0    0    1.0000

>> qs = p560.ikine6s(Ts);
>> figure(1)
>> p560.plot(qs)

```

Workspace

Name	Value	Class
TA	<4x4 double>	double
TB	<4x4 double>	double
Ts	<4x4x50 double>	double
ans	<4x4 double>	double
deg	0.0175	double
p560	<1x1 SerialLink>	SerialLink
qA	[3.2631,2.0791,0.5992,-...	double
qB	[2.3358,2.0791,0.5992,-...	double
qn	[0,0.7854,3.1416,0,0.78...	double
qr	[0,1.5708,-1.5708,0,0,0]	double
qs	<50x6 double>	double
qz	[0,0,0,0,0,0]	double
tg	<50x6 double>	double



Command Window

```

>> Ts(:, :, 3)

ans =

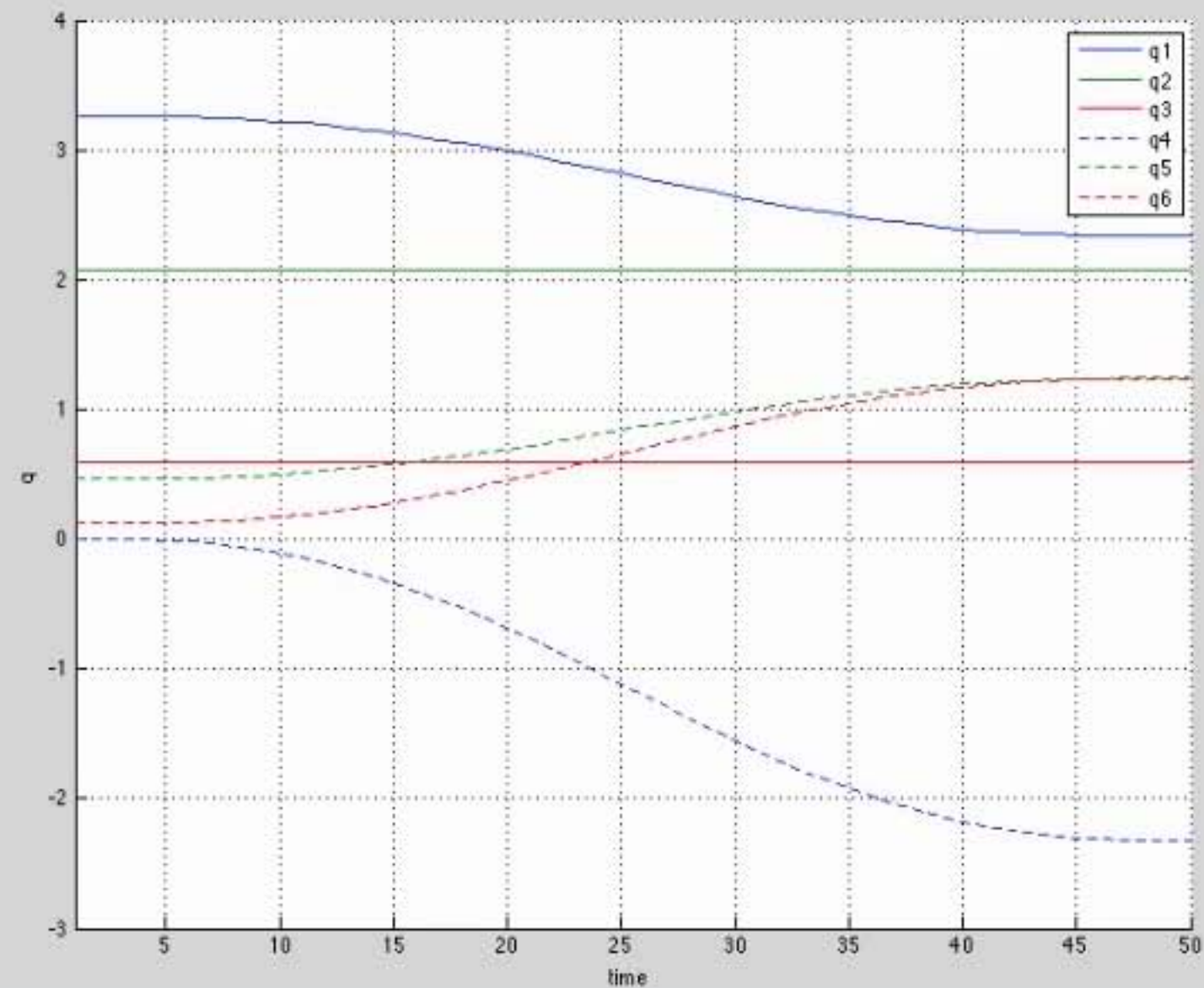
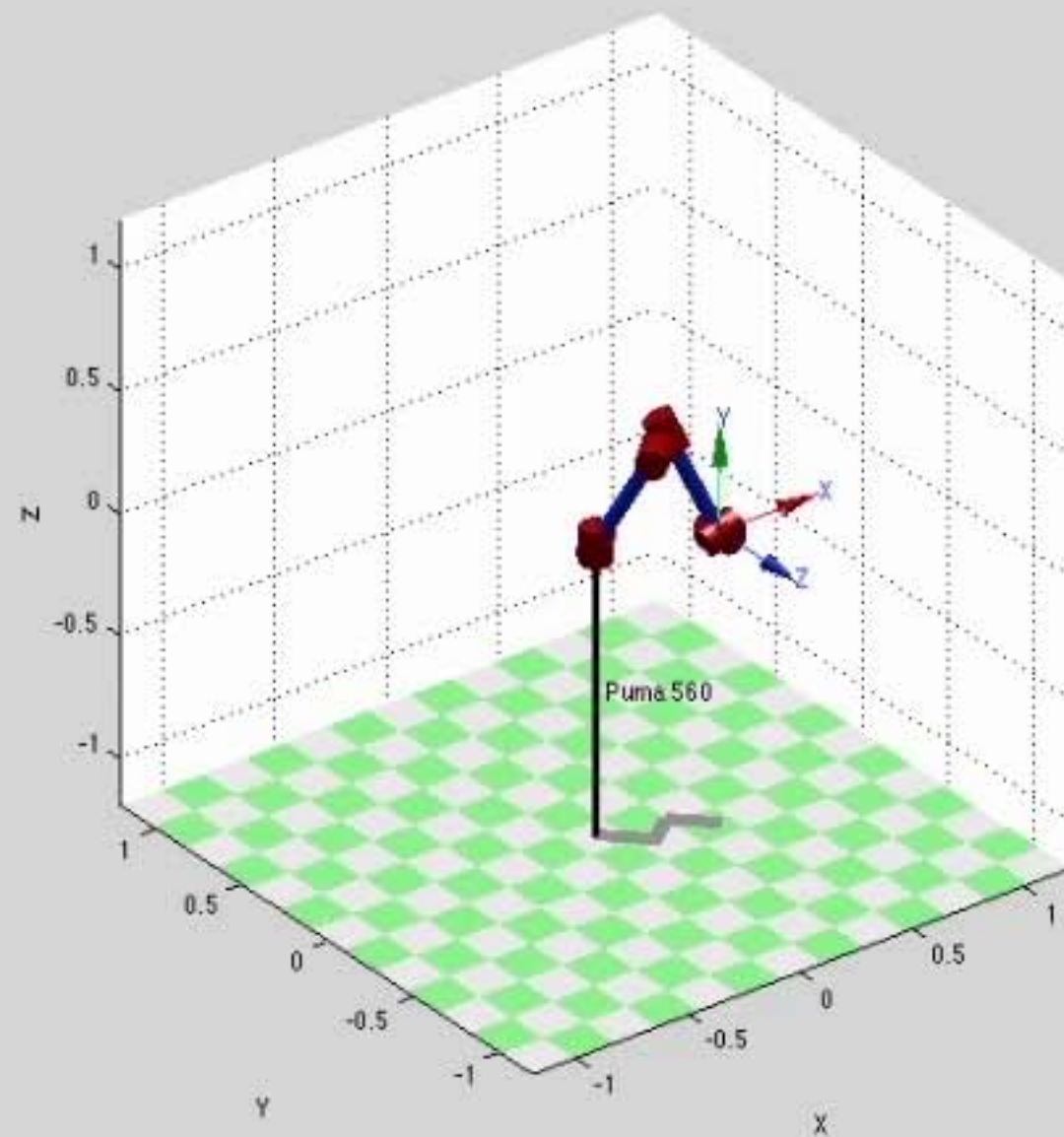
    1.0000         0         0    0.4000
         0   -1.0000   -0.0059    0.1985
         0    0.0059   -1.0000         0
         0         0         0    1.0000

>> qs = p560.ikine6s(Ts);
>> figure(1)
>> p560.plot(qs)
fx >>

```

Workspace

Name	Value	Class
TA	<4x4 double>	double
TB	<4x4 double>	double
Ts	<4x4x50 double>	double
ans	<4x4 double>	double
deg	0.0175	double
p560	<1x1 SerialLink>	SerialLink
qA	[3.2631,2.0791,0.5992,-...	double
qB	[2.3358,2.0791,0.5992,-...	double
qn	[0,0.7854,3.1416,0,0.78...	double
qr	[0,1.5708,-1.5708,0,0,0]	double
qs	<50x6 double>	double
qz	[0,0,0,0,0,0]	double
tg	<50x6 double>	double



Command Window

```
>> Ts(:,:,3)

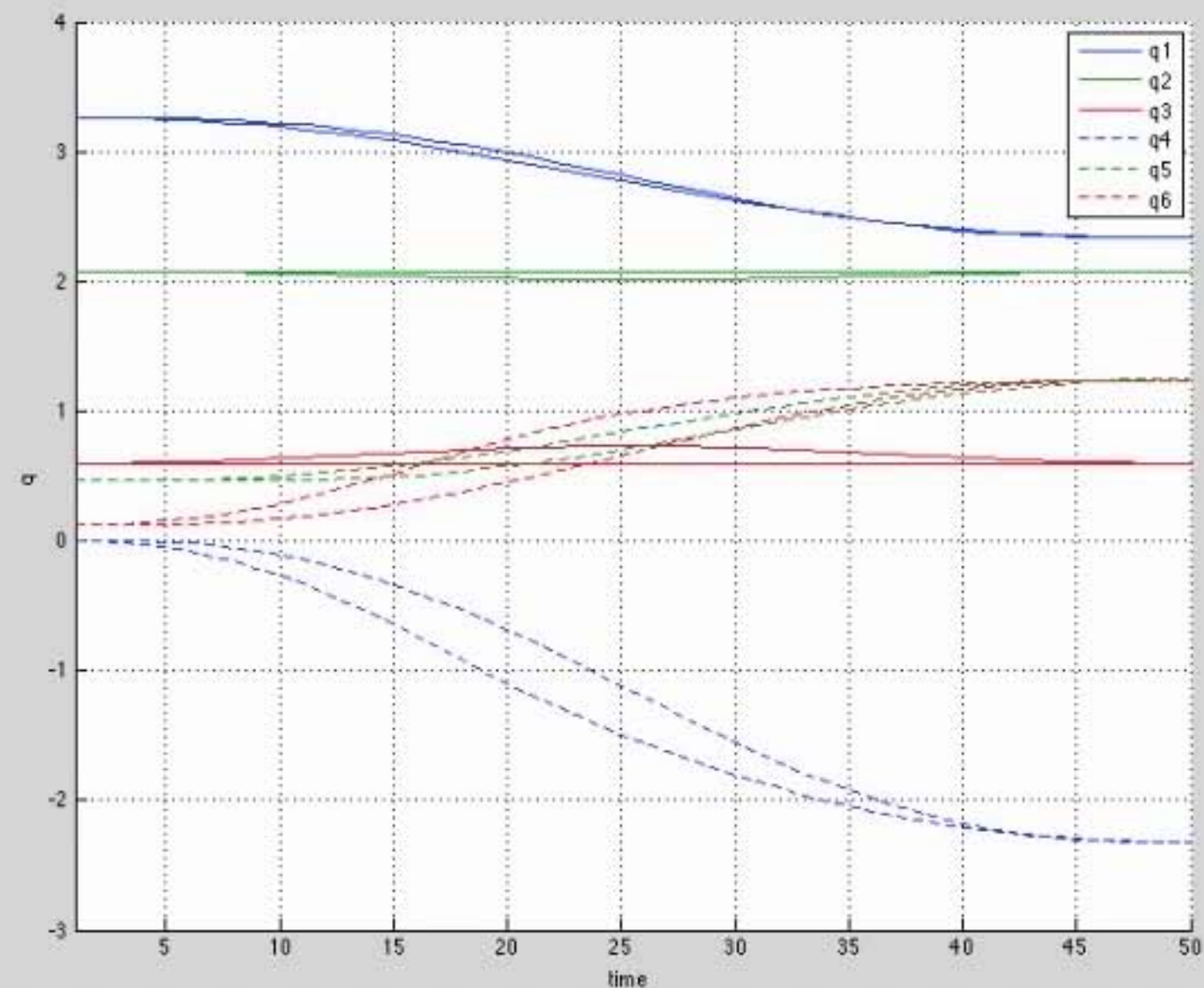
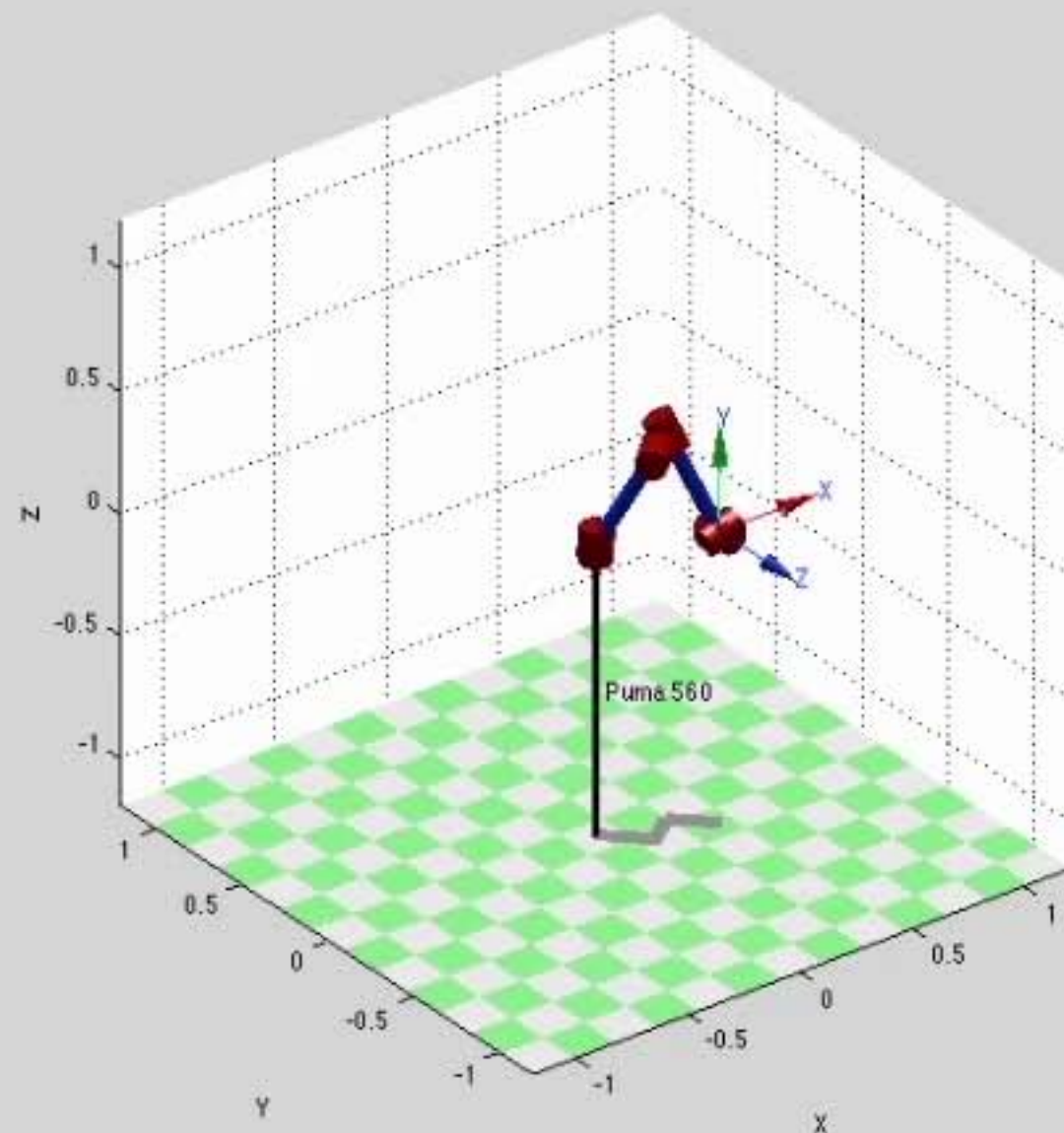
ans =

    1.0000    0    0    0.4000
         0   -1.0000   -0.0059    0.1985
         0    0.0059   -1.0000    0
         0    0    0    1.0000

>> qs = p560.ikine6s(Ts);
>> figure(1)
>> p560.plot(qs)
>> figure(2)
```

Workspace

Name	Value	Class
TA	<4x4 double>	double
TB	<4x4 double>	double
Ts	<4x4x50 double>	double
ans	<4x4 double>	double
deg	0.0175	double
p560	<1x1 SerialLink>	SerialLink
qA	[3.2631,2.0791,0.5992,-...	double
qB	[2.3358,2.0791,0.5992,-...	double
qn	[0,0.7854,3.1416,0,0.78...	double
qr	[0,1.5708,-1.5708,0,0,0]	double
qs	<50x6 double>	double
qz	[0,0,0,0,0,0]	double
tg	<50x6 double>	double



Command Window

```
>> Ts(:,:,3)

ans =

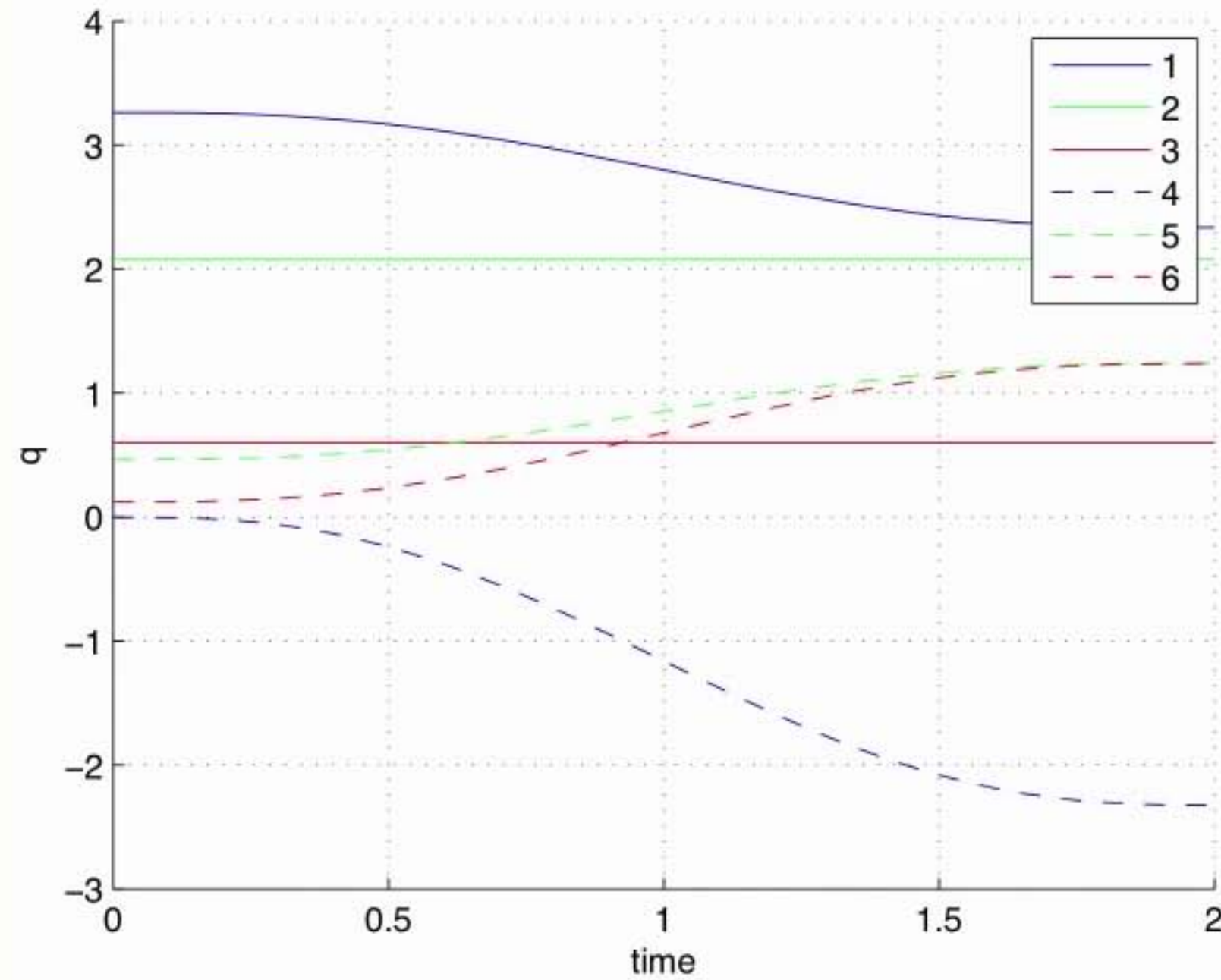
    1.0000         0         0    0.4000
         0   -1.0000   -0.0059    0.1985
         0    0.0059   -1.0000         0
         0         0         0    1.0000

>> qs = p560.ikine6s(Ts);
>> figure(1)
>> p560.plot(qs)
>> figure(2)
>> qplot(qs)
fx >>
```

Workspace

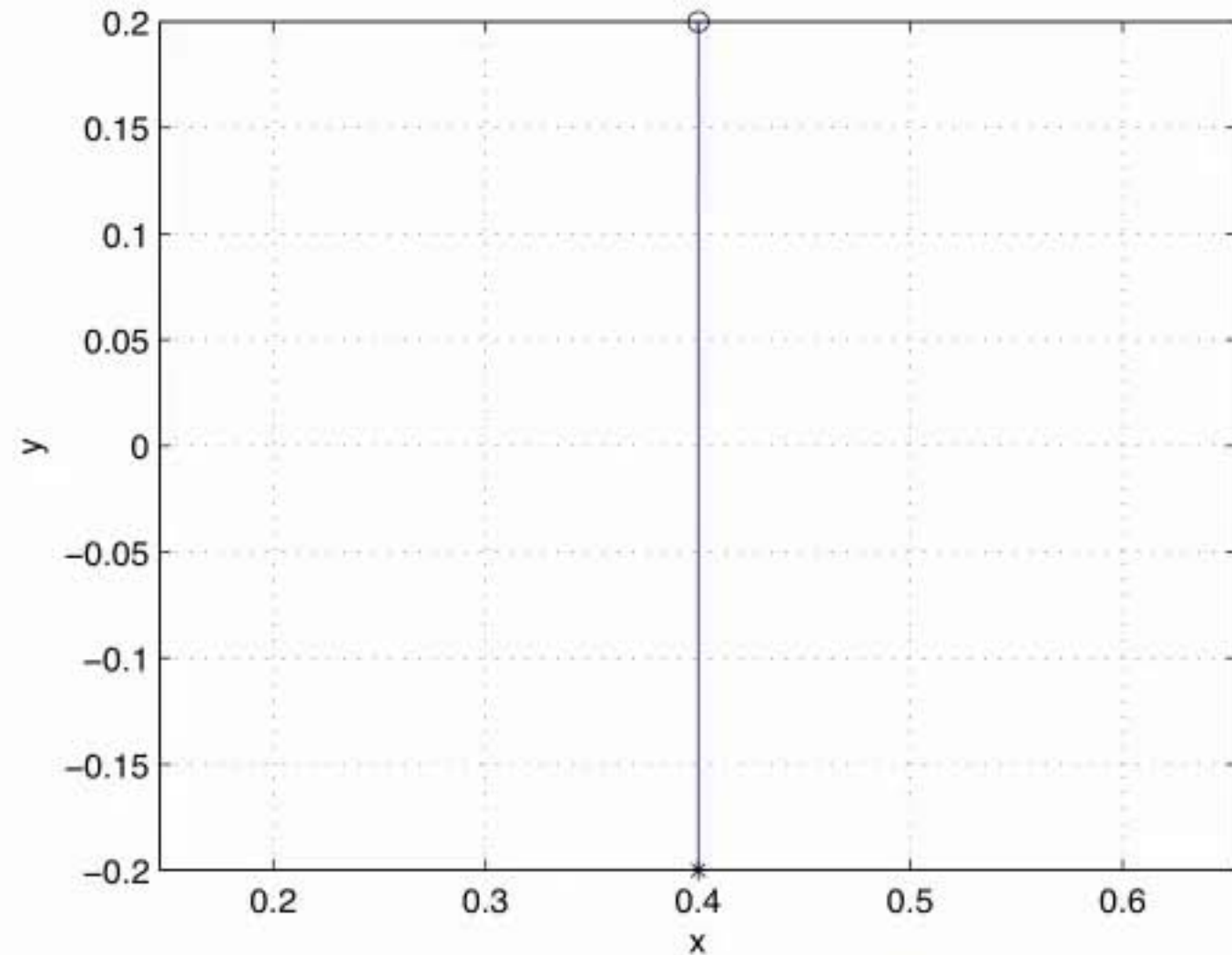
Name	Value	Class
TA	<4x4 double>	double
TB	<4x4 double>	double
Ts	<4x4x50 double>	double
ans	<4x4 double>	double
deg	0.0175	double
p560	<1x1 SerialLink>	SerialLink
qA	[3.2631,2.0791,0.5992,-...	double
qB	[2.3358,2.0791,0.5992,-...	double
qn	[0,0.7854,3.1416,0,0.78...	double
qr	[0,1.5708,-1.5708,0,0,0]	double
qs	<50x6 double>	double
qz	[0,0,0,0,0,0]	double
tg	<50x6 double>	double

Joint angle trajectories



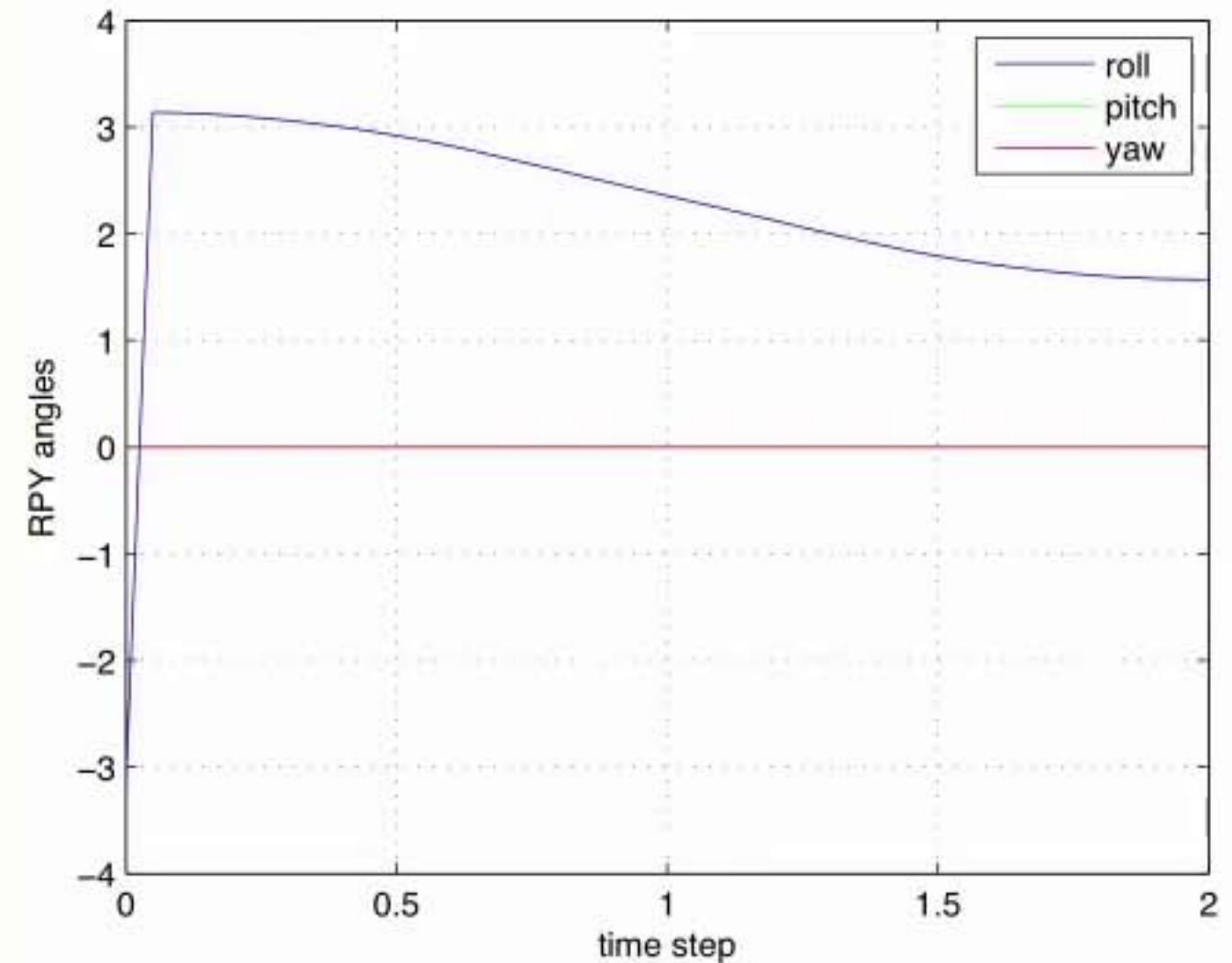
With kind permission of Springer Science+Business Media

Cartesian **position**



With kind permission of Springer Science+Business Media

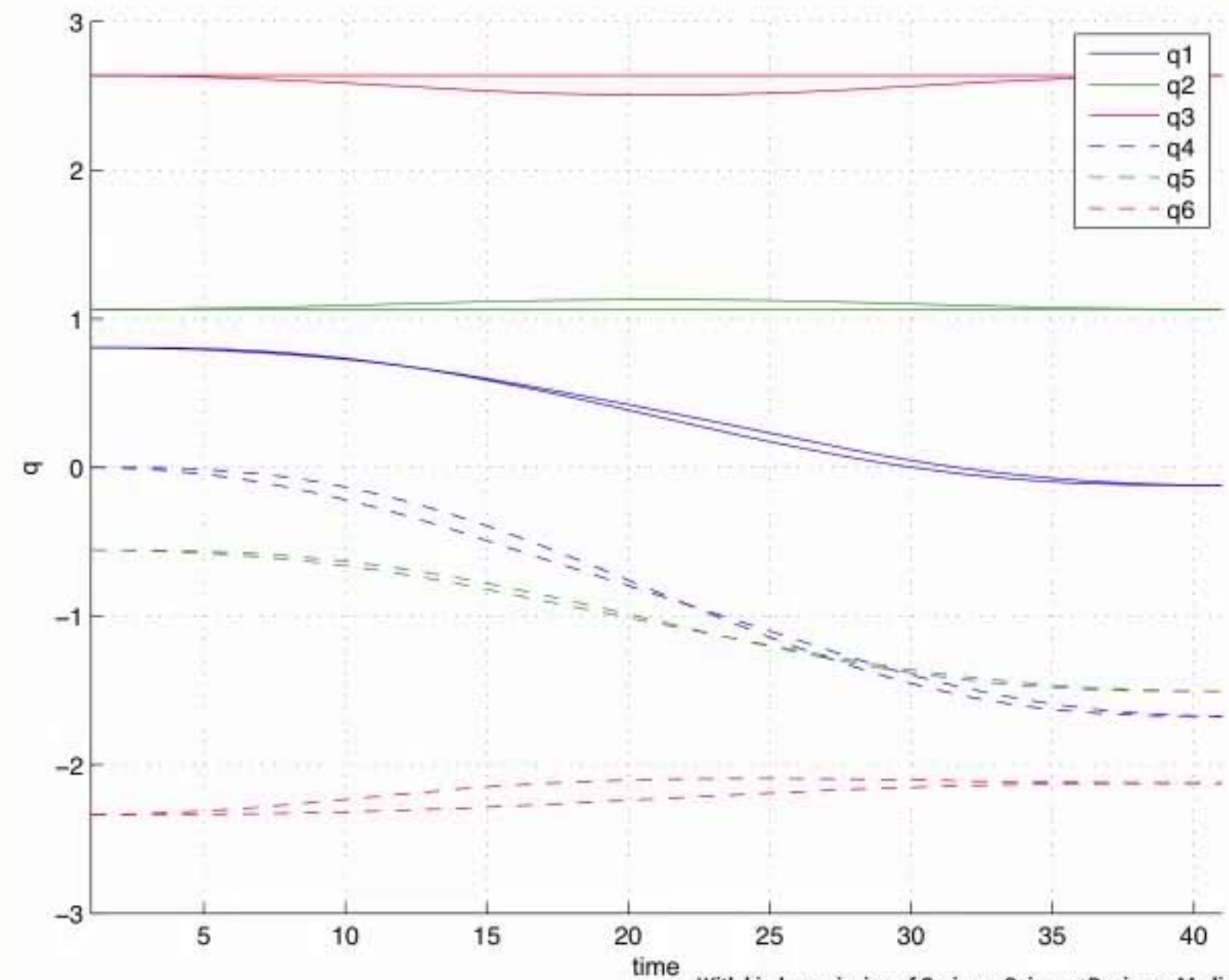
- Now the tool tip moves in a straight line



With kind permission of Springer Science+Business Media

- Tool orientation remains constant along the trajectory

Comparing the **joint trajectories**



With kind permission of Springer Science+Business Media

■ Very little difference

Section 12 Summary

Inverse **kinematics**

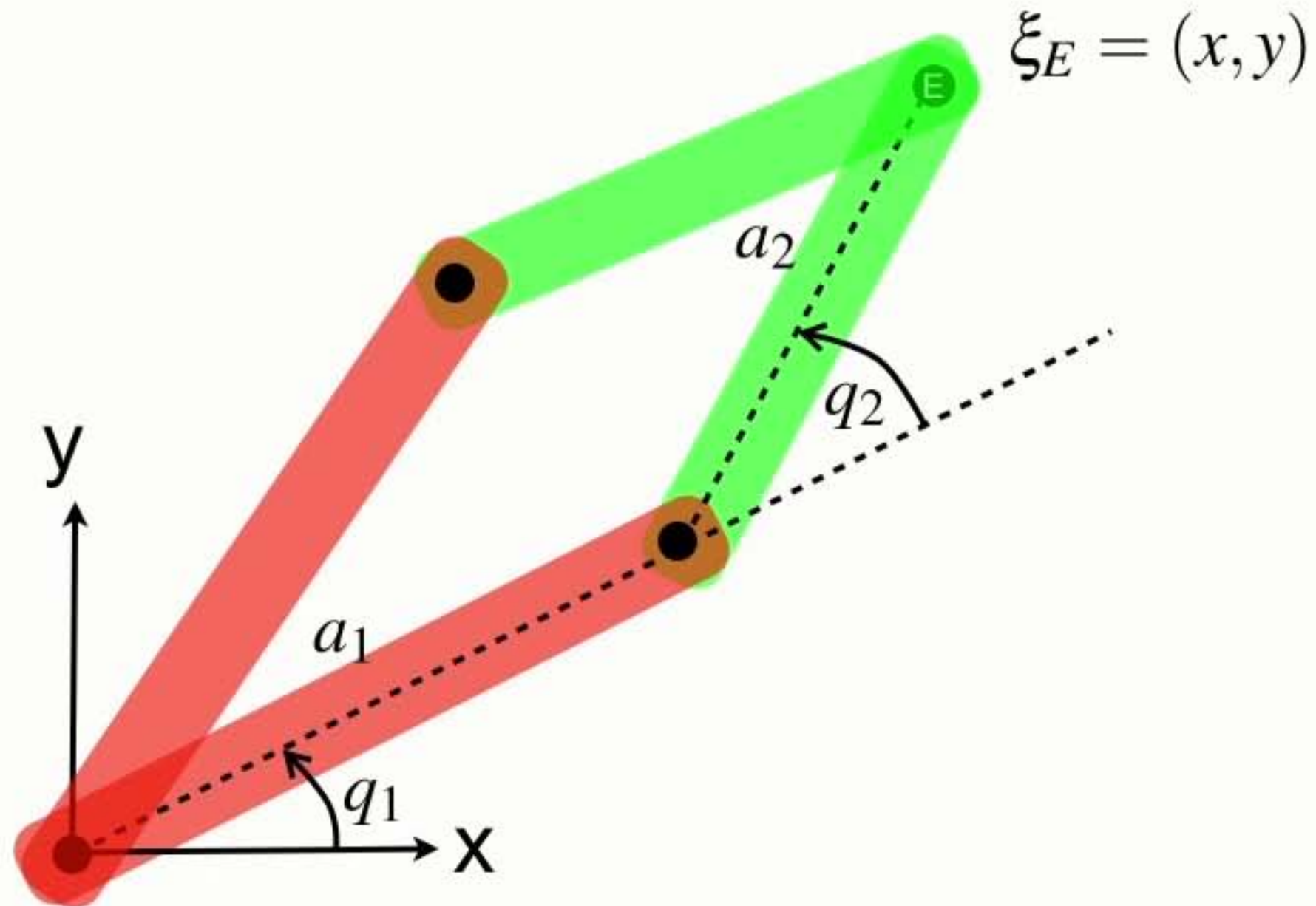
- Set of joint angles given the robot tip pose

$$\xi_N = \mathcal{K}(\boldsymbol{q})$$

$$\boldsymbol{q} = \{q_j, j \in [1 \cdots N]\}$$

$$\boldsymbol{q} = \mathcal{K}^{-1}(\xi_N)$$

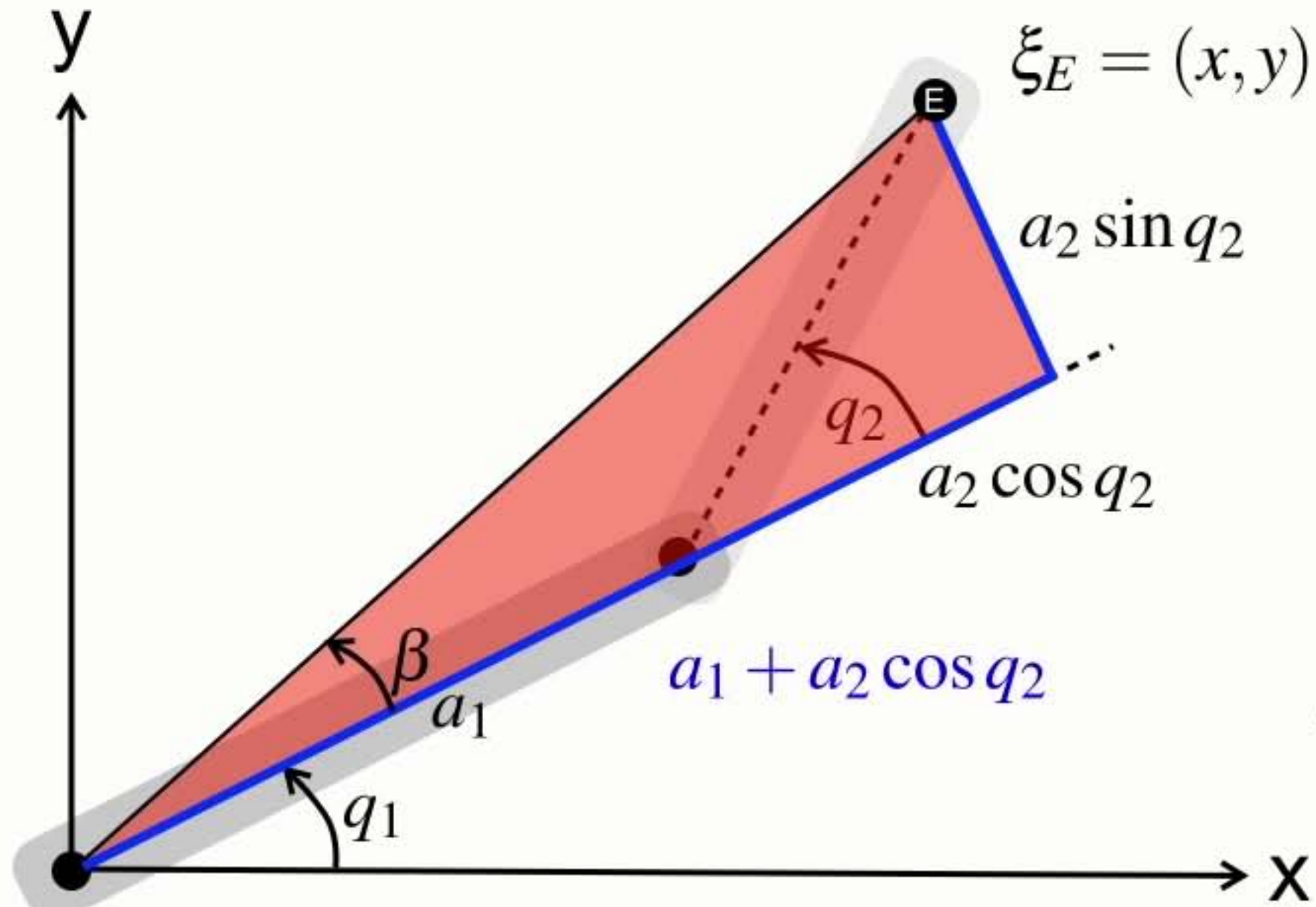
There can be **multiple solutions**



- These are all 2D transforms



Geometric solution



$$\beta = \tan^{-1} \frac{a_2 \sin q_2}{a_1 + a_2 \cos q_2}$$

- Apply simple trigonometry

Analytic **solution**

- create a model of the robots joints and links
- write the forward kinematics
- write down lots of equations and attempt to solve for the unknown joint variables
 - ➡ needs some human insight
 - ➡ can be automated
- explicitly shows the multiple configurations
- solution are compact and fast to execute

- increasingly difficult as the number of joints increases
 - ➡ however many solutions have been derived and published

Numerical solution

- What if the analytic solution
 - ➔ does not exist ?
 - ➔ is just too hard to figure out ?
- We can **always** figure out the forward kinematics $\xi = \mathcal{K}(\boldsymbol{q})$
- We know the pose we want ξ^*
- Just have to adjust $\boldsymbol{q}$ until $\xi \rightarrow \xi^*$
- This can be posed as an **optimization problem**

$$\boldsymbol{q}^* = \arg \min_{\boldsymbol{q}} |\mathcal{K}(\boldsymbol{q}) \ominus \xi^*|$$

Issues with **numerical solution**

- Choosing good initial joint coordinates
- Can't guarantee a particular configuration
- Can be computationally expensive

Smooth joint **interpolated** paths

- We wish to move from pose A to pose B $\xi_A \rightarrow \xi_B$
- We can determine the joint angles at each pose

$$\mathbf{q}_A = \mathcal{K}^{-1}(\xi_A)$$

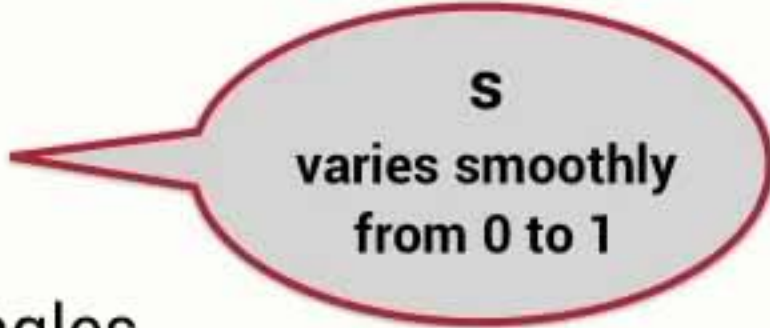
$$\mathbf{q}_B = \mathcal{K}^{-1}(\xi_B)$$

- Now we need to move **smoothly** $\mathbf{q}_A \rightarrow \mathbf{q}_B$ $\mathbf{q}_A, \mathbf{q}_B \in \mathbb{R}^6$
- We've previously discussed how to do this

Smooth Cartesian **interpolated** paths

- We wish to move from pose A to pose B $\xi_A \rightarrow \xi_B$
- We determine the pose at each intermediate point

$$\xi_s = f(\xi_A, \xi_B, s), \quad 0 \leq s \leq 1$$



s
varies smoothly
from 0 to 1

- Then use inverse kinematics to find the joint angles

$$q_s = \mathcal{K}^{-1}(\xi_s)$$

- At every time step we
 - ➡ interpolate the pose
 - ➡ perform inverse kinematics